Accident Reconstruction, the Autonomous Vehicle and Advanced Driver Assistance Systems

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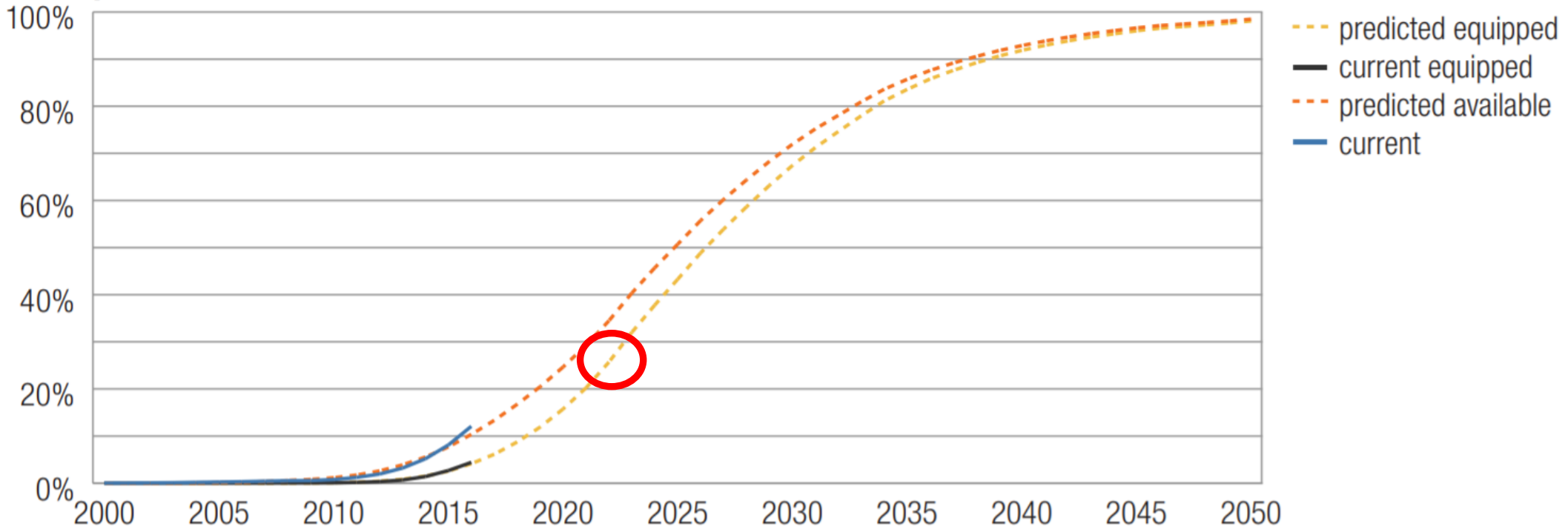
- Mr. Moore is a mechanical engineer and principal of A.B.Moore Forensic Engineering. He specializes in vehicle accident reconstruction, vehicle design analysis, and mechanical engineering consulting.

- Mr. Moore holds a Bachelor of Science in Mechanical Engineering from Michigan State University and a Masters degree in Business Administration from the University of Florida. He is a licensed Professional Engineer, a Board Certified Forensic Engineer, and an ACTAR-certified accident reconstructionist. His past experience includes two decades of accident reconstruction and automotive engineering. He previously worked at Ford Motor Company as a design engineer on the Ford Expedition and Excursion. Mr. Moore also serves as a high-performance driving coach for aspiring race car drivers through the Porsche Club of America.

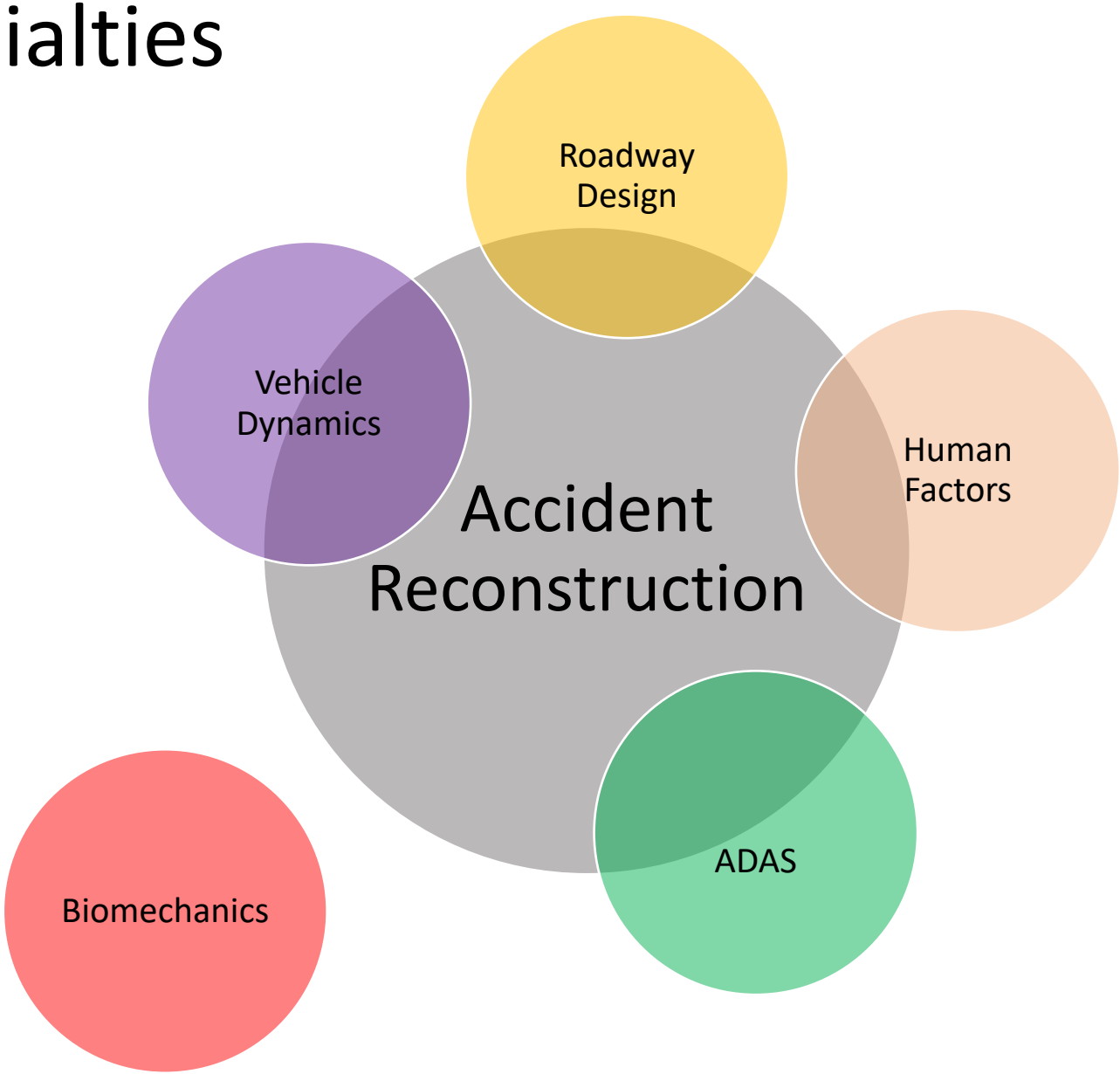
The Future of ADAS and Accident Reconstruction

Figure 6: Predicted percentage of registered vehicles with front crash prevention

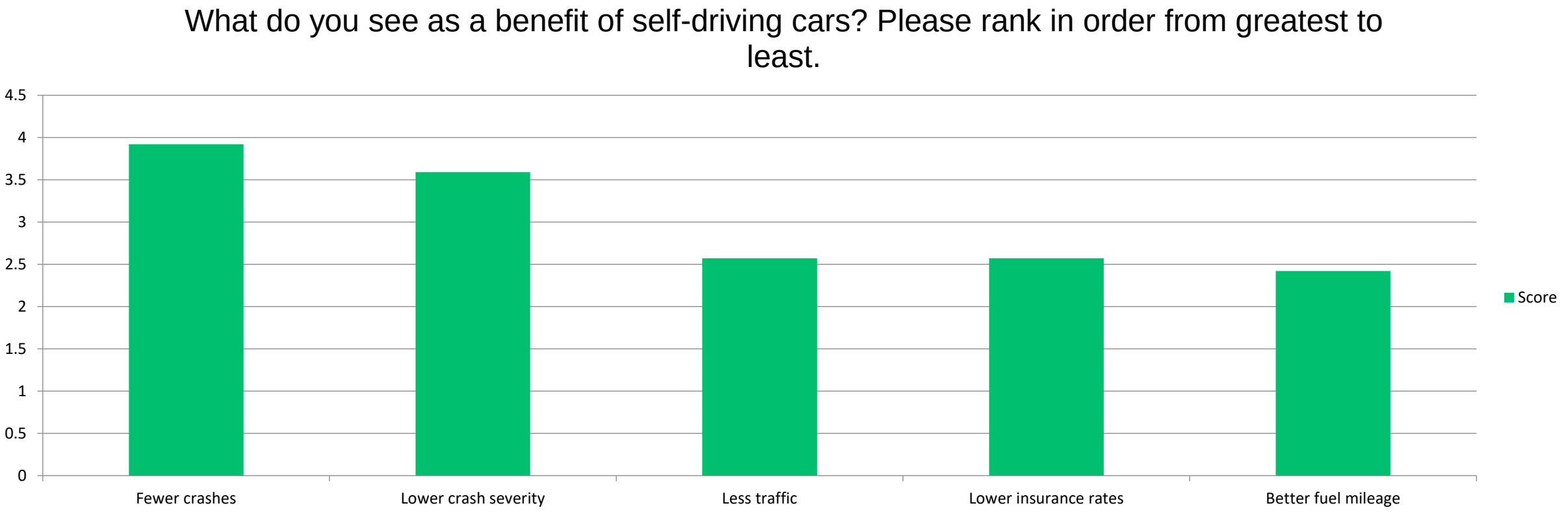


Predicted availability and fitment of safety features on registered vehicles, IIHS/HLDI, 2018

Overlapping specialties

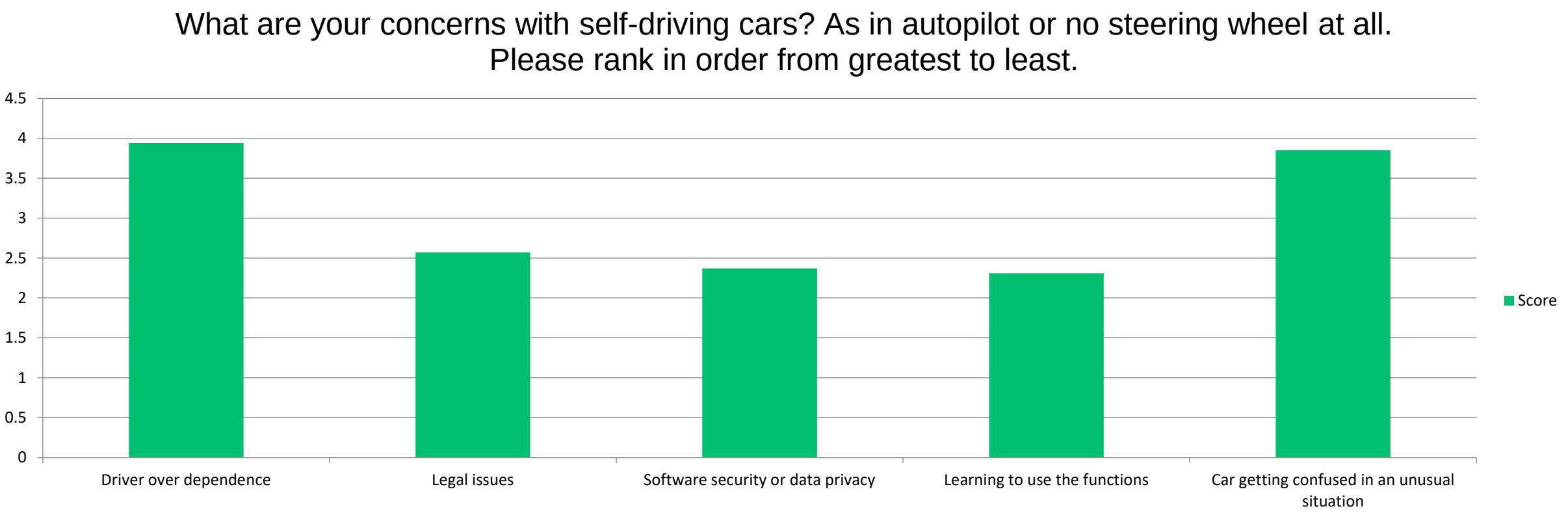


2020 INCR Informal Survey Results



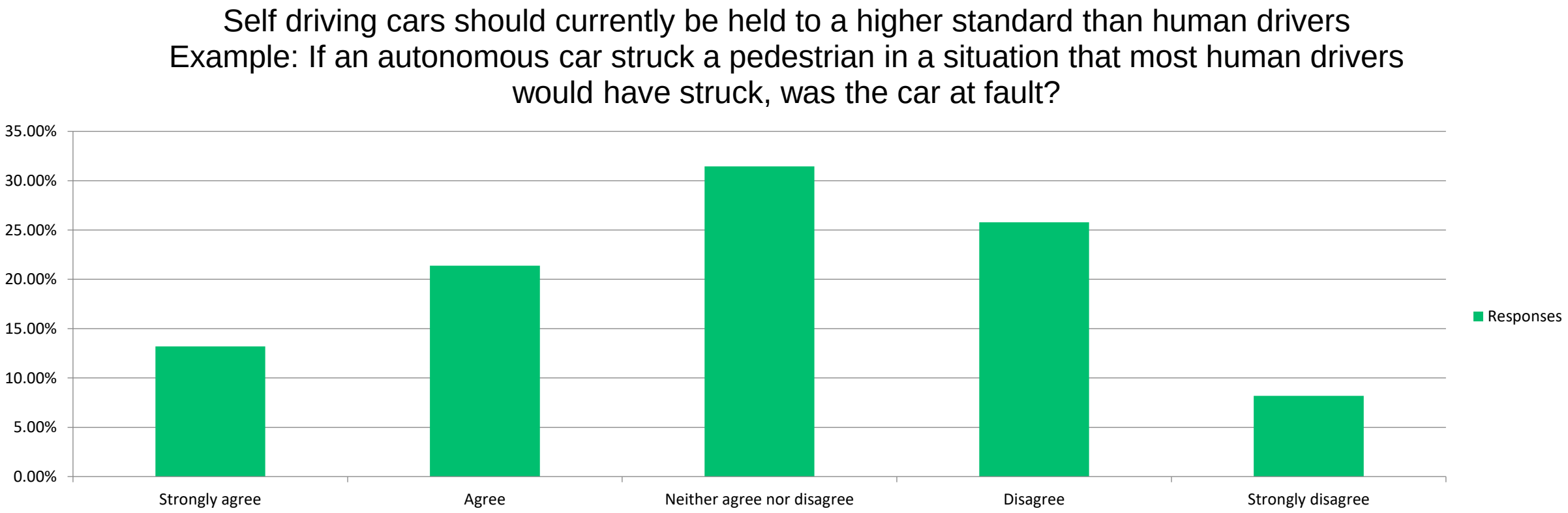
Survey of INCR Mailing List Members, Gary Johnson, VEAR, 2020

2020 INCR Informal Survey Results



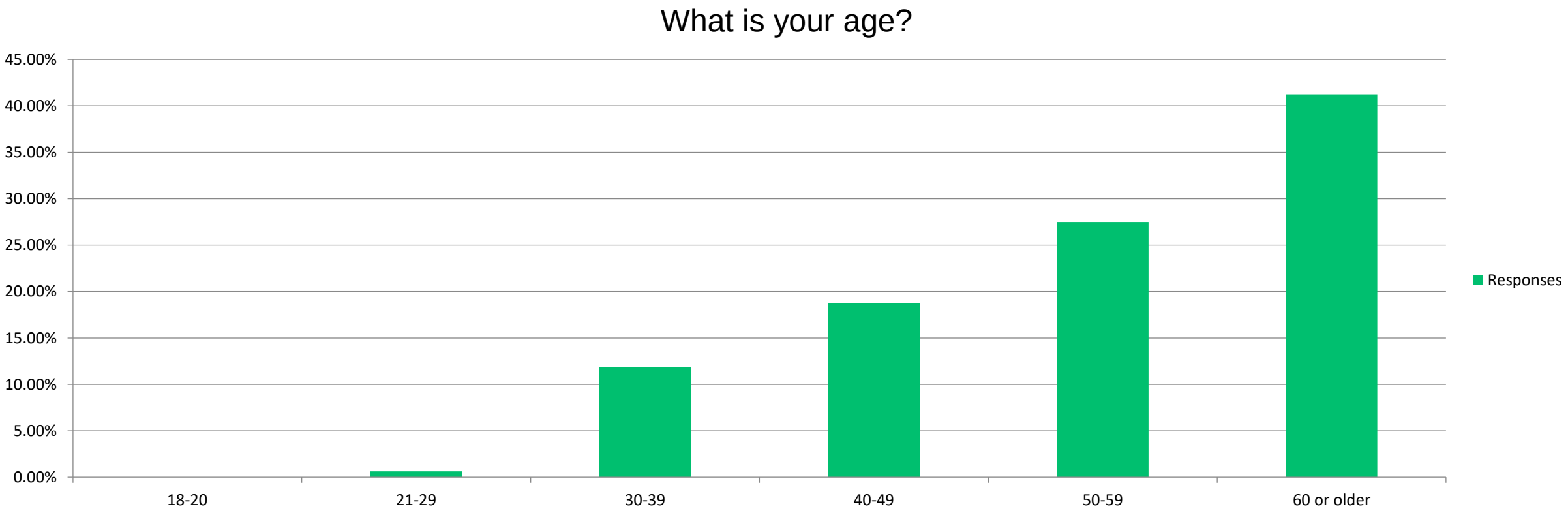
Survey of INCR Mailing List Members, Gary Johnson, VEAR, 2020

2020 INCR Informal Survey Results



Survey of INCR Mailing List Members, Gary Johnson, VEAR, 2020

2020 INCR Informal Survey Results



Survey of INCR Mailing List Members, Gary Johnson, VEAR, 2020

Overview of the available technology

By the end of this module you will be able to:

- Describe what features comprise ADAS
- Describe the safety implications of the technologies

Overview of the available technology

- Adaptive cruise control (ACC)
- Forward Collision Warning (FCW)
- Automatic Emergency Braking (AEB) with Brake Assist
- Blind spot monitoring and rear cross traffic alert
- Forward Cross-Traffic Alert
- Post-Impact Automatic Braking (PIB)

Overview of the available technology

- Lane departure warning and lane keep assist (LDW, LKA)
- Lane centering
- Traffic signal awareness
- SAE Level 4 “Full Self Driving”

Overview of the available technology

- **Automakers commit to voluntary adoption of AEB by 2022.**
- **Details of the commitment**: Participating automakers commit to make AEB standard on virtually all light-duty cars and trucks with a gross vehicle weight of 8,500 pounds or less no later than Sept. 1, 2022, and on virtually all trucks with a gross vehicle weight between 8,501 pounds and 10,000 pounds no later than Sept. 1, 2025.

NHTSA/IIHS press release, December 2017
https://www.nhtsa.gov/press-releases/nhtsa-iihs-announcement-aeb

Effectiveness of FCW and AEB

- FCW alone, low-speed AEB, and FCW with AEB *reduced* rear-end striking crash involvement rates by 27%, 43%, and 50%, respectively[1]
- FCW with AEB *increased* rates of rear-end struck crash involvements by 20%[1]
- Model years 2010–2014

[1]Effectiveness of Forward Collision Warning and autonomous Emergency Braking Systems in reducing front-to-rear crash rates, Cicchino, Jessica B., Accident Analysis and Prevention February 2017

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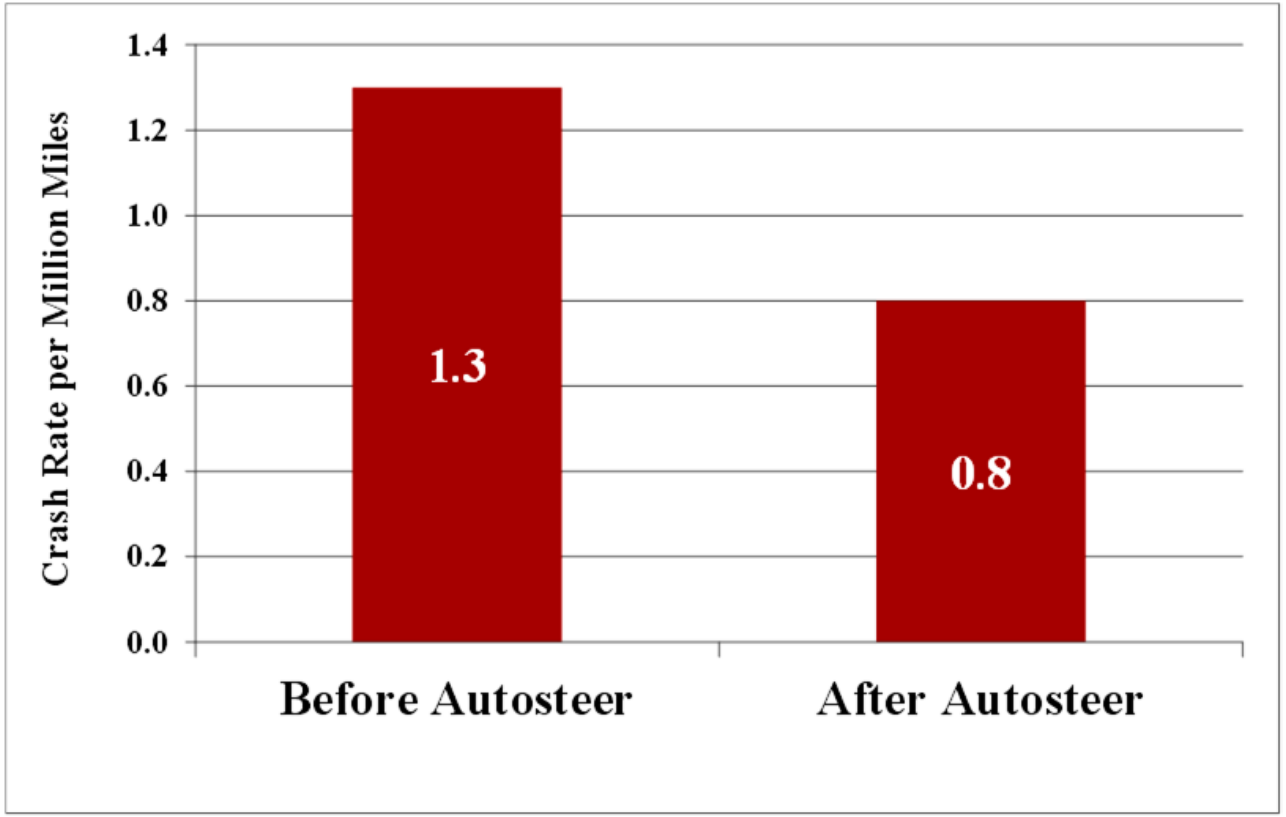
Overview of the available technology



Tesla crash rates dropped 40% after introduction of Autopilot

[1]NHTSA Office of Defects Investigation #PE 16-007, 2017

Overview of the available technology

- Q3 2018, Tesla vehicles:
  - 1 crash/near-crash per 3.34 million miles on Autopilot
  - 1 crash/near-crash per 1.92 million miles without Autopilot
- 1 crash per 0.49 million miles all makes, per NHTSA

million miles between crashes	# crashes in million miles	
3.34	0.30	AP
1.92	0.52	no AP
	43%	% AP improvement

Q3 2018 Vehicle Safety Report, October 4, 2018, Tesla https://www.tesla.com/blog/q3-2018-vehicle-safety-report

Effectiveness of Lane Departure Warning

Inadvertent lane departure represented 5% of all car accidents but almost 10% of fatalities
Causes in descending order of frequency:

- Physical problems and health problems such as overfatigue and faintness or loss of consciousness (66%); average age was 67.
- Distraction/inattentiveness, e.g. caused by adjusting the navigation system or radio or engaging in some other activity in the car (26%)
- Alcohol/drugs (6%)
- Weather conditions, such as heavy rain or fog (2%).

"For the future there is a need to consider systems… that bring the vehicle to a standstill safely when the driver is no longer in a fit state to drive."

Study limitations:
German-insured vehicles only
Excluded single-vehicle accidents

Analysis of Car Accidents Caused by Unintentional Run-Off Road, Matthias Kuehn, Thomas Hummel, Jenö Bende - German Insurers Accident Research, Paper Number 15-0245

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Effectiveness of Lane Departure Warning

Table 1: Multiple Imputation Parameter Estimates *‡		
Variables	**Estimate**	**Odds Ratio**
Driver performance-related factor: sleepy vs. others	1.1675 ‡	3.21
Driver with alcohol : BAC=.01+ g/dL vs. BAC=0	0.6605 ‡	1.94
Road alignment: curve vs. straight	0.5535 ‡	1.74
Vehicle speeding status: speeding vs. non-speeding	0.4966 ‡	1.64
Vehicle type: passenger car vs. other LTs	0.4780 ‡	1.61
Road function: rural vs. urban	0.3637 ‡	1.44
Roadway # of lanes: one or two (undivided) vs. three & more (undivided)	0.3346 ‡	1.41
Road speed limit: ≥ 60 mph vs. <60 mph	0.2659 ‡	1.30
Weather: adverse vs. good	0.2130 ‡	1.24
Driver performance-related factor: avoiding vs. others	0.2007 ‡	1.22
Weather ×Speeding status: adverse & speeding vs. good & non-speeding	0.1056 ‡	1.11
Driver Age × Speeding status: 25-44 & speeding vs. 65+ & non-speeding	0.1003 ‡	1.11
Driver Age × Speeding status: 15-24 & speeding vs. 65+ & non-speeding	0.0578 ‡	1.06

Study limitations: Study preceded texting (1991-2007) and lane departure systems, Single-vehicle crashes only

Factors Related to Fatal Single-Vehicle Run-Off-Road Crashes, Cejun Liu, Ph.D. and Rajesh Subramanian , DOT HS 811 232, November 2009

Overview of the available technology

Summary:

- ADAS is compromised of a variety of driver assistance systems
- ADAS provides safety improvements, but data is sparse due to recent adoption

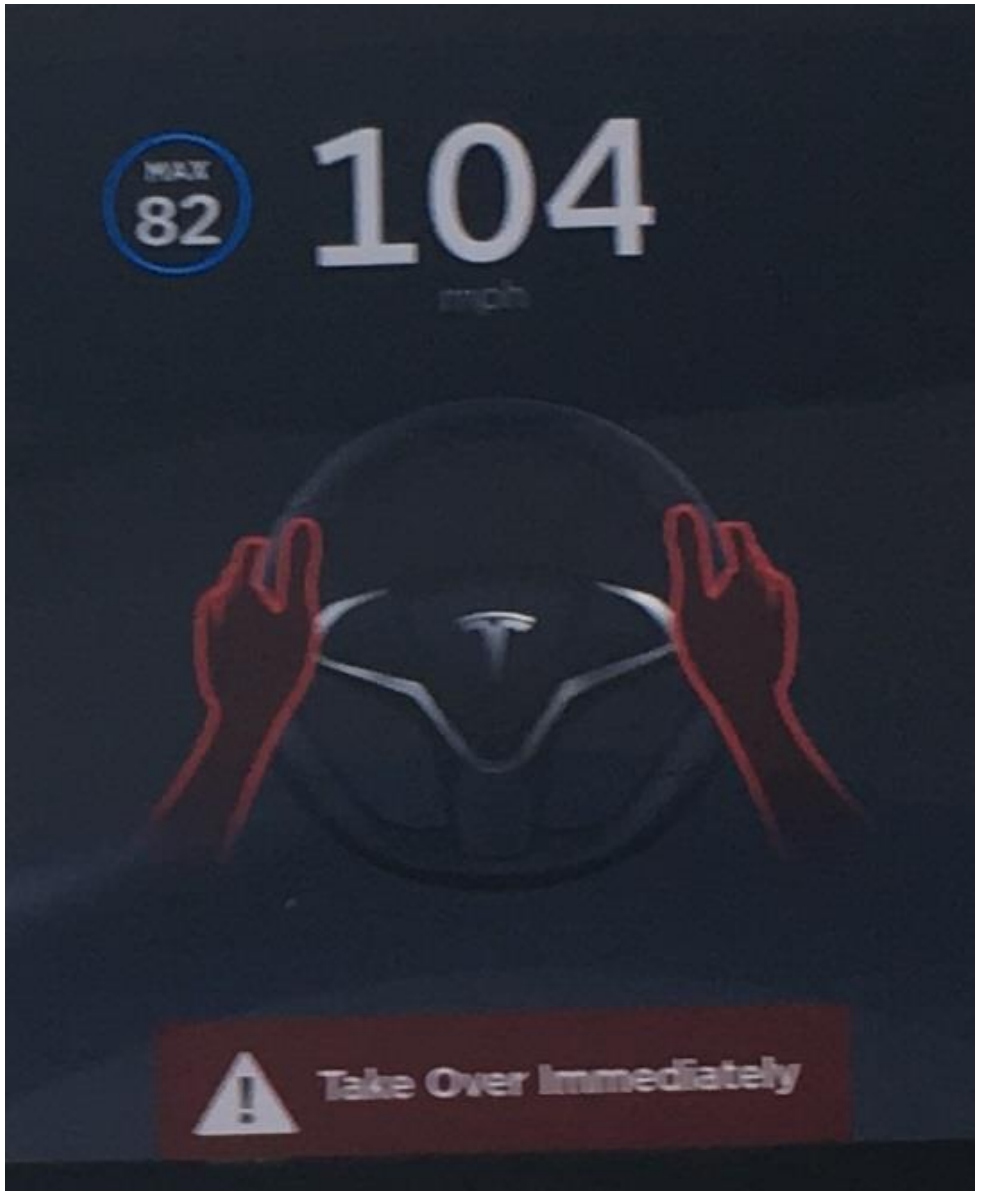
Levels of Self-Driving Vehicles

By the end of this module you will be able to:

- Describe the levels of autonomy
- Evaluate fallback performance

SAE Levels of Automation

- Level 1: Cruise Control
- Levels 2-3: Adaptive cruise control and lane centering
- Levels 4-5: Hands-off operation
- The difference is the fallback performance; how much time is the driver given to respond?



Tesla Fallback performance

Slow to a safe stop under Level 4 automation

Avoiding this "handoff problem" is part of the reason why Waymo is working on *fully* self-driving vehicles. Our technology takes care of all of the driving, allowing passengers to stay passengers.

Waymo Fallback performance

SAE level	Name	Narrative Definition	Execution of Steering and Acceleration/ Deceleration	*Monitoring* of Driving Environment	Fallback Performance of *Dynamic Driving Task*	System Capability *(Driving Modes)*
***Human driver* monitors the driving environment**						
0	**No Automation**	the full-time performance by the *human driver* of all aspects of the *dynamic driving task*, even when enhanced by warning or intervention systems	Human driver	Human driver	Human driver	n/a
1	**Driver Assistance**	the *driving mode*-specific execution by a driver assistance system of either steering or acceleration/deceleration using information about the driving environment and with the expectation that the *human driver* perform all remaining aspects of the *dynamic driving task*	Human driver and system	Human driver	Human driver	Some driving modes
2	**Partial Automation**	the *driving mode*-specific execution by one or more driver assistance systems of both steering and acceleration/ deceleration using information about the driving environment and with the expectation that the *human driver* perform all remaining aspects of the *dynamic driving task*	**System**	Human driver	Human driver	Some driving modes
***Automated driving system* ("system") monitors the driving environment**						
3	**Conditional Automation**	the *driving mode*-specific performance by an *automated driving system* of all aspects of the dynamic driving task with the expectation that the *human driver* will respond appropriately to a *request to intervene*	System	**System**	Human driver	Some driving modes
4	**High Automation**	the *driving mode*-specific performance by an automated driving system of all aspects of the *dynamic driving task*, even if a *human driver* does not respond appropriately to a *request to intervene*	System	System	**System**	Some driving modes
5	**Full Automation**	the full-time performance by an *automated driving system* of all aspects of the *dynamic driving task* under all roadway and environmental conditions that can be managed by a *human driver*	System	System	System	**All driving modes**

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SAE J3016 Levels of Automation

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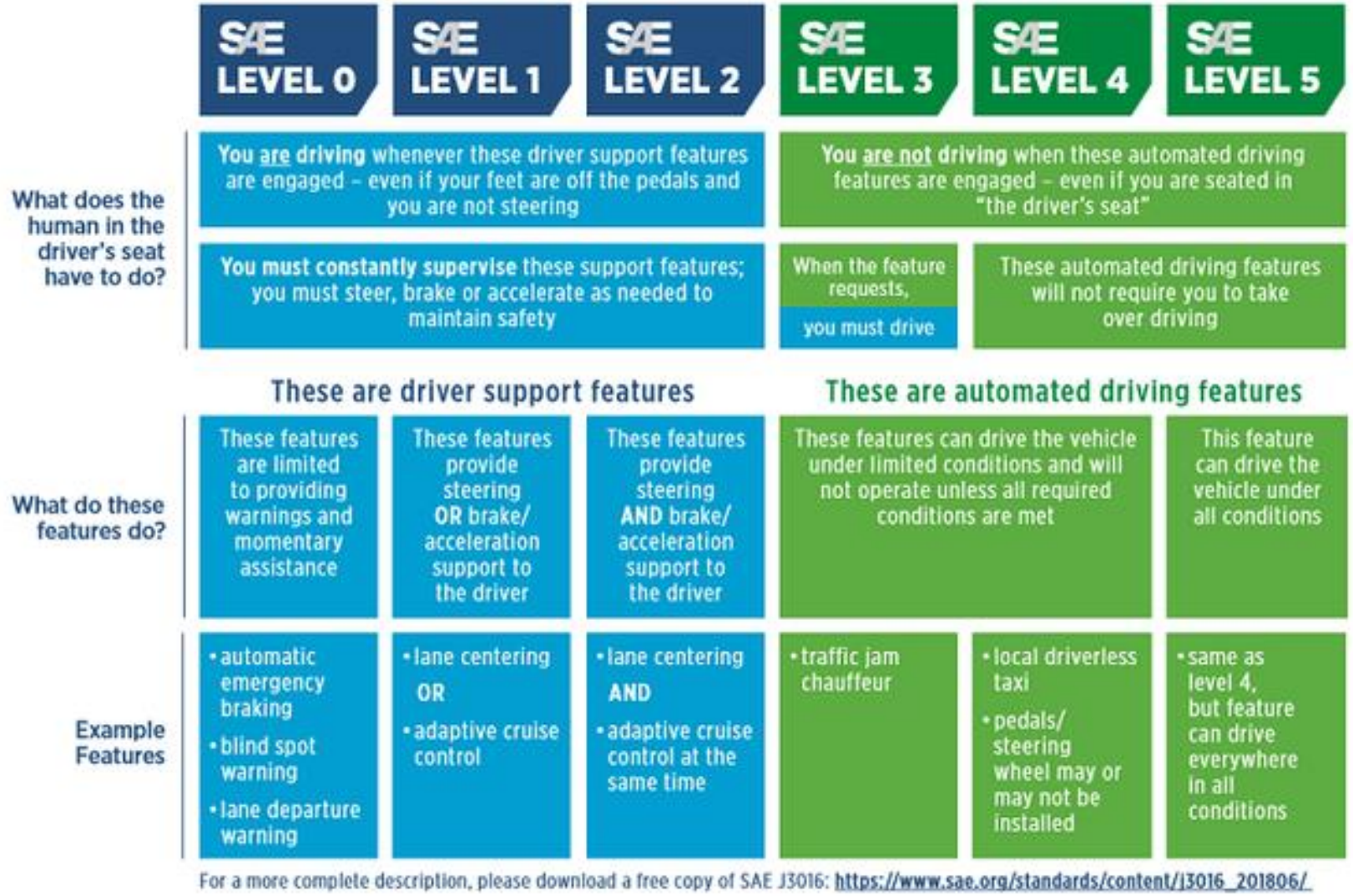
SAE J3016™ LEVELS OF DRIVING AUTOMATION



For a more complete description, please download a free copy of SAE J3016: https://www.sae.org/standards/content/j3016_201806/

SAE J3016
Levels
of Automation

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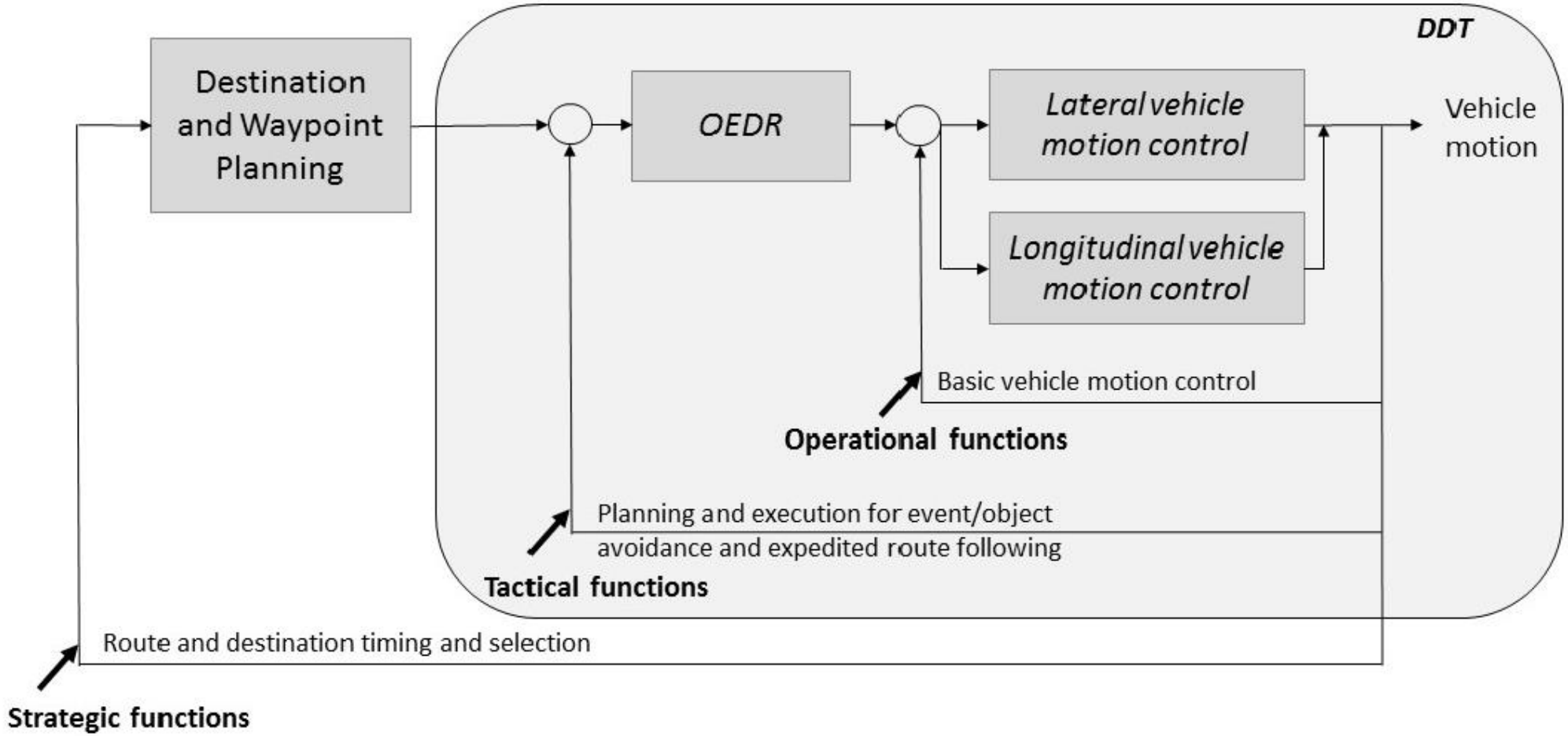
Updates to SAE J3016, 2018

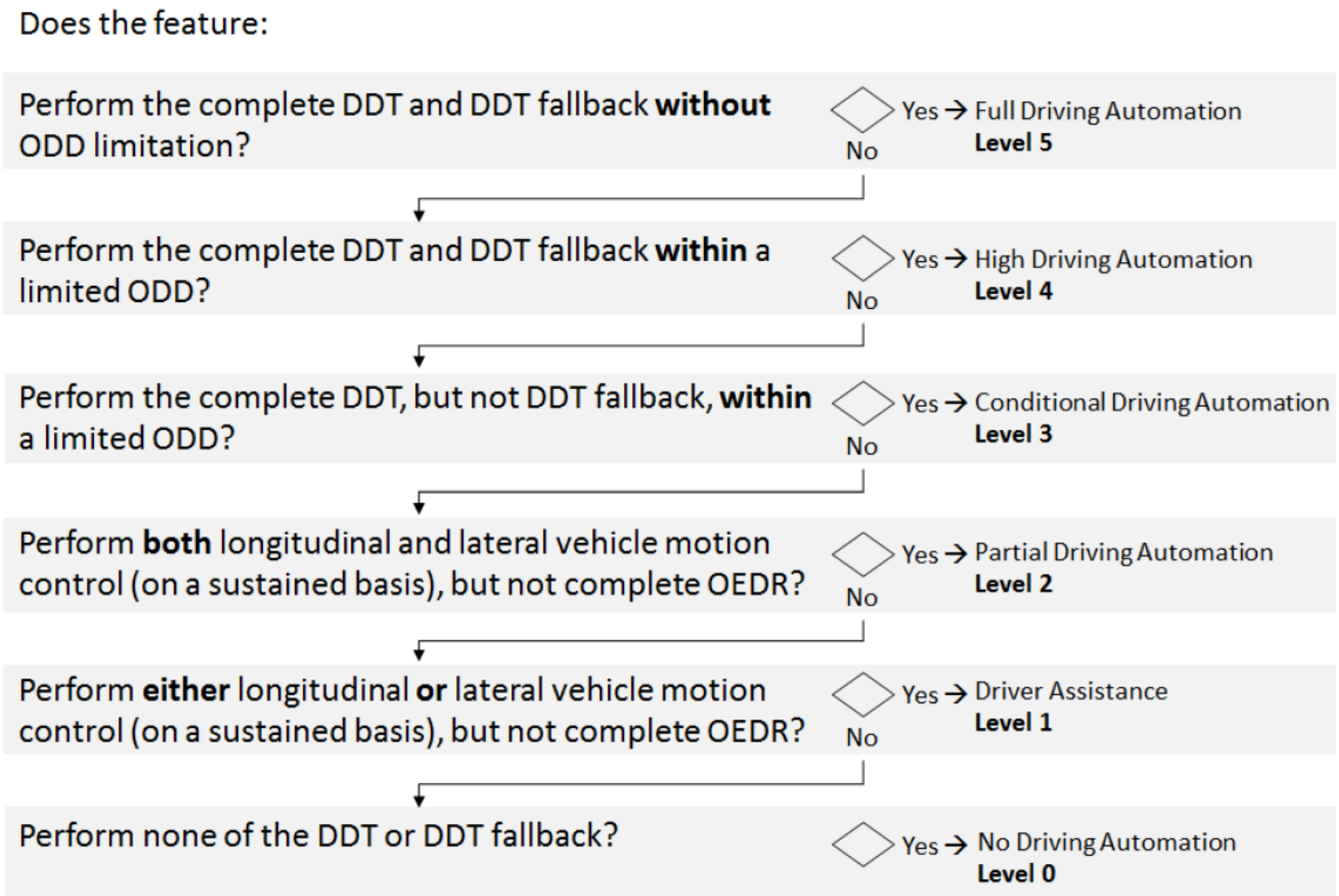
- The level of driving automation exhibited in any given instance is determined by the feature(s) that are engaged.

- Active safety systems, such as AEB and LKA, are **not** considered driving automation because they do not perform on a sustained basis, but rather only in momentary interventions.

Updates to SAE J3016, 2018



Updates to SAE J3016, 2018



Safety Elements of self-driving vehicles[1]

[1]NHTSA's voluntary guidance, Automated Driving Systems 2.0 – A Vision for Safety

Levels of Self-Driving Vehicles

Summary:

- The driver must be vigilant if needed as the fallback
- Levels 2-3 require driver involvement
- Levels 4-5 do not require driver attention

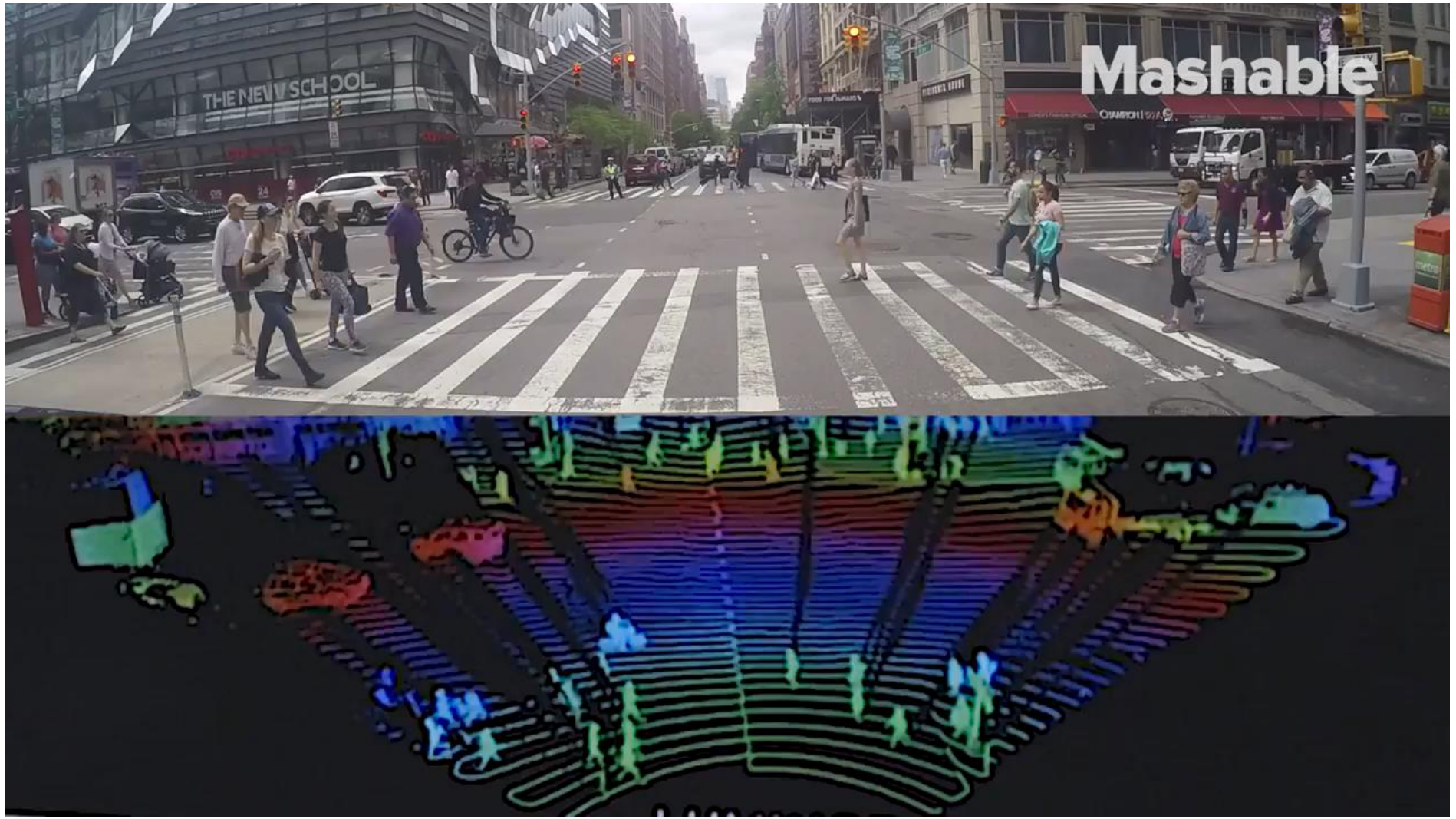
The enabling technologies

- By the end of this module you will be able to describe the technologies that enabled ADAS:
  - Sensors
  - Actuators
  - Computing

Enabling Technologies: Sensors

- Cameras (including eye-tracking)
- GPS
- LIDAR*
- Radar*
- V2V/V2I*

LIDAR



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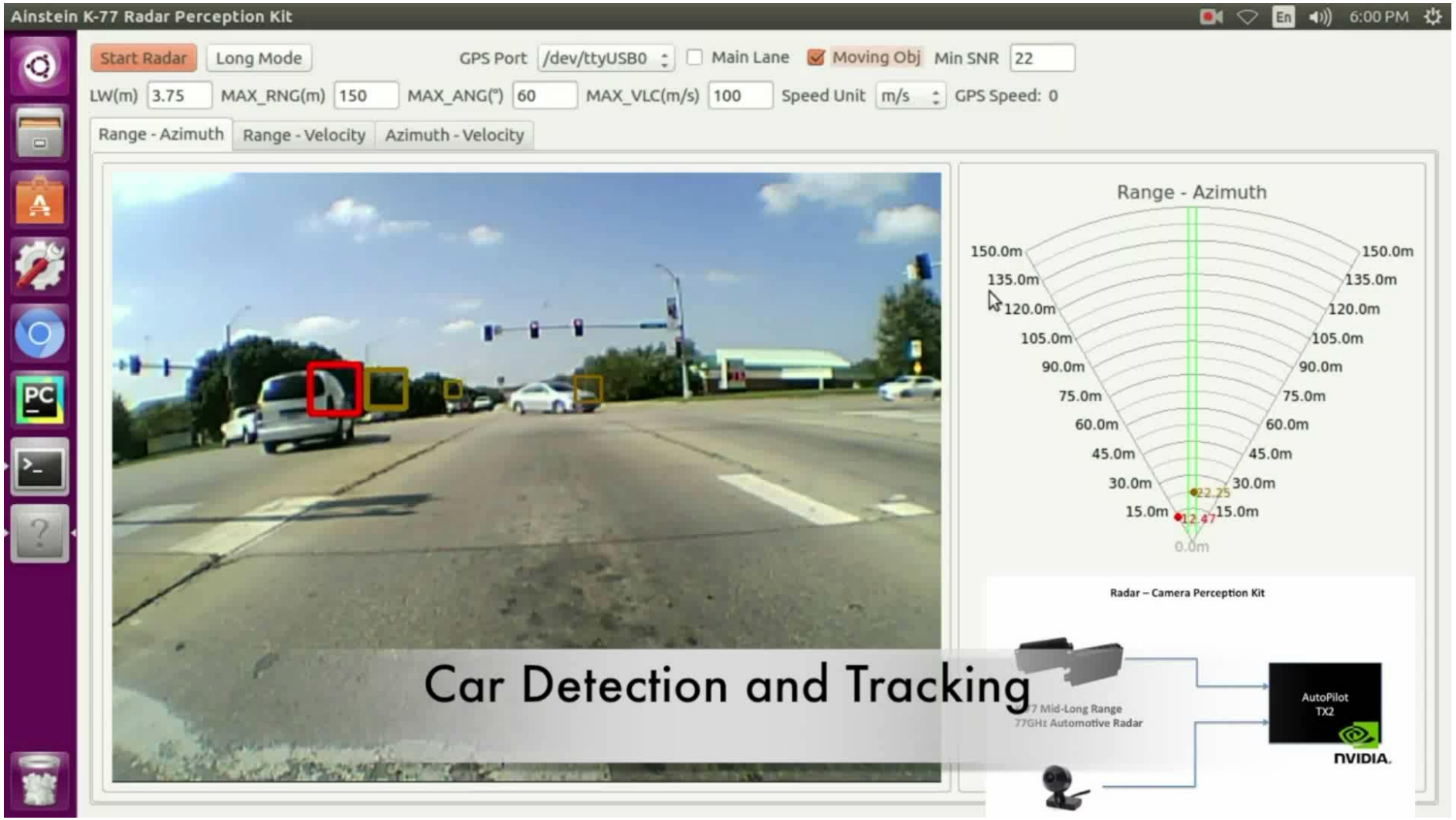
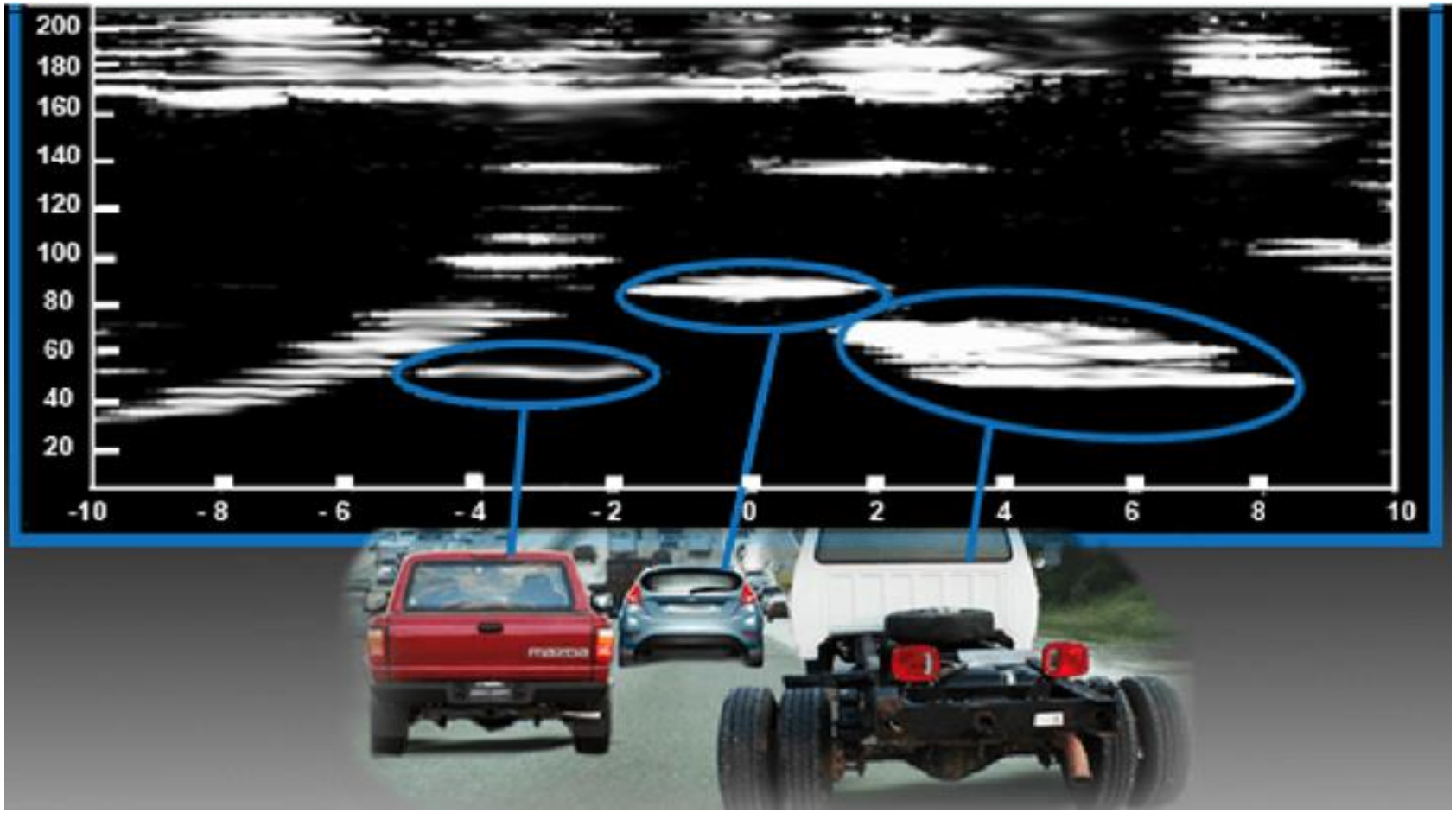
Radar – Ainstein Video



Radar Visualization



Collision Avoidance and Driver Support [34S34W1], Fordservicetraining.com, 2019

Example Radar Specifications – Delphi ESR

- Maximum range – 175m w/ 10 dB target
- Range accuracy – "< +/-0.5m noise component with +/- 5% bias component"
  - At 50m, range accuracy calculates to 3m
  - Azimuth (lateral) field of view
  - >+/-20 deg long range
  - >+/-90 deg medium range
- Range Rate (Speed) Accuracy – +/- 0.12 m/s
- Azimuth angle centroid accuracy +/- 1deg medium range (1.7m at 100m range)
- Number of targets tracked – 64

Delphi ESR Startup Guide Version 1.11, 26 March 2015

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Example Radar Specifications – Delphi ESR

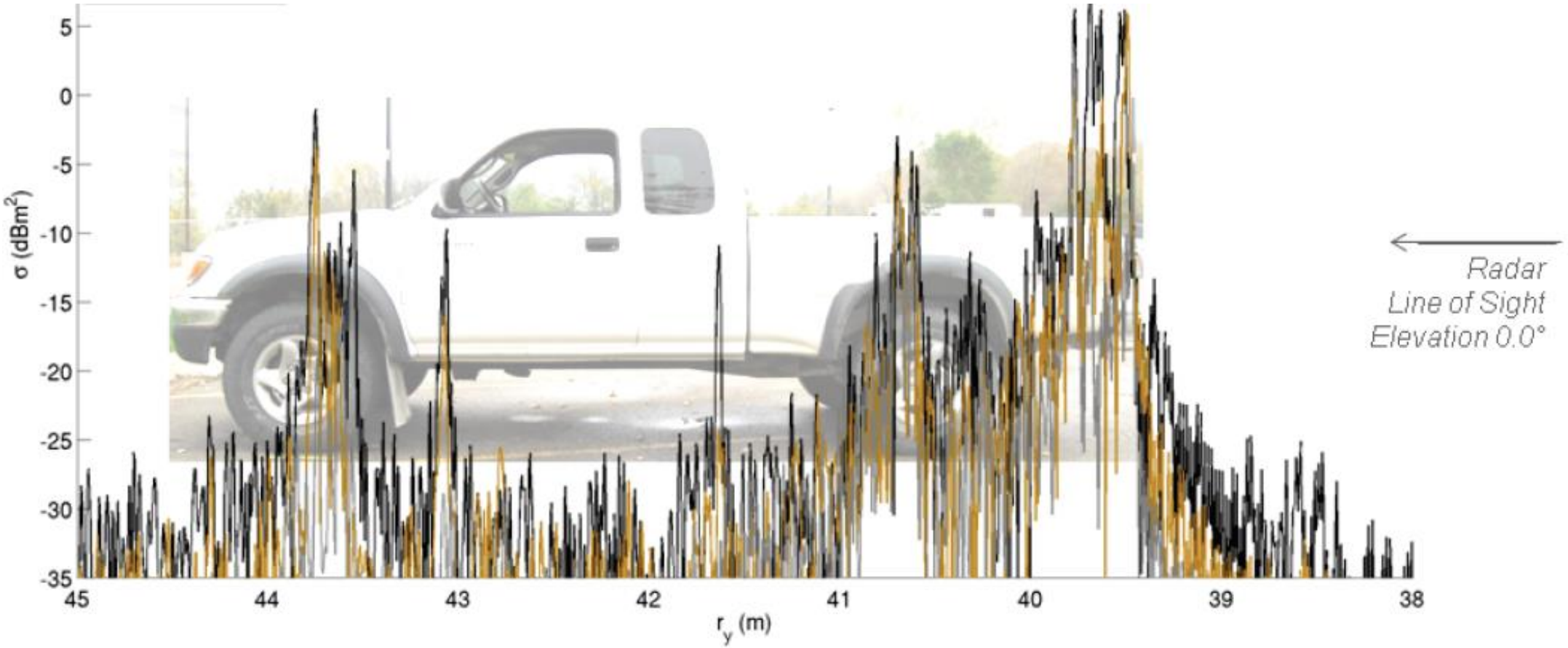
- Response time
  - Track confirmation within 150ms for an object entering detection zone
  - Track drop within 250ms after an object leaves detection zone
  - In-path classification of previously detected target that enters path of host vehicle within 100ms
- Movement Classification (including the sub-classifications of Oncoming and *Crossing*):
  - Stationary and never moving
  - Stationary and previously moving
- Bridge Classification
  - The probability of rejecting overhead bridges (by correct classification as a bridge) shall be 95% at 80m range increasing to 100% at 45m range. Minimum bridge height assumed to be 3.65m

Delphi ESR Startup Guide Version 1.11, 26 March 2015

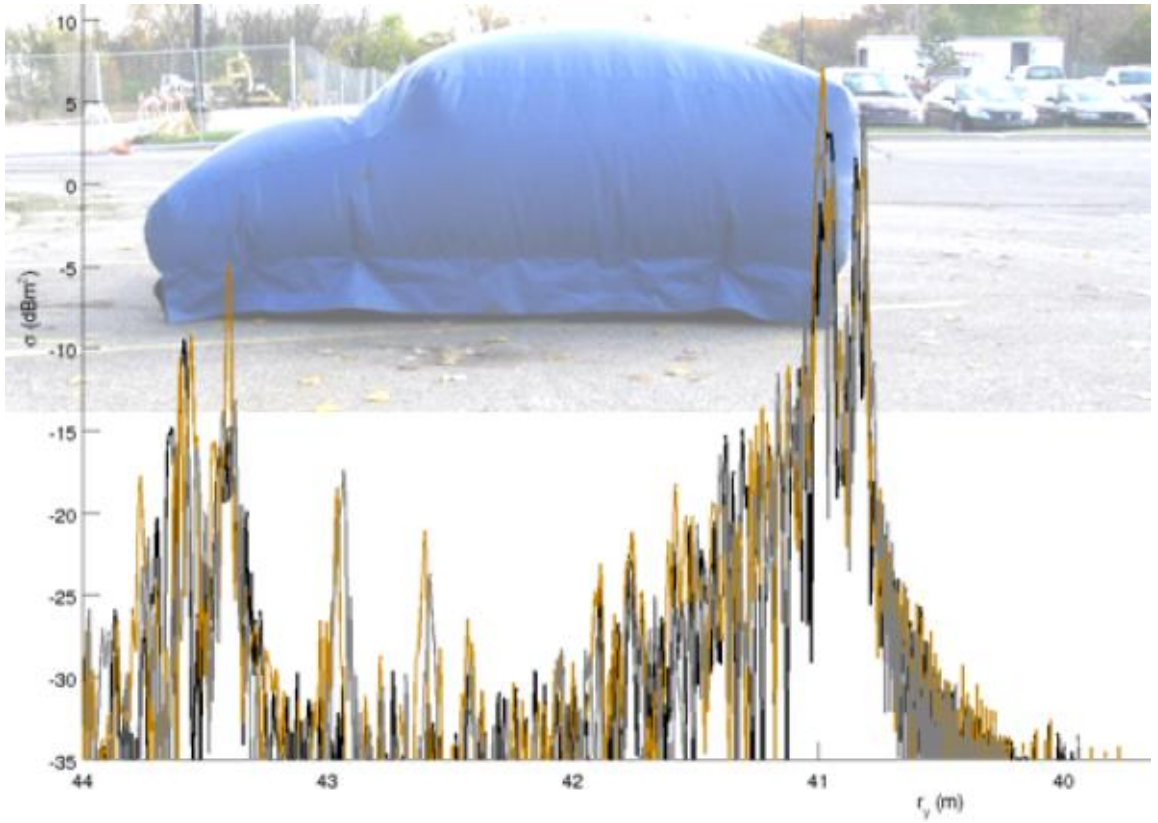
Test Results – Delphi ESR Radar

- Max range – >180m with RMS error 1.8m and lateral RMS error 0.4m
- Detection unreliable at angles beyond 40 deg
- Min range 2-3m
- Range error at 25m <.35m with tree, pole, or human
- Azimuth error at 25m <1deg with tree, pole, or human
- Lateral separation needed to separately identify 2 humans ~10deg at 15m

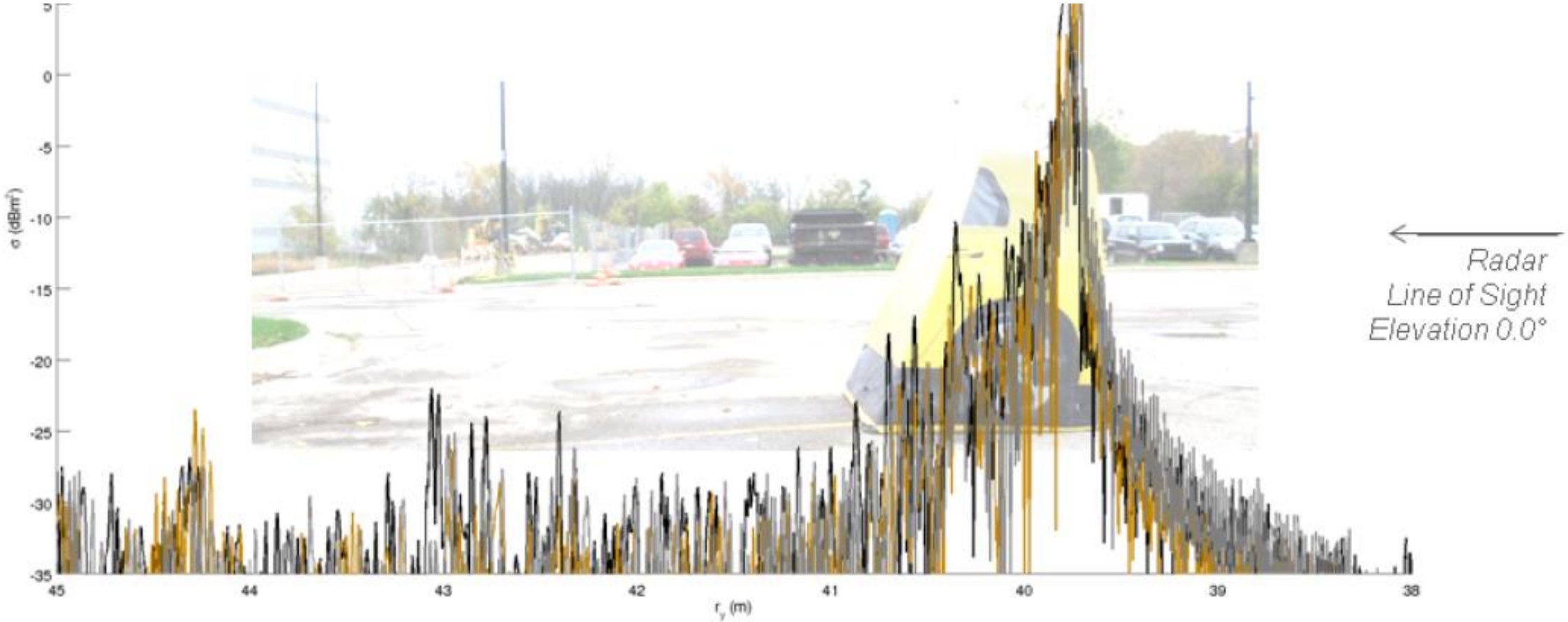
Characterisation of the Delphi Electronically Scanning Radar for Robotics Applications, Leo Stanislas and Thierry Peynot, Queensland University of Technology and Polytech Paris-UPMC.


5
0
-5
-10
-15
-20
-25
-30
-35
σ (dBm²)
45
44
43
42
41
40
39
38
r_y (m)
Radar
Line of Sight
Elevation 0.0°
Range increases Right to Left
Length =5.15 m

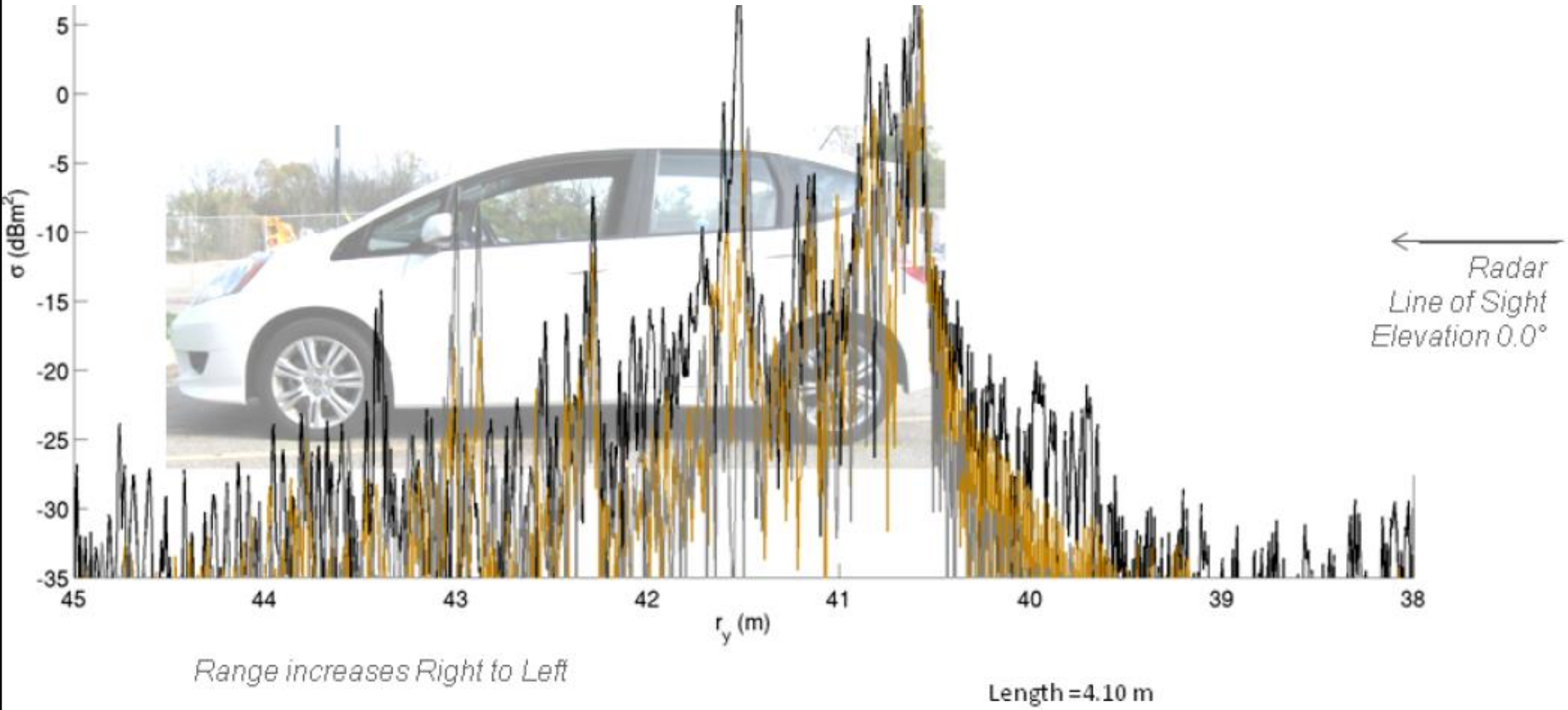
Radar – 2D vs 3D reflection



Radar Measurements of NHTSA's Surrogate Vehicle SS-V, William Buller et al, 2012



Range increases Right to Left



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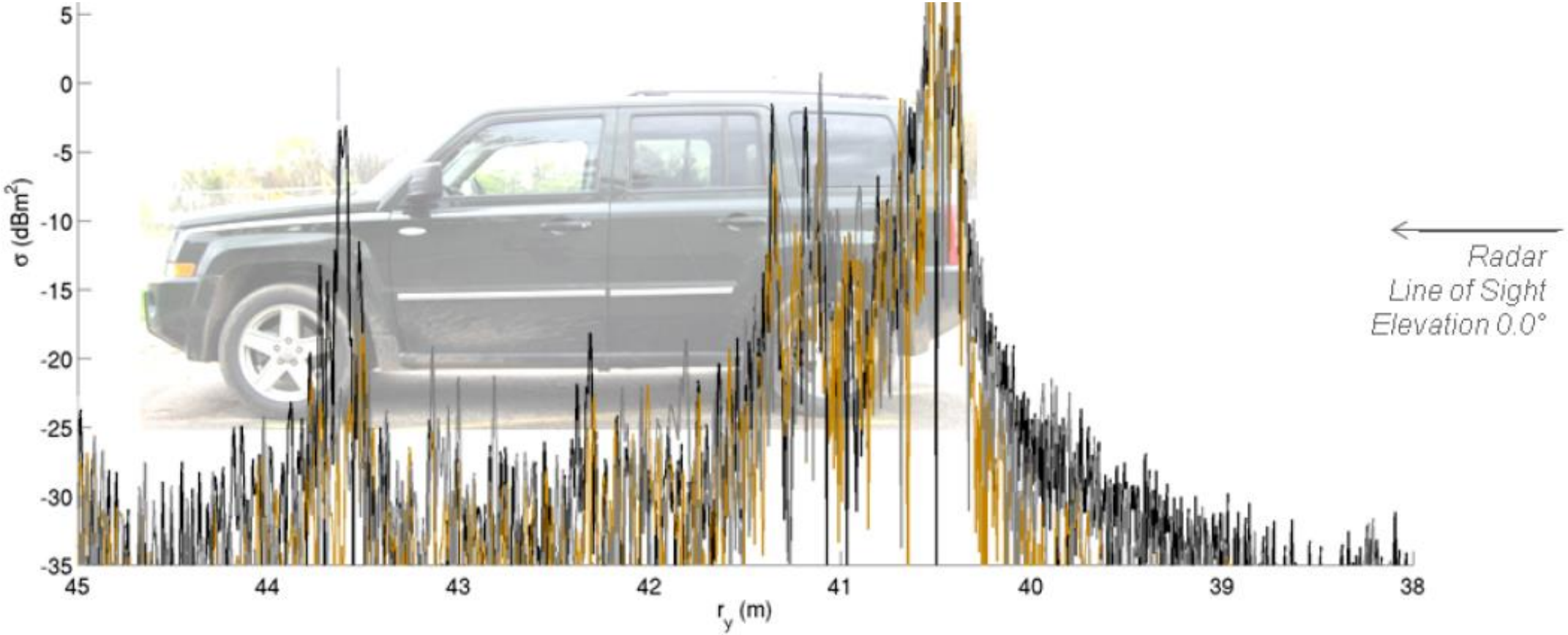
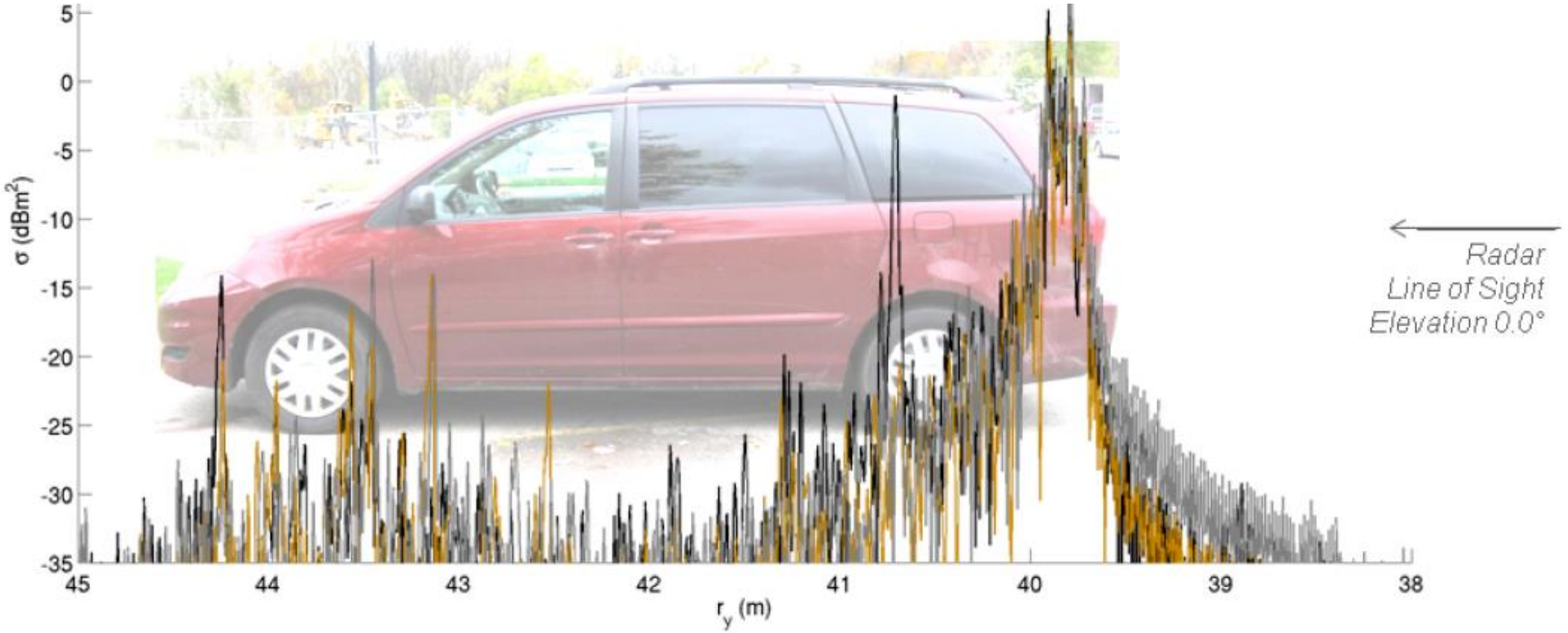
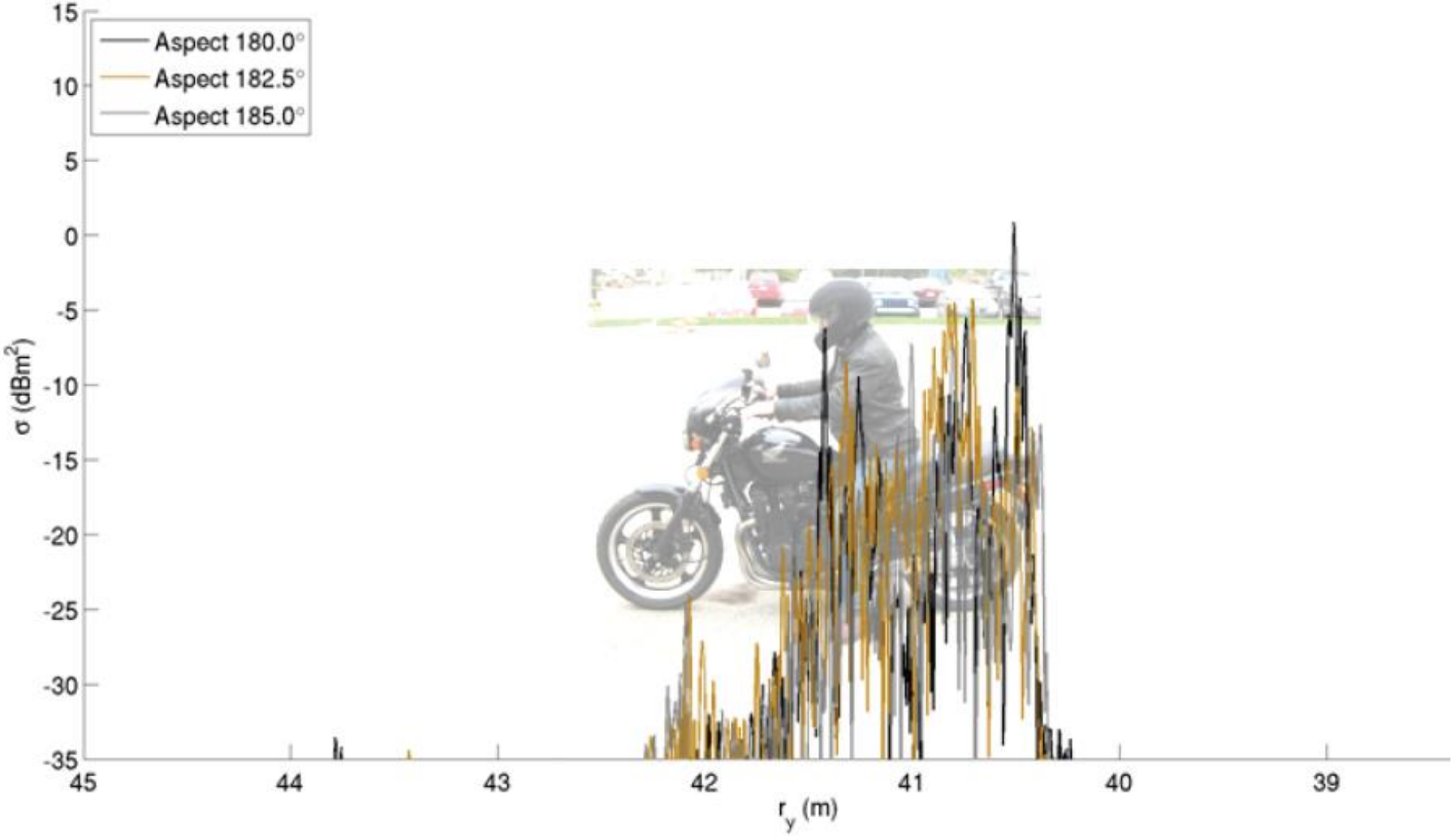

5
0
-5
-10
-15
-20
-25
-30
-35
σ (dBm²)
45
44
43
42
41
40
39
38
r_y (m)
Radar
Line of Sight
Elevation 0.0°
Range increases Right to Left
Length = 4.41 m


5
0
-5
-10
-15
-20
-25
-30
-35
σ (dBm²)
45
44
43
42
41
40
39
38
r_y (m)
Radar
Line of Sight
Elevation 0.0°
Range increases Right to Left
Length =5.10 m

car18: Elevation 0.0°
HondaNighthawk 7501993



Range increases Right to Left

Length =2.21m

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V2X Communication

Vehicle-to-Vehicle (V2V)

- Platooning
- Advance warning of traffic

Communication Methods:

- DSRC (Dedicated Short Range Communication)
- Cellular (LTE and upcoming 5G)

Vehicle-to-Infrastructure (V2I)

- Traffic alerts
- First responders
- Predictive signal timing

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Honda DSRC Demo

Honda Demonstrates New "Smart Intersection" Technology that Enables Vehicles to Virtually See Through and Around Buildings, Honda News, Oct 04, 2018

NHTSA predicted with DSRC a 30% reduction in crash reports, using a series of large assumptions:

•All light vehicles are fully integrated with V2V technology and corresponding safety applications.
•All vehicles (i.e., light vehicles, medium and heavy trucks, busses, and motorcycles) are fully equipped with V2V technology and can communicate with fully integrated light vehicles.
• V2V safety applications are always available to perform their safety functions.
• V2V safety applications have 100 percent accuracy; i.e., identification of a true driving conflict and accurate warning activation.
• Simple driving conflicts are modeled using basic kinematic equations, where the host vehicle only responds to the conflict while the other vehicle stays the course.
• No external conflicts or unintended consequences are modeled, such control loss or secondary impacts.
• Crash impacts are inelastic; i.e., center-of-mass collisions, including conservation of momentum, without incorporating any additional resulting impacts.
• The “host” vehicle is the light vehicle, the host vehicle initiates the applicable maneuver.
• Driver of the host vehicle responds to a driving conflict (or warning) with a single appropriate response (brake or steer) that depends on the nature of the conflict. The host driver is not impaired.
• All motion and reaction occur without intermittent delays or interference, and are constant until otherwise acted upon.

Status of the Dedicated Short-Range Communications Technology and Applications, FHWA-JPO-15-218, USDOT, July 2015, quoted from:

http://www.nhtsa.gov/DOT/NHTSA/NVS/Crash%20Avoidance/Technical%20Publications/2010/811381.pdf

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SAE J2735 DSRC Message list, partial

HIGH PRIORITY	MEDIUM PRIORITY	NOT IN BSM
Timestamp	**SteeringWheel Angle**	*Fuel type*
Position (lat/long/elev)	**Positional Accuracy**	*Fuel consumption*
Speed and heading	ABS, Traction status	*Emissions*
Acceleration	Stability control	*Fuel level*
Brake system status	Differential GPS	*Road grade*
Vehicle size	Lights status	*Engine drive cycle*
Recent braking	Wiper status	*Operating mode*
Path prediction	Brake level	*Engine temperature*
Throttle position	Coefficient of friction	
Vehicle mass	Rain type	
Trailer weight	Air temperature	
Vehicle type	Air pressure	
Vehicle description	Vehicle identification	
	Cargo weight	
	GPS status	

Vehicle Based Data and Availability - Brian Cronin, Intelligent Transportation Systems and USDOT
https://www.its.dot.gov/itspac/october2012/PDF/data_availability.pdf

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SAE J2735 DSRC Message list, partial

- Emergency Vehicle Alert
- Intersection Collision Avoidance – “warning of a potential collision with a vehicle that is likely to be entering an intersection without the right of way. The sender may be either an equipped vehicle or another source such as the infrastructure. “
- Personal Device Usage – geared to smart phones and indicates level of pedestrian distraction. Reports human interaction – typing, talking, video playback.
- Disabled Vehicle
- Work zone status and worker activity
- Coefficient of friction, Rain Sensor, Wiper speed
- “VINString” data element – “a unique identifying string about the vehicle. This may be the vehicle's VIN value assignment “

V2I – Traffic sign Detection

Tesla March 2019 Software update

- Traffic sign recognition
- Stopping at red light on some models and hardware configurations

2018+ Lexus Road Sign Assist (RSA)

- When the RSA system recognizes a do not enter sign and determines that your vehicle has entered a no-entry area, the displayed sign will flash and a buzzer will sound.

2019 Lexus LS500 Owner's Manual: https://drivers.lexus.com/t3Portal/document/om-s/OM50G70U/pdf/OM50G70U.pdf

Enabling Technologies: Actuators

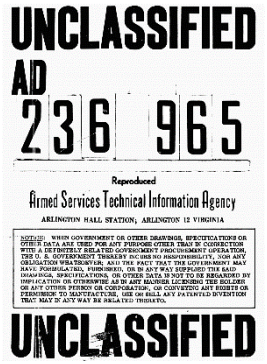
- Electric Power Assist Steering (EPAS, EPS)
- Throttle by wire
- Electronic Stability Control (ESC) brake control

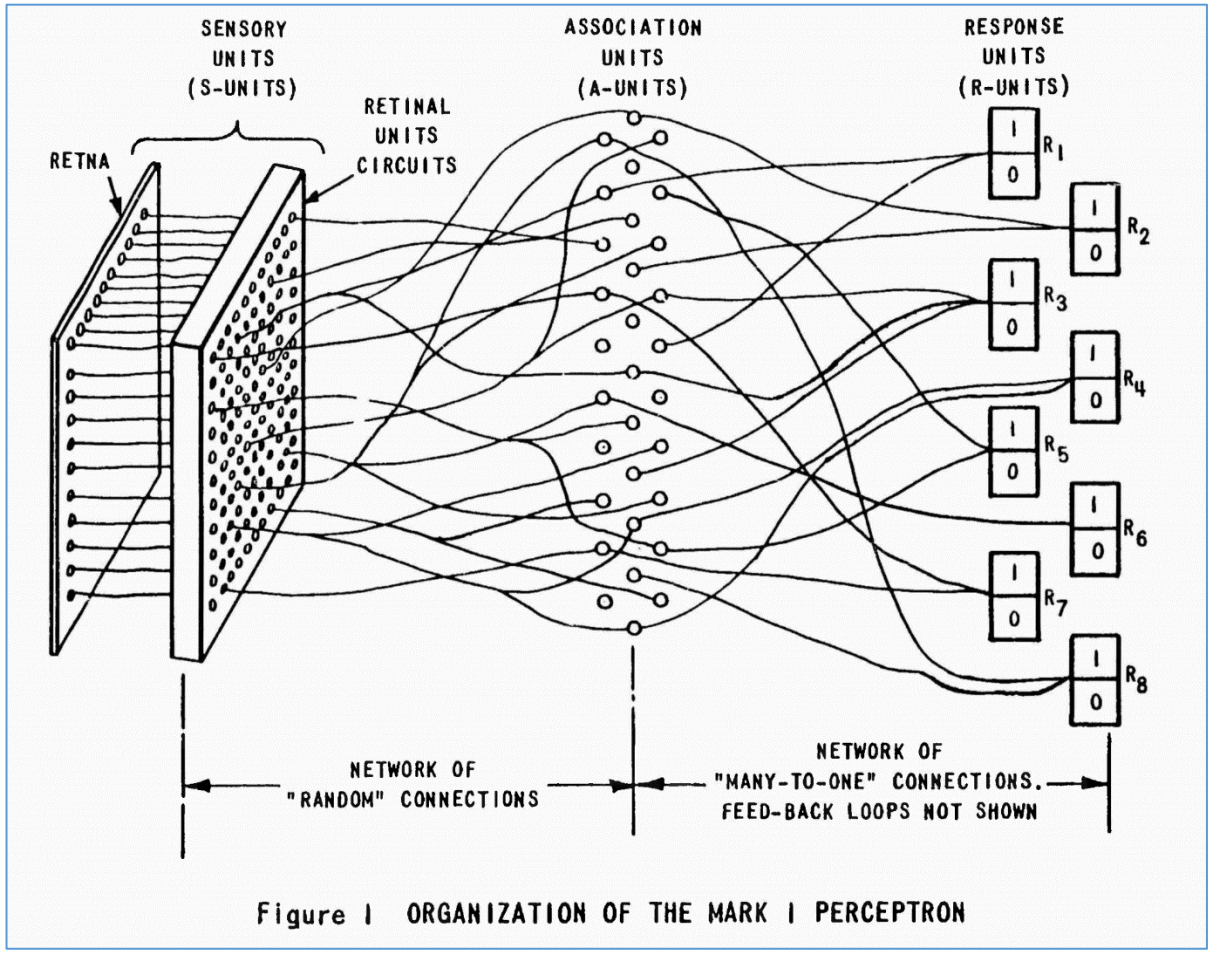
Enabling Technologies: Computing

- Image classification – MobilEye and Bayes theorem
- Artificial intelligence and neural nets
- Probabilistic decision making

Image Classification

1960s Image Classifier

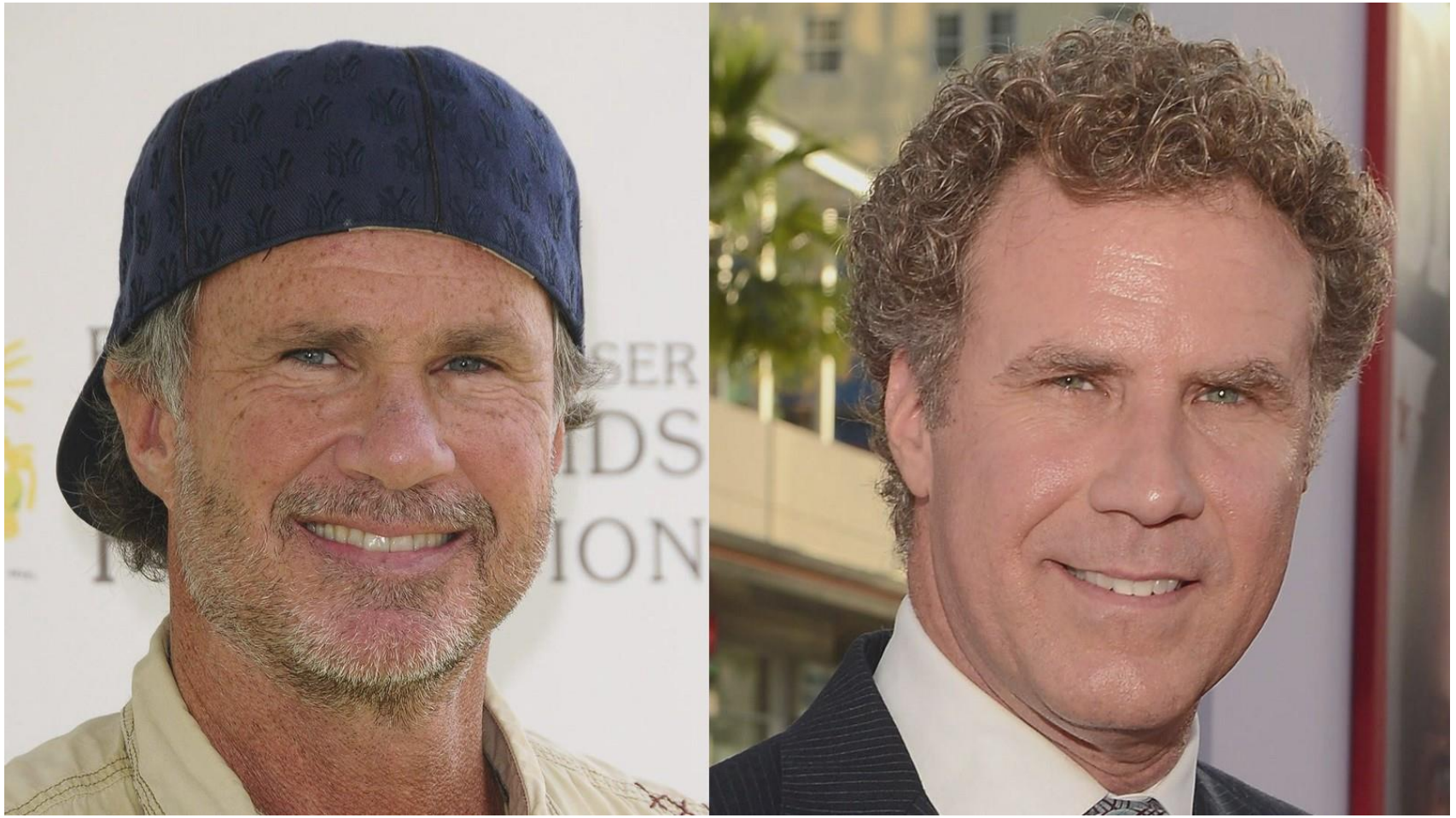




Mark 1 Perceptron Operator's Manual, Cornell Aeronautical Laboratory, 1960
https://apps.dtic.mil/dtic/tr/fulltext/u2/236965.pdf

Image Classification and Artificial Intelligence

Chad Smith



Will Farrell

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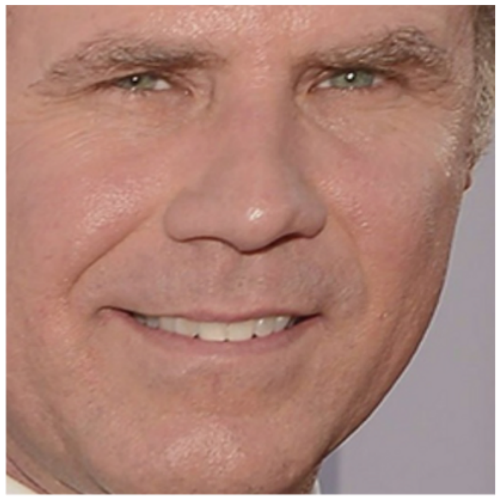
Image Classification and Artificial Intelligence

Input Image



128 Measurements Generated from Image

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-0.024210374802351	-0.11443792283535	0.071997955441475	-0.012062266469002
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Will Ferrell (93%)
Chad Smith (56%)
(83%)

Image Classification Errors

Altered images mis-classified as “Speed Limit 45” with ~70% confidence

Robust Physical-World Attacks on Deep Learning Visual Classification, Eykholt et al, 2018. https://arxiv.org/pdf/1707.08945.pdf

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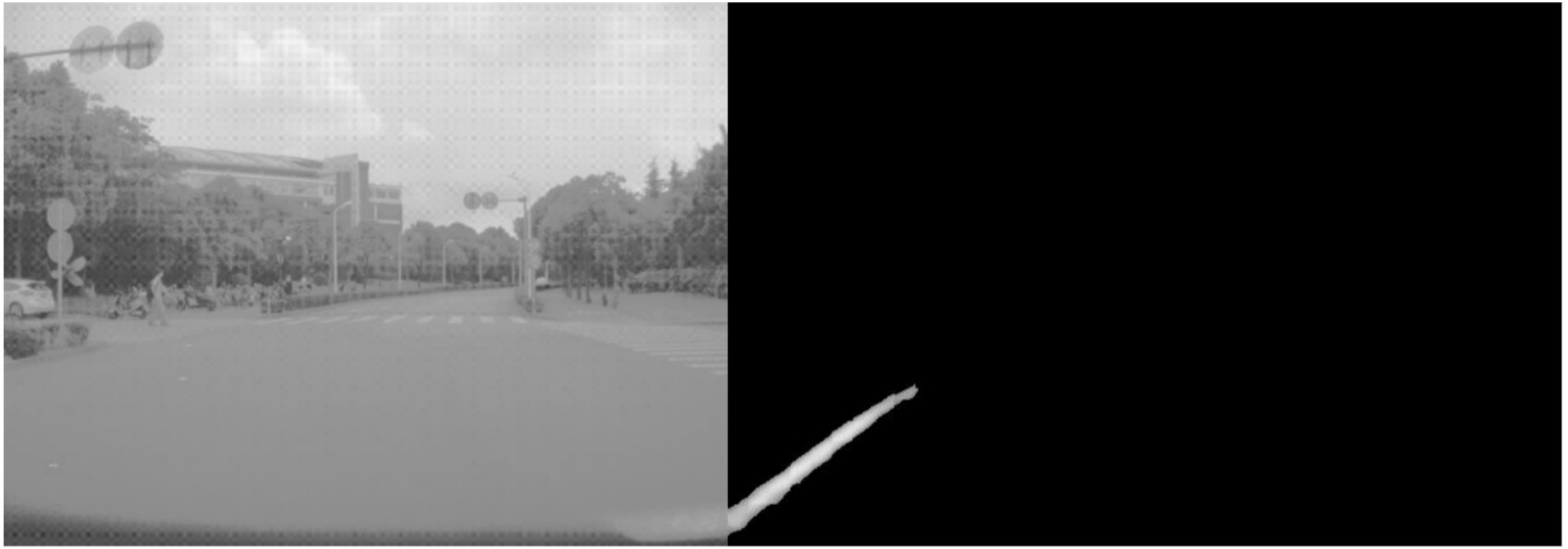
Image Classification Errors

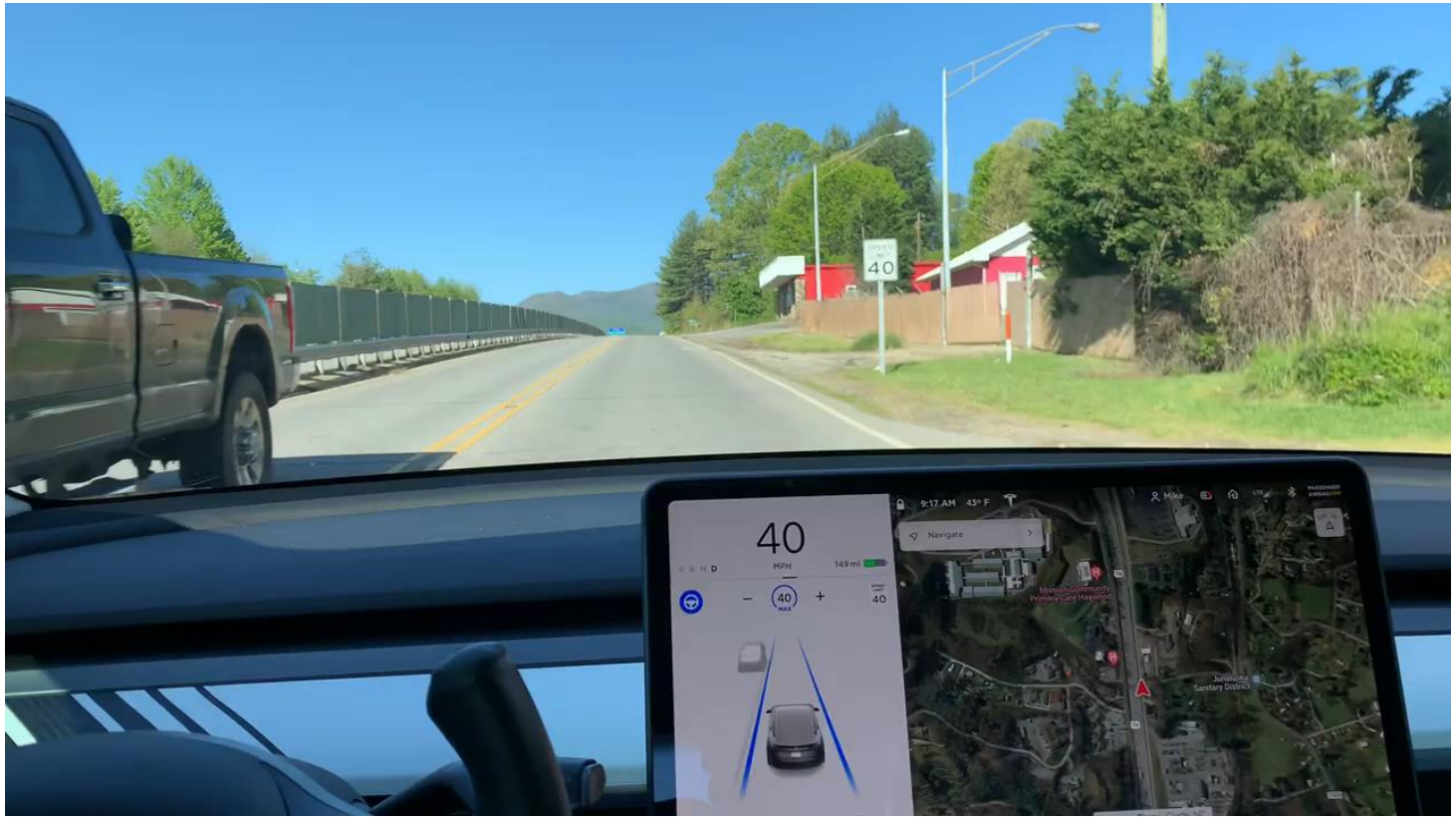
Pavement targets caused Tesla Autopilot to render a lane line



Experimental Security Research of Tesla Autopilot Tencent Keen Security Lab, 2019-03

https://keenlab.tencent.com/en/whitepapers/Experimental_Security_Research_of_Tesla_Autopilot.pdf

Tesla Autopilot stops for Burger King



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In-Path Stationary Object (IPSO)

An object is classified as an IPSO if it is associated with a stationary radar target—specific distance measurement from the radar system—and the camera system does not detect a recognized object at that distance, but does detect lack of *free space* or clear path in travel direction of the Tesla. For example, a highway overpass would likely not be classified as IPSO, even though a radar system may detect the overpass as a stationary object, but camera system *would* detect available free space in the vehicle's path underneath the overpass.

NTSB – Human Performance Factors Group Chairman's Supplemental Report: Driver Assistance Systems Williston, FL HWY16FH018

Tesla Autopilot Visualization

SAE INTERNATIONAL

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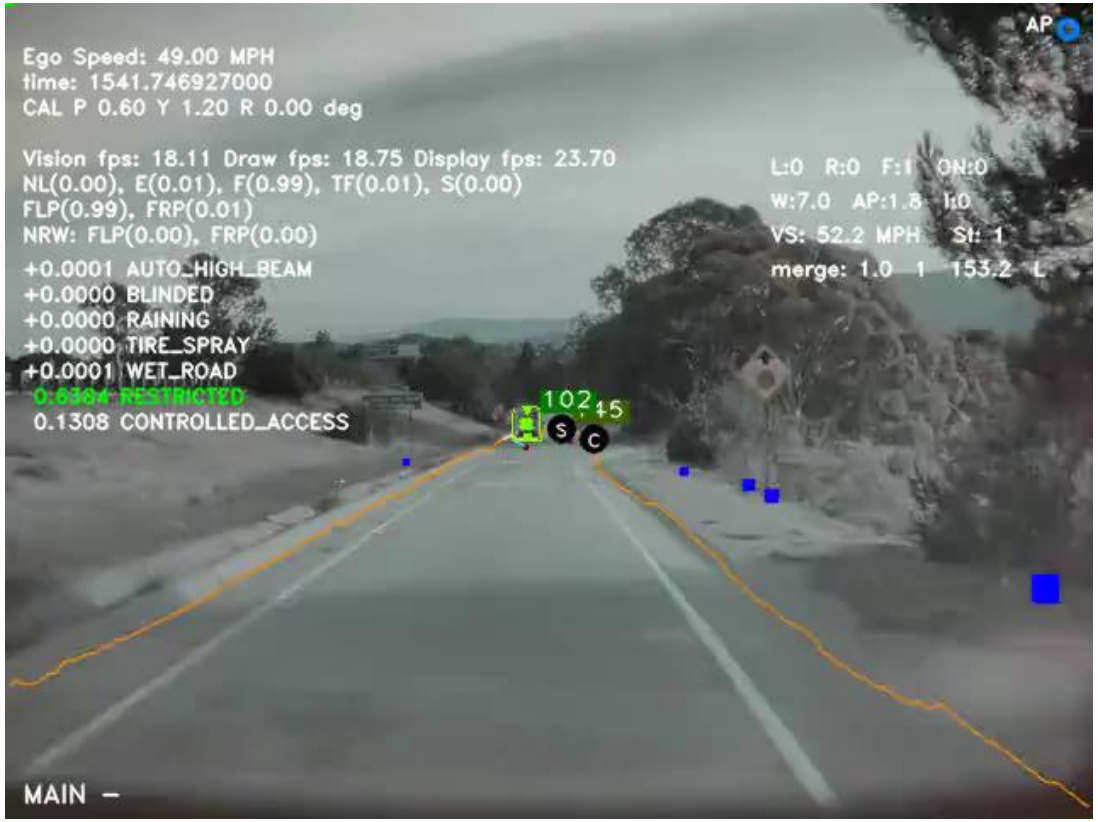
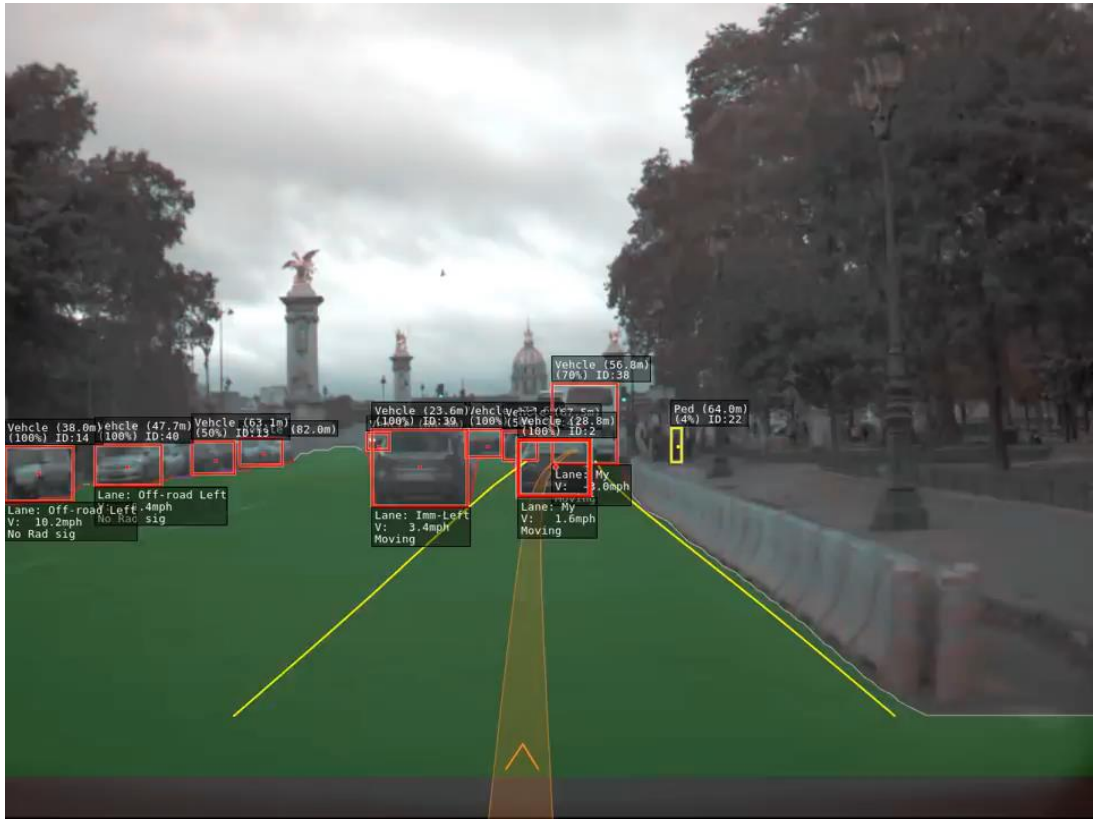
Tesla “Full Self Driving” Visualization



Tesla Neural Network Recruiting Demo
https://www.tesla.com/autopilotAI

Tesla Autopilot Visualization



Hacker's visualization of Tesla Autopilot sensing data – vehicle under manual control
Version 18.34, September 2018
https://youtu.be/_1MHGUC_BzQ

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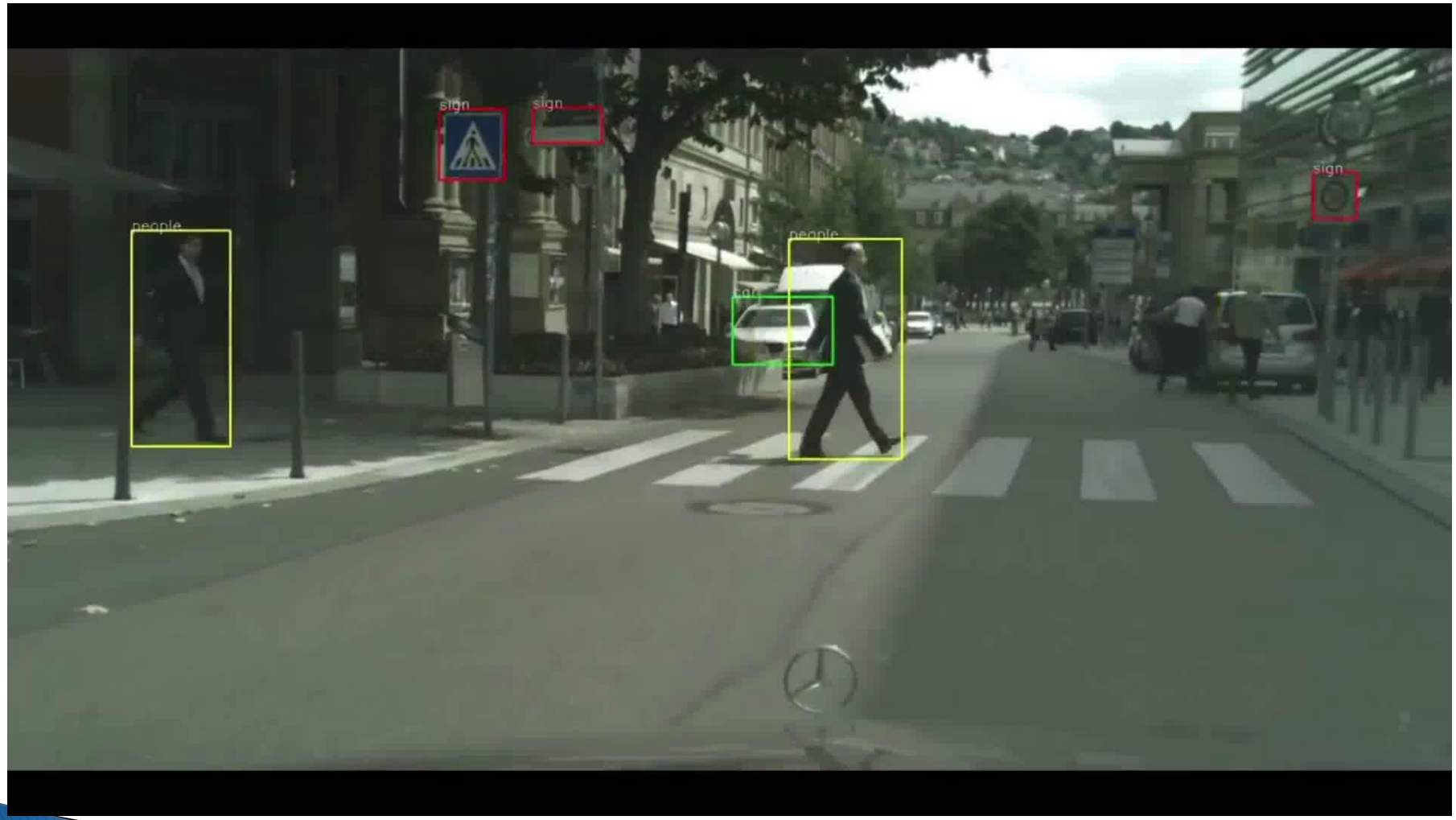
CES 2016 Nvidia DriveNet demo video



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The enabling technologies

- Summary:
- Technology in three areas – sensors, actuators, and computing have allowed ADAS to enter the market

What is on the road today?

By the end of this module you will:

Know what technology is in current production vehicles, and what some of the limitations are.

Subaru “Eyesight” stereo cameras

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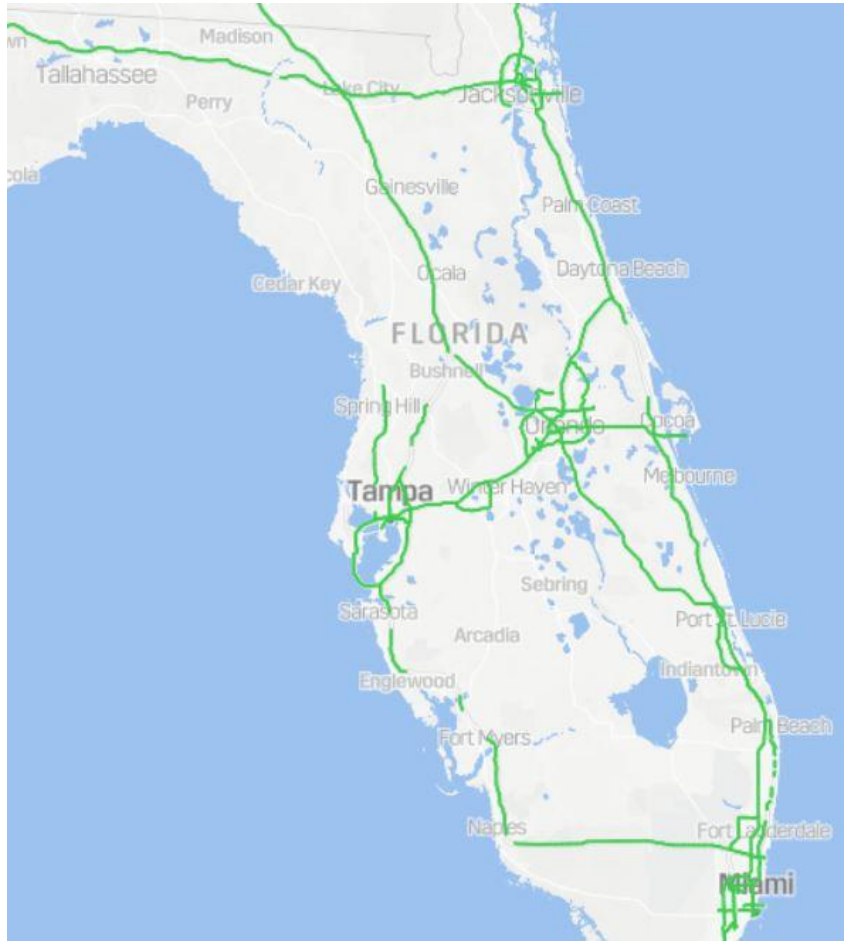
Cadillac SuperCruise

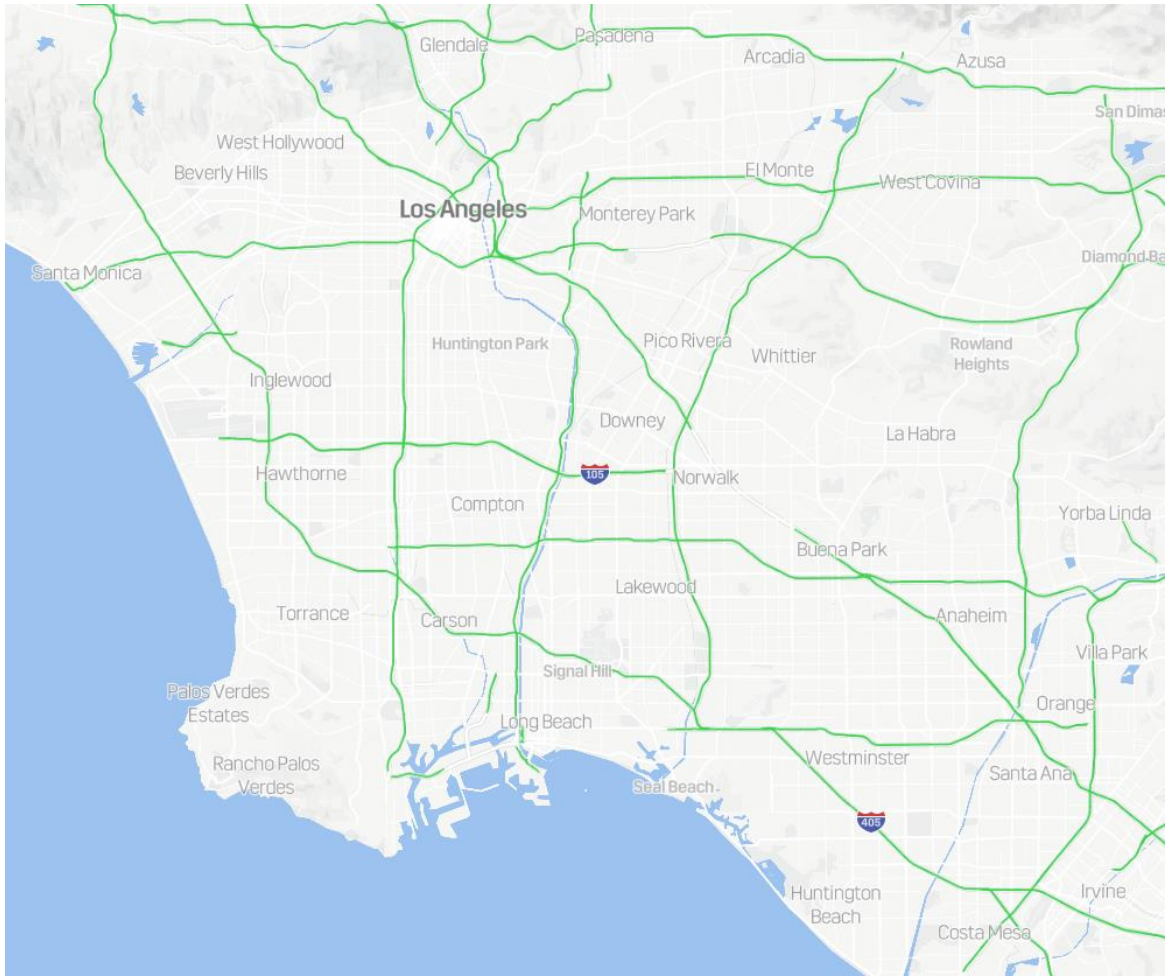
- Adaptive cruise and lane centering
- “Selfie” camera monitors driver attention
- Only on CT6, expanding 2020 to other models

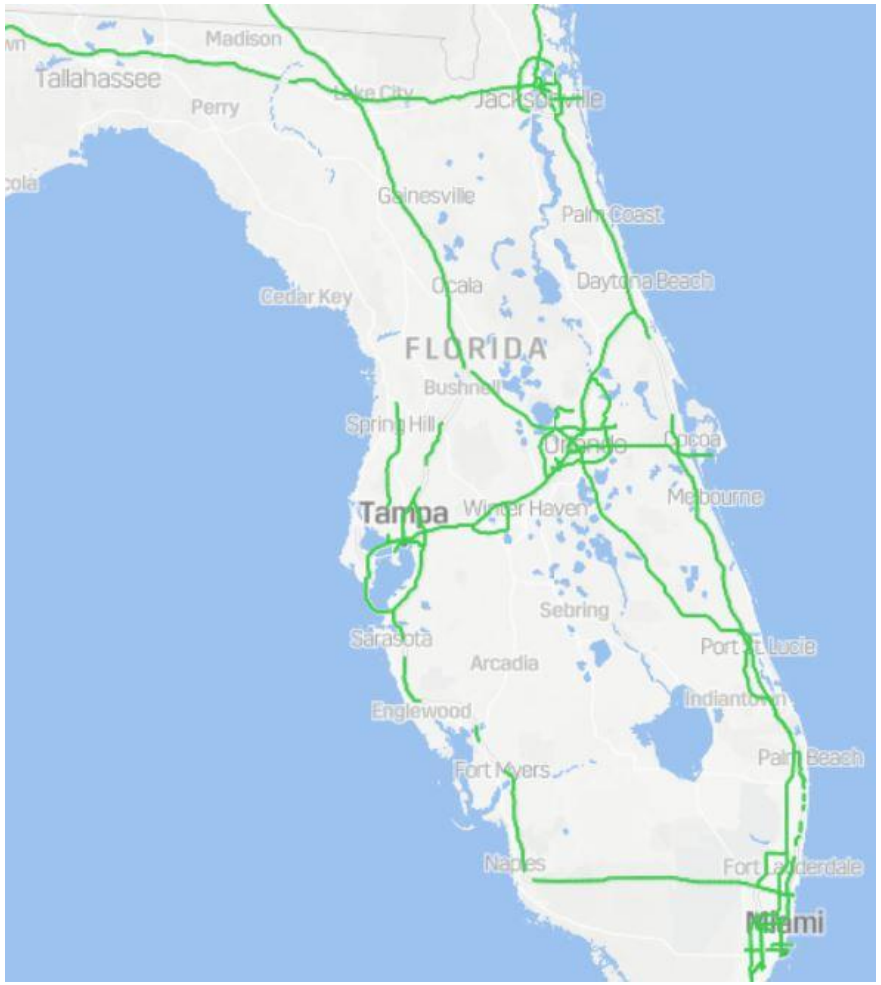
Cadillac SuperCruise

- Adaptive cruise and lane centering
- “Geo-fenced” to mapped limited-access roadways



Cadillac SuperCruise





https://supercruise-map-viewer.cp.gm.com/

Volvo Pilot Assist II

- Adaptive Cruise and lane centering
- Requires hands on the wheel
- “It is important to be aware that steering assistance may toggle between off and on at any time, without prior warning”

Mercedes Distronic

- Stereo cameras and radar
- Adaptive cruise and lane centering
- “Attention Assist”
- Evasive Steering Assist
- Active Lane Change Assist
- Car-to-X Communication
- Good summary of features:
- https://www.youtube.com/watch?v=Nc_w1S1Ks0U

Tesla

- Adaptive Cruise and lane centering
- Lane change assist
- Requires steering wheel feedback at irregular intervals
- Audible and visual alarm at fallback (sometimes)

Toyota

90% of U.S. models 2018+ equipped with Forward Collision Warning

Records copious amounts of data available through Vehicle Control History (VCH) and Freeze Frame Data (FFD)

Lexus Forward Cross Traffic Alert, 2018+

Operating conditions:

- A shift position other than P or R is selected.
- Vehicle speed is approximately 9 mph (15 km/h) or lower.
- A vehicle is approaching from the left or right in front of your vehicle at a speed between approximately 6 mph (10 km/h) and 37 mph (60 km/h).
- There are no preceding vehicles ahead of your vehicle.
- The accelerator pedal is not strongly depressed.
- The brake pedal is not being strongly depressed.

2019 Lexus LS500 Owner's Manual: https://drivers.lexus.com/t3Portal/document/om-s/OM50G70U/pdf/OM50G70U.pdf
https://www.youtube.com/watch?v=A61JvDJcW7I

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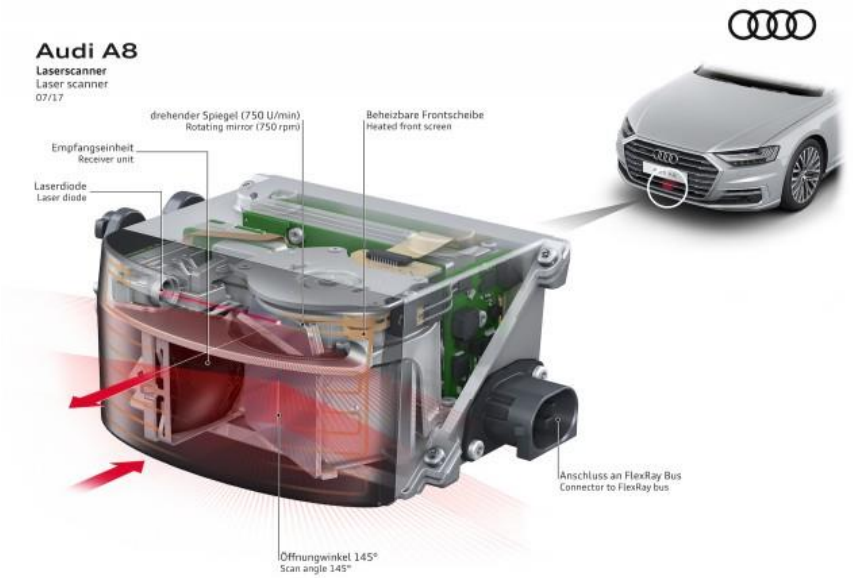
Audi Traffic Jam Assist – A8 Model

- Drivers “can take their hands off the steering wheel for longer periods and can focus on other activities.”
- Less than 37 mph, limited access roadway with median
- Europe only*
- First Level 3 production vehicle



Nissan Pro Pilot

- Adaptive Cruise and lane centering
- Requires continuous steering wheel feedback
- Audible and visual alarm – criticized as too active

Nissan Pro Pilot

Ford – Evasive Steering Assist

Comma.ai “Openpilot”

Comma.ai “Openpilot”

Supported Cars

Make	Model	Supported Package	Lateral	Longitudinal	No Accel Below	No Steer Below	Giraffe
Acura	ILX 2016-17	AcuraWatch Plus	Yes	Yes	25mph[1]	25mph	Nidec
Acura	RDX 2018	AcuraWatch Plus	Yes	Yes	25mph[1]	12mph	Nidec
Chevrolet[3]	Malibu 2017	Adaptive Cruise	Yes	Yes	0mph	7mph	Custom[7]
Chevrolet[3]	Volt 2017-18	Adaptive Cruise	Yes	Yes	0mph	7mph	Custom[7]
Holden[3]	Astra 2017	Adaptive Cruise	Yes	Yes	0mph	7mph	Custom[7]
Honda	Accord 2018	All	Yes	Stock	0mph	3mph	Bosch
Honda	Civic 2016-18	Honda Sensing	Yes	Yes	0mph	12mph	Nidec
Honda	Civic 2017-18 *(Hatch)*	Honda Sensing	Yes	Stock	0mph	12mph	Bosch
Honda	CR-V 2015-16	Touring	Yes	Yes	25mph[1]	12mph	Nidec
Honda	CR-V 2017-18	Honda Sensing	Yes	Stock	0mph	12mph	Bosch
Honda	Odyssey 2017-19	Honda Sensing	Yes	Yes	25mph[1]	0mph	Inverted Nidec
Honda	Pilot 2016-18	Honda Sensing	Yes	Yes	25mph[1]	12mph	Nidec
Honda	Pilot 2019	All	Yes	Yes	25mph[1]	12mph	Inverted Nidec
Honda	Ridgeline 2017-18	Honda Sensing	Yes	Yes	25mph[1]	12mph	Nidec
Hyundai	Santa Fe 2019	All	Yes	Stock	0mph	0mph	Custom[6]
Hyundai	Elantra 2017	SCC + LKAS	Yes	Stock	19mph	34mph	Custom[6]
Hyundai	Genesis 2018	All	Yes	Stock	19mph	34mph	Custom[6]
Kia	Sorento 2018	All	Yes	Stock	0mph	0mph	Custom[6]
Kia	Stinger 2018	SCC + LKAS	Yes	Stock	0mph	0mph	Custom[6]
Lexus	RX Hybrid 2016-18	All	Yes	Yes[2]	0mph	0mph	Toyota
Toyota	Camry 2018[4]	All	Yes	Stock	0mph[5]	0mph	Toyota
Toyota	C-HR 2017-18[4]	All	Yes	Stock	0mph	0mph	Toyota
Toyota	Corolla 2017-18	All	Yes	Yes[2]	20mph	0mph	Toyota
Toyota	Highlander 2017-18	All	Yes	Yes[2]	0mph	0mph	Toyota
Toyota	Highlander Hybrid 2018	All	Yes	Yes[2]	0mph	0mph	Toyota
Toyota	Prius 2016	TSS-P	Yes	Yes[2]	0mph	0mph	Toyota
Toyota	Prius 2017-18	All	Yes	Yes[2]	0mph	0mph	Toyota
Toyota	Prius Prime 2017-18	All	Yes	Yes[2]	0mph	0mph	Toyota
Toyota	Rav4 2016	TSS-P	Yes	Yes[2]	20mph	0mph	Toyota
Toyota	Rav4 2017-18	All	Yes	Yes[2]	20mph	0mph	Toyota
Toyota	Rav4 Hybrid 2017-18	All	Yes	Yes[2]	0mph	0mph	Toyota

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What is on the road today?

Summary:

OEMs have implemented ADAS technology in different manners, all with the same goal: keep the driver involved while providing assistance in the driving task.

Standards & Protocols

By the end of this module you will:

- Know what relevant standards have been promulgated by SAE, ISO, NHTSA, IIHS, and Euro NCAP
- Have perspective on the rapid changes occurring in the field

SAE J3048 – LKAS Driver Interface

- No specific symbol for LKAS exists in ISO 2575, but efforts are underway to address active safety system symbols, generally.
- “Systems may or may not provide explicit feedback to the driver”
- “Haptic LKAS alerts … will not be masked by road feedback through the vehicle chassis.”

SAE J3087 – AEB Performance Testing

- Limited to front-to-rear, in-lane scenarios
- 3 scenarios tested for both DBS and CIB:
  - Lead vehicle stopped
  - Lead vehicle moving slowly
  - Lead vehicle decelerating
- Brake burnishing required at GVWR - 100 stops at 0.31g from 50mph.

SAE J3087 – AEB Performance Testing

Specific requirements

- Brake applicator configured to 0.4 +/- .025 g deceleration
- Throttle released <500ms after FCW alert
- In DBS testing, brakes applied at TTC 1.1s
- Target vehicle deceleration 0.3 +/- .03 g

SAE J3087 – AEB Performance Testing

Specific requirements

- The AEB setting for “Earliest” activation should be set
- “Process auditory and/or haptic data using a phaseless, forward-reverse pass, elliptical (Cauer) digital bandpass filter, having filter parameters [5th order filter, 3dB ripple, 60dB minimum attenuation, passband at center frequency +/- 20%] to determine the alert onset programmatically”

IIHS and Euro NCAP AEB Target

The ADAC Advanced Emergency Braking System (AEBS), which is produced by Messring Systembau GmbH, is used as the stationary target (http://www.messring.de/test-facilities-and-components/adac/). The stationary carriage and balloon car are shown in Figure 2. The balloon car is equipped with a U.S.-specific license plate. At the beginning of testing, the target will be inflated to 250 mbar and maintained at that pressure throughout testing. The target cover should also be free of wrinkles during testing.

Figure 2

ADAC Advanced Emergency Braking System Stationary Target

IIHS Autonomous Emergency Braking Test Protocol (Version I), 2013

Euro NCAP Test Protocol – AEB Systems, Version 1.1, 2015

NHTSA AEB Target – Strikeable Surrogate Vehicle (SSV)

Accident Reconstruction Test Barriers

MEA Forensic Engineers & Scientists

ISO 22839 - Forward vehicle collision mitigation systems

6.3.6.4.1.1 Light vehicles

MB shall not be initiated for TTC or ETTC above 3,0 s to achieve the deceleration requirement of 6.3.6.4.2.

6.3.6.4.2.1 Light vehicles

FVCMS shall generate a deceleration at least 5,0 m/s^2 (0,51 g) for a duration chosen to achieve a speed reduction of at least 2,0 m/s. FVCMS with both MB and SRB combined (Type 3) shall generate a minimum speed reduction of at least 4,0 m/s. This requirement does not constrain the time when mitigation braking is activated.

0.51G

9 mph

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ISO 22839 – Automatic Emergency Braking

6.3.6.5.2 Maximum deceleration in SRB

The average deceleration generated by SRB when active shall not exceed the line d_{SV} = 5,33 m/s^2 - 0,067/s*v_{SV} (averaged over the first time period T_1_SRB sec) for any SV speed between 5,0 m/s and 20 m/s; where d_{SV} is the deceleration of the SV, and v_{SV} is the speed of the SV, and d_{SV} = -a_{SV}.

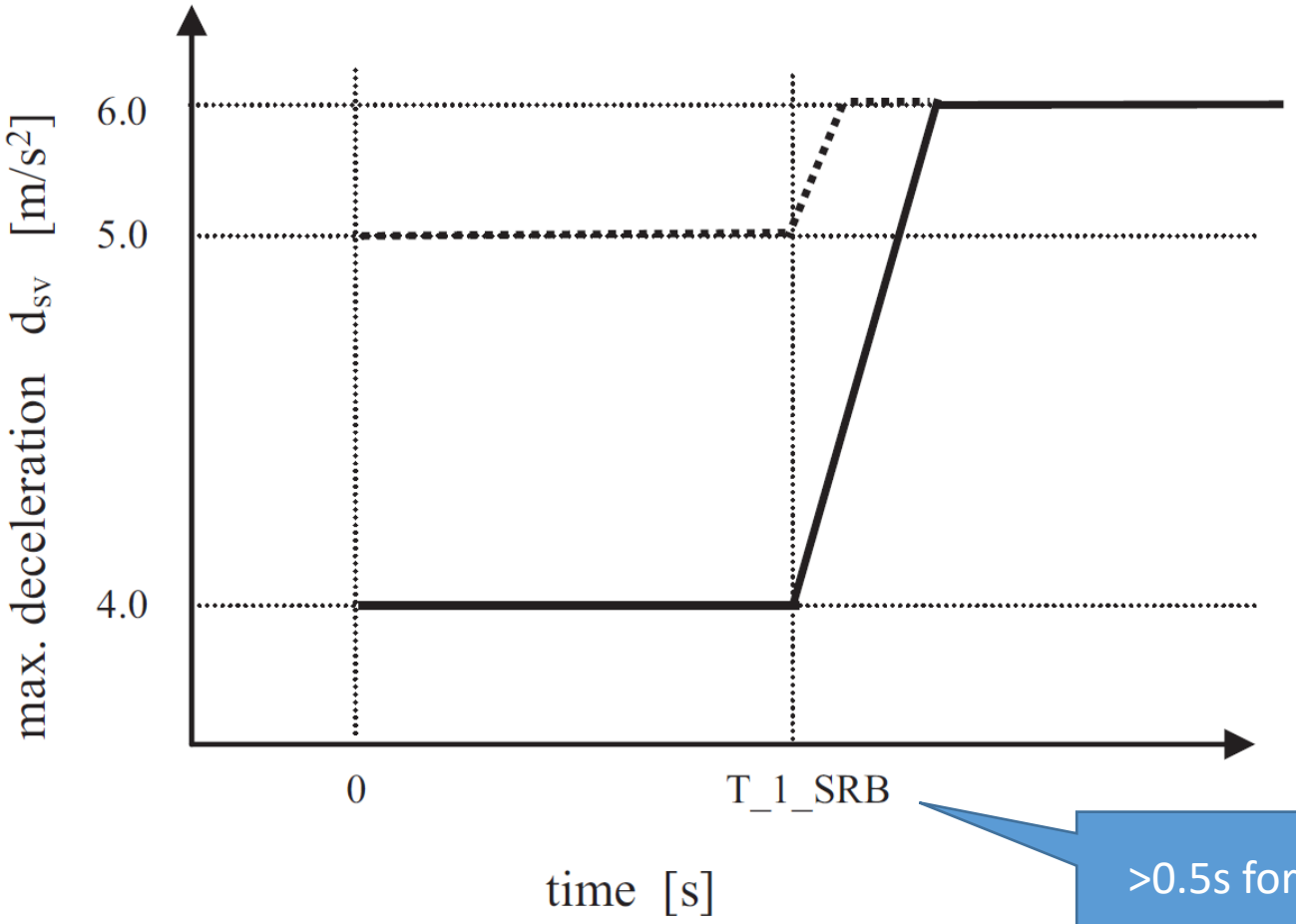
The average deceleration generated by SRB shall not exceed, in the first time period T_1_SRB of the SRB, 4,0 m/s^2 (averaged over the first time period T_1_SRB sec) for v_{SV} > 20 m/s or 5 m/s^2 for v_{SV} < 5 m/s (as depicted in Figure 6).

After a time period of at least T_1_SRB, the maximum allowed deceleration of the SRB may be continuously increased up to 6 m/s^2 (averaged over 1 sec). For Type 3 and 1: T_1_SRB > = 0,5 sec.

The average jerk achieved during SRB, while increasing the maximum deceleration, shall not exceed 6,0 m/s^3 (averaged over 0,5 s).

Refer to Figure 6 Maximum SRB Deceleration for an illustration of this requirement. Refer to Figure 7 for an illustration of the continuous deceleration increase.

ISO 22839 – Automatic Emergency Braking (Speed Reduction Braking mode, SRB)



>0.5s for many cases

ISO 22839 – Automatic Emergency Braking

3.20
lateral offset
lateral distance between the longitudinal centerlines of a subject vehicle (SV) and a target vehicle (TV), measured as a percentage of the width of the SV, such that if the centers of the two vehicles are aligned, the value is zero

6.3.3.2.3 Maximum lateral offset (lateral discrimination)

FVCMS shall function when the lateral offset is 20 % or lower in either direction. FVCMS may function for lateral offsets greater than 20 % at the discretion of the manufacturer.

NHTSA AEB Procedure (draft, 2014)

Table 2-2 CIB and DBS Draft Assessment Reference Values (ARVs)

AEB System	LVS 25_0	LVM 45_20	LVM 25_10	LVD1 35_35	LVD2 25_25	STP_45 (FP)	STP_25 (FP)
CIB	Speed Reduction ≥ 9.8 mph (15.8 km/h)	Speed Reduction ≥ 9.8 mph (15.8 km/h)	Crash Avoidance	Speed Reduction ≥ 10.5 mph (16.9 km/h)	Speed Reduction ≥ 9.8 mph (15.8 km/h)	No Activation[1]	
DBS	Crash Avoidance					No Activation[2]	

[1]STP = Steel trench plate

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NHTSA 2014 AEB Evaluations

Table 3-1 SV Speed Reductions Per Scenario (mph).

Vehicle	AEB System	SV Speed Reduction Per Scenario (mph)				
		LVS 25_0	LVM 45_20	LVM 25_10	LVD1 35_35	LVD2 25_25
2014 Acura MDX	CIB	7.5 - 7.9[1]	8.6 - 10.3	7.9 - 15.3	11.0 - 14.0	6.6 - 9.5
2014 Jeep Grand Cherokee		7.7 - 8.7	9.1 - 12.9	7.9 - 10.5	12.2 - 13.4	12.4 - 12.8
2014 BMW i3		7.6 - 9.5	Not performed[4]	8.2 - 10.4	8.3 - 11.7	7.5 - 9.0
2015 Hyundai Genesis		17.4 - 25.5	23.8 - 24.8	14.1 - 15.3	Not performed[5]	Not completed[6]
2014 Acura MDX[2]	DBS	24.3 - 24.9	23.7 - 24.9	14.1 - 15.3	12.9 - 19.5	19.6 - 25.1
2014 Jeep Grand Cherokee[3]		24.5 - 25.1	24.5 - 25.2	14.2 - 15.1	17.4 - 23.5	24.2 - 25.1
2014 BMW i3[3]		9.4 - 11.6	Tests not performed[7]			
2015 Hyundai Genesis[2]		16.3 – 25.2	21.8 - 25.0	13.9 – 15.2	23.7 – 31.3	21.0 – 25.1

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IIHS AEB Rating Scale

	12 mph test			25 mph test				Forward collision warning
Speed reduction (mph)	less than 5	5 to 9	10 or more	less than 5	5 to 9	10 to 21	22 or more	n/a
Points	0	1	2	0	1	2	3	1

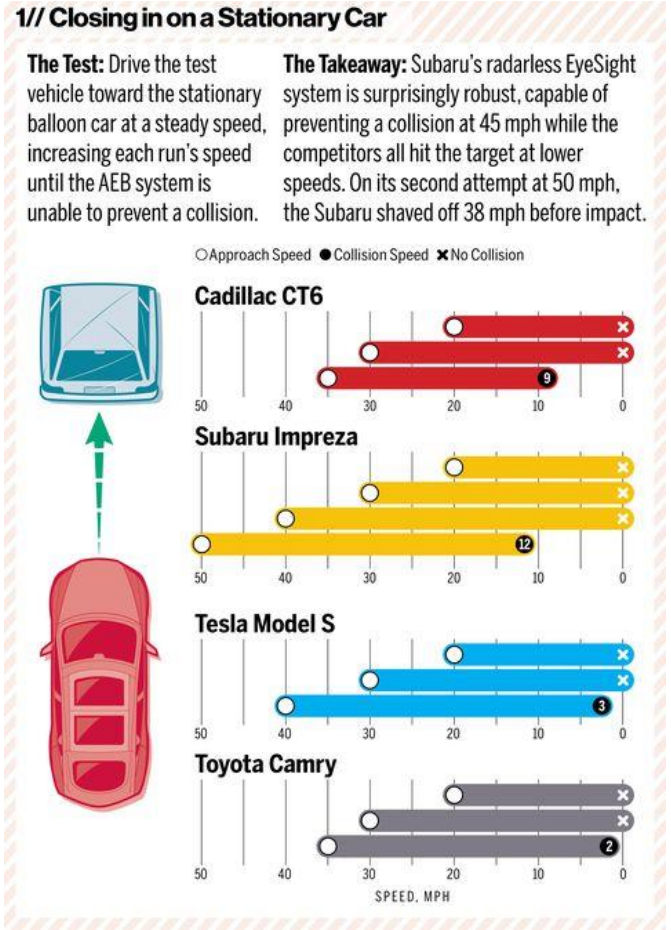
IIHS Frontal Crash Prevention Ratings, 2016

Car & Driver AEB Testing, 2018

- Tested vehicles can avoid a stopped vehicle when traveling 20-30mph, and up to 40mph in some cases.

- Maximum braking of tested vehicles:
  - .4-.7g on Adaptive Cruise
  - .9-1g Autobrake

- Inconsistent performance



We Crash Four Cars Repeatedly to Test the Latest Automatic Braking Safety Systems, Tingwall, 2018
https://www.caranddriver.com/features/a24511826/safety-features-automatic-braking-system-tested-explained/

Standards & Protocols

Summary:

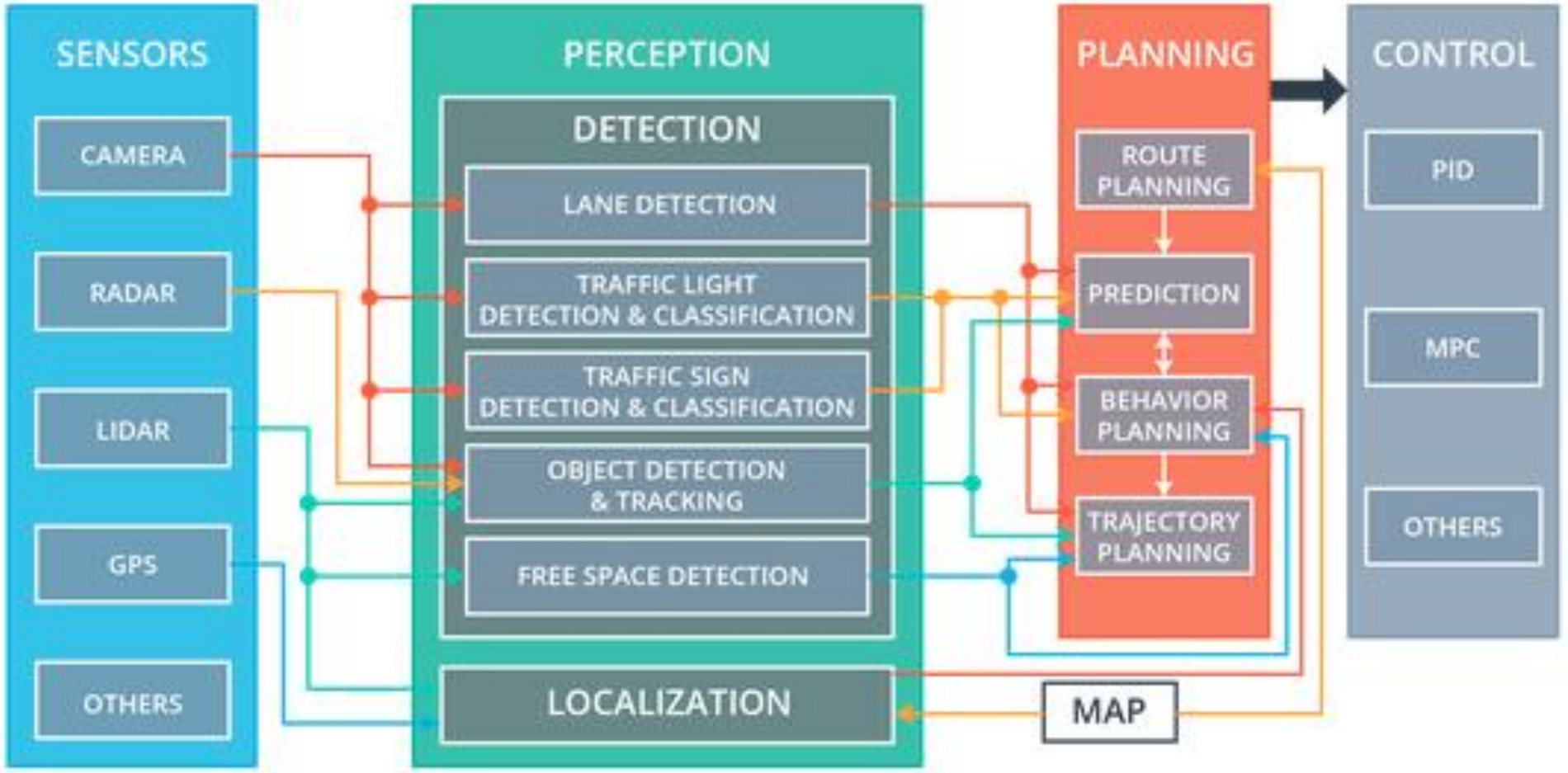
- Standards have been promulgated by SAE, ISO, NHTSA, IIHS, and Euro NCAP
- The standards provide a preliminary framework in a rapidly changing field

Human Factors

By the end of this module you will:

- Understand that human-machine interaction is key to public trust in autonomy
- Gain further knowledge of necessary driver involvement in Level 2-3 vehicles

A model of the Autonomous Vehicle… or a human



David Silver – Udacity https://medium.com/udacity/how-the-udacity-self-driving-car-works-575365270a40

Human-Machine-Interface

Augmented Reality will be Key for Driver Acceptance

Continental

Division Chassis & Safety
Public

9 November 2015
A. Eckert, © Continental AG

17

From Assisted to Automated Driving, Alfred Eckert, Continental Chassis & Safety Division, 2015

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HMI - Tesla

MAX
65
50
mph
SPEED
LIMIT
55
300
150
3
Avg. 347 W
Past 30 m
7:39 PM
P
71°F

HMI - Nissan

12:28 PM
89 °F
45 MPH
CRUISE
Steering Assist Setting
D
312
190.9
RPMx1000
MPH

Human factors – sharing the road

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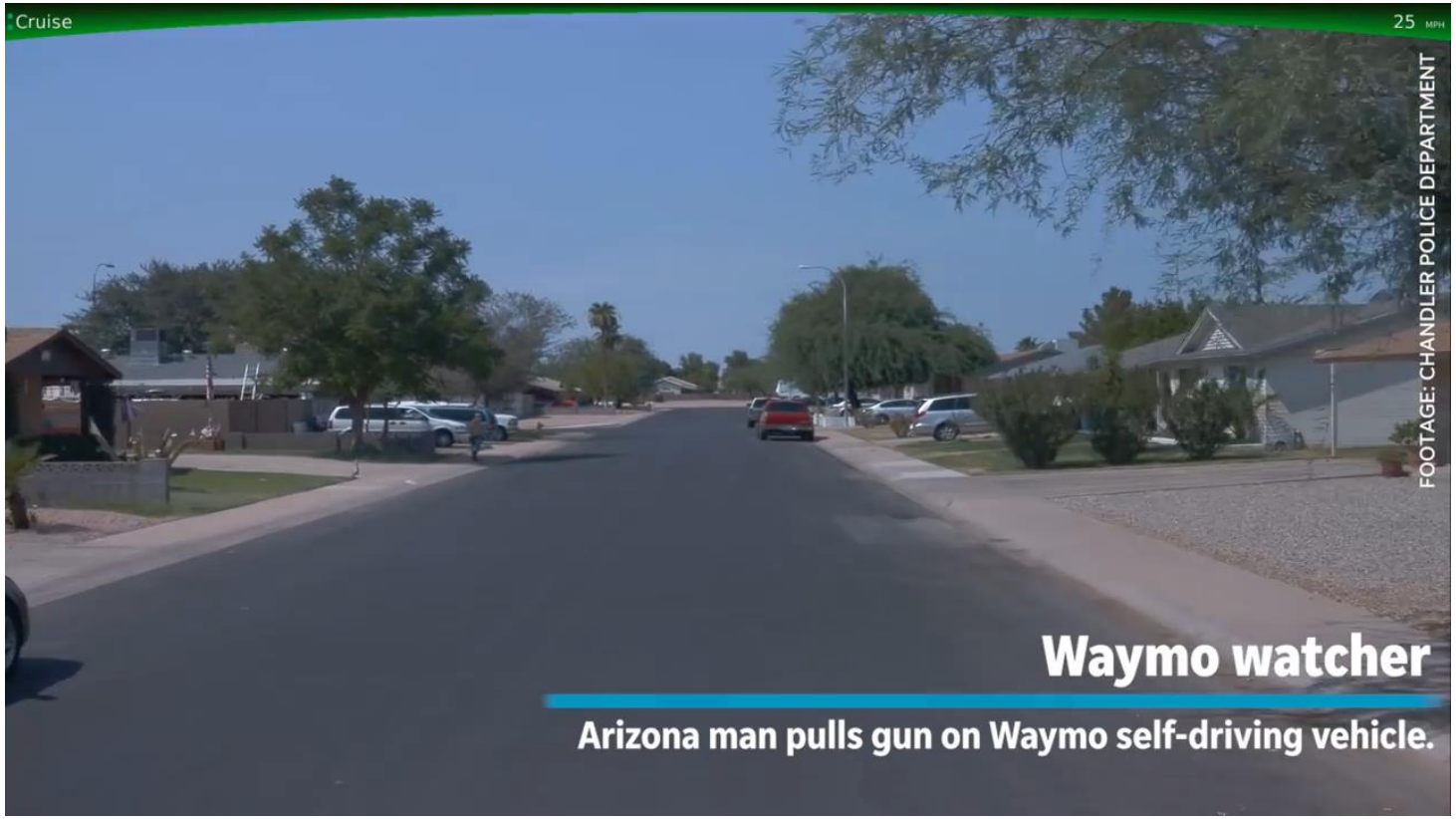
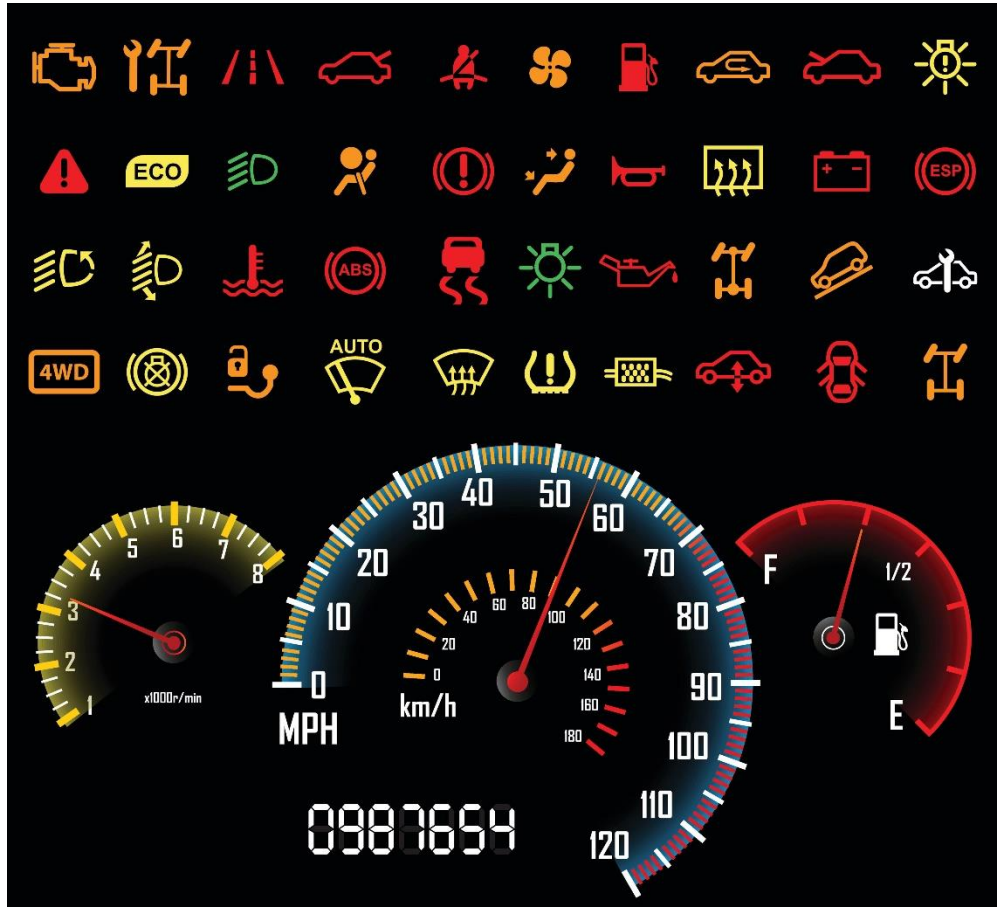
Human factors – sharing the road

Autonomous Road Rage



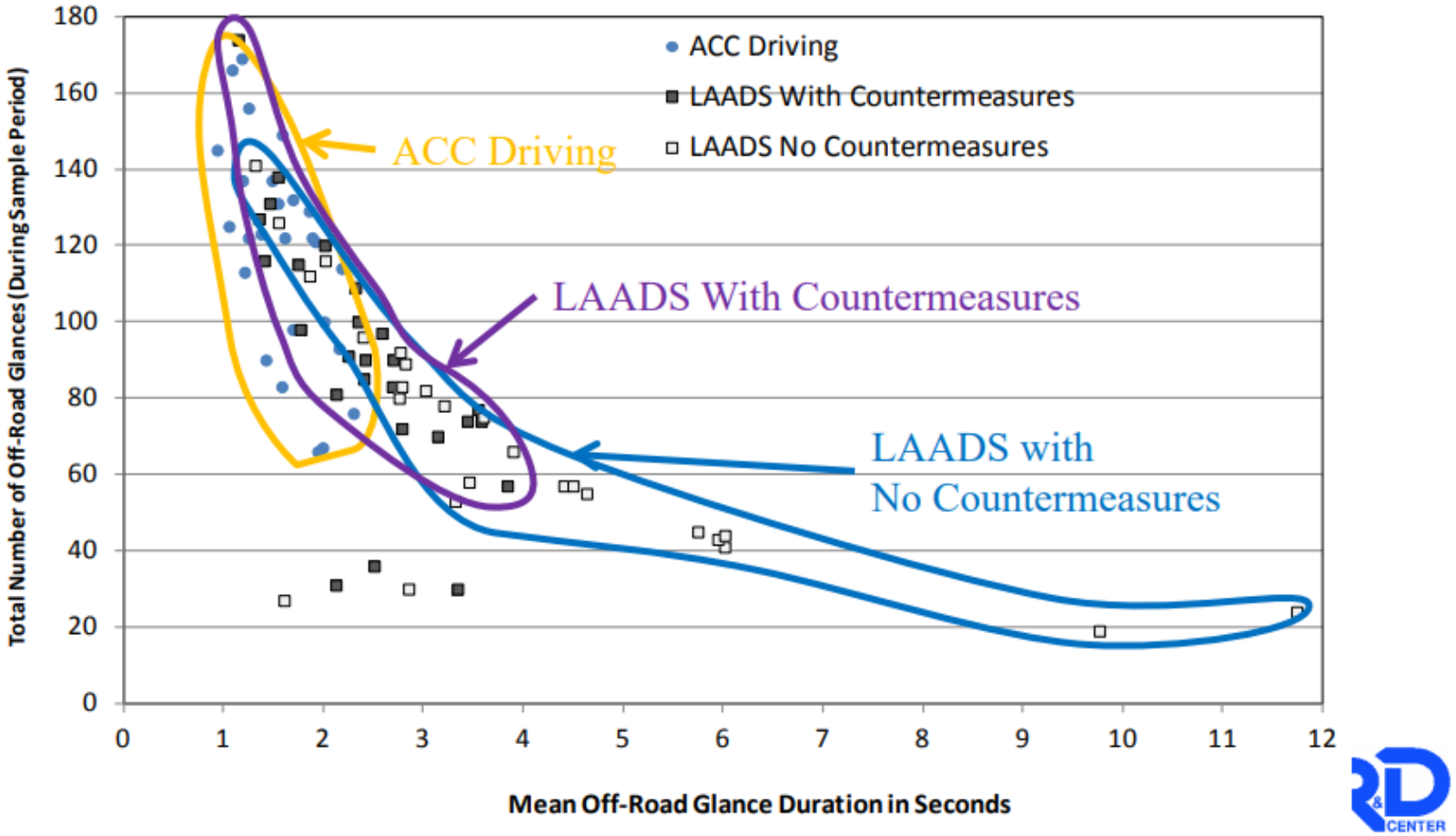
Driver braking in response to audible/visual alert

0.8s from known sound to known response (foot on the brake pedal).



IDRR Software and Related Research, Jeff Muttart and Swaroop Dinakar, 2019

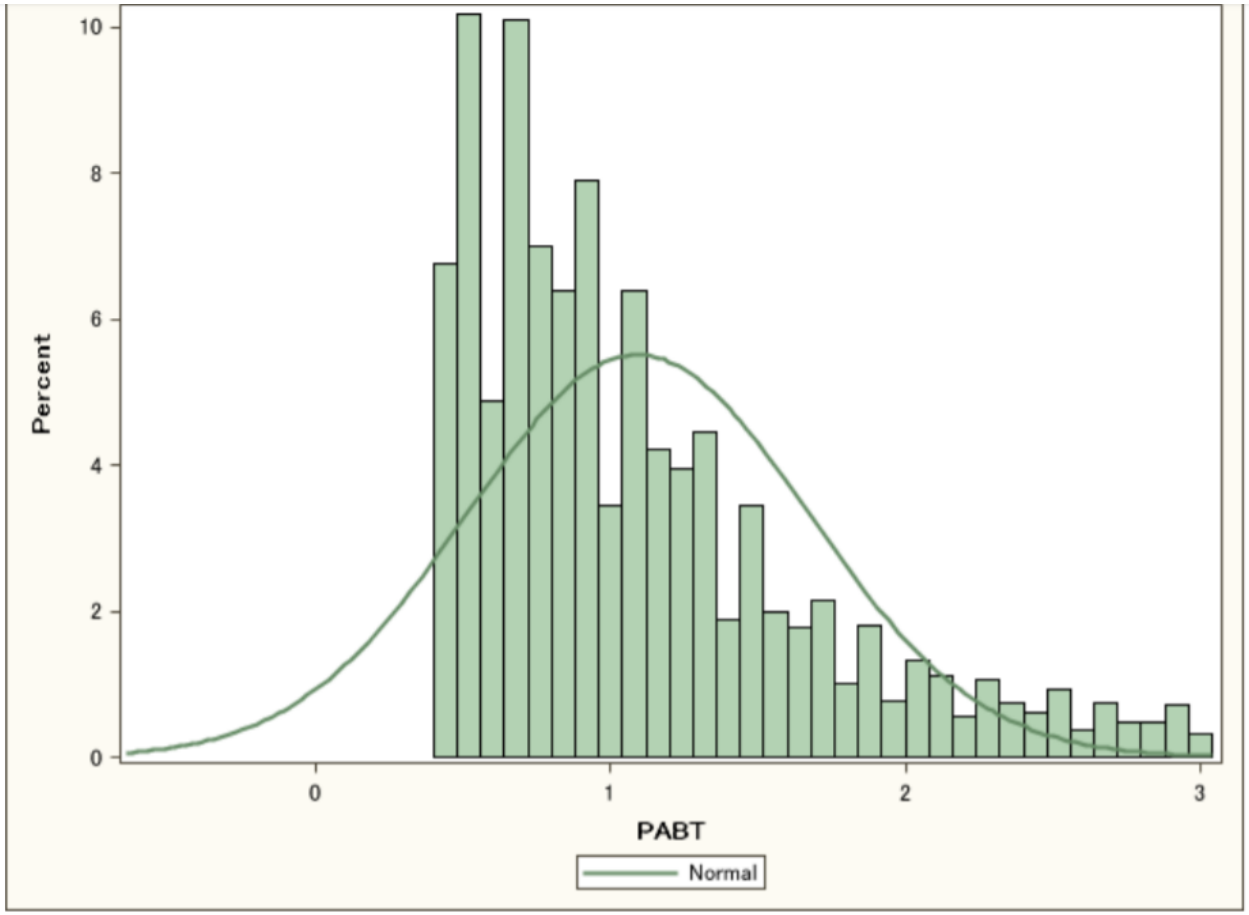
Glance Behavior with Automation



LAADS – Limited Ability Autonomous Driving Systems (i.e., SAE Level 2)

Human Factors for Limited-Ability Autonomous Driving Systems, Jeremy Salinger, GM Research and Development

FCW – Driver Response Time



Large-Scale Field Test of Forward Collision Alert And Lane Departure Warning Systems, DOT HS 812 247, 2016
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812247b-lanedeparturewarningsystems.pdf

Driver Response Time to FCW, Lead Vehicle Scenarios Only

Effect	Level	Mean PABT (s)
Vehicle Model*	SRX	0.741
	XTS	0.790
	Equinox	0.731
Time of Day	Night	0.811
	Day	0.701
Setting	Medium	0.709
	Near	0.737
	Off	0.827
	Far	0.747

Large-Scale Field Test of Forward Collision Alert And Lane Departure Warning Systems, DOT HS 812 247, 2016
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812247b-lanedeparturewarningsystems.pdf

Level 2 takeover requests – how much time does a driver need to regain control?

When engaged in a secondary, non-driving task:

Visual and haptic – 1.3s

Visual alert – 2.3-4.8s (with occasional lack of response)

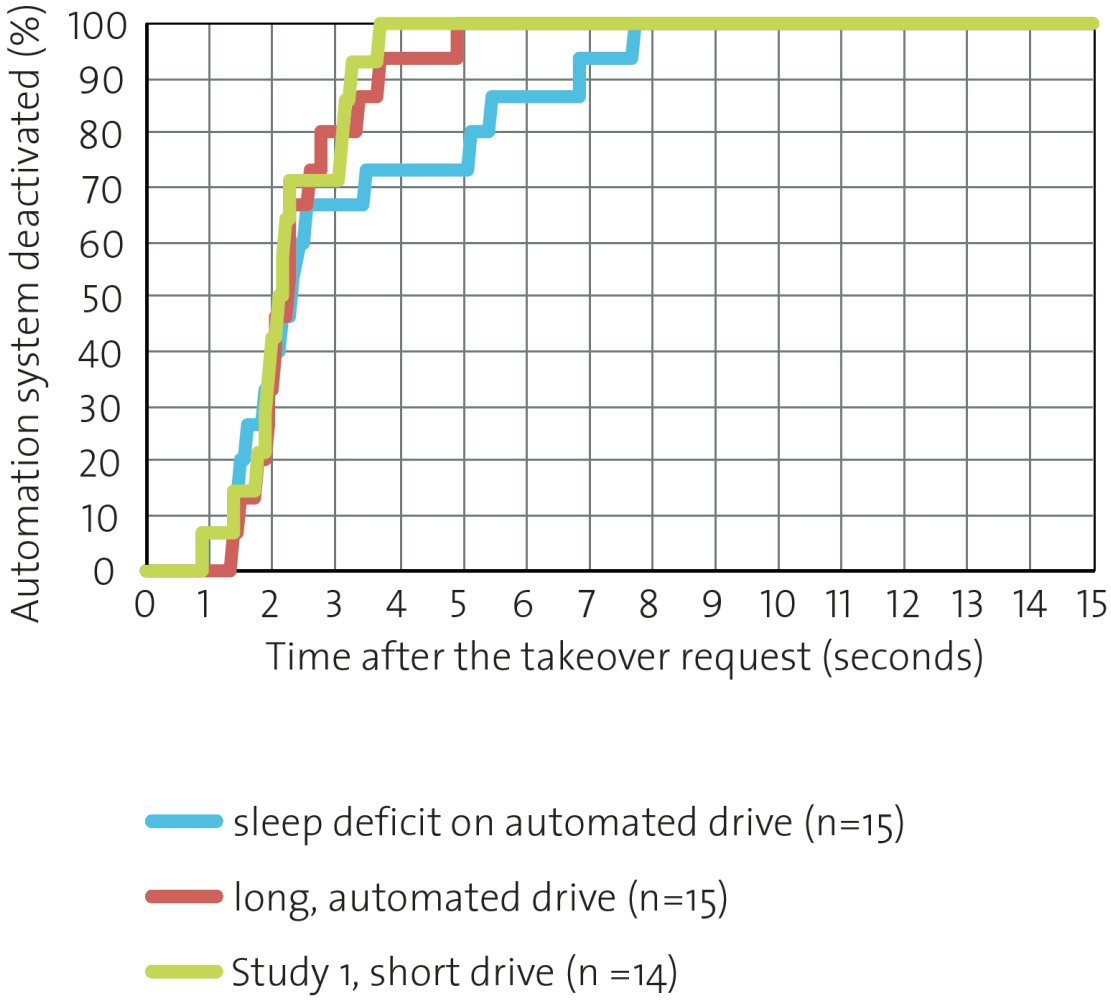
Human Factors Evaluation of Level 2 and Level 3 Automated Driving Concepts – NHTSA & Virginia Tech
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812182_humanfactorseval-l2l3-automdrivingconcepts.pdf

Level 2 takeover requests – how much time does a driver need?

Participants “prioritized the completion of the non-driving task over the operation of the partially automated vehicle”.

Human Factors Evaluation of Level 2 and Level 3 Automated Driving Concepts – NHTSA & Virginia Tech
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812182_humanfactorseval-l2l3-automdrivingconcepts.pdf

Level 3 takeover requests – how much time does a driver need?



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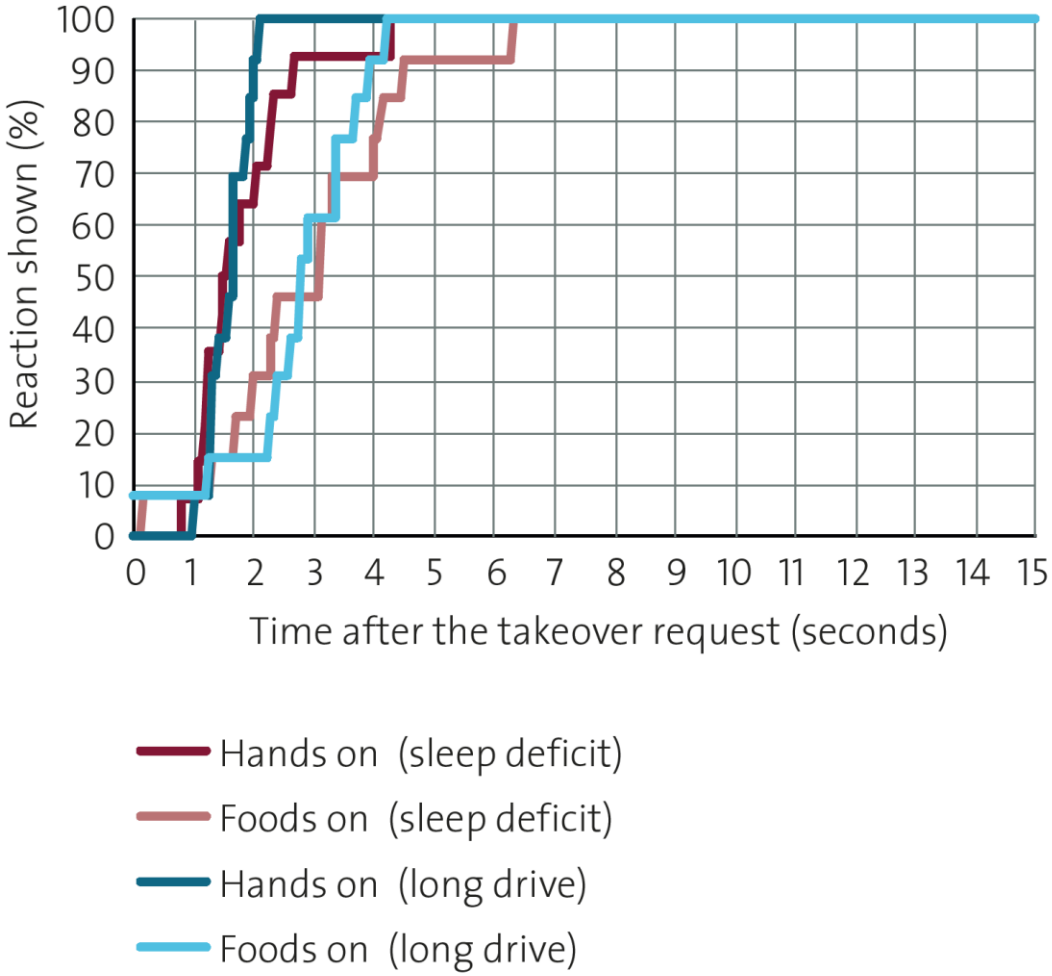
Tiredness and level 3 – automated driving,
German Insurers Association/ Insurers Accident Research, 2017
https://udv.de/download/file/fid/10261

Level 3 takeover requests – how much time does a driver need?



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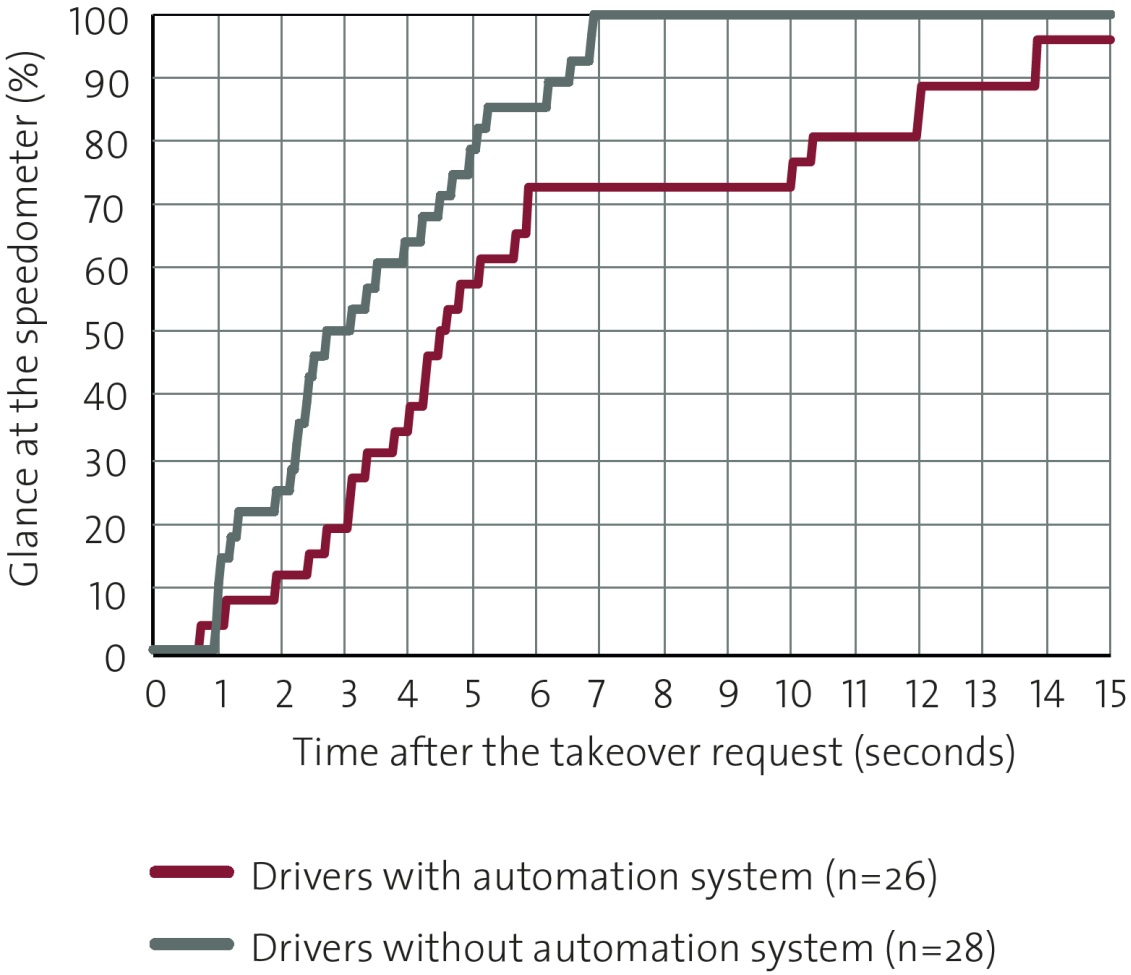
Tiredness and level 3 – automated driving,
German Insurers Association/ Insurers Accident Research, 2017
https://udv.de/download/file/fid/10261

Level 3 takeover requests – how much time does a driver need?



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Tiredness and level 3 – automated driving,
German Insurers Association/ Insurers Accident Research, 2017
https://udv.de/download/file/fid/10261

Level 3 takeover requests – how much time does a driver need?

Study conclusions:

- Drivers are able to take control of a Level 3 system with a warning of 12 to 15 seconds (paraphrased).
- A Level 3 drive without secondary activities for the driver should not exceed 15 minutes
- In contrast to drivers of conventional vehicles, it may make sense for drivers with an automation system to be distracted by a secondary task that is controlled by the vehicle to prevent tiredness setting in.

Tiredness and level 3 – automated driving,
German Insurers Association/ Insurers Accident Research, 2017
https://udv.de/download/file/fid/10261

Mode Confusion

A situation when the operator is uncertain about the status or behavior of the automation.

Example: Due to a lack of feedback from the ACC (the system accelerates and decelerates without any indication it will do so), drivers could only react after the system had performed the action or when they realized the system was not taking action as expected.

http://www.cargroup.org/wp-content/uploads/2017/02/IMPACT-OF-AUTOMATED-VEHICLE-TECHNOLOGIES-ON-DRIVER-SKILLS.pdf

SAE J3087 –Automatic Emergency Braking HMI

Suggested rating system for AEB encourages “Deactivation of AEB system is NOT accomplished via a single command (e.g. button push)”

…and discourages “Deactivation of AEB system is accomplished via a single command (e.g., button push)”

Mis-use of automation – SAE J3114

Abuse: The intentional mis-use of a driving automation system

Eutactic Behavior – Use outside of design intent

Disuse – usually the result of a complete lack of trust

Over-reliance, bias & complacency

Lack of receptivity – failure to respond to automation warning

Human Factors

Summary:

- Human-machine interaction is key to public trust in autonomy
- Driver involvement is necessary in Level 2-3 vehicles
- Drivers need to time to re-engage with the vehicle and environment after a period of autonomous operation

Liability and Litigation

By the end of this module you will:

- Know the current and future status of ADAS reconstruction
- Have a method for evaluating ADAS-involved accidents

Liability and litigation – current & future status

- Current status: Drivers can blame technology for what may be driver error
  - "To a user who experiences it, a driving automation system performance limitation may be indistinguishable from a driving automation system failure"[1]

- Manufacturers can blame the driver for abuse and eutactic behavior (not complying with design intent and/or owner's manual)

- Future status: Manufacturers may accept liability if vehicle was operating within design parameters. Insurance rates may lower.

1 Human Factors Definitions, SAE J3114, 2016

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Forensic Approach to ADAS Accidents

Investigation: Determine if vehicle was equipped with ADAS technology

Start by looking for windshield camera and bumper radar

Analysis: Correlate EDR data to expected ADAS performance. Compare design parameters to crash scenario (ODD)

Testing: Replicate crash conditions with exemplar vehicle; how does it respond?

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Forensic Approach to ADAS Accidents

Investigation: Was the vehicle equipped with ADAS technology?

Analysis: Did it perform as designed?

Effectiveness: Could it have mitigated or prevented the accident?

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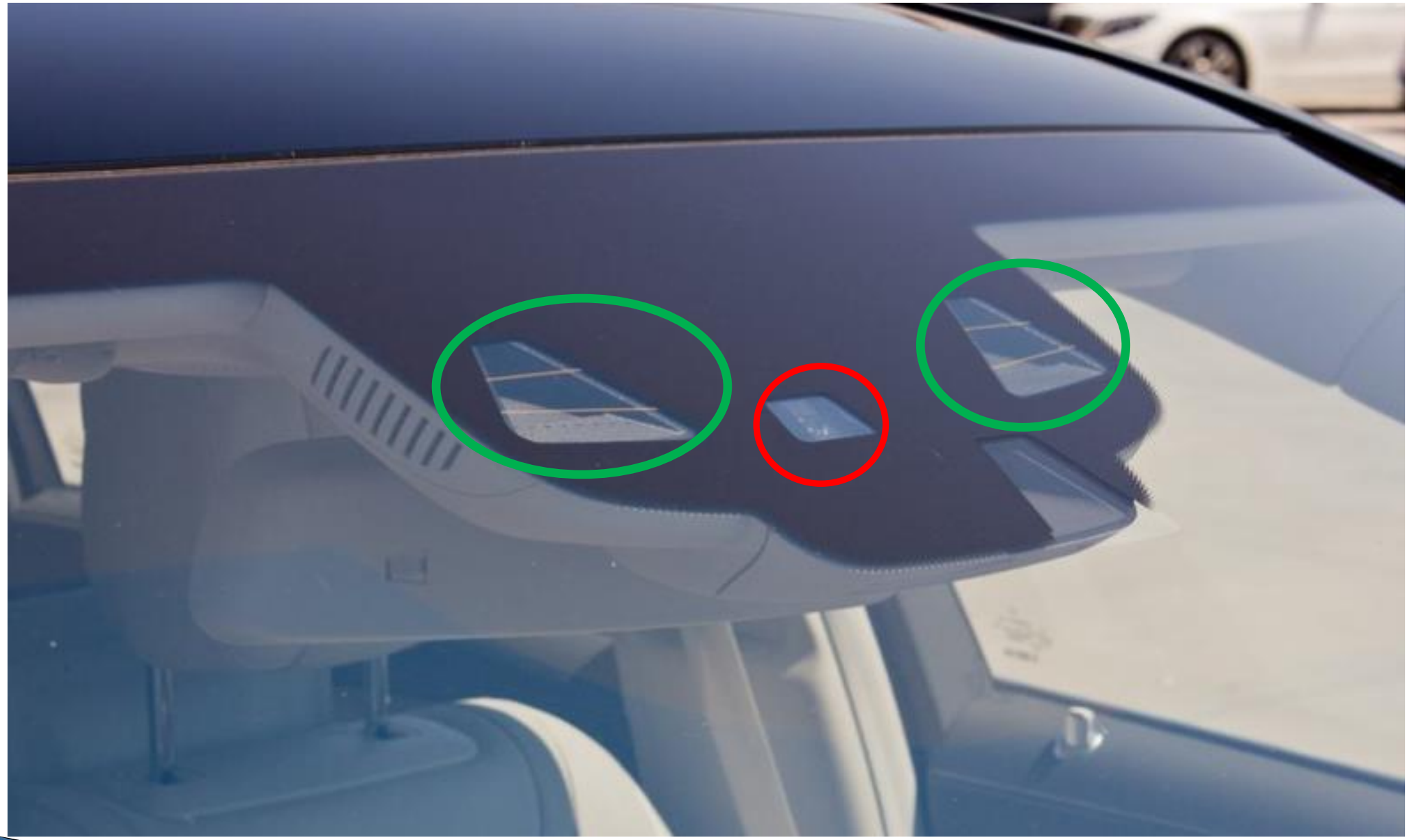

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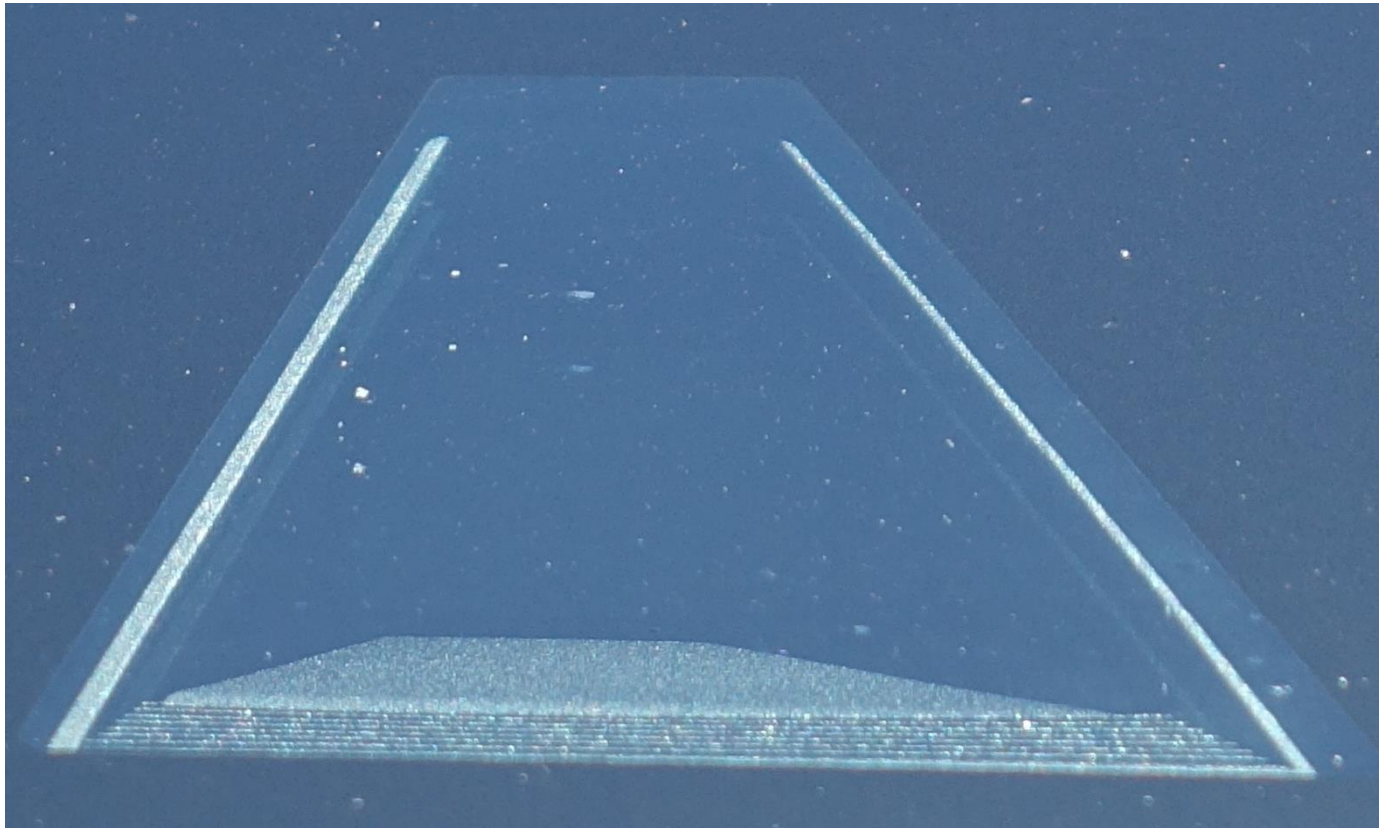
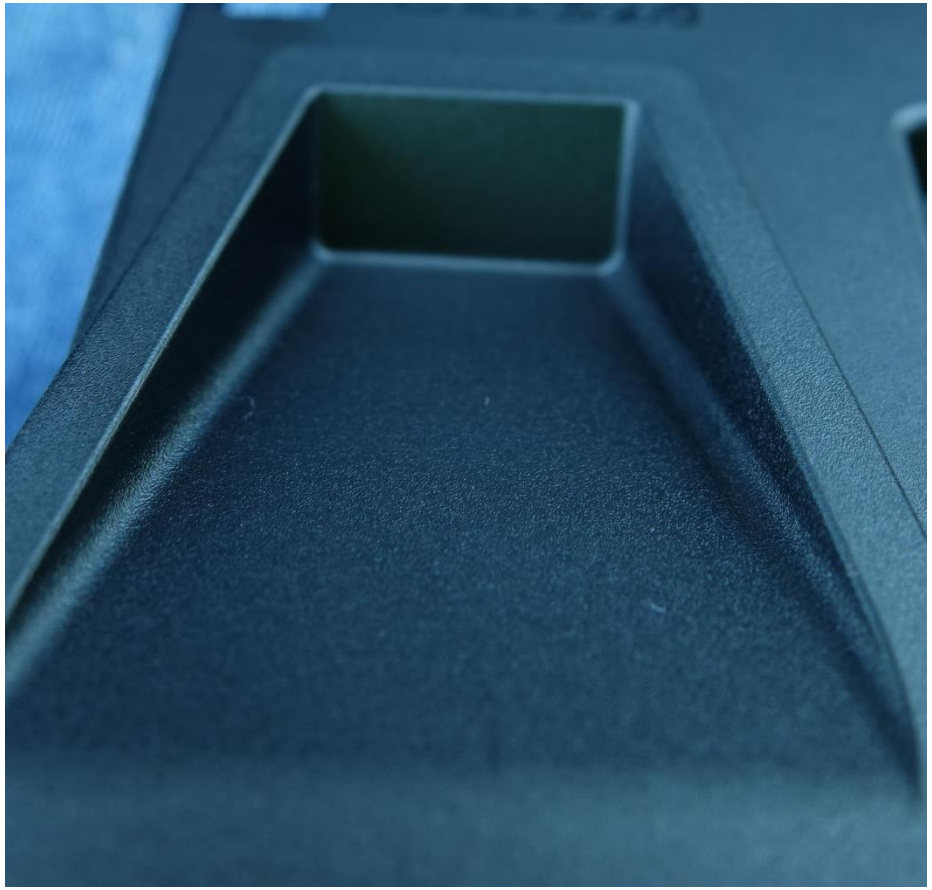
2019 Ford Transit – Camera opening with plastic insert





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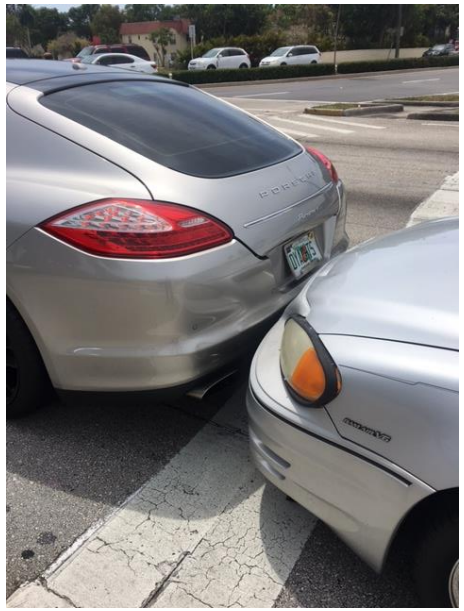
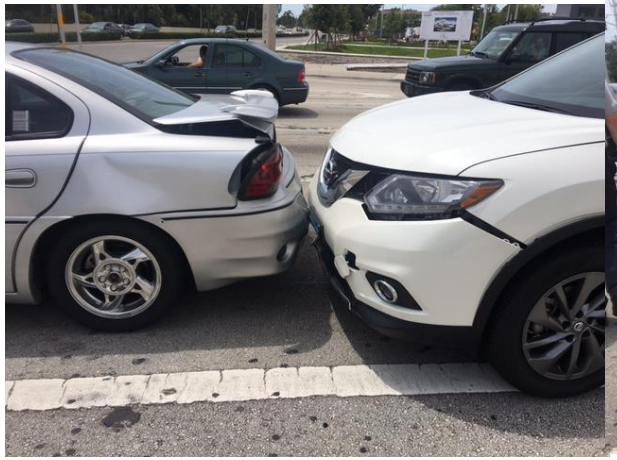
Liability and litigation – forensic approaches

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Liability and litigation – forensic approaches

Driver Awareness Package

- Safety Alert Seat
- Forward Collision Alert
- Low Speed Forward Automatic Braking
- Lane Keep Assist with Lane Departure Warning
- Lane Change Alert
- Rear Cross Traffic Alert
- Side Blind Zone Alert[14]
- IntelliBeam automatic high-beam On/Off

If the vehicle has Forward Collision Alert (FCA), it also has FAB, which includes Intelligent Brake Assist (IBA).

FAB and IBA can be disabled through vehicle personalization. See "Collision/

Liability and litigation – forensic approaches

Pre-Crash Data -5.0 to -0.5 sec (Event Record 1)

Times (sec)	Accelerator Pedal, % Full (Accelerator Pedal Position)	Service Brake (Brake Switch Circuit State)	Engine RPM (Engine Speed)	Engine Throttle, % Full (Throttle Position)	Speed, Vehicle Indicated (Vehicle Speed) (MPH [km/h])
-5.0	0	Off	1408	8	35 [57]
-4.5	0	Off	1344	8	35 [56]
-4.0	0	On	1280	11	35 [56]
-3.5	0	On	1152	8	34 [55]
-3.0	0	On	1152	8	34 [54]
-2.5	0	On	1152	8	33 [53]
-2.0	0	On	1088	8	32 [52]
-1.5	0	Off	1088	8	32 [51]
-1.0	0	On	1024	8	31 [50]
-0.5	0	On	832	8	24 [39]

0.64G, 8mph dV

2017 Cadillac Escalade ESV Platinum

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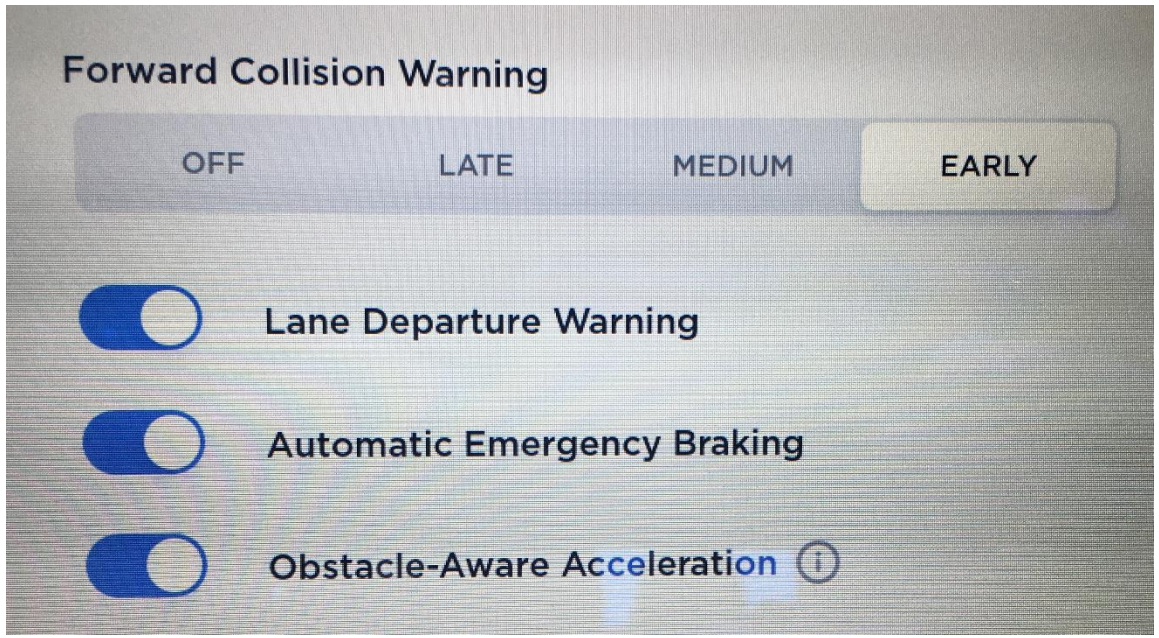
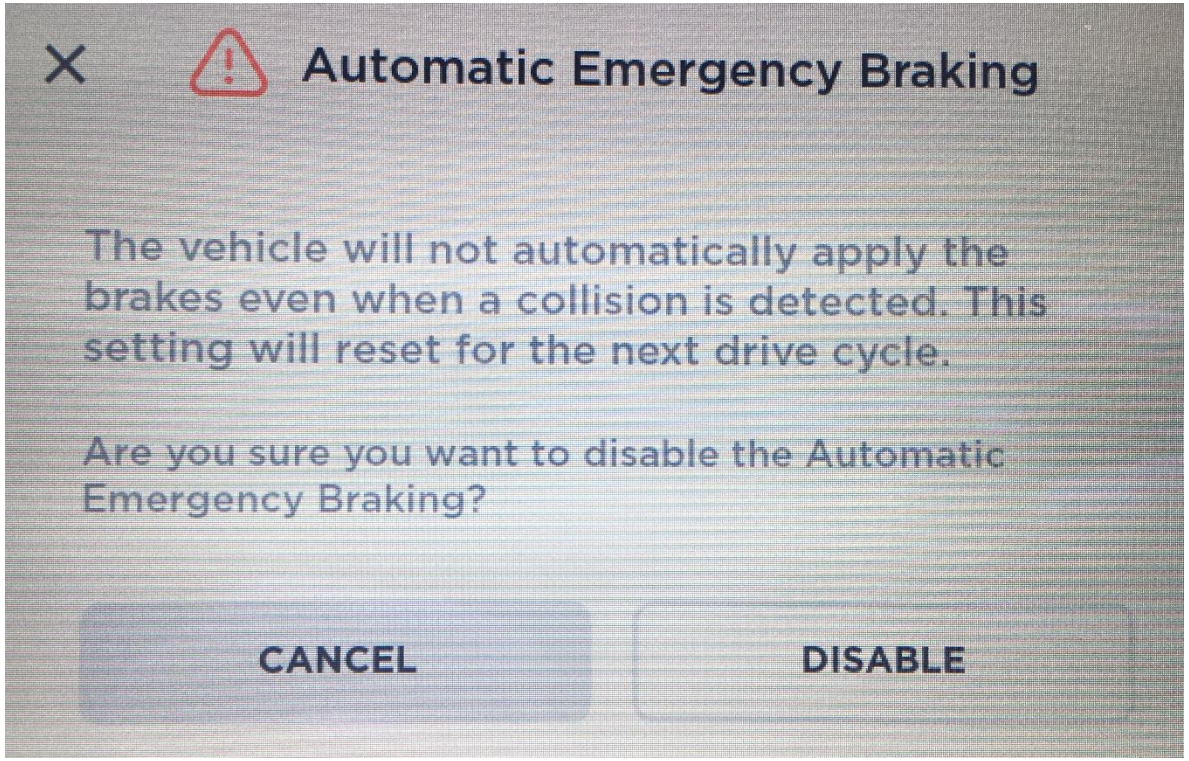
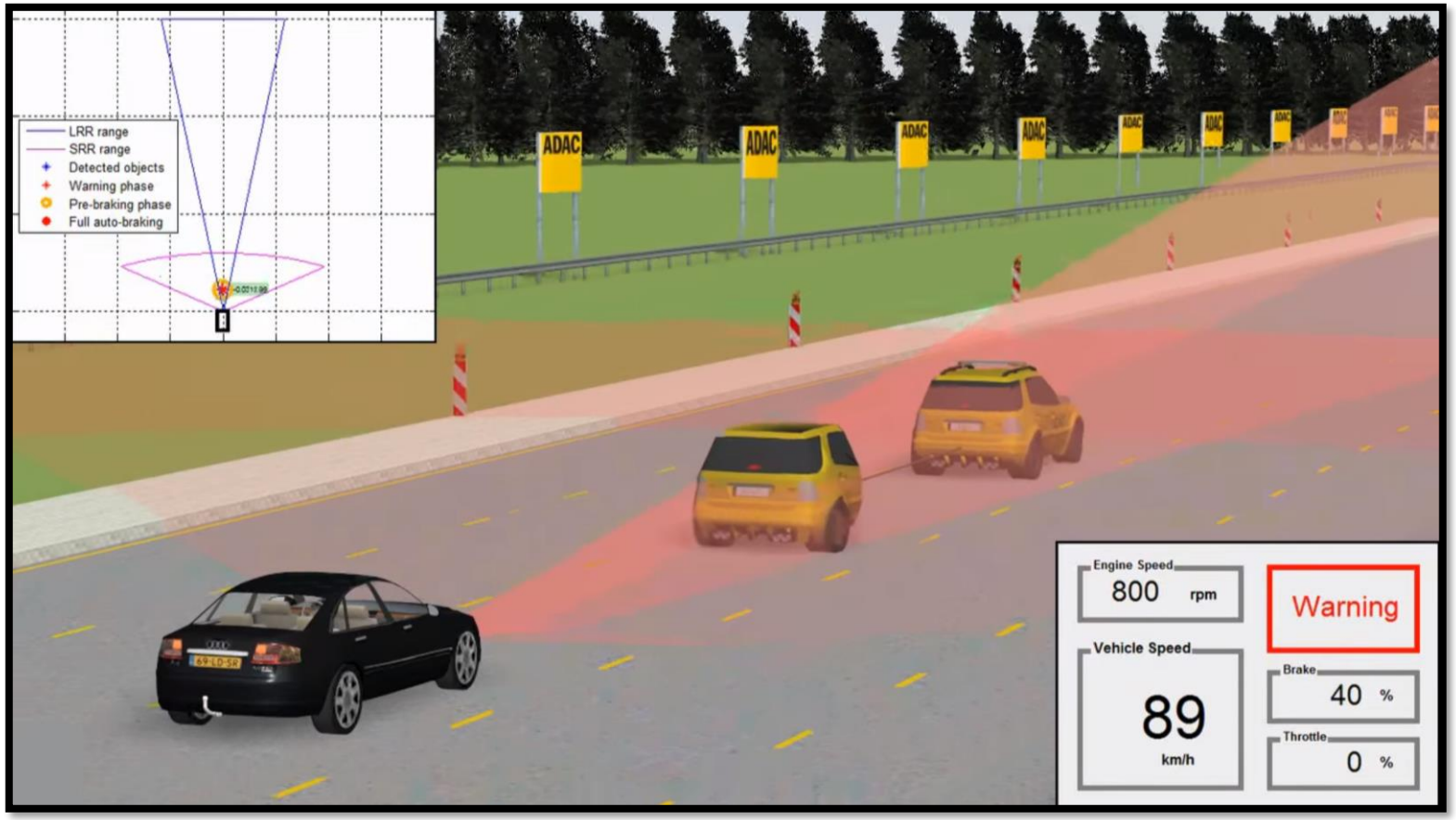
Autobrake disabled - Tesla





Simulation – PreScan by Tass



https://tass.plm.automation.siemens.com/prescan

MobilEye's Responsibility-Sensitive Safety Plan for AVs

1. Keep a safe distance from the car in front of you, so that if it will brake abruptly you will be able to stop in time.
2. Keep a safe distance from cars on your side, and when performing lateral manoeuvres and cutting-in to another car's trajectory, you must leave the other car enough space to respond.
3. You should respect "right-of-way" rules, but "right-of-way" is given not taken.
4. Be cautious of occluded areas, for example, a little kid might be occluded behind a parked car.

On a Formal Model of Safe and Scalable Self-driving Cars - Shai Shalev-Shwartz, Shaked Shammah, Amnon Shashua, Mobileye, 2017

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MobilEye's Responsibility-Sensitive Safety Plan for AVs

Consider a vehicle *c* whose policy complies with the following constraints:

1. The policy can react to the environment in less than *T* seconds
2. The policy never accelerates longitudinally by more than $a_{maxaccel}$, never decelerates longitudinally by more than $a_{maxbrake}$, and never drives faster than v_{limit}.
3. The policy never accelerates laterally by more than $a_{latmaxaccel}$ and in unstructured scenarios, it never changes the yaw rate by more than $h''_{maxaccel}$
4. If the situation at time t is dangerous w.r.t. some other road user, then the policy adheres to the constraints of the proper response (as in Definition 9 or Definition 14).

Then *c* is guaranteed to never be responsible for accidents.

On a Formal Model of Safe and Scalable Self-driving Cars - Shai Shalev-Shwartz, Shaked Shammah, Amnon Shashua, Mobileye, 2017

MobilEye's Responsibility-Sensitive Safety Plan for AVs

The type of errors a sensing system might make:

- False negative: the sensing system misses an object
- False positive: the sensing system indicates a "ghost" object
- Inaccurate measurements: the sensing system correctly detects an object but incorrectly estimates its position or speed
- Inaccurate semantic: the sensing system correctly detects an object but misinterpret its semantic meaning, for example, the color of a traffic light

Sensing error risk can be reduced with redundant sensors (camera, radar, & lidar)

On a Formal Model of Safe and Scalable Self-driving Cars - Shai Shalev-Shwartz, Shaked Shammah, Amnon Shashua, Mobileye, 2017

Liability and Litigation

- Summary:
- Currently, drivers may attempt to blame ADAS features for causing or not preventing an accident
- You have a method for evaluating ADAS-involved accidents

Electronic Data (EDR)

By the end of this module, you will:

- Know what EDR is available, and how it can be used in ADAS-involved accidents
- Have a workflow for manufacturer-specific EDRs

2017 Lincoln Data Elements

Pre-Crash Data -5 to 0 sec [5 samples/sec] (Second Record) - Table 2 of 2

Time (sec)	Brake Pre-Charge Request	Brake Assist Sensitivity Level	Brake Deceleration Request (m/s^2)	Large Driver Steering or Accel Pedal Input	Collision Mitigation System Fault	Collision Mitigation System Enabled
- 5.0	N/A	N/A	N/A	N/A	N/A	N/A
- 4.8	N/A	N/A	N/A	N/A	N/A	N/A
- 4.6	N/A	N/A	N/A	N/A	N/A	N/A
- 4.4	N/A	N/A	N/A	N/A	N/A	N/A
- 4.2	N/A	N/A	N/A	N/A	N/A	N/A
- 4.0	N/A	N/A	N/A	N/A	N/A	N/A
- 3.8	N/A	N/A	N/A	N/A	N/A	N/A
- 3.6	N/A	N/A	N/A	N/A	N/A	N/A
- 3.4	N/A	N/A	N/A	N/A	N/A	N/A
- 3.2	N/A	N/A	N/A	N/A	N/A	N/A
- 3.0	N/A	N/A	N/A	N/A	N/A	N/A
- 2.8	N/A	N/A	N/A	N/A	N/A	N/A
- 2.6	N/A	N/A	N/A	N/A	N/A	N/A
- 2.4	N/A	N/A	N/A	N/A	N/A	N/A
- 2.2	N/A	N/A	N/A	N/A	N/A	N/A
- 2.0	N/A	N/A	N/A	N/A	N/A	N/A
- 1.8	N/A	N/A	N/A	N/A	N/A	N/A
- 1.6	N/A	N/A	N/A	N/A	N/A	N/A
- 1.4	N/A	N/A	N/A	N/A	N/A	N/A
- 1.2	N/A	N/A	N/A	N/A	N/A	N/A
- 1.0	N/A	N/A	N/A	N/A	N/A	N/A
- 0.8	N/A	N/A	N/A	N/A	N/A	N/A
- 0.6	N/A	N/A	N/A	N/A	N/A	N/A
- 0.4	N/A	N/A	N/A	N/A	N/A	N/A
- 0.2	N/A	N/A	N/A	N/A	N/A	N/A
0.0	N/A	N/A	N/A	N/A	N/A	N/A

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2018 Ford Edge Data Elements

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 1 of 2

Time (sec)	Brake Pre-Charge Request	Brake Assist Sensitivity Level	Brake Deceleration Request (m/s^2)	Brake Deceleration Request Enable	Large Driver Steering or Accel Pedal Input	Collision Mitigation System Fault
- 5.0	No PreCharge Request	Normal	-20	No	No	No
- 4.8	No PreCharge Request	Normal	-20	No	No	No
- 4.6	No PreCharge Request	Normal	-20	No	No	No
- 4.4	No PreCharge Request	Normal	-20	No	No	No
- 4.2	No PreCharge Request	Normal	-20	No	No	No
- 4.0	No PreCharge Request	Normal	-20	No	No	No

2019 Ford F-450 Data Elements

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record)

Time (sec)	Brake Pre-Charge Request	Brake Assist Sensitivity Level	Brake Deceleration Request (m/s^2)	Brake Deceleration Request Enable	Large Driver Steering or Accel Pedal Input	Collision Mitigation System Fault	Collision Mitigation System Enabled
- 5.0	No PreCharge Request	Normal	0	No (no CMbB deceleration request)	No	No (not denied)	No
- 4.8	No PreCharge Request	Normal	0	No (no CMbB deceleration request)	No	No (not denied)	No

2014 Jeep EDR data elements

- Adaptive Cruise Control (ACC) Status (if equip.)- “Off” indicates that all cruise control functionality is disabled; “NCC_On” indicates that the Normal Cruise Control system is turned on; “NCC_Set” indicates the Normal Cruise Control is actively controlling vehicle speed; “ACC_On” indicates that ACC is turned on; “ACC_Set” indicates that the ACC is actively controlling vehicle speed. If the value is SNA for all time stamps, then the vehicle is not equipped with ACC.
- ACC Speed Set (if equip.)- This indicates the desired speed in mph that was input by the driver for the ACC system. If the value is SNA for all time stamps, then the vehicle is not equipped with ACC.

- Forward Collision Warning (FCW) (if equip.):
    - Object of Interest Distance - This indicates the actual forward distance to the main object being tracked by the FCW system. “FCW present but not tracking” indicates that the FCW system is not currently tracking an object. If the value is SNA for all time stamps, then the vehicle is not equipped with FCW.
    - FCW System Status - “Off” indicates that the FCW system is off and the FCW Warning Lamp will be “On”. “On-braking” indicates that the FCW system is on with active braking enabled but there will no FCW audible or visual warnings in an FCW event. “On-warning” indicates that the FCW system is on but active braking is disabled. In an FCW event, the driver will only receive FCW audible and visual warnings. “On-full” indicates that the FCW system is fully on with active braking as well as the audible and visual warnings enabled. SNA indicates that the vehicle is not equipped with FCW.

NHTSA NASS Case #N-2015-81-095-V1-ACM

2015 Dodge Charger EDR data elements

Time Stamp (sec)	Tire Pressure, LR (psi)	Tire Pressure, RR (psi)	Cruise Control Status	Cruise Control Engaged	Adaptive Cruise Control Status (if equip.)	ACC Speed Set (MPH) (if equip.)	Object of Interest Distance (if equip.)	Forward Collision Warning Lamp Status (if equip.)
-5.0	43	43	Off	Not Engaged	Off	SNA	SNA	Off
-4.9	43	43	Off	Not Engaged	Off	SNA	SNA	Off
-4.8	43	43	Off	Not Engaged	Off	SNA	SNA	Off
-4.7	43	43	Off	Not Engaged	Off	SNA	SNA	Off
-4.6	43	43	Off	Not Engaged	Off	SNA	SNA	Off
-4.5	43	43	Off	Not Engaged	Off	SNA	SNA	Off
-4.4	43	43	Off	Not Engaged	Off	SNA	SNA	Off
-4.3	43	43	Off	Not Engaged	Off	SNA	SNA	Off

NHTSA NASS Case #N-2015-73-023-V2-ACM

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2018 Chrysler 300 CDR Data Limitations

Master Cylinder Pressure - This indicates the brake pressure applied to the brakes through the brake pedal.

Service Brake - "On" indicates that the brake pedal is physically depressed. Braking from the ABS or FCW systems will not be reported in this data element.

2017 Honda CR-V Data Elements

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 2 of 3

Time Stamp (sec)	PCM Derived Accelerator Pedal Position, % full	Forward Collision Warning (Not Warning/ Warning)	Collision Mitigation Braking System (Not Engaged/ Engaged)	Collision Mitigation Braking System, Forward Collision Warning (On/Off)	Lane Departure Warning (Not Warning/ Warning)	Road Departure Mitigation (Not Engaged/ Engaged)	Road Departure Mitigation, Lane Departure Warning (On/Off)
-5.0	38	Not warning	Not engaged	On	Not warning	Not engaged	On
-4.5	37	Not warning	Not engaged	On	Not warning	Not engaged	On

Time Stamp (sec)	Adaptive Cruise Control (Not Engaged/ Engaged)	Adaptive Cruise Control (On/Off)	Lane Keeping Assist (Not Engaged/ Engaged)	Lane Keeping Assist (On/Off)	Cruise Control (Not Engaged/ Engaged)	Cruise Control (On/Off)
-5.0	Not engaged	Off	Not engaged	Off	Not Engaged	Off
-4.5	Not engaged	Off	Not engaged	Off	Not Engaged	Off

2018 Honda CR-V Data Elements

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 2 of 3

Time Stamp (sec)	PCM Derived Accelerator Pedal Position, % full	Forward Collision Warning (Not Warning/ Warning)	Collision Mitigation Braking System (Not Engaged/ Engaged)	Collision Mitigation Braking System, Forward Collision Warning (On/Off)	Lane Departure Warning (Not Warning/ Warning)	Road Departure Mitigation (Not Engaged/ Engaged)	Road Departure Mitigation, Lane Departure Warning (On/Off)
-5.0	16	Not warning	Not engaged	On	Not warning	Not engaged	Off
-4.5	8	Not warning	Not engaged	On	Not warning	Not engaged	Off
-4.0	11	Not warning	Not engaged	On	Not warning	Not engaged	Off
-3.5	13	Not warning	Not engaged	On	Not warning	Not engaged	Off
-3.0	18	Not warning	Not engaged	On	Not warning	Not engaged	Off
-2.5	18	Not warning	Not engaged	On	Not warning	Not engaged	Off
-2.0	17	Not warning	Not engaged	On	Not warning	Not engaged	Off
-1.5	0	Warning	Engaged	On	Not warning	Not engaged	Off
-1.0	0	Warning	Engaged	On	Not warning	Not engaged	Off
-0.5	0	Warning	Engaged	On	Not warning	Not engaged	Off
0.0	0	Warning	Engaged	On	Not warning	Not engaged	Off

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2018 Honda CR-V Data Elements

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 3 of 3

Time Stamp (sec)	Adaptive Cruise Control (Not Engaged/ Engaged)	Adaptive Cruise Control (On/Off)	Lane Keeping Assist (Not Engaged/ Engaged)	Lane Keeping Assist (On/Off)	Cruise Control (Not Engaged/ Engaged)	Cruise Control (On/Off)
-5.0	Not engaged	On	Not engaged	On	Not Engaged	On
-4.5	Not engaged	On	Not engaged	On	Not Engaged	On
-4.0	Not engaged	On	Not engaged	On	Not Engaged	On
-3.5	Not engaged	On	Not engaged	On	Not Engaged	On
-3.0	Not engaged	On	Not engaged	On	Not Engaged	On
-2.5	Not engaged	On	Not engaged	On	Not Engaged	On
-2.0	Not engaged	On	Not engaged	On	Not Engaged	On
-1.5	Not engaged	On	Not engaged	On	Not Engaged	On
-1.0	Not engaged	On	Not engaged	On	Not Engaged	On
-0.5	Not engaged	On	Not engaged	On	Not Engaged	On
0.0	Not engaged	On	Not engaged	On	Not Engaged	On

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2018 Audi Data Elements

Pre-Crash Data -5 to 0 sec (Record 2)

Time (sec)	Engine RPM (Combustion Engine) (RPM)	ABS Activity	Stability Control	Steering Input (deg)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal (%)	Service Brake Activation
-5.0	2304	No ABS Activity	No ESC Activity	-2	73 [117]	20	Off
-4.5	2240	No ABS Activity	No ESC Activity	-2	73 [117]	0	Off
-4.0	2048	No ABS Activity	No ESC Activity	-2	72 [116]	14	Off
-3.5	1728	No ABS Activity	No ESC Activity	-2	73 [117]	33	Off
-3.0	1792	No ABS Activity	No ESC Activity	-2	73 [118]	42	Off
-2.5	1792	No ABS Activity	No ESC Activity	-2	74 [119]	42	Off
-2.0	1792	No ABS Activity	No ESC Activity	-2	75 [120]	1	Off
-1.5	1728	No ABS Activity	No ESC Activity	-2	75 [121]	0	Off
-1.0	1792	No ABS Activity	No ESC Activity	0	75 [120]	0	Off
-0.5	1728	No ABS Activity	No ESC Activity	-2	75 [120]	0	On (Driver)
0.0	1472	ABS Activity	No ESC Activity	-8	64 [103]	0	On (Driver)

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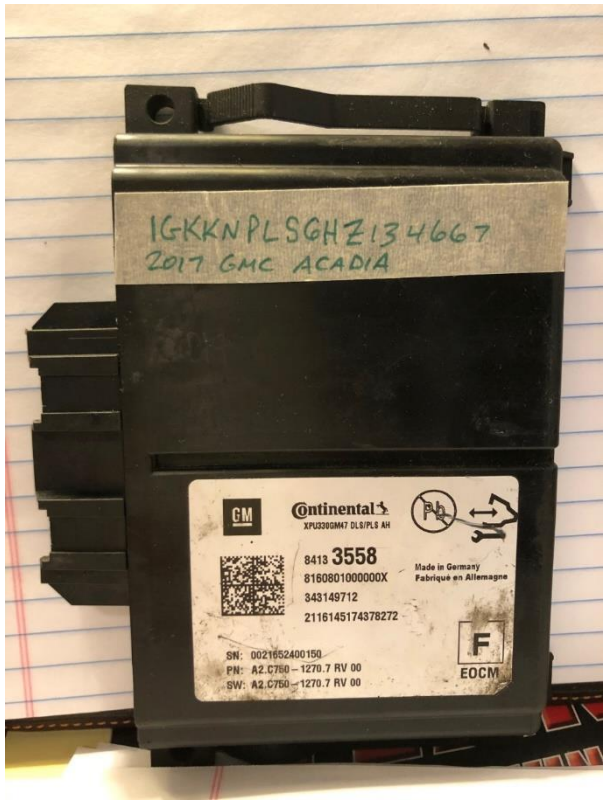
2015 BMW X5 EDR

- CDR Data Limitations: “Service Brake Status only indicates driver initiated braking. An automatic braking (e.g. brake intervention by Adaptive Cruise Control) will not be recorded.”

- NHTSA NASS Case #N-2015-04-092-V2-ACM

GM Active Safety Control Module (ASCM) and External Object Calculating Module (EOCM)

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4, 6, 13	2013 Cadillac	ATS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, left side, near rear, on outboard side of brace, behind trim
4, 6, 13	2013 Cadillac	CTS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
4, 6, 13	2013 Cadillac	Escalade (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center Tunnel
4, 6, 13	2013 Cadillac	Limo/Funeral Coach	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
4, 6, 13	2013 Cadillac	SRX (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the passenger compartment, under the left rear seat
4, 6, 13	2013 Cadillac	XTS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, under the left side trim panel



Bosch CDR v19.3 (December 2019) Help File, Supported Vehicles

Bosch CDR Help File – Cadillac EOCM Data

The GM Active Safety Control Module (ASCM) provides alerts and driver assistance through various active safety systems.

The ASCM is equipped with a data recorder capable of storing one event. Event data recording can be triggered by collision imminent braking or airbag deployments.

Recorded Data

Data can be captured every 80 milliseconds and can record up to 50 records for a total of 4 seconds of data. The data recording window can be either 3.92 seconds before the end of an auto braking event or 3.92 seconds before an airbag deployment.

Data is recorded in non-volatile memory at the next proper power-down of the module. Data may not be stored if the battery power is lost before the power-down.

Bosch CDR v19.3 (December 2019) Help File, Supported Vehicles – EOCM1A, 2013 Cadillac ATS and similar

ASCM/EOCM Data Elements (partial list)

VEHICLE ACTION DATA, -3.92 to 0.00 sec

- Master Clock
- Vehicle Longitudinal Acceleration
- Vehicle Lateral Acceleration
- Primary IMU Lateral Acceleration
- Secondary IMU Lateral Acceleration
- Primary IMU Longitudinal Acceleration
- Secondary IMU Longitudinal Acceleration
- Primary IMU Yaw Rate
- Secondary IMU Yaw Rate
- Vehicle Path Longitudinal Velocity
- Vehicle Path_Data2 Yaw Rate
- Vehicle Stability Enhancement Status
- Commanded Transmission Gear
- Driver Intended Axel Torque
- Driver Intended Axel Torque MIN
- Actual Axel Torque

DRIVER SELECTION AND ACTION DATA, -3.92 to 0.00 sec

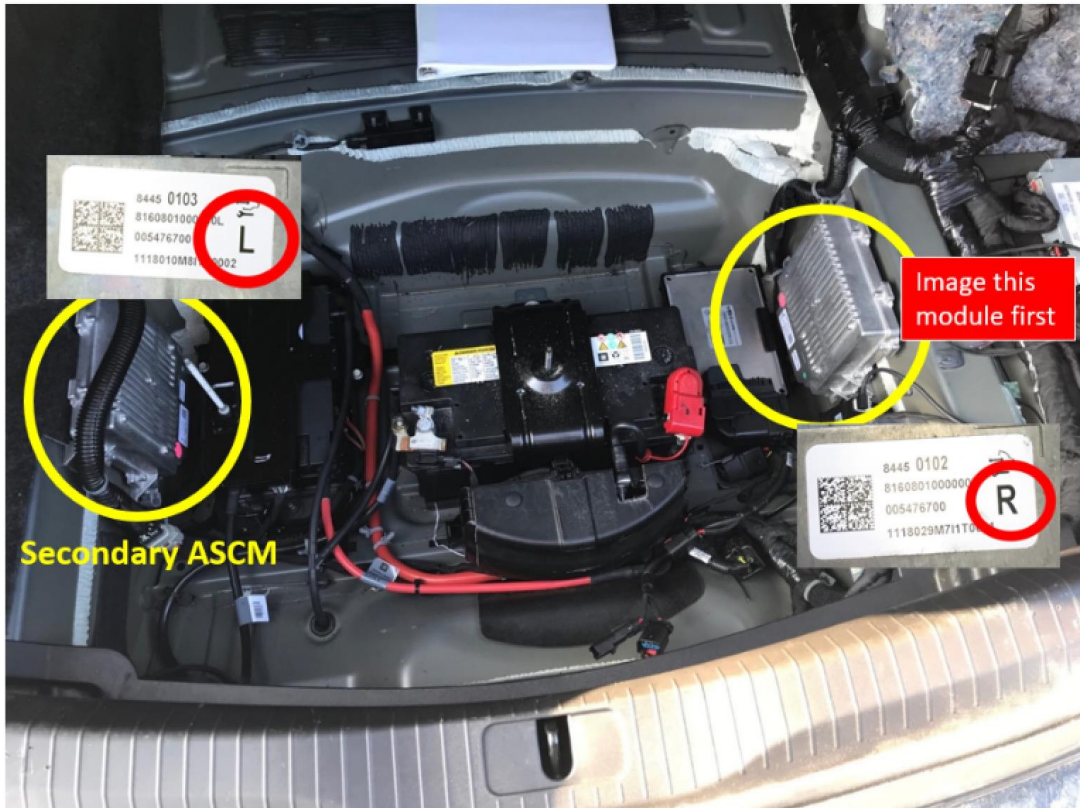
- Driver Initiated Braking
- Brake Pedal Initial Travel Achieved
- Brake Pedal Position
- Accelerator Actual Position
- Vehicle Stability Enhancement System Active
- FAB Current Setting Value
- Steering Wheel Angle
- Steering Wheel Angle Gradient
- FAB Driver Override Status - Pre-Automatic Braking Accelerator Override
- FAB Driver Override Status - Steering Wheel Motion Inhibit
- FAB Driver Override Status - Post-Autonomous Braking Accelerator Pedal Override
- FAB Driver Override Status - Driver Braking

Bosch CDR v19.3 (December 2019) Help File, Supported Vehicles – EOCM1A, 2013 Cadillac ATS and similar

Patent #US9090265B2 - Active safety systems of vehicles with graphical microprocessors

"The example active safety system also includes a redundant EOCM. The redundant EOCM is identical to the primary EOCM. In the event that a fault is detected in the primary EOCM, control is transferred from the primary EOCM to the redundant EOCM…"

ASCM/EOCM Locations



Bosch CDR Help File, v19.3 (December 2019) - 2020 Cadillac CT6

GM Active Safety Control Module (ASCM) and External Object Calculating Module (EOCM)

CDR File Information

User Entered VIN	1G6AV5S86H0154278
User	
Case Number	
EDR Data Imaging Date	01/16/2020
Crash Date	
Filename	TEST ASCM 1G6AV5S86H0154278_ASCM.CDRX
Saved on	Thursday, January 16 2020 at 17:08:29
Imaged with CDR version	Crash Data Retrieval Tool 19.3
Imaged with Software Licensed to (Company Name)	Jones Analytics, LLC
Reported with CDR version	Crash Data Retrieval Tool 19.3
Reported with Software Licensed to (Company Name)	A.B.Moore Forensic Engineering
EDR Device Type	Active Safety Control Module
Event(s) recovered	Active Safety Event

Comments

No comments entered.

Data Limitations

Recorded Crash Events:

The Active Safety Control module (ASCM) can record one event. There are two event triggers. The first type is the Crash Imminent Braking (CIB) event. The second type is the Airbag Deployment event. CIB events can be overwritten by the next CIB event and can be overwritten by a Deployment event. Deployment Events cannot be overwritten or cleared by the ASCM for 40 ignition cycles. After 40 ignition cycles, the recorded Deployment can be overwritten by a CIB or another Deployment event.

Data:

-Data parameters are recorded every 80 milliseconds for 50 samples for a total of 4 seconds.
-Pre-event data is recorded asynchronously. The 0.08 second pre-event data value (most recent recorded data point) is the data point last sampled before the event trigger. That is to say, the last data point may have been captured just before the event trigger but no more than 0.08 second before the event trigger. All subsequent pre-event data values are referenced from this data point.
-All data should be examined in conjunction with other available physical evidence from the vehicle and scene.
-The ignition cycle counter will increment when the power mode cycles from OFF/Accessory to RUN. Applying and removing of battery power to the module will not increment the ignition cycle counter.

Hexadecimal Data:

Data that the vehicle manufacturer has specified for data retrieval is shown in the hexadecimal data section of the CDR report. The hexadecimal data section of the CDR report may contain data that is not translated by the CDR program. The control module contains additional data that is not retrievable by the CDR tool.

GM Active Safety Control Module (ASCM)

Vehicle Action Data, -3.92 to 0.00 sec - Table 1 of 3

Time (sec)	Master Clock (Sec)	Odometer (miles [km])	GPS Latitudinal Position (degrees)	GPS Longitudinal Position (degrees)	Vehicle Speed (MPH [km/h])	Vehicle Longitudinal Acceleration (m/s^2)	Vehicle Lateral Acceleration (m/s^2)
-3.92	103,900	1,071 [1,723]	27.9776	-81.6484	9.9 [16]	0.000	0.000

2017 Cadillac CTS-V

GM Active Safety Control Module (ASCM)

Driver Selection and Action Data, -3.92 to 0.00 sec - Table 2 of 3

Time (sec)	Steering Wheel Angle (degrees)	Steering Wheel Angle Gradient (deg/sec)	FAB Driver Override Status - Pre-Automatic Braking Accelerator Override	FAB Driver Override Status - Steering Wheel Motion Inhibit	FAB Driver Override Status - Post-Autonomous Braking Accelerator Pedal Override	FAB Driver Override Status - Driver Braking
-3.92	0.0000	0	False	False	False	False

2017 Cadillac CTS-V

GM Active Safety Control Module (ASCM)

Tracked Target and Alert Data, -3.92 to 0.00 sec - Table 1 of 3

Time (sec)	FAB Target Time to Collision (Sec)	FAB Target Dynamic Property	FAB Target Lateral Position (meter)	FAB Target Longitudinal Position (meter)	FAB Target Relative Longitudinal Velocity (meter/sec)	FAB Target Lateral Velocity (meter/sec)	FAB Target Relative Longitudinal Acceleration (m/s^2)
-3.92	99.900	Unknown	127.875	-56.000	0.000	0.000	0.000

2017 Cadillac CTS-V

GM Active Safety Control Module (ASCM)

Automated System Control Data, -3.92 to 0.00 sec - Table 3 of 7

Time (sec)	FAB Diagnostic Status - Left Roadway Partially or Fully Limited	FAB Situational Status - Construction Zone	FAB Situational Status - Close/Slow Crossing	FAB Situational Status - Full Speed Range Adaptive Cruse Control Engaged	FAB Situational Status - Motorcycle Target	FAB Situational Status - Pedal Misapplication

2017 Cadillac CTS-V

GM Active Safety Control Module (ASCM)

Automated System Control Data, -3.92 to 0.00 sec - Table 4 of 7

Time (sec)	FAB Situational Status - Oncoming	FAB Situational Status - Left Roadway	FPB Suppression Reason: Above time to collision threshold	FPB Suppression Reason: Pedestrian Moving Away	FPB Suppression Reason: Out of Range	FPB Suppression Reason: Vehicle Speed too Low or too High

2017 Cadillac CTS-V

GM Active Safety Control Module (ASCM)

Automated System Control Data, -3.92 to 0.00 sec - Table 5 of 7

Time (sec)	FPB Suppression Reason: Out of Range or Above Time to Collision Threshold	FPB Suppression Reason: Pedestrian Not Seen	FCA/FPB Suppression Reason: Accelerator or Clutch Pedal Applied	FCA/ FPB Suppression Reason: Brake Pedal Applied	FCA Suppression Reason: Above Time to Collision Threshold	FCA Suppression Reason: Out of Range or Moving Away

2017 Cadillac CTS-V

GM Active Safety Control Module (ASCM)

Automated System Control Data, -3.92 to 0.00 sec - Table 6 of 7

Time (sec)	FCA Suppression Reason: Imminent Alert Open Scenario	FCA Suppression Reason: Speed too Low or Hi Threat Pedestrian	Headway Suppression Reason: High Range, Low Speed, or Opening Scenario	FCA/Headway Suppression Reason: Vehicle out of Path	FCA/Headway Suppression Reason: Oncoming Vehicle	FCA Suppression Reason: Never Before Seen Moving Object and Low Host Speed

2017 Cadillac CTS-V

GM Active Safety Control Module (ASCM)

1G6KT5R66KU****** Time (sec)	Adaptive Cruise Control Active	Adaptive Cruise Control Disengage Reason
-1.68	False	Brake Pressed
-1.60	False	Brake Pressed
-1.52	False	Brake Pressed
-1.44	False	Brake Pressed
-1.36	False	Brake Pressed
-1.28	False	Crash Imminent Braking Auto Braking

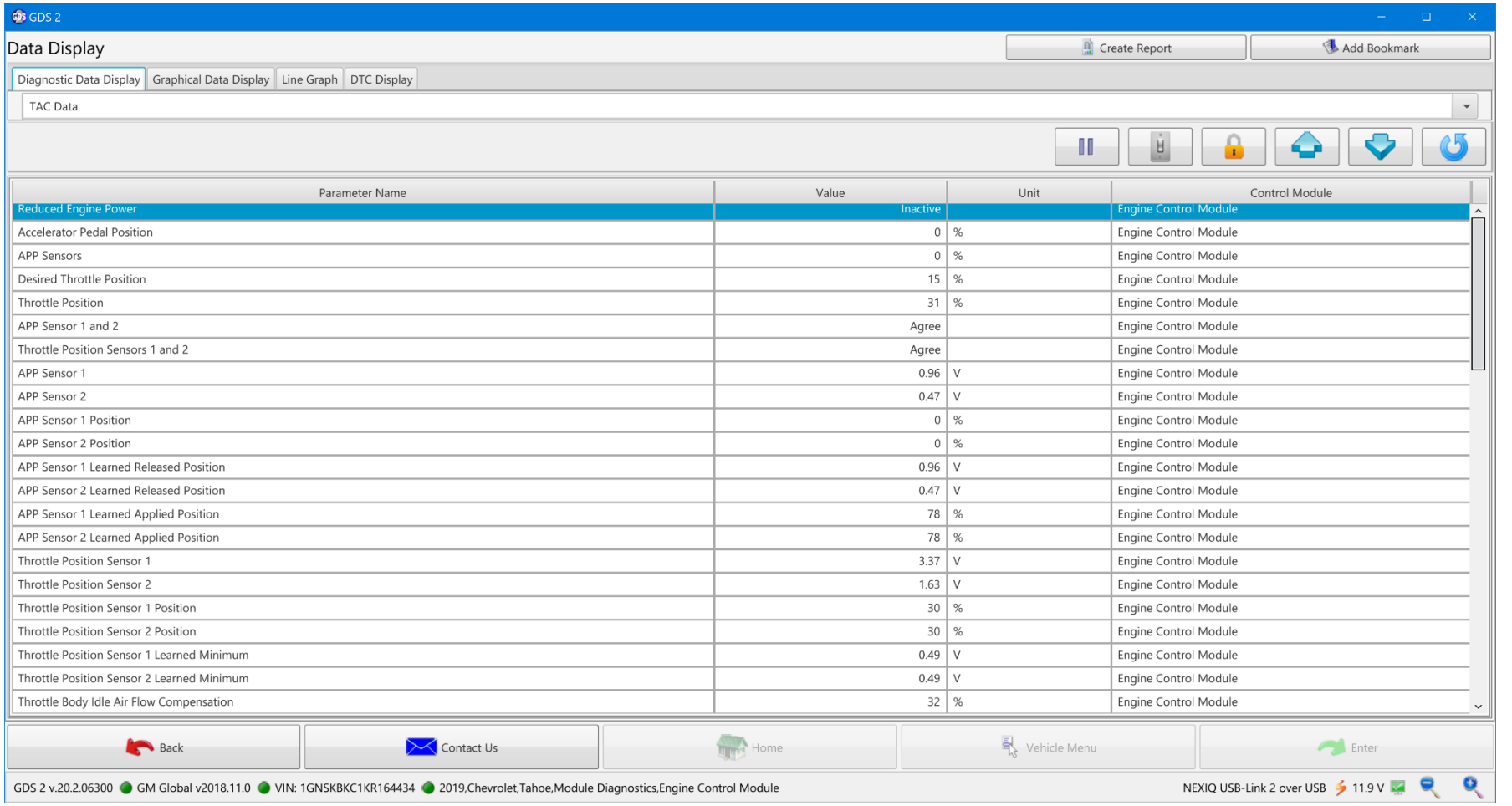
2019 Cadillac CT6

GM Active Safety Control Module (ASCM)

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report. Directional references to sign notation are allfrom the perspective of the driver when seated in the vehicle facing the direction of forward vehicle travel.

Data Element Name	Positive Sign Notation Indicates
Longitudinal Acceleration	Forward
Longitudinal Velocity Change	Forward
Lateral Acceleration	Right to Left
Lateral Velocity Change	Right to Left
Steering Wheel Angle	Counter-clockwise Rotation
Axle Torque	Forward
Lane Centering Control Steering Torque Command	Counter-clockwise Rotation
Yaw Rate	Counter-clockwise Rotation
Vehicle Speed	Forward
Adaptive Cruise Control Auto Brake Request	ISO
Steering Torque	Counter-clockwise Rotation
Target Object Selection Closest in Path - Relative Path Distance	ISO
Collision Preparation System Target Object Selection Highest Threat Time to Collision	ISO

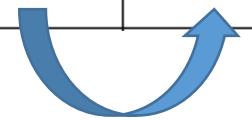
GM's GDS2 Software and Sudden Acceleration



Parameter Name	Value	Unit	Control Module
Reduced Engine Power	Inactive		Engine Control Module
Accelerator Pedal Position	0	%	Engine Control Module
APP Sensors	0	%	Engine Control Module
Desired Throttle Position	15	%	Engine Control Module
Throttle Position	31	%	Engine Control Module
APP Sensor 1 and 2	Agree		Engine Control Module
Throttle Position Sensors 1 and 2	Agree		Engine Control Module
APP Sensor 1	0.96	V	Engine Control Module
APP Sensor 2	0.47	V	Engine Control Module
APP Sensor 1 Position	0	%	Engine Control Module
APP Sensor 2 Position	0	%	Engine Control Module
APP Sensor 1 Learned Released Position	0.96	V	Engine Control Module
APP Sensor 2 Learned Released Position	0.47	V	Engine Control Module
APP Sensor 1 Learned Applied Position	78	%	Engine Control Module
APP Sensor 2 Learned Applied Position	78	%	Engine Control Module
Throttle Position Sensor 1	3.37	V	Engine Control Module
Throttle Position Sensor 2	1.63	V	Engine Control Module
Throttle Position Sensor 1 Position	30	%	Engine Control Module
Throttle Position Sensor 2 Position	30	%	Engine Control Module
Throttle Position Sensor 1 Learned Minimum	0.49	V	Engine Control Module
Throttle Position Sensor 2 Learned Minimum	0.49	V	Engine Control Module
Throttle Body Idle Air Flow Compensation	32	%	Engine Control Module

2015 Camry EDR data elements

Time (sec)	-0.5	0 (TRG)
Vehicle Speed (MPH [km/h])	33.6 [54]	26.7 [43]
Accelerator Pedal, % Full (%)	0.0	0.0
Percentage of Engine Throttle (%)	0.0	0.0
Engine RPM (RPM)	1,500	1,300
Motor RPM (RPM)	Invalid	Invalid
Service Brake, ON/OFF	ON	ON
Brake Oil Pressure (Mpa)	3.94	6.77
Longitudinal Acceleration , VSC Sensor (m/sec^2)	-5.312	-6.963
Yaw Rate (deg/sec)	-14.64	-20.01
Steering Input (degrees)	-67.5	-94.5



0.63G, 7mph dV

NHTSA NASS Case #N-2015-41-037-V2-ACM

Toyota Vehicle Control History

Partial availability in 2013, many models by 2015.

Requires Toyota Techstream subscription

Event Data Recorder (EDR) Developed by Toyota Motor Corporation, SAE 016-01-1495, Motomi Iyoda et al

Toyota Vehicle Control History

TRIGGER ITEM		RECORD DATA				
		BEFORE DETECTION		AFTER DETECTION		SAMPLING CYCLE (SEC.)
ID NO.	ITEM	RECORDING TIME (SEC.)	NUMBER OF POINTS	RECORDING TIME (SEC.)	NUMBER OF POINTS	
2	Accelerator pedal opening angle signal is high during low speed	7.5	15	7.5	16	0.5
3	Accelerator high position in mid to high speed	7.5	15	7.5	16	0.5
4	Accelerator pedal opening angle signal is high immediately after brake pedal is released	7.5	15	7.5	16	0.5
5-1	Accelerator pedal opening angle is medium or higher immediately after shifting to R	5	10	5	11	0.5
5-2	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position					
5-3	Accelerator pedal opening angle is medium or higher immediately after shifting to driving position					
6-1	R position signal input during medium or higher accelerator signal input	5	10	5	11	0.5
6-2	Forward position signal input during medium or higher accelerator signal input					
6-3	Driving position signal input during medium or higher accelerator signal input					
8	Accelerator signal and brake signal input simultaneously	5	10	5	11	0.5
9	Medium or higher accelerator signal input immediately after switching to D or R	5	10	5	11	0.5
10	Medium or higher accelerator signal input during N	5	10	5	11	0.5
12	VSC operation history	5	34	5	35	0.15
13	TRC operation history	5	34	5	35	0.15
14	ABS operation history	5	34	5	35	0.15
15	Sudden braking history	5	34	5	35	0.15
16	Sudden turning history	5	34	5	35	0.15
17	Accelerator pedal opening angle signal is high during cruise control operation	5	10	5	11	0.5
18-1	PCS operation history (warning buzzer operation)	5	10	5	11	0.5
18-2	PCS operation history (warning brake operation)					
18-3	PCS operation history (pre-collision brake assist operation)					
18-4	PCS operation history (prior brake operation)					
18-5	PCS operation history (pre-collision brake operation)					
18-6	PCS operation history (pre-collision seat belt operation)					
18-7	PCS operation history (when acceleration above certain amount is detected)					
19	Torque sensor signal above certain amount is detected during LKA/LDA operation	8	40	0	1	0.2

Event Data Recorder (EDR) Developed by Toyota Motor Corporation, SAE 016-01-1495, Motomi Iyoda et al

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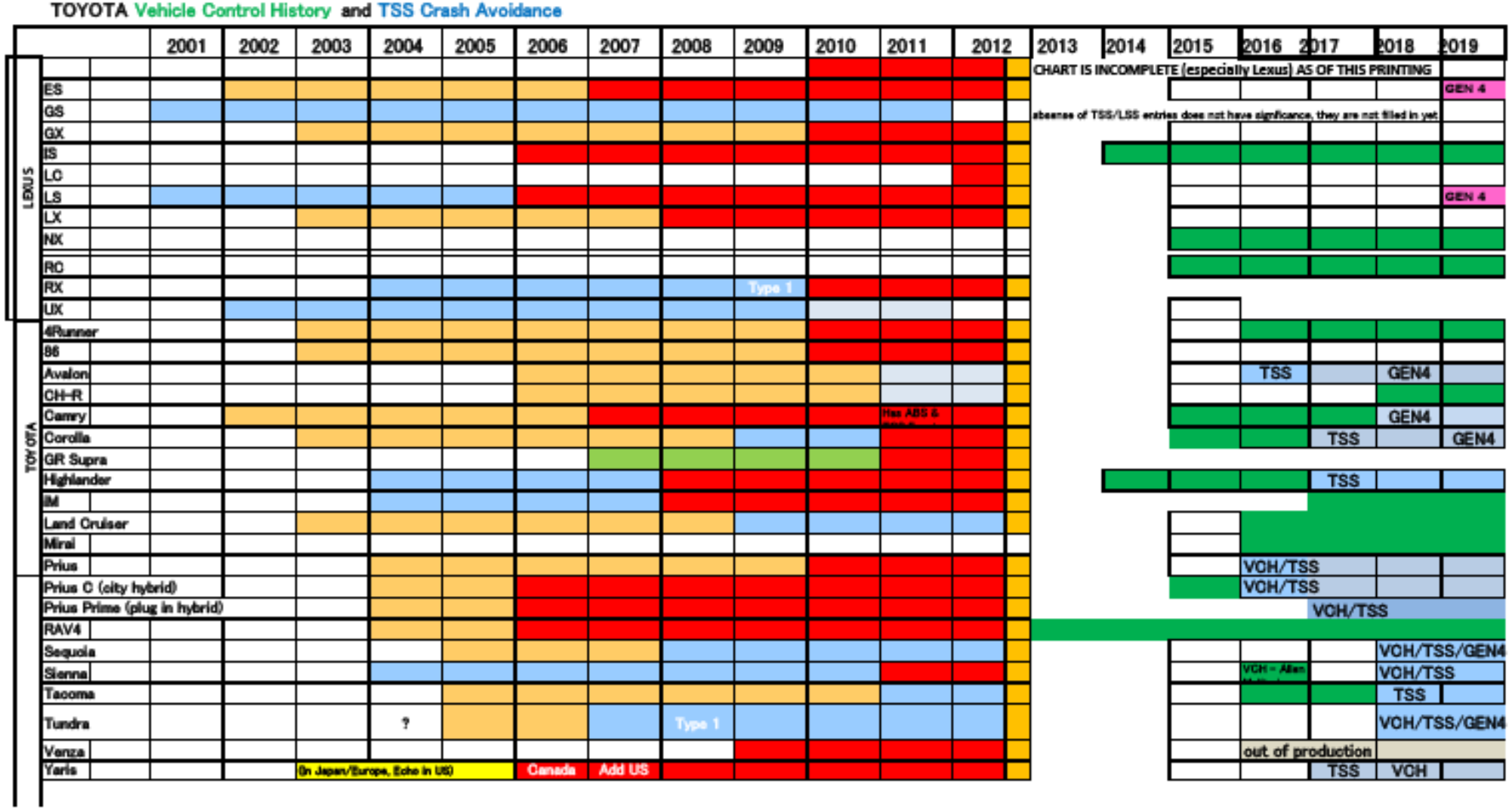

Preliminary

Courtesy of Rick Ruth and Wes VanDiver

Toyota TechStream

techstream LITE

Techstream LITE
OEM Level Diagnostics support for 1996 and later model year Toyota/Lexus/Scion vehicles.

Technical Capabilities Comparison:

	Full TIS Techstream 2.0 TS2UNITM3AMP $6,199.00	Techstream LITE Kit TSLP2S[illegible]01 $1[illegible]45.00
		 (PC not included)
Legacy Model Support with DLC3 (OBD II) • *DTC* • *Datalist* • *Active Test* • *Monitor* • *Utility*	1996 - Present	1996 - Present
Reprogramming	Yes	Yes
Health Check	Yes	Yes
Customize (C-BEST)	Yes	Yes
TIS Professional Diagnostics [illegible]bscription • Library (Repair Manu[illegible]EWD, TSB, and more) • One Techstrea[illegible]agnostic Software License • ECU Repr[illegible]amming Calibrations • IDENTIFIX FIND FIXES FASTER Direct-Hit Access for Toyota/Scion/Lexus vehicles • Visit https://techinfo.toyota.com for details	Two Years *$2,590 Value* (Includes Direct-Hit Access)	One Year *$1,295 Value* (Includes Direct-Hit Access)

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Toyota TechStream

Subscription, $65 for 2 days:

https://techinfo.toyota.com or

https://techinfo.snapon.com/TIS/termbill.aspx?action=new

No need for the Mongoose interface Toyota recommends; the Nexiq USB Link and CDR 900 work fine

Nexiq OBD cable: Diesellaptops.com

- USB Link 1: Item # 448013
- USB Link 2: Item # 493013

Click on the TIS tab:

Click on Full Install:

Find the install file. Right-click, run as administrator

First install the Mongoose Pro MFC driver, even if not used

You may also need, in Windows 10, to temporarily disable Virus & Threat Protection

After install, open Techstream which is now installed on your computer

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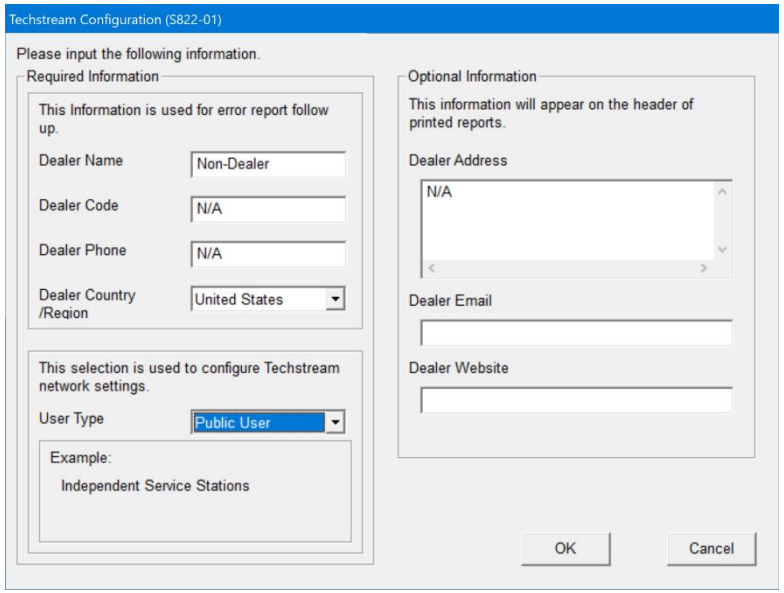
In Techstream:

Go to Setup, VIM Select, and select your Nexiq

Configure Techstream – Set User Type to “Public User”



Click on Software Registration, then again on Software Registration. Follow the directions to create a key.

TIS TECHNICAL INFORMATION SYSTEM

TOYOTA SCION LEXUS

Subscriber Login

Your Account has been locked.

Email ID: Remember Me

Password:

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Clicking 'Subscribe' will take you to a payment website hosted by a Toyota third party vendor. Please refer to the specific privacy and security policies of this third-party hosted website for information regarding the privacy and data security of your activities on this payment website.

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Click the button above to manage your current TIS account.

TIS Featured Information

Click here for a tutorial on our Technical Information System.

Access to vehicle security information is provided for security professionals registered with the NASTF Vehicle Security Professional Registry. For more information click here.

What Is TIS? | **Manuals** | **Keycode Check**

The Technical Information System or TIS is your service support source for all Toyota produced vehicles marketed in North America. TIS includes all of the vital information you'll need to effectively service most **1990 and later** Toyota products.

Emergency Responder and Dismantler Information is available HERE free of charge. Select the Manuals tab above to view available Emergency Response Guides, Dismantling Guides and Owner's Manuals.

TIS is available in three different levels. The table below summarizes the features of each subscription level.

Subscription Level Features	Standard	Professional Diagnostic	Security Professional
TIS Library			
Service Bulletins	●	●	●
Repair Manuals	●	●	●
Wiring Diagrams	●	●	●
Technical Training	●	●	●
Other Technical Information	●	●	●
Diagnostics / Reprogramming			
Techstream Scantool Software		●	●
ECU Calibrations		●	●
Identifix Direct-Hit		●	●
Security (U.S. and Mexico VINs only)			
Key Codes			●
Immobilizer / Smart Reset			●

>>Learn More

Beyond service information, several additional resources are available to assist in the swift and accurate service of Toyota, Scion, and Lexus vehicles. Follow the links below for more details.

>>Special Service Tools
>>Technical Training & Printed Materials
>>Emergency Response Presentation
>>Emergency Response Presentation PDF Printout

Professional Resources

Safety Recall and Service Campaign Information
Get information on open safety recalls and service campaigns released since 1999 for Toyota, Lexus and Scion vehicles intended for sale or sold in the U.S., U.S. Territories, and Mexico using our quick and easy recall and campaign lookup:

>>Lookup Safety Recalls & Service Campaigns by VIN

Factory Service Information and Diagnostics for your shop!
Need an inexpensive solution for diagnosing & repairing Toyota, Scion, and Lexus vehicles?
The NEW Techstream Lite Kit allows you to use your own PC with Techstream Software.

>>Learn More About Techstream Lit
>>Scantools & Vehicle Reprogrammin

Find a Dealership near you for a full selection of Genuine Parts and Accessories.

Learn all about factory authorized parts and repair at Toyota's new Wholesale Parts website. Or use the locator form to find a STAR (Support To Automotive Repair) Dealer.

>>Wholesale Part
>>Star Dealer

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To renew a subscription, don't login on the left. Click on the circled "My Account" button and follow from there.

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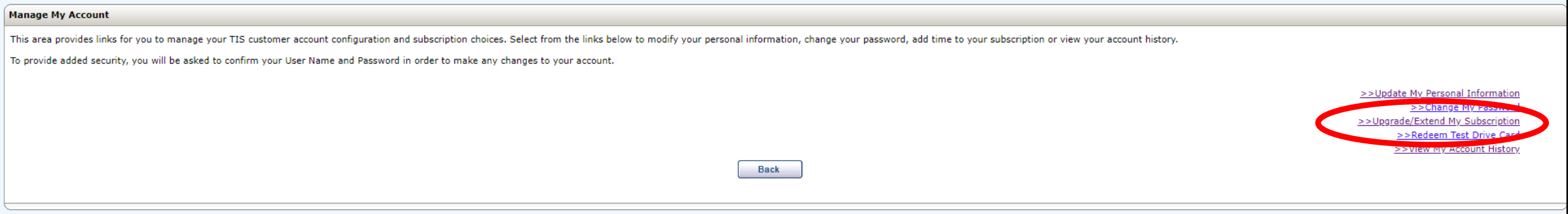
Toyota Techstream



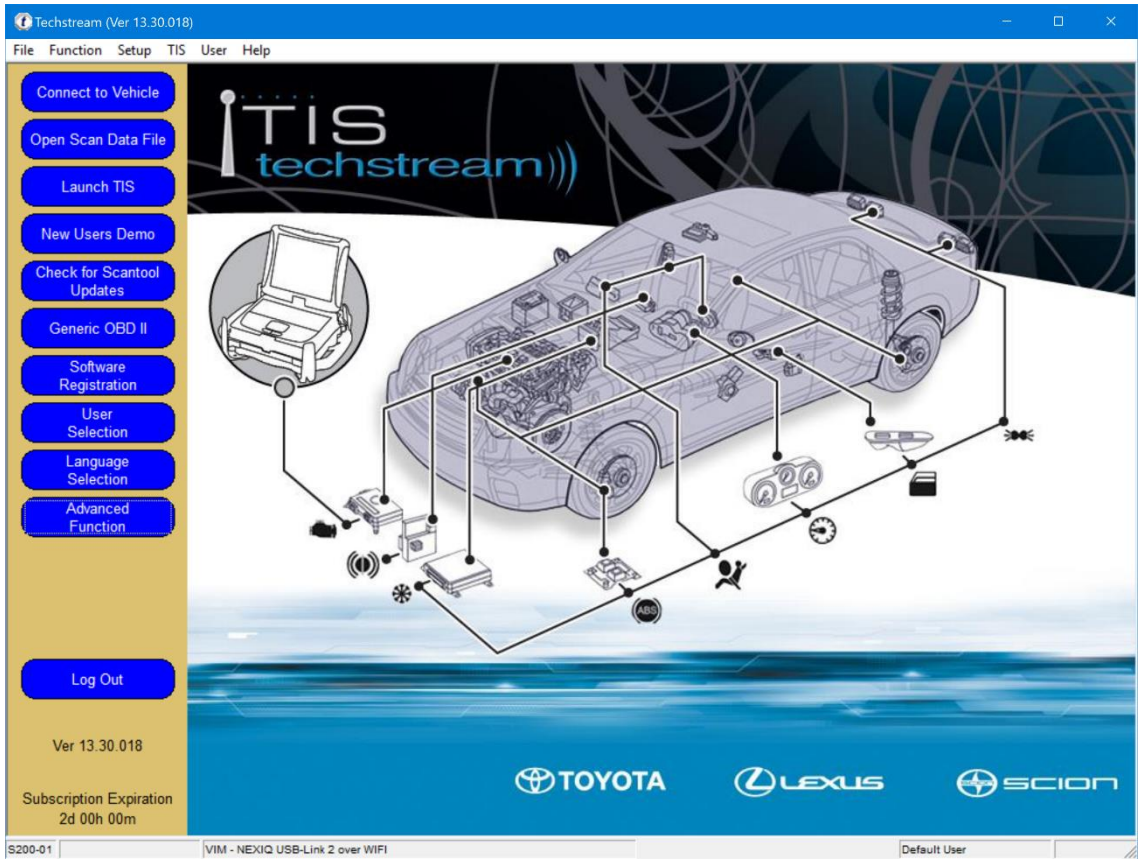
Click on Upgrade/Extend My Subscription

After you do this, wait for an email for account re-enable, and click on the link in it.

Then you should be able to log in using the upper left login on the prior slide.

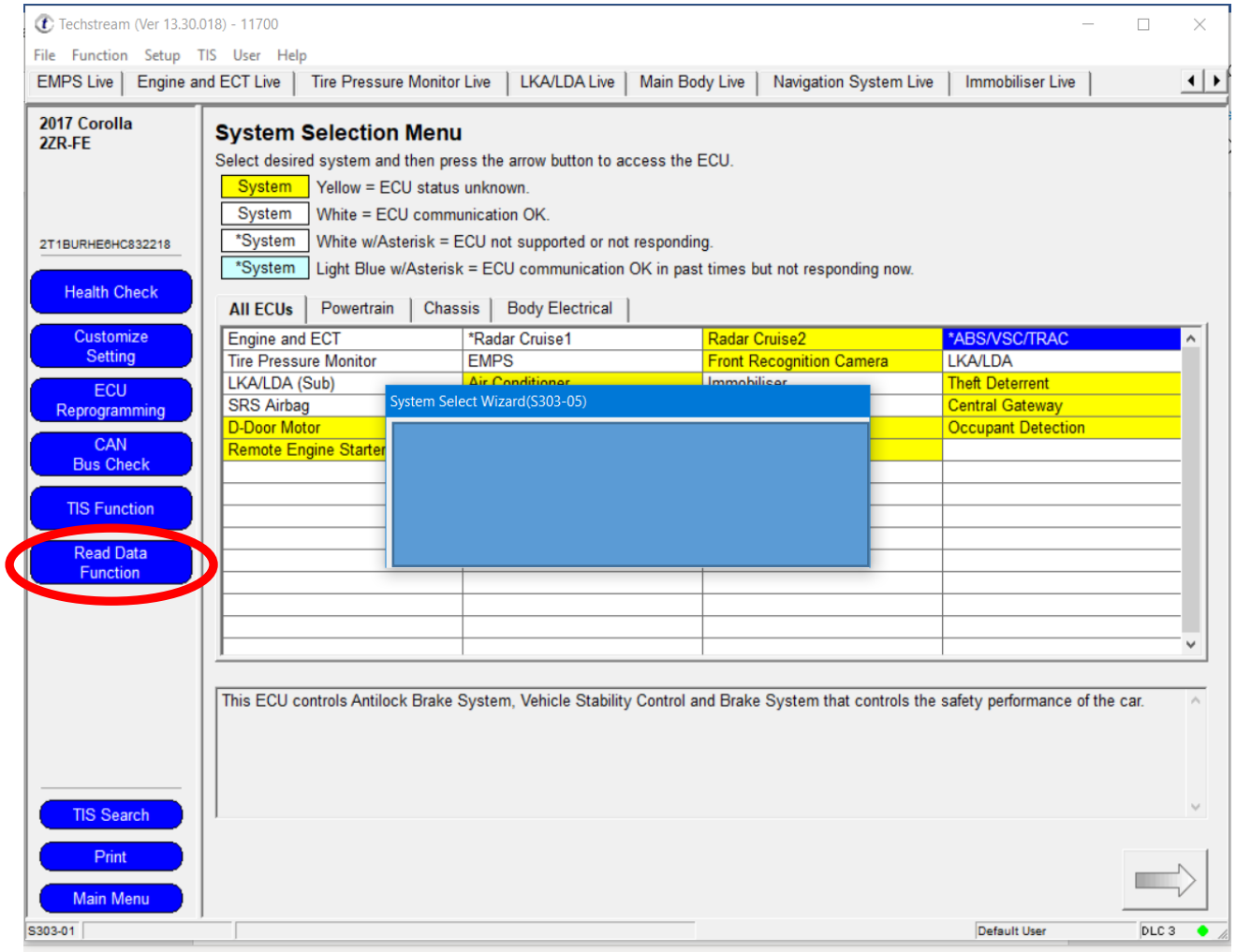
Techstream Main Window

Click on “Connect To Vehicle”



How to get Toyota VCH Data

From the main Techstream screen, click on “Read data”, then “Vehicle Control History”



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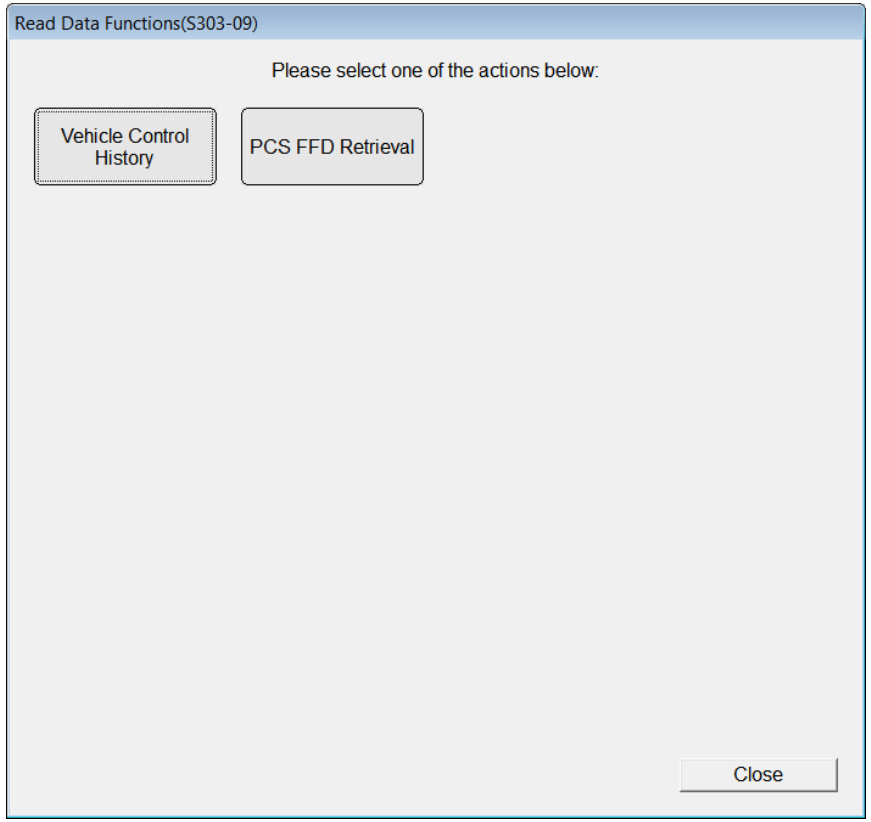
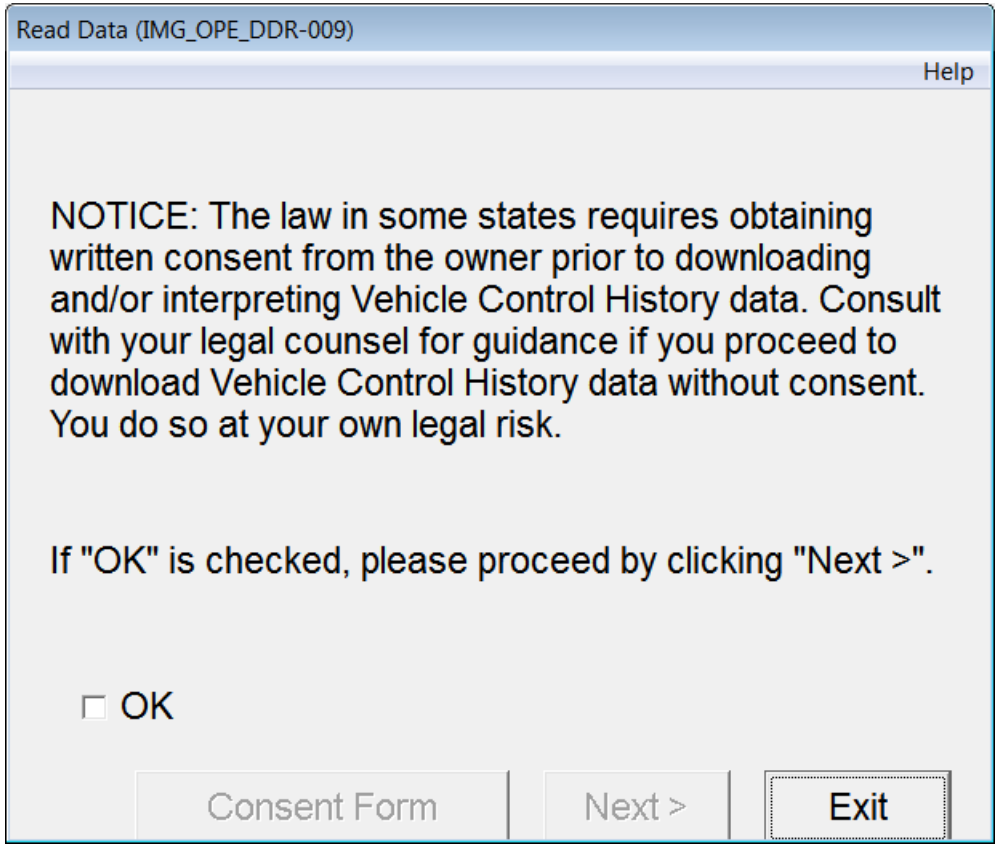
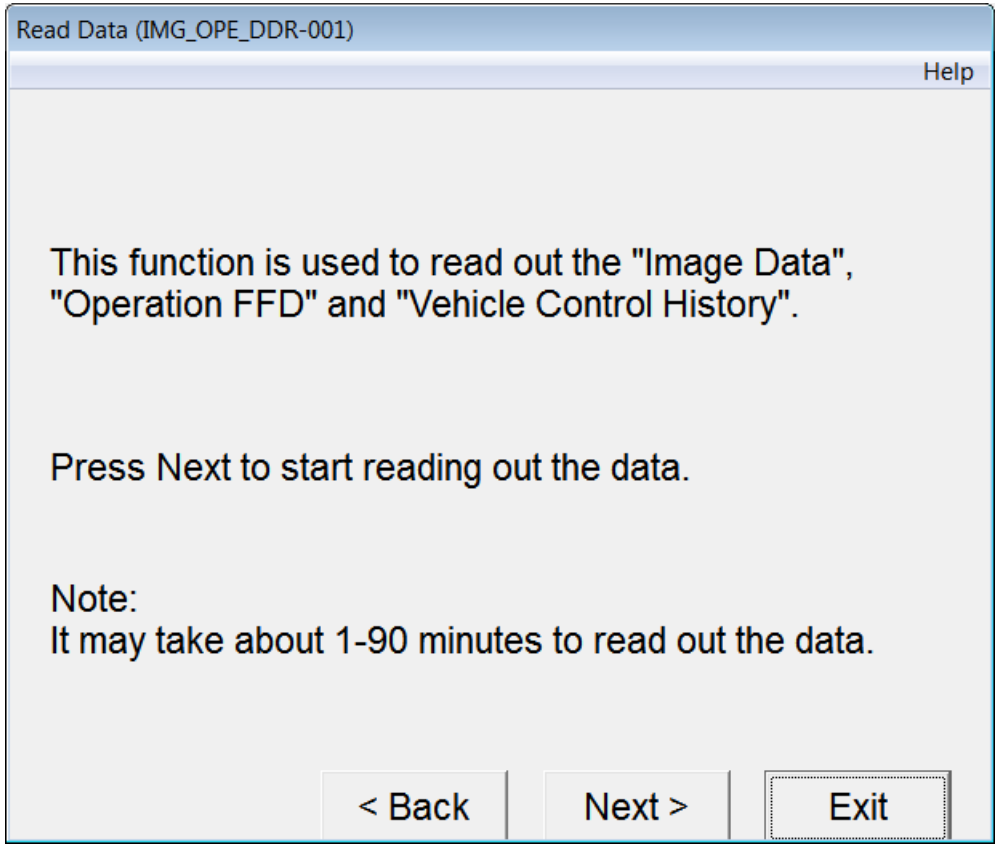
How to get Toyota VCH Data



From the main Techstream screen, click on “Read data”, then “Vehicle Control History”

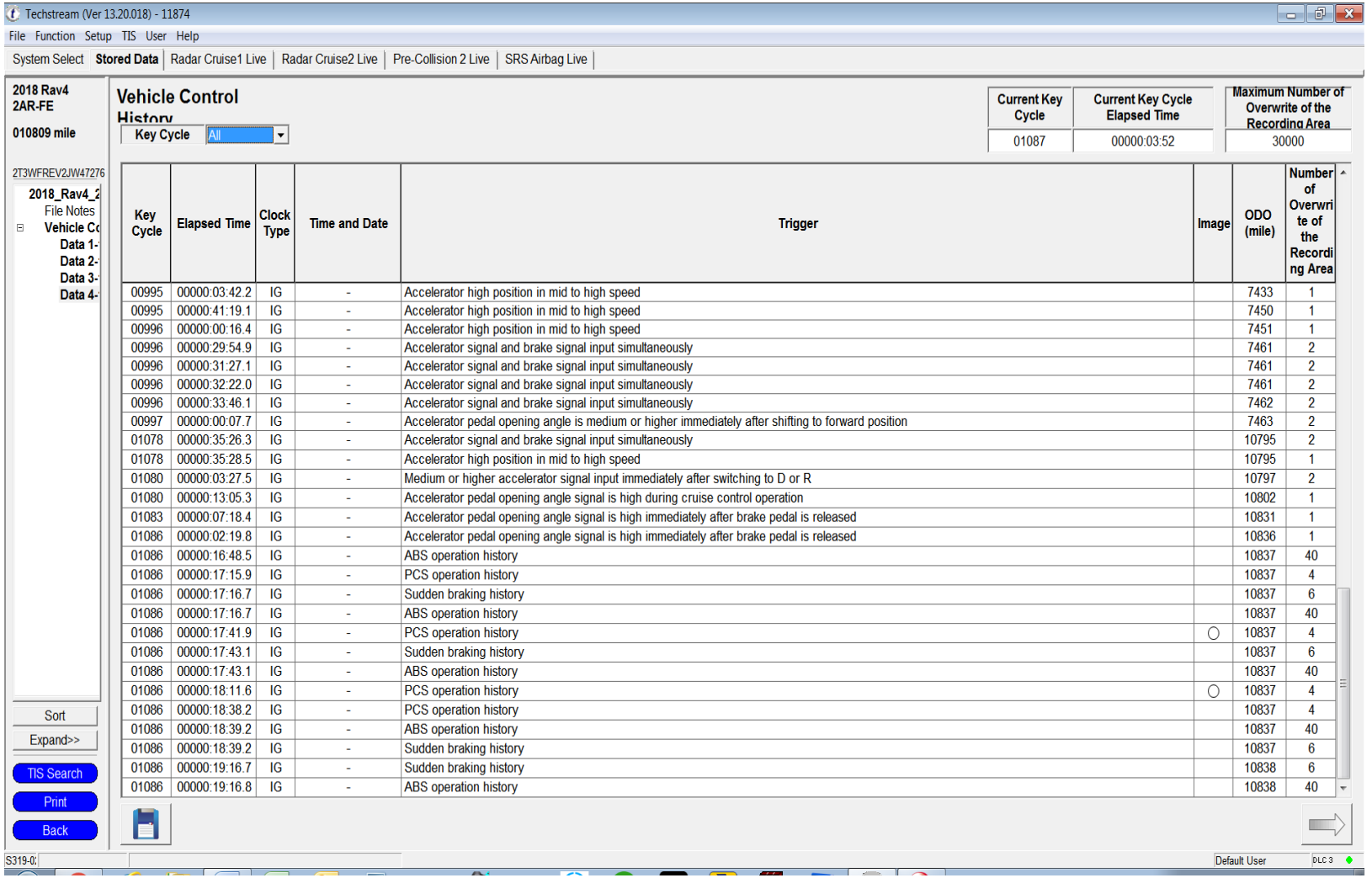
How to get Toyota VCH Data





How to get Toyota VCH Data

2 events indicated with images



Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area
00995	00000:03:42.2	IG	-	Accelerator high position in mid to high speed		7433	1
00995	00000:41:19.1	IG	-	Accelerator high position in mid to high speed		7450	1
00996	00000:00:16.4	IG	-	Accelerator high position in mid to high speed		7451	1
00996	00000:29:54.9	IG	-	Accelerator signal and brake signal input simultaneously		7461	2
00996	00000:31:27.1	IG	-	Accelerator signal and brake signal input simultaneously		7461	2
00996	00000:32:22.0	IG	-	Accelerator signal and brake signal input simultaneously		7461	2
00996	00000:33:46.1	IG	-	Accelerator signal and brake signal input simultaneously		7462	2
00997	00000:00:07.7	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		7463	2
01078	00000:35:26.3	IG	-	Accelerator signal and brake signal input simultaneously		10795	2
01078	00000:35:28.5	IG	-	Accelerator high position in mid to high speed		10795	1
01080	00000:03:27.5	IG	-	Medium or higher accelerator signal input immediately after switching to D or R		10797	2
01080	00000:13:05.3	IG	-	Accelerator pedal opening angle signal is high during cruise control operation		10802	1
01083	00000:07:18.4	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released		10831	1
01086	00000:02:19.8	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released		10836	1
01086	00000:16:48.5	IG	-	ABS operation history		10837	40
01086	00000:17:15.9	IG	-	PCS operation history		10837	4
01086	00000:17:16.7	IG	-	Sudden braking history		10837	6
01086	00000:17:16.7	IG	-	ABS operation history		10837	40
01086	00000:17:41.9	IG	-	PCS operation history	○	10837	4
01086	00000:17:43.1	IG	-	Sudden braking history		10837	6
01086	00000:17:43.1	IG	-	ABS operation history		10837	40
01086	00000:18:11.6	IG	-	PCS operation history	○	10837	4
01086	00000:18:38.2	IG	-	PCS operation history		10837	4
01086	00000:18:39.2	IG	-	ABS operation history		10837	40
01086	00000:18:39.2	IG	-	Sudden braking history		10837	6
01086	00000:19:16.7	IG	-	Sudden braking history		10838	6
01086	00000:19:16.8	IG	-	ABS operation history		10838	40

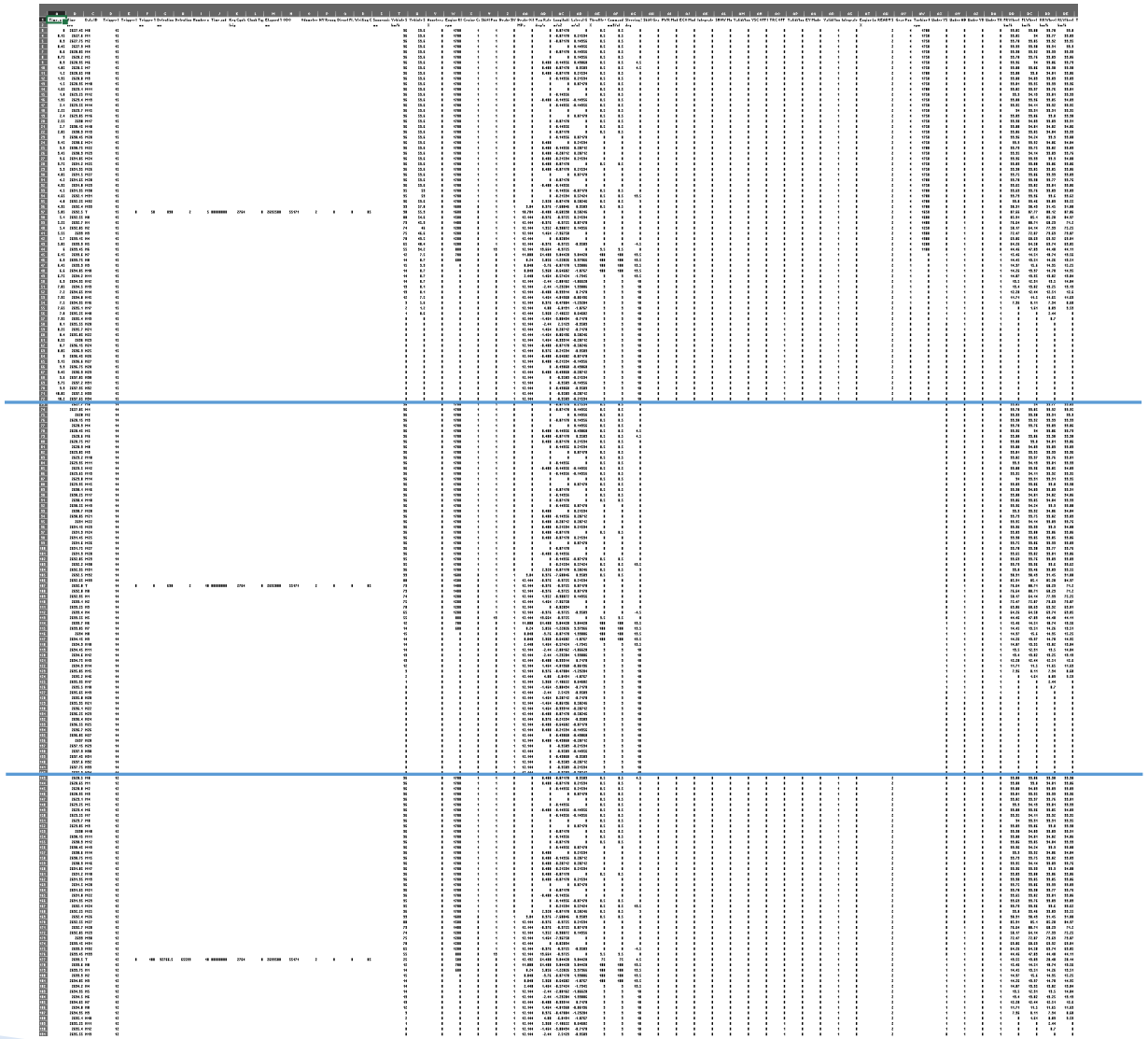
Export as CSV file example #1 – VIN only

2016 TOYOTA 4Runner w/ Smart Key 1GR-FE

055170 mile

JTEBU5JR1G5XXXXXX

Export as CSV file example #2 – all events



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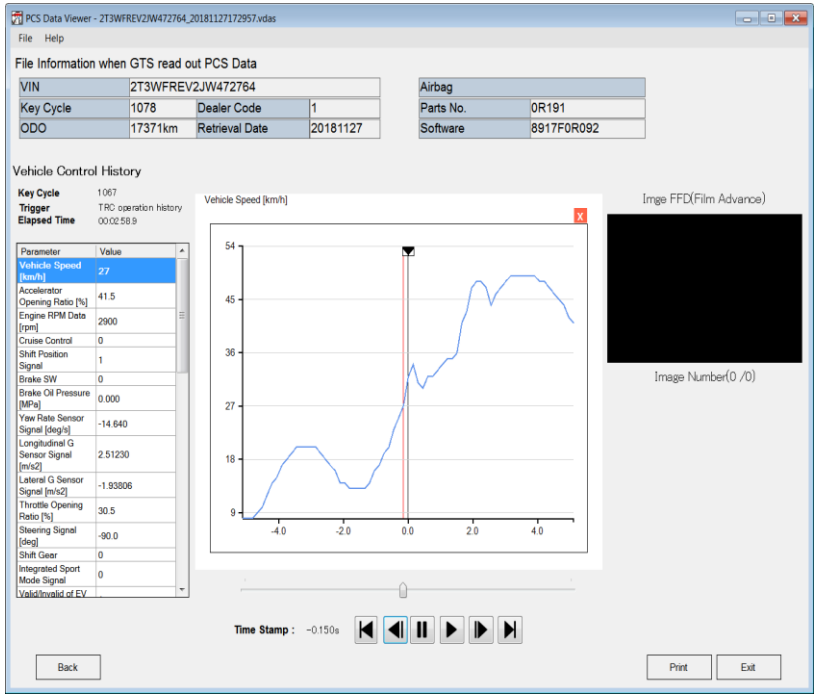
Toyota Vehicle Control History - Sampling rate

"Data within the PCS events are recorded at 2 Hz and are supplemented with asynchronous data when PBA, FPB, and PB flags occur. The resolution of the times associated with these flags was 0.01 s. The Corolla VCH can store up to 4 PCS events before overwriting the earliest event. For ABS events, the VCH is triggered by the onset of the ABS and data are recorded at 6.67 Hz."



The Accuracy of Toyota Vehicle Control History Data during Autonomous Emergency Braking, SAE 2018-01-1441, Peter Xing et al

Vehicle Control History can sample at .5s or .15s.

FFD (Freeze Frame Data) samples at approximately .343s with some variability

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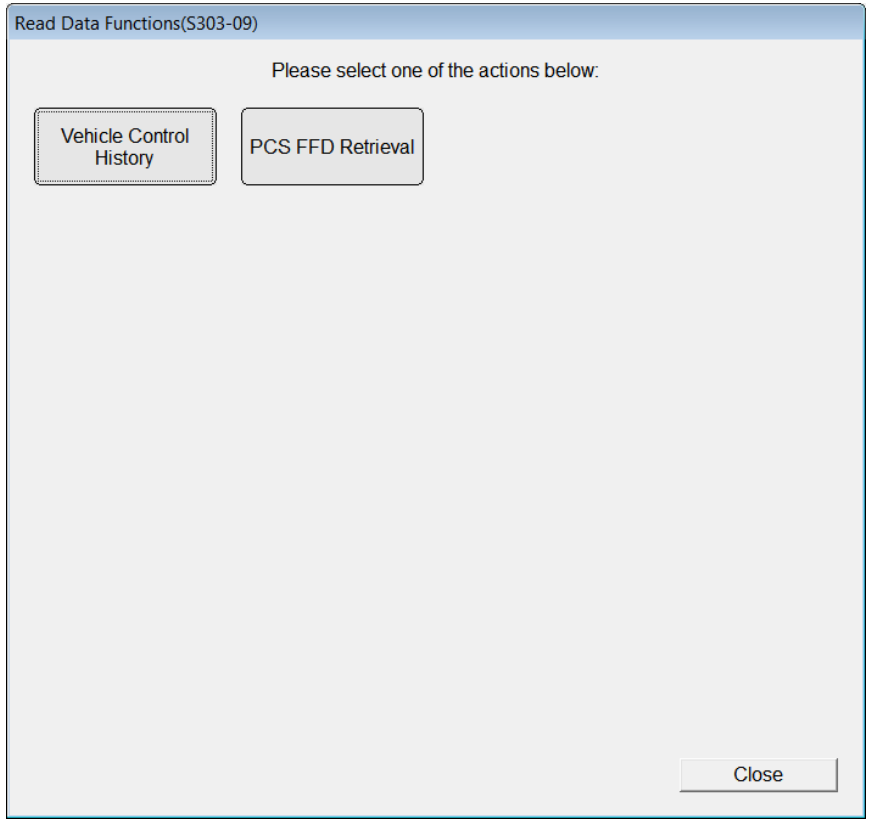
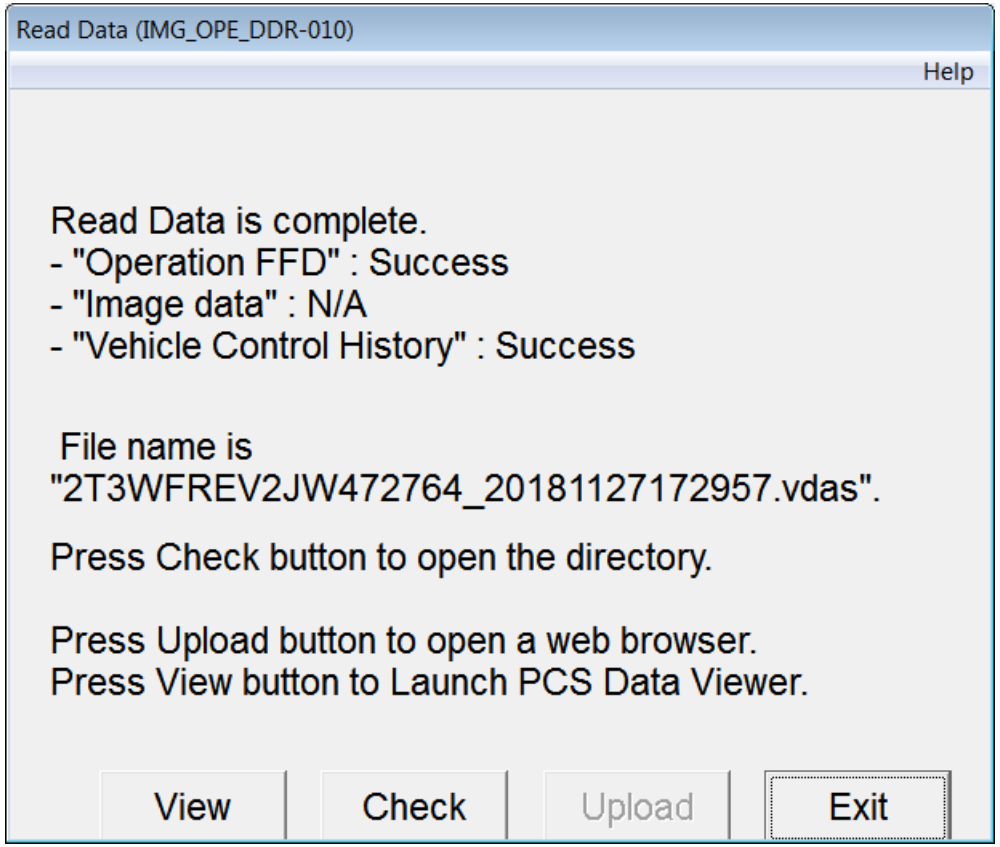
How to get Toyota Freeze Frame (FFD) Data



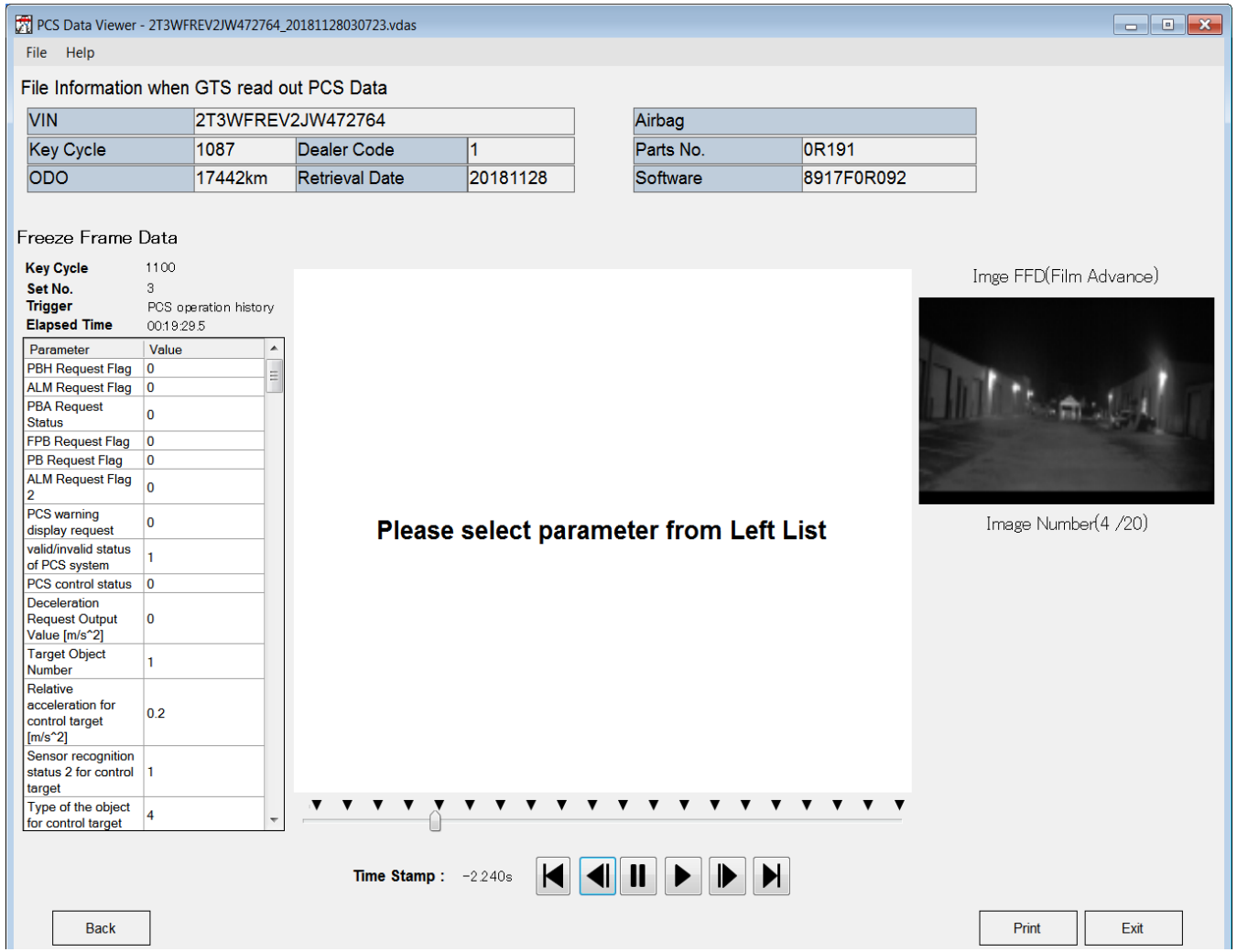
From the main Techstream screen, click on “Read data”, then “PCS FFD Retrieval”

How to get Toyota Freeze Frame (FFD) Data



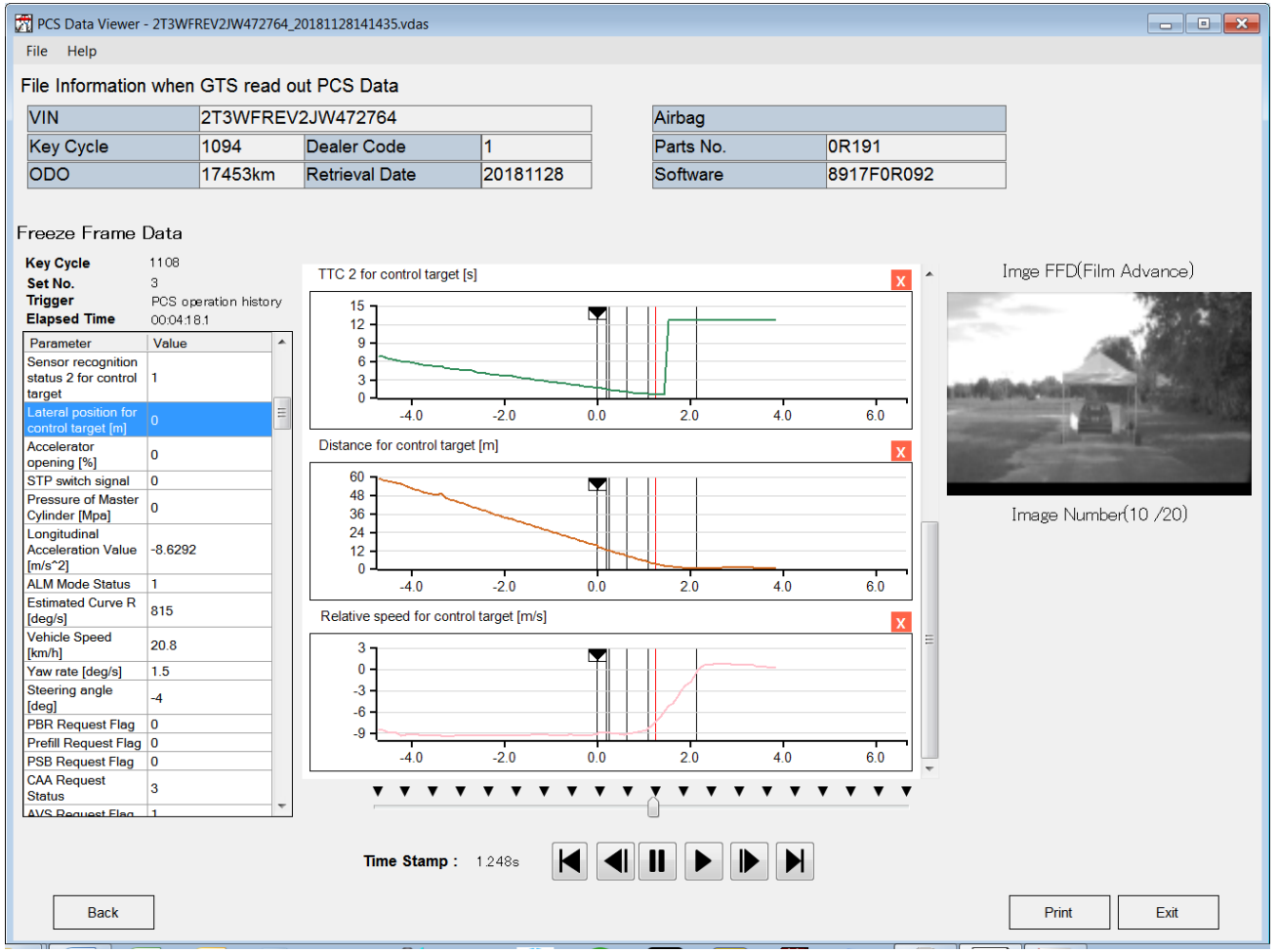
Click on “Check” to see the data files in a folder, and copy them to your case file. “View” launches the PCS Data Viewer from within which *you have to open the files that were saved.*

Example Toyota Freeze Frame (FFD) Data



Example Toyota Freeze Frame (FFD) Data

To save the images, go to File, Output PCS Data, and select a folder. It will dump a lot of data including very low-res images.



Example Toyota Freeze Frame (FFD) Data - Images

Example Toyota Freeze Frame (FFD) Data

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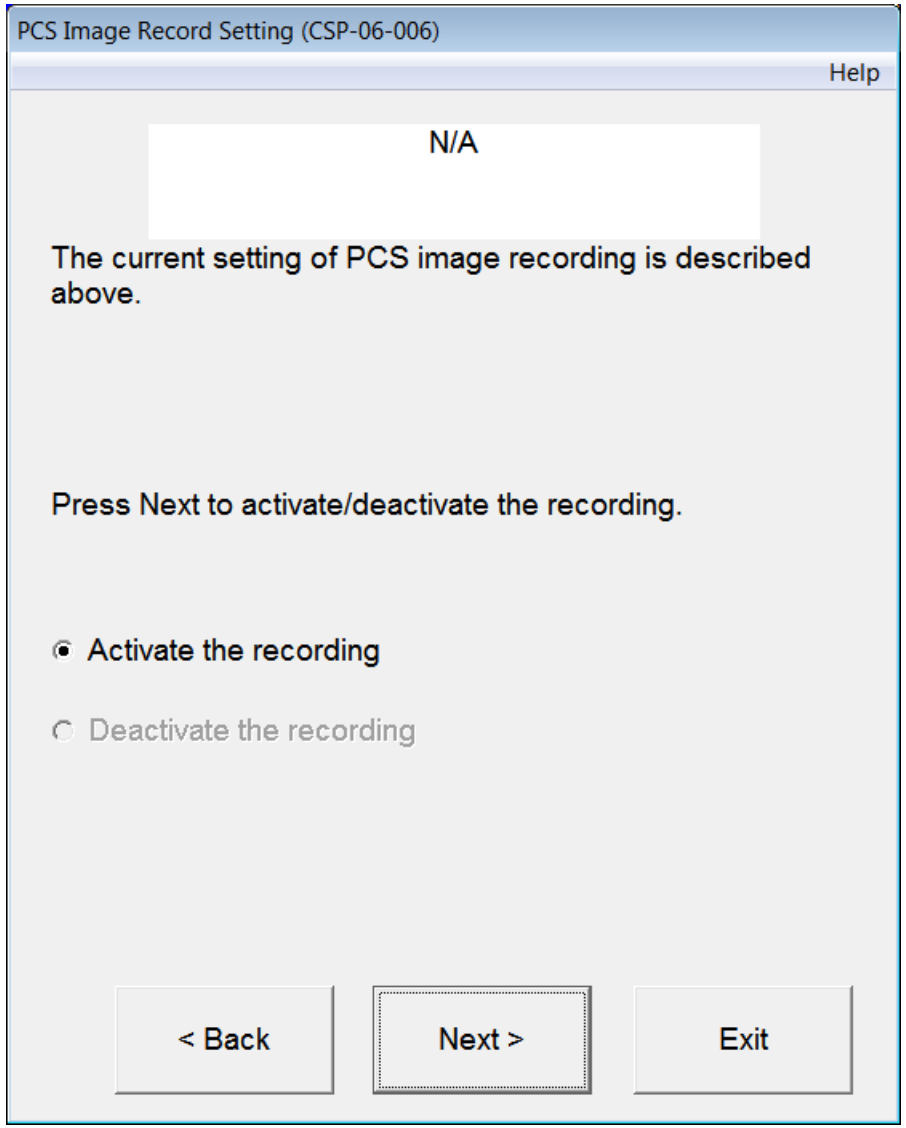


Image recording disabled by default – 2018 RAV4

Find image recording enabled/disabled status

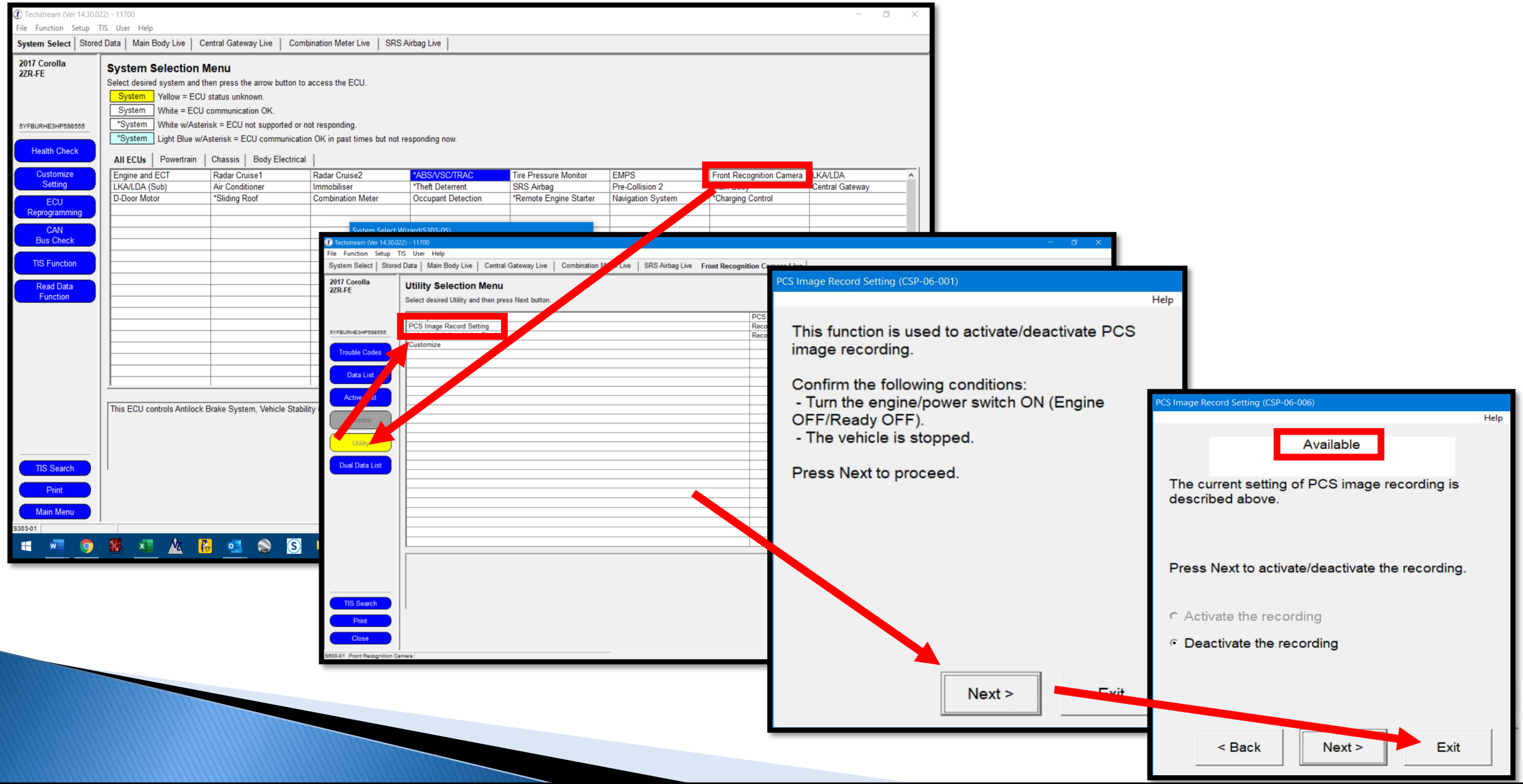


Image Timing – Toyota Freeze Frame Data

- PCS data (Image FFD) is recorded upon the pre-collision brake operation (PBA, PB).
  - In addition to the above, system records image information (Image FFD) of the camera sensor.
- Image information, Image FFD, is recorded 20 shots×3 events in TSS-P/LSS+, 2 shots×3events in TSS-C and 20 shots×10 events in LSS+A, and also maximum 10 shots at every trigger including the first trigger each event if another trigger occurs after the first trigger for six seconds.

Image FFD

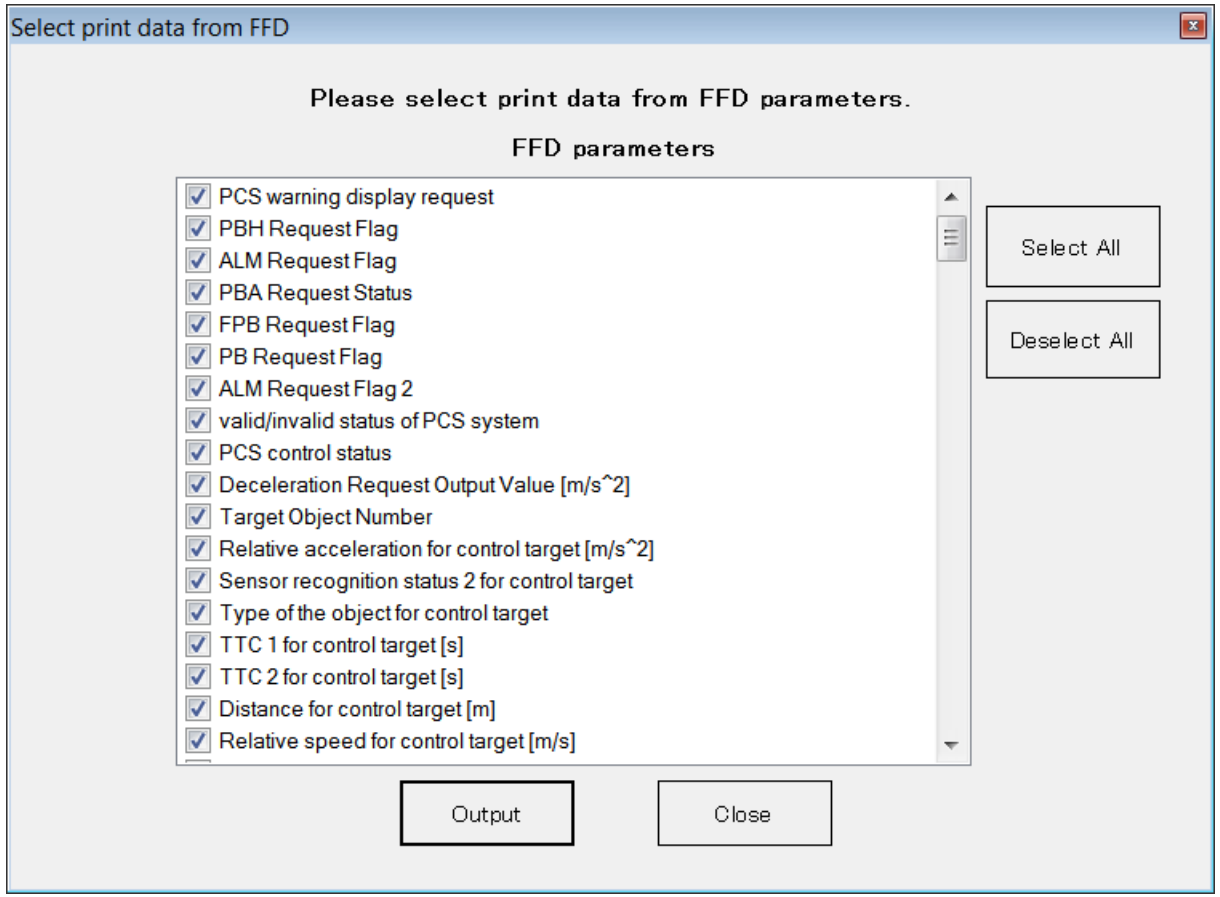
- TSS-P/LSS+ PCS : 20shots
- LSS+A PCS : 20 shots+ Maximum 10 shots at every trigger including the first trigger

Toyota PCS Data Viewer – Data Limitations

Image Timing – Toyota Freeze Frame Data

Toyota PCS Data Viewer – Data Limitations

Example Toyota Freeze Frame (FFD) Data



Print a report with all items selected; impressive amount of data!

Pre-trigger

-1.124s

0.076s

Post-trigger

Image Timing example – Toyota Vehicle Control History

-0.524s

0.676s

Sign Conventions – Toyota Vehicle Control History

Data Element Name	Positive Sign Notation Indicates
Longitudinal Acceleration Value	Forward
Lateral Acceleration Value	Left to Right
Yaw rate	Left Turn
Steering angle	Left Turn
Lateral position for control target	Right Side
PB	Pre-collision Brake
PBA	Pre-collision Brake Assist

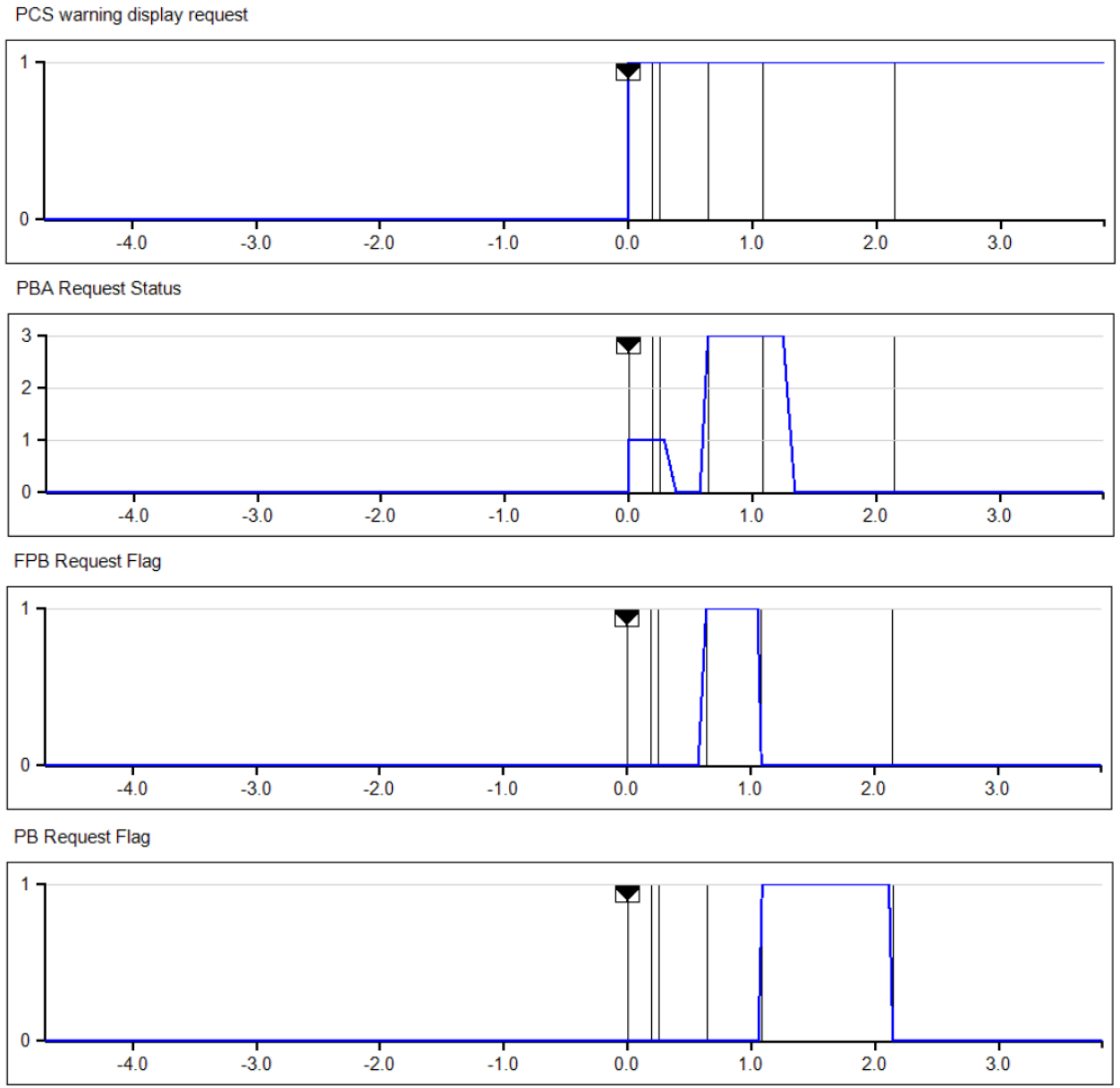
Toyota PCS Data Viewer – Data Limitations

AEB Test Results – 2018 Toyota RAV4

Parameter	Value	Unit
Vehicle Speed	21	MPH
Brake SW	OFF	
Longitudinal G Sensor Signal	-0.28712	m/s2
ALM Request Flag	OFF	
PBA Request Flag	OFF	
FPB Request Flag	OFF	
PB Request Flag	OFF	
Target Distance	133.60	feet

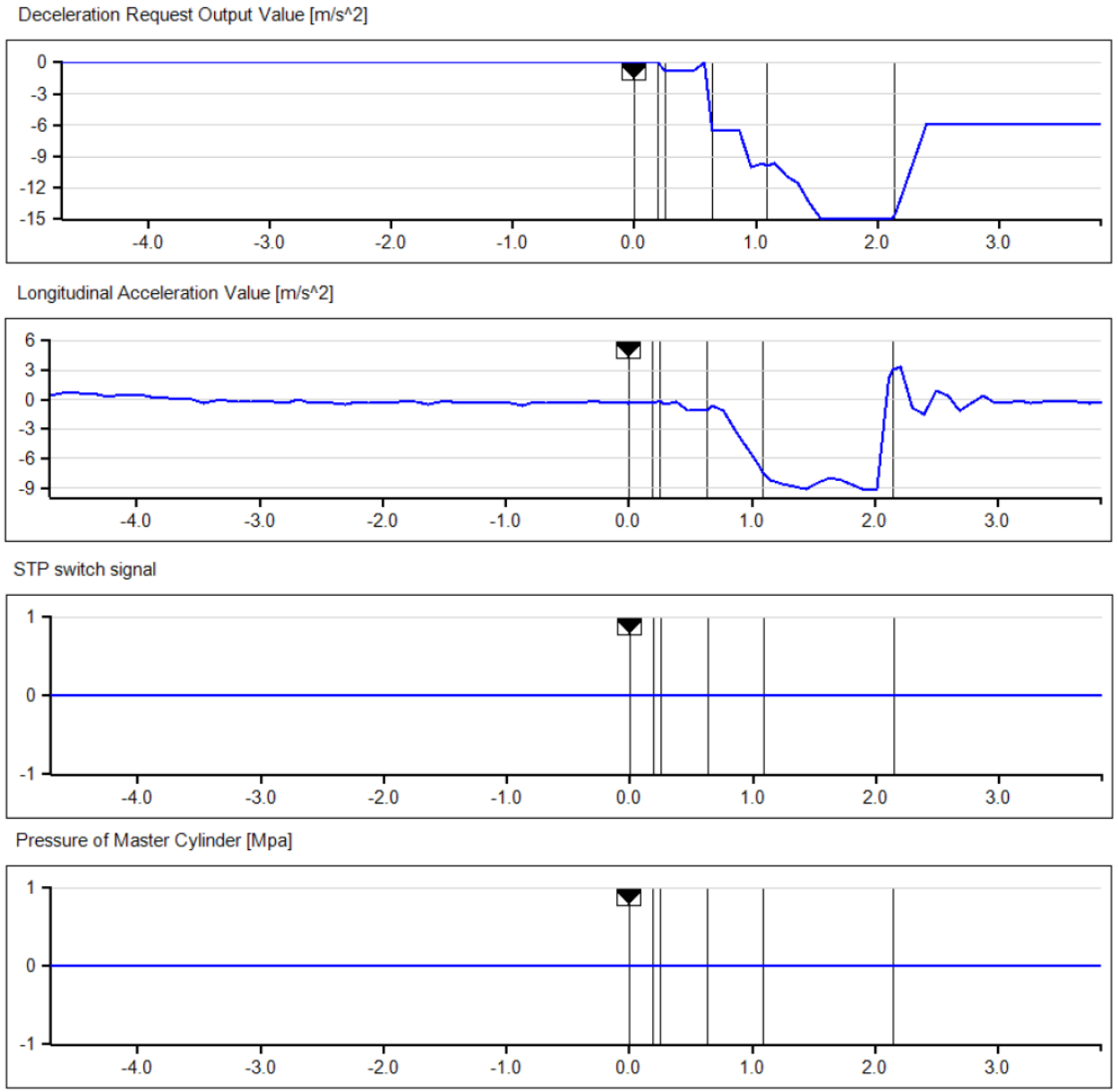
AEB Test Results – 2018 Toyota RAV4

Warnings and AEB status



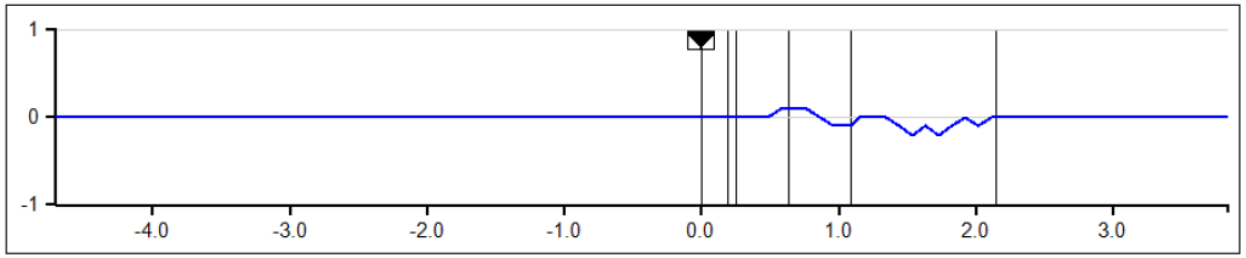
AEB Test Results – 2018 Toyota RAV4

Vehicle deceleration

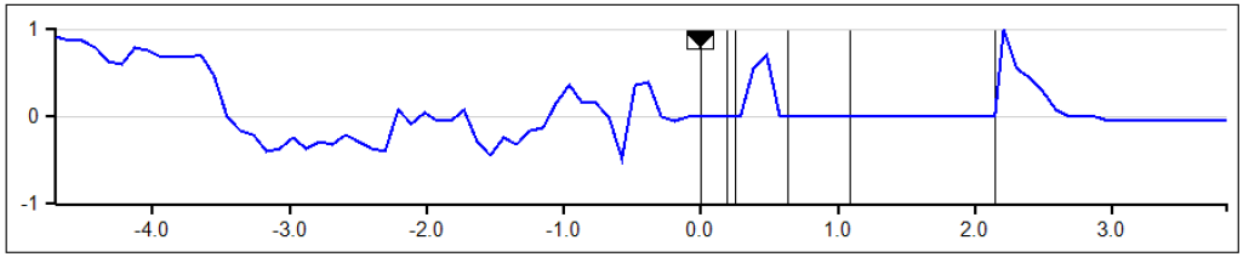


AEB Test Results – 2018 Toyota RAV4

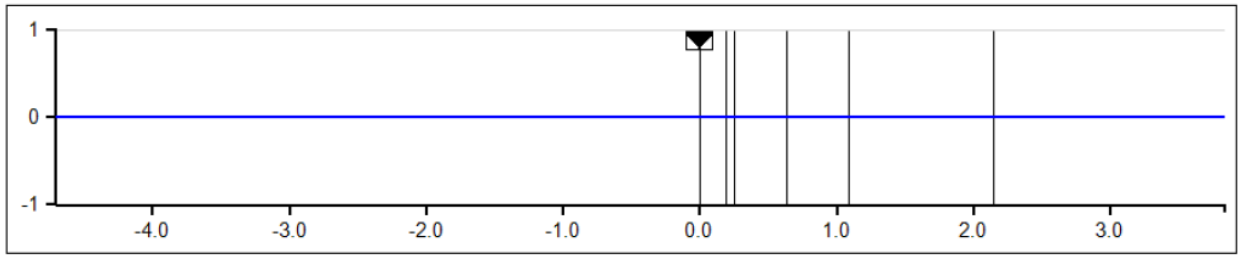
Lateral speed for control target [m/s]



Lateral position for control target [m]

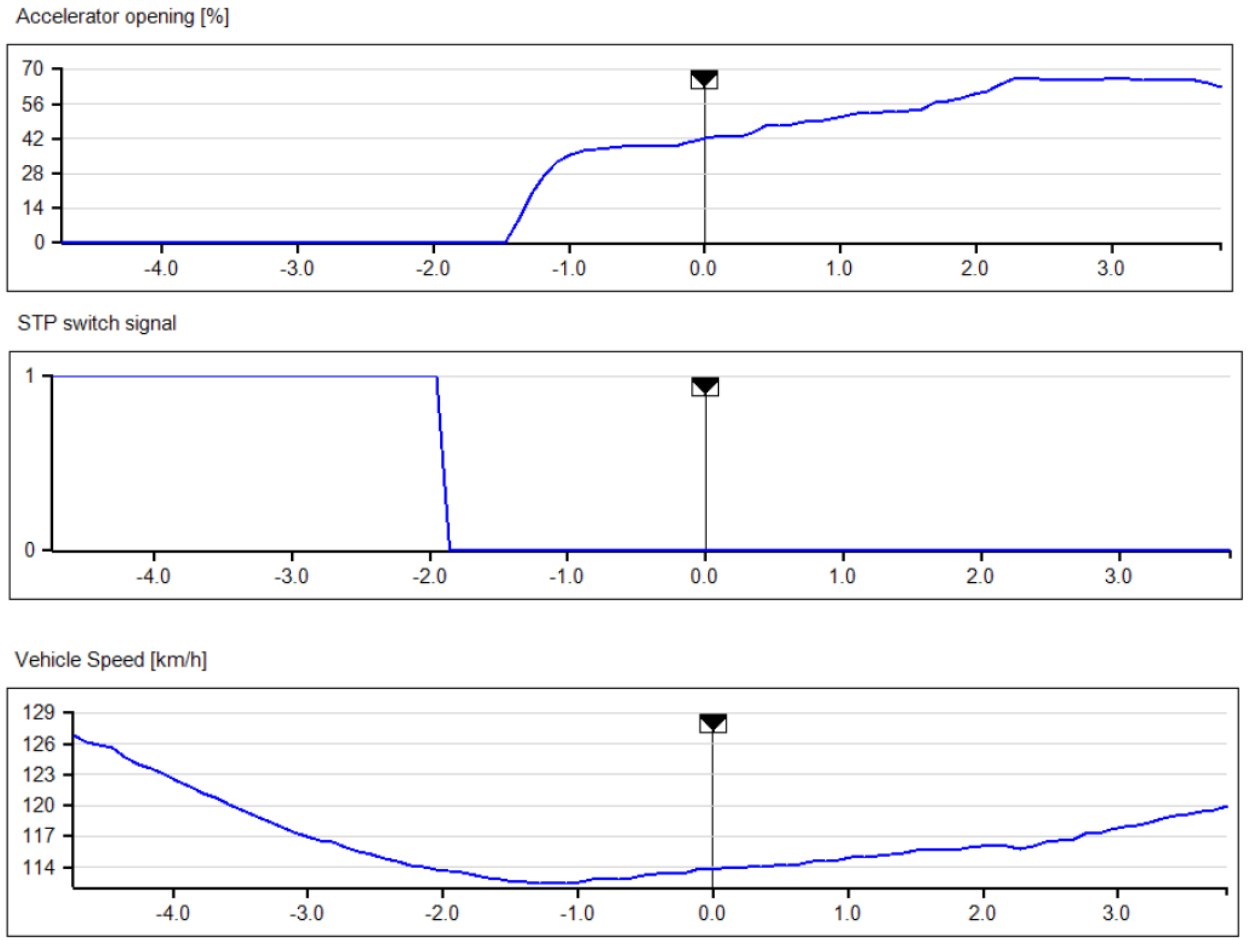


Shift position R signal

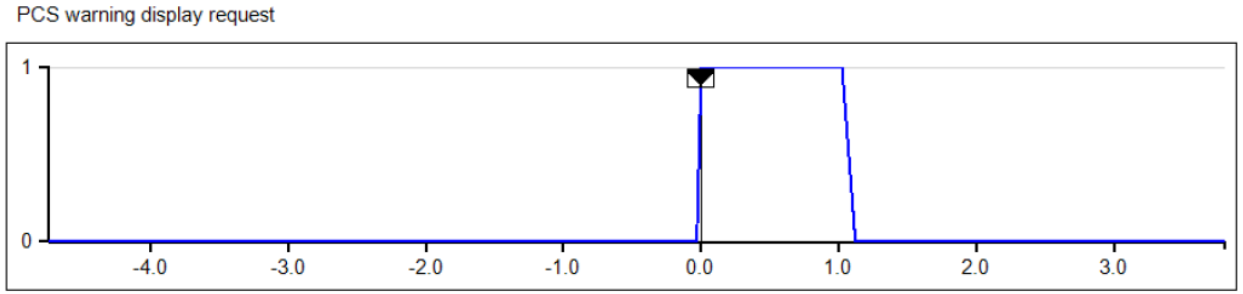


Other interesting data.

False positive

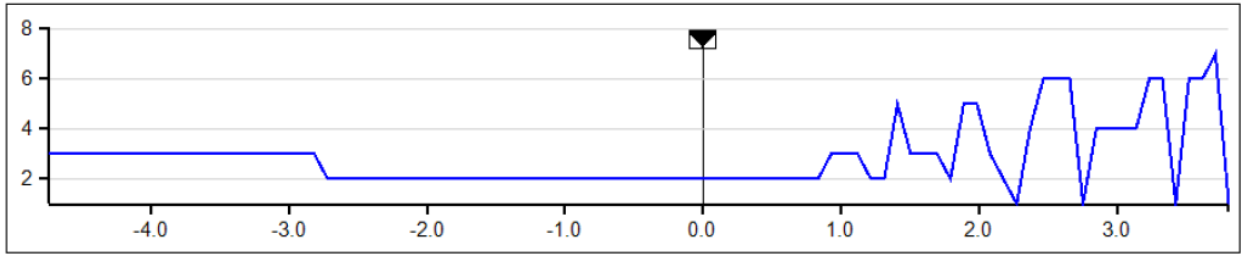


False Positives

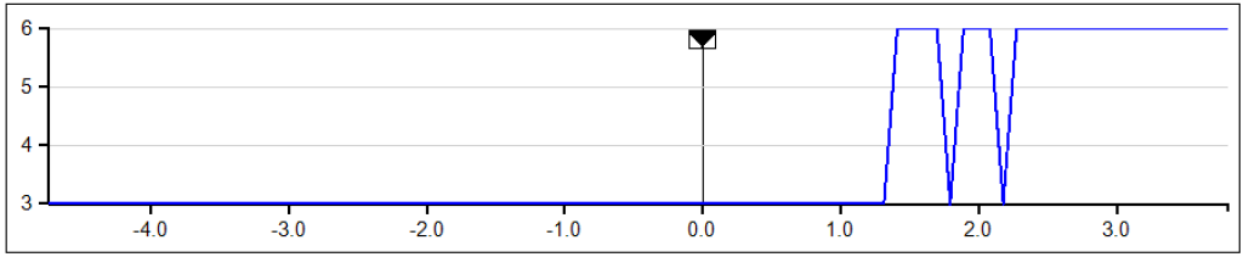


False positive

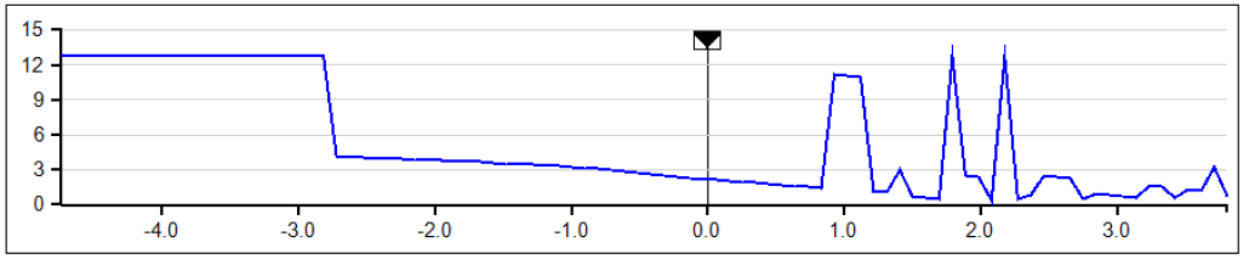
Target Object Number



Type of the object for control target

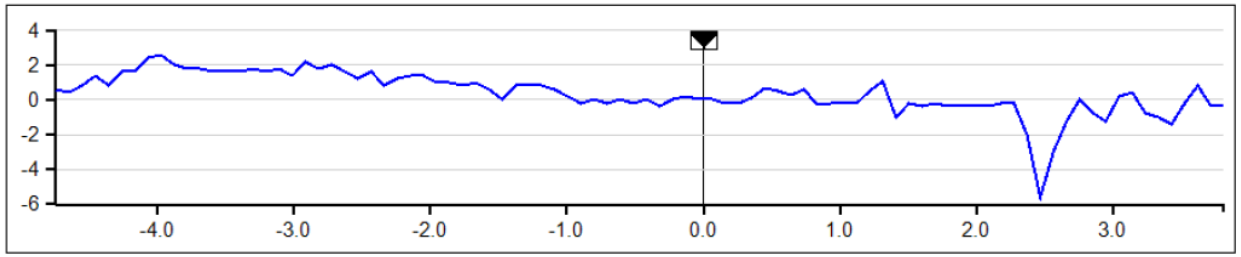


TTC 1 for control target [s]

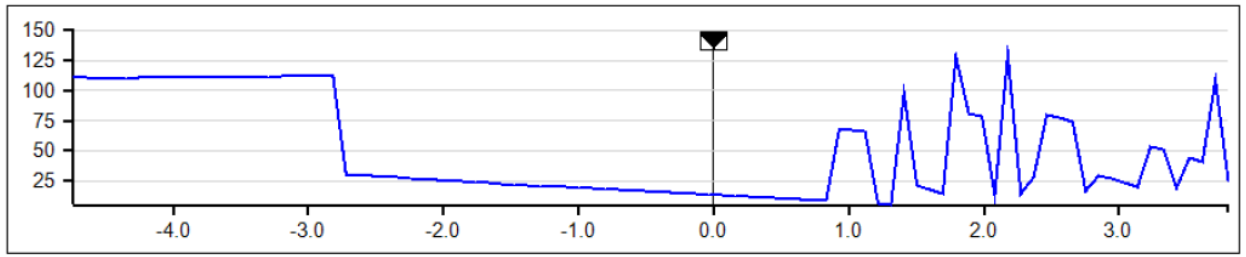


False positive

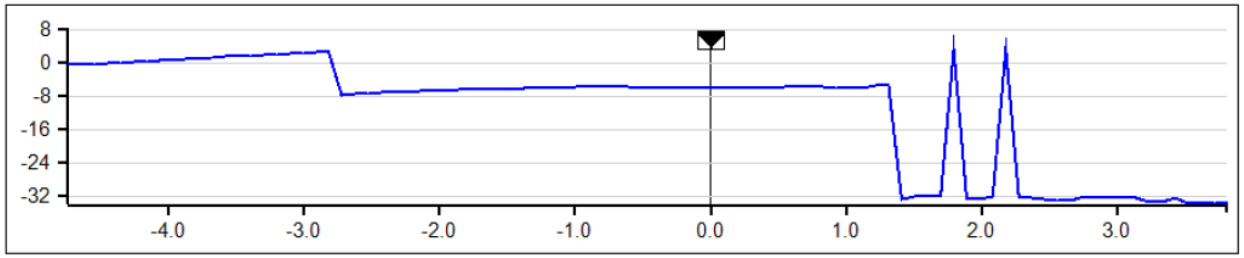
Relative acceleration for control target [m/s^2]



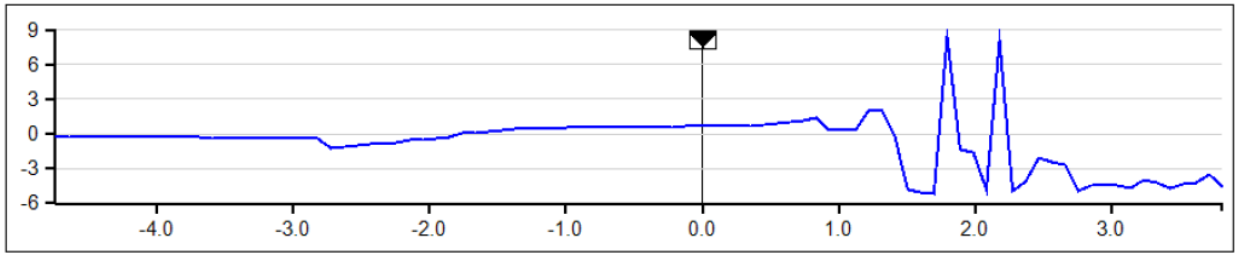
Distance for control target [m]



Relative speed for control target [m/s]



Lateral position for control target [m]



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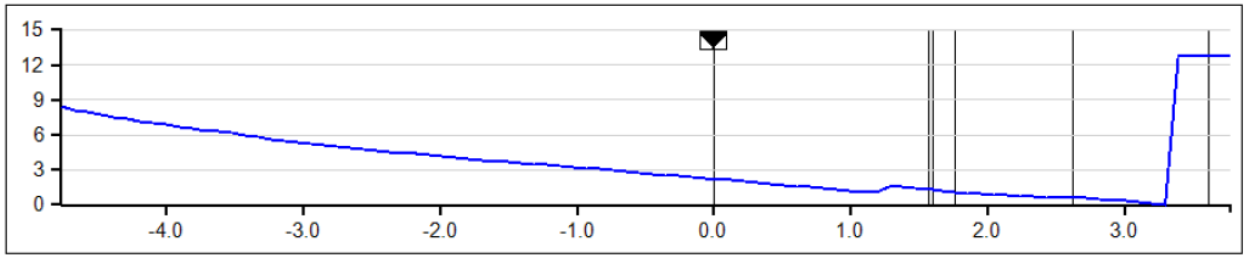
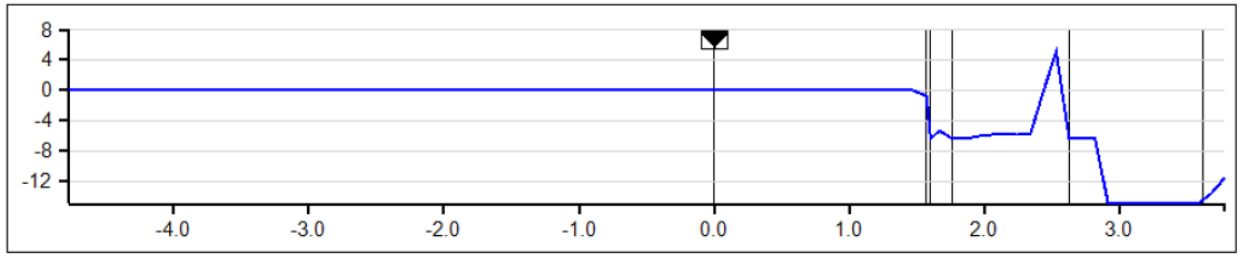
Toyota AEB

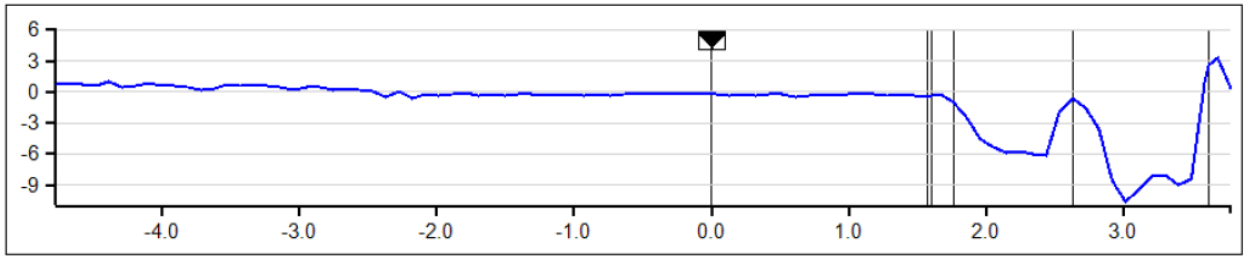
TTC 1 for control target [s]



Deceleration Request Output Value [m/s^2]

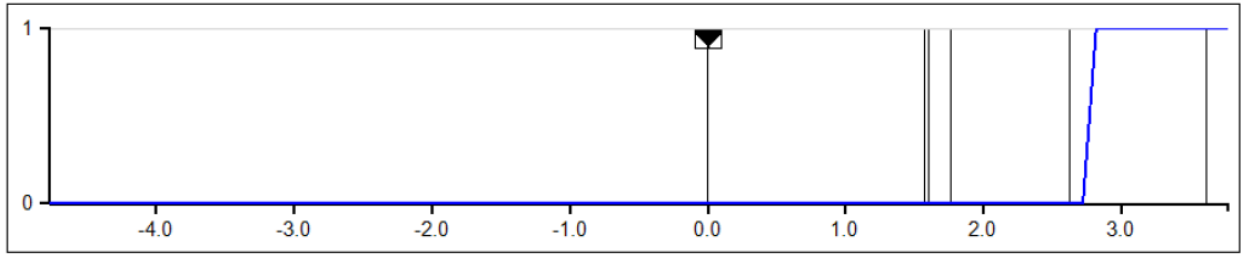


Longitudinal Acceleration Value [m/s^2]

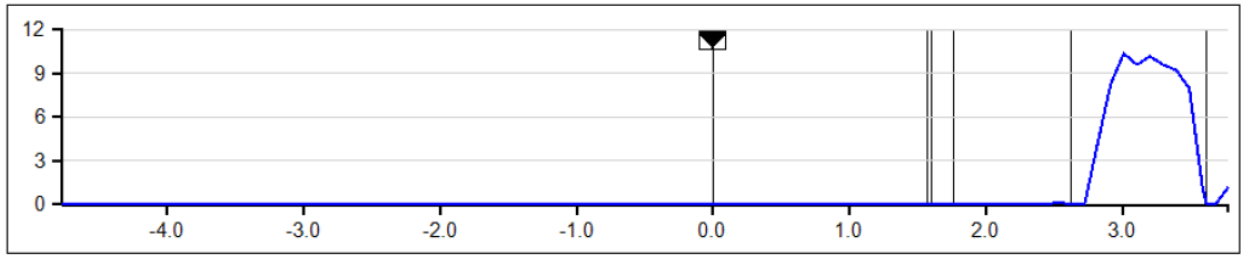


Toyota AEB

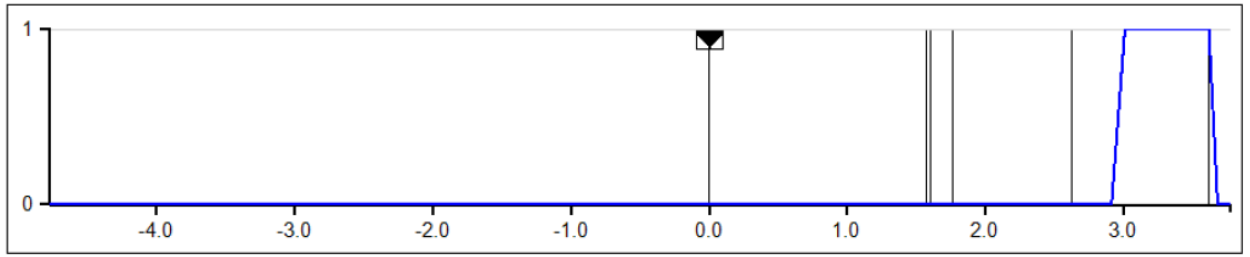
STP switch signal



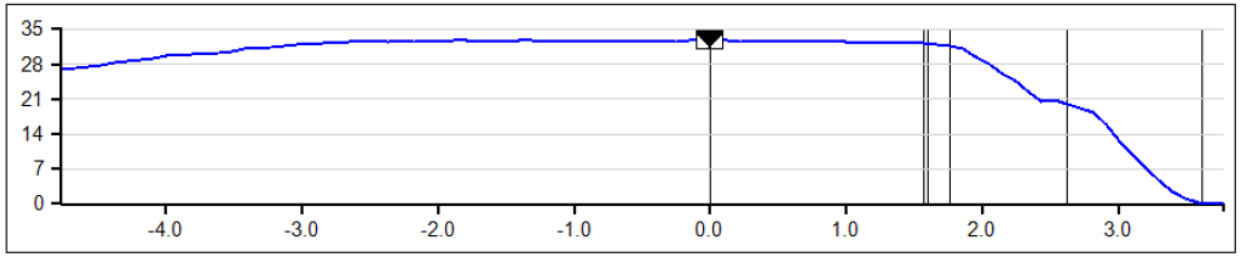
Pressure of Master Cylinder [Mpa]



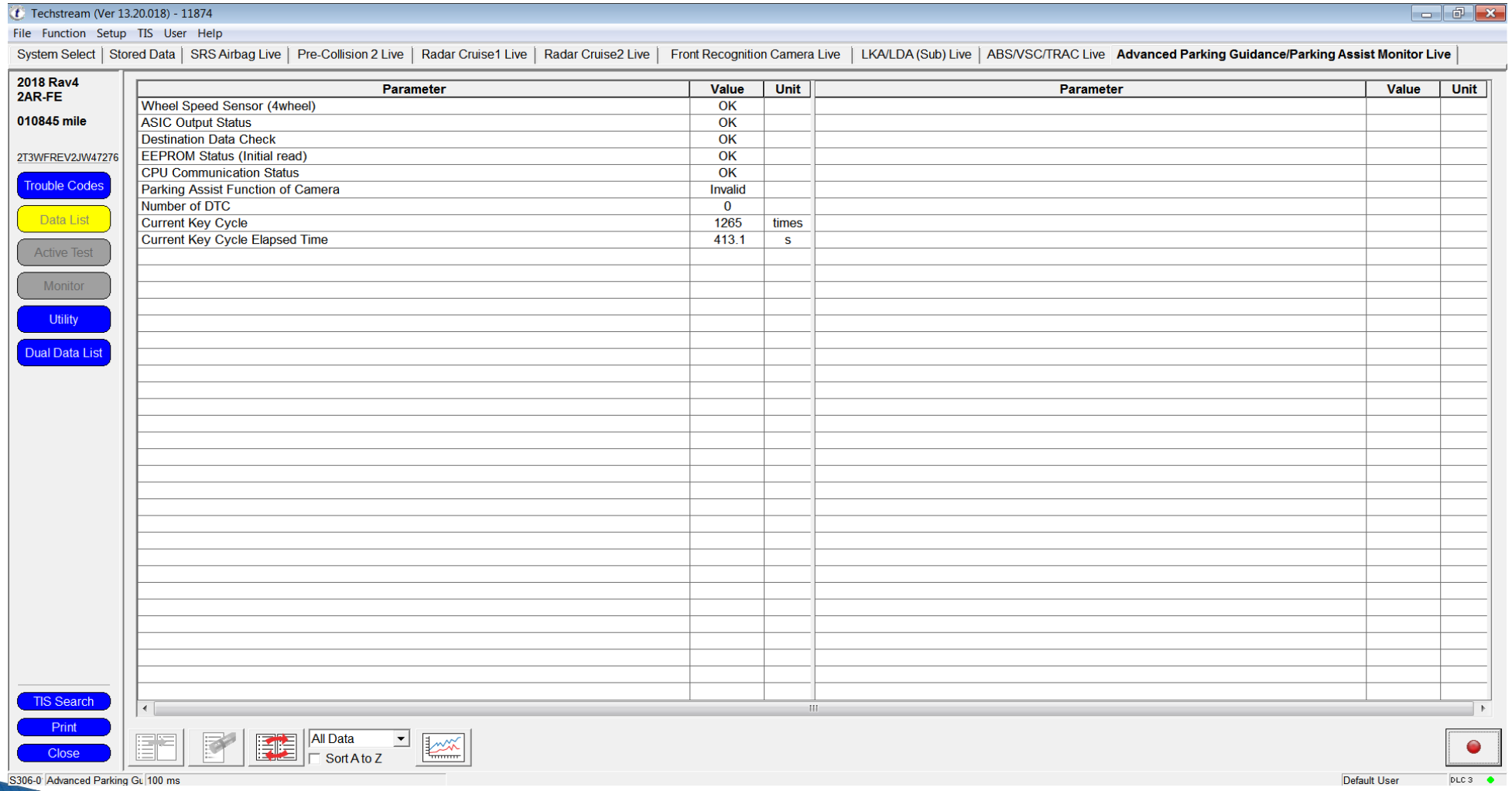
Under ABS control



Vehicle Speed [km/h]



Example Toyota Vehicle Control History Data

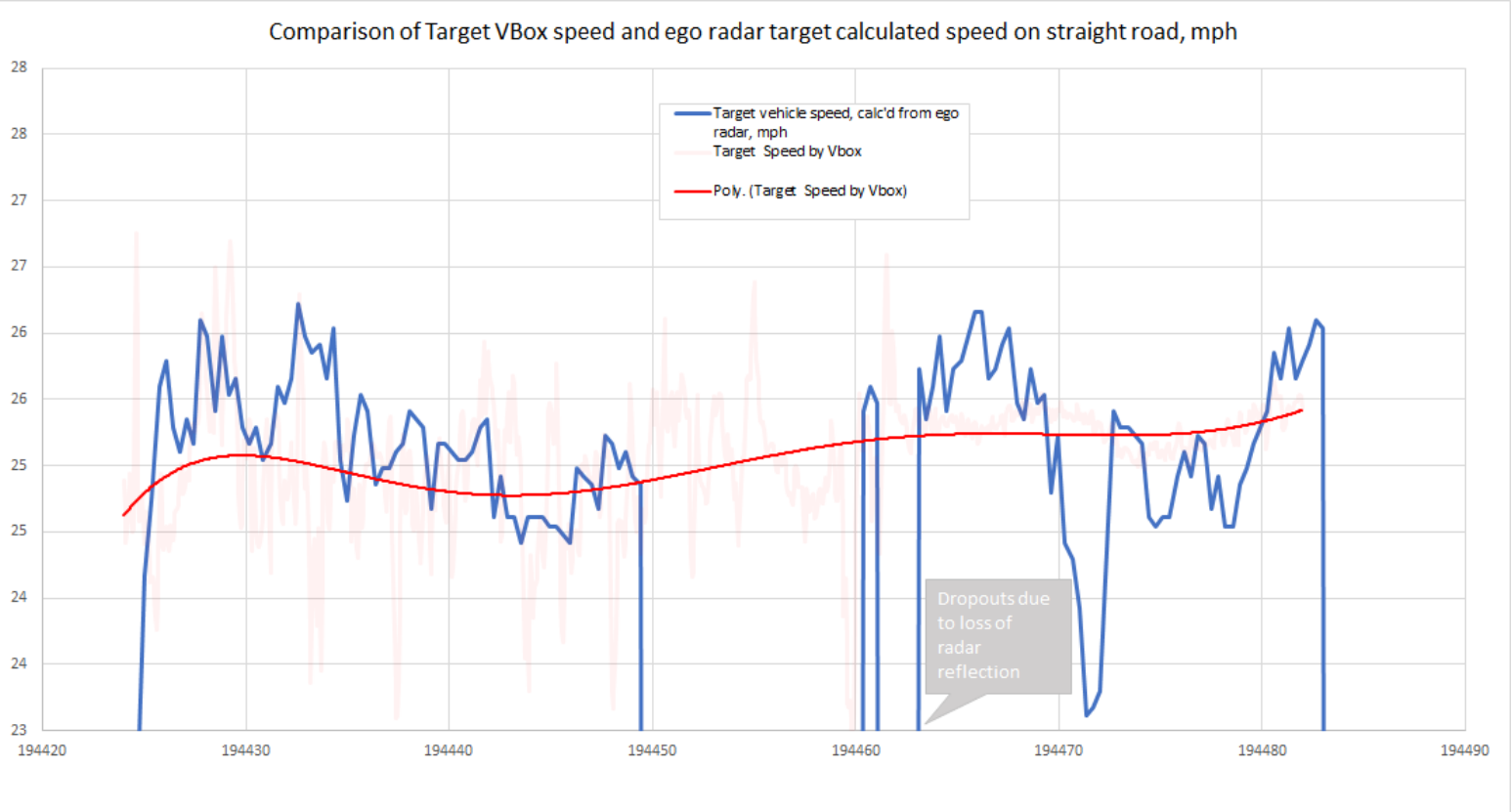


AEB Test Results – 2017 Toyota Corolla

- Approached a barrier at ~20mph
- Alarm triggered at Time To Contact (TTC) of 2s
- Autobrake triggered 0.9s later
- AEB jerk 0.6-0.8 g/s
- Radar reflectivity of target did not significantly affect AEB performance
- Some inconsistent performance

The Effect of Target Features on Toyota's Autonomous Emergency Braking System SAE 2018-01-0533 – Yang, MEA, et al

Comparison of target vehicle speed with target speed as calculated by host radar



Overall is a reliable representation of the target vehicle speed

Radar lateral offset distance – An Example

2018 Toyota RAV4 with offset reference overlay, following a 25mph target vehicle.

Radar Tracking On Curves – An Example

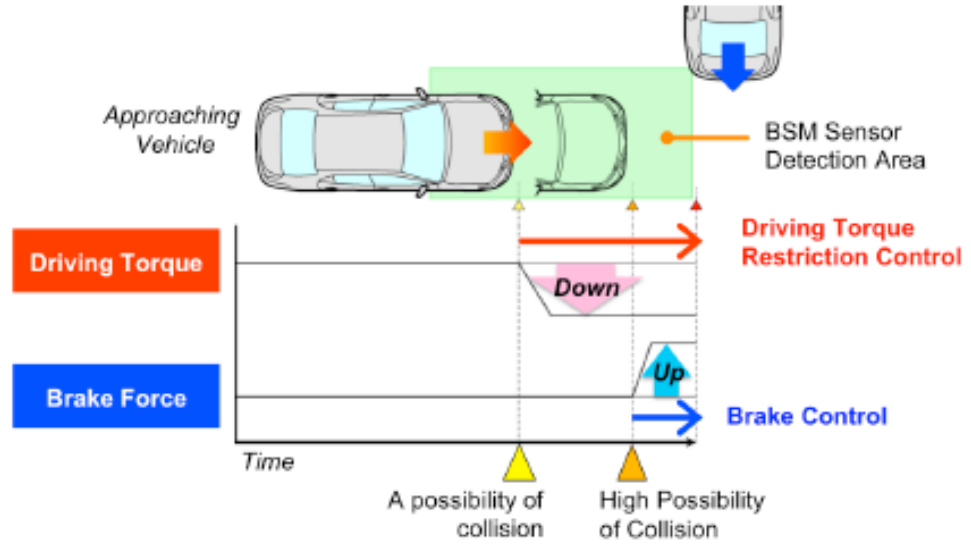
2018 Toyota RAV4 with Techstream snapshot overlay, following a 25mph target vehicle.

VCH – Rear Cross Traffic Events

ICS System
Rear Cross Traffic Auto Brake (RCTAB)

RCTAB is a function of the ICS system.

- RCTAB restricts the driving torque and controls the brake force when a BSM sensor detects an approaching vehicle while backing up.



TOYOTA Handbook

TIN517B Fall 2017 New Technology Update

ICS System
Freeze Frame Data

When the brake control operates, the ICS system records the following items.

Tester Display	Measurement Item	Range
Distance to an object detected by ICS control	The calculated distance to an object detected by the front/rear sensors when the vehicle is moving forward/ backward	Unknown, Not Detected or 0 to 7.65 m
ICS Temporarily Not Available (Shift Position Invalid)	ICS system temporarily not useable as shift position indeterminable	Available or Not Available
Front Left Center Sensor Obstructed	Intuitive parking assist system temporarily not useable due to foreign matter on ultrasonic sensor	No or Yes
Front Left Sensor Dirty	Intuitive parking assist system temporarily not useable due to dirty ultrasonic sensor	No or Yes
RCTAB Controlled Object Relative Position (X)	Displays the distance of an object relative to the position of the vehicle when reversing	-12.8 to 12.6 m

Freeze Frame Data

Freeze Frame data is recorded in the Clearance Warning ECU when ICS brake control is activated, in addition to Vehicle Control History data recorded in the SRS ECU. This information is related to the control function of ICS as opposed to Freeze Frame data that is stored when a system DTC is set related to a system failure. DTC-related Freeze Frame information is stored in a different area and can be accessed through the DTC menu of the Techstream.

To access Freeze Frame Data for brake control of the ICS system, enter the following Techstream menus:

Body Electrical > Advanced Parking Guidance/ICS/Intuitive P/A > Utility > Operative/Inoperative Condition History

A maximum of 30 sets of FFD is stored. Old FFD is overwritten by new FFD.

Note: The FFD that is stored and updated each time the brake control is operated is stored in a different area from that of DTC FFD.

TIN517B Fall 2017 New Technology Update, Toyota Technical Training

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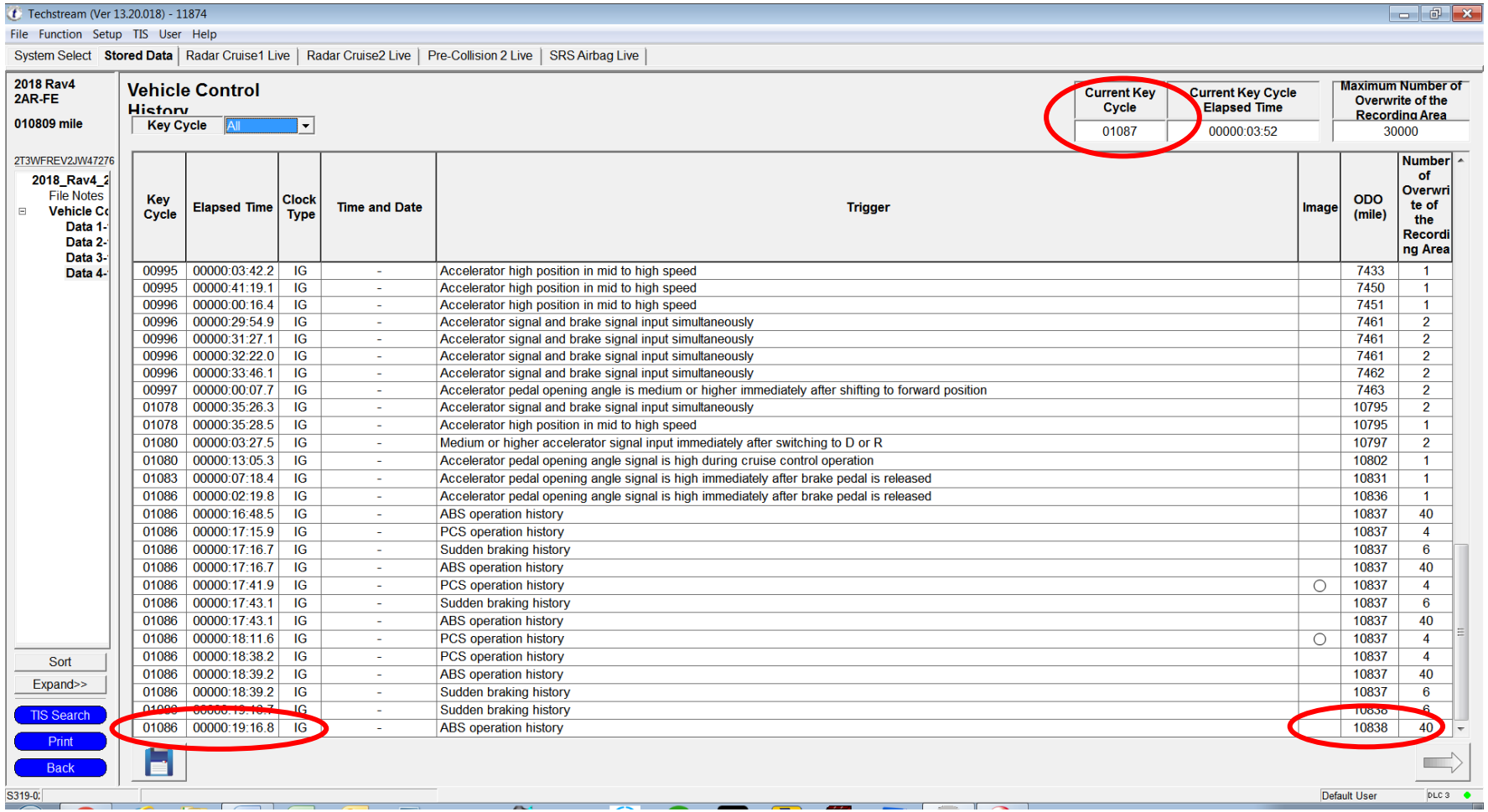
Toyota Vehicle Control History - Comparison to CDR Data



Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area
00995	00000:03:42.2	IG	-	Accelerator high position in mid to high speed		7433	1
00995	00000:41:19.1	IG	-	Accelerator high position in mid to high speed		7450	1
00996	00000:00:16.4	IG	-	Accelerator high position in mid to high speed		7451	1
00996	00000:29:54.9	IG	-	Accelerator signal and brake signal input simultaneously		7461	2
00996	00000:31:27.1	IG	-	Accelerator signal and brake signal input simultaneously		7461	2
00996	00000:32:22.0	IG	-	Accelerator signal and brake signal input simultaneously		7461	2
00996	00000:33:46.1	IG	-	Accelerator signal and brake signal input simultaneously		7462	2
00997	00000:00:07.7	IG	-	Accelerator pedal opening angle is medium or higher immediately after shifting to forward position		7463	2
01078	00000:35:26.3	IG	-	Accelerator signal and brake signal input simultaneously		10795	2
01078	00000:35:28.5	IG	-	Accelerator high position in mid to high speed		10795	1
01080	00000:03:27.5	IG	-	Medium or higher accelerator signal input immediately after switching to D or R		10797	2
01080	00000:13:05.3	IG	-	Accelerator pedal opening angle signal is high during cruise control operation		10802	1
01083	00000:07:18.4	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released		10831	1
01086	00000:02:19.8	IG	-	Accelerator pedal opening angle signal is high immediately after brake pedal is released		10836	1
01086	00000:16:48.5	IG	-	ABS operation history		10837	40
01086	00000:17:15.9	IG	-	PCS operation history		10837	4
01086	00000:17:16.7	IG	-	Sudden braking history		10837	6
01086	00000:17:16.7	IG	-	ABS operation history		10837	40
01086	00000:17:41.9	IG	-	PCS operation history	○	10837	4
01086	00000:17:43.1	IG	-	Sudden braking history		10837	6
01086	00000:17:43.1	IG	-	ABS operation history		10837	40
01086	00000:18:11.6	IG	-	PCS operation history	○	10837	4
01086	00000:18:38.2	IG	-	PCS operation history		10837	4
01086	00000:18:39.2	IG	-	ABS operation history		10837	40
01086	00000:18:39.2	IG	-	Sudden braking history		10837	6
01086	00000:19:16.8	IG	-	ABS operation history		10838	40

Church Impact – Radar Data

Church Impact – Radar Data

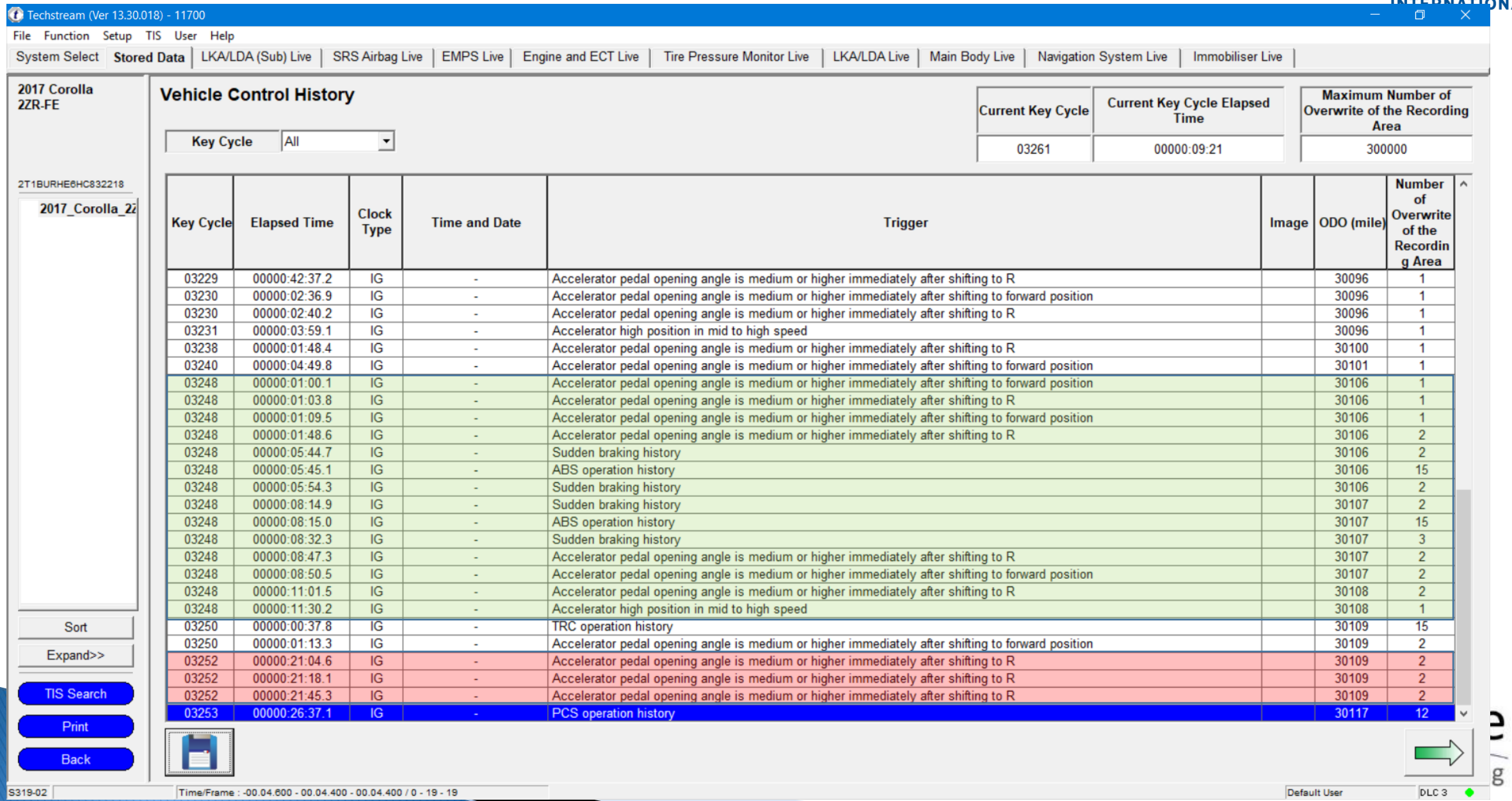
Time, s	Speed, mph	Accelerator, %	Brake SW	Steering, deg	Target distance, ft	Target speed, mph	Target lateral position, ft
1595	48	18.5	0	4.5	253	-49	-4.6
1595.5	48	20	0	4.5	232	-49	-7.7
1596	48	20.5	0	4.5	189	-50	-4.5
1596.5	48	19	0	4.5	161	-49	-2.8
1597	49	19	0	4.5	118	-50	-1.0
1597.1	49	19	0	4.5	118	-50	-1.0
1597.5	48	0	0	-4.5	92	-50	0.1
1598	48	0	0	9	49	-49	-0.3
1598.5	49	0	0	4.5	19	-48	2.5

Bosch CDR Crash Data From Toyota

Pre-Crash Data, -5 to 0 seconds (2nd Prior Event, TRG 3)

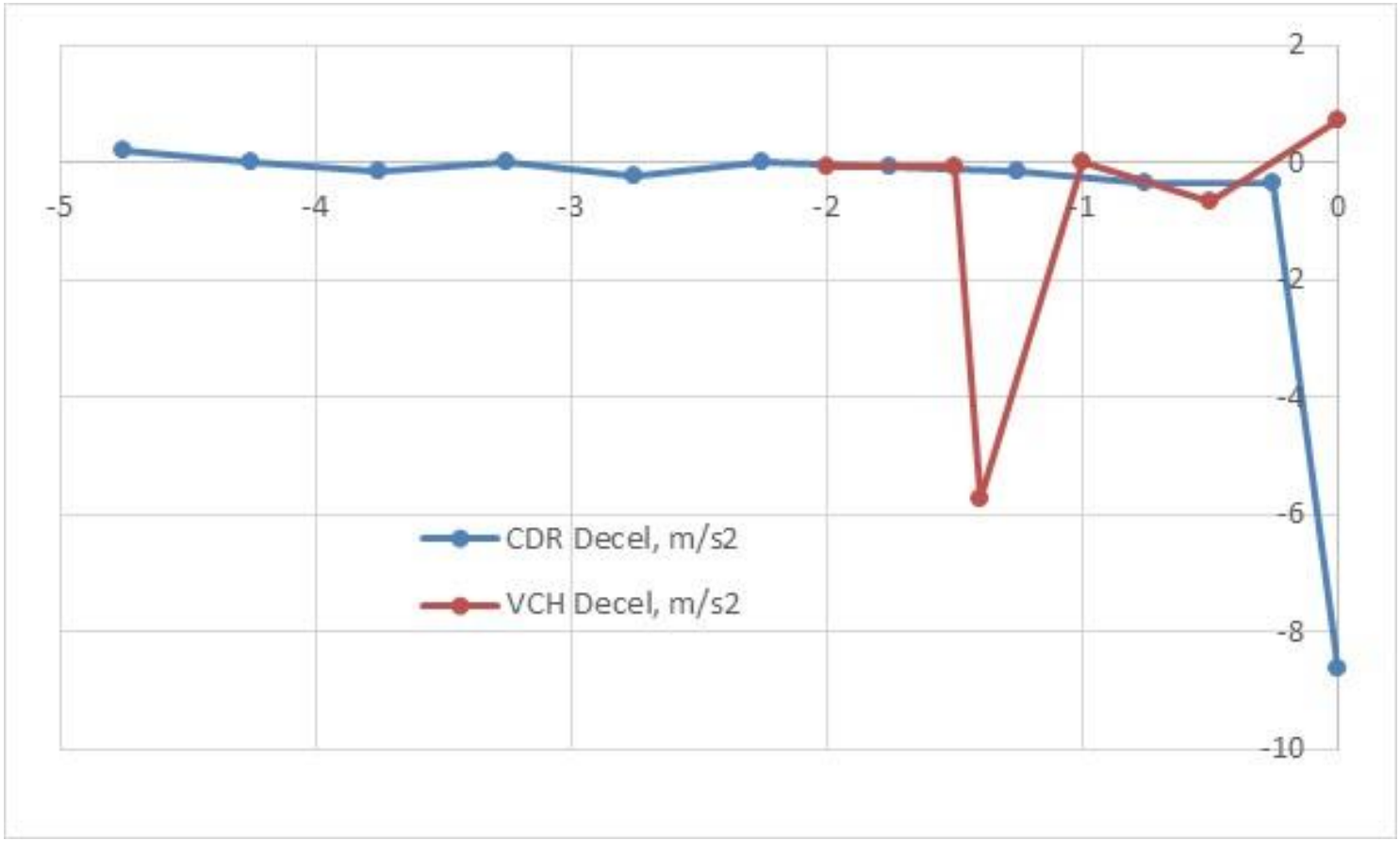
Time (sec)	-4.75	-4.25	-3.75	-3.25	-2.75	-2.25	-1.75	-1.25	-0.75	-0.25	0 (TRG)
Vehicle Speed (MPH [km/h])	46.6 [75]	47.2 [76]	47.8 [77]	47.8 [77]	48.5 [78]	48.5 [78]	48.5 [78]	49.1 [79]	48.5 [78]	47.8 [77]	47.2 [76]
Accelerator Pedal, % Full (%)	35.5	25.0	22.0	18.5	19.5	20.5	19.0	19.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	21.0	16.5	15.0	12.0	13.0	13.0	13.0	13.0	0.0	0.0	0.0
Engine RPM (RPM)	2,200	2,300	2,000	1,700	1,600	1,700	1,700	1,700	1,400	1,200	1,200
Motor RPM (RPM)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Longitudinal Acceleration , VSC Sensor (m/sec^2)	0.215	0.000	-0.144	0.000	-0.215	0.000	-0.072	-0.144	-0.359	-0.359	-8.614
Yaw Rate (deg/sec)	0.00	0.49	0.49	0.49	0.98	0.98	1.46	0.49	-1.95	0.49	0.00
Steering Input (degrees)	4.5	4.5	7.5	6.0	6.0	6.0	6.0	6.0	-7.5	10.5	10.5
Shift Position	D	D	D	D	D	D	D	D	D	D	D
Sequential Shift Range	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined
Cruise Control Status	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, PWR	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, ECO	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, Sport	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, Snow	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, EV	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF

Toyota Techstream Crash Data

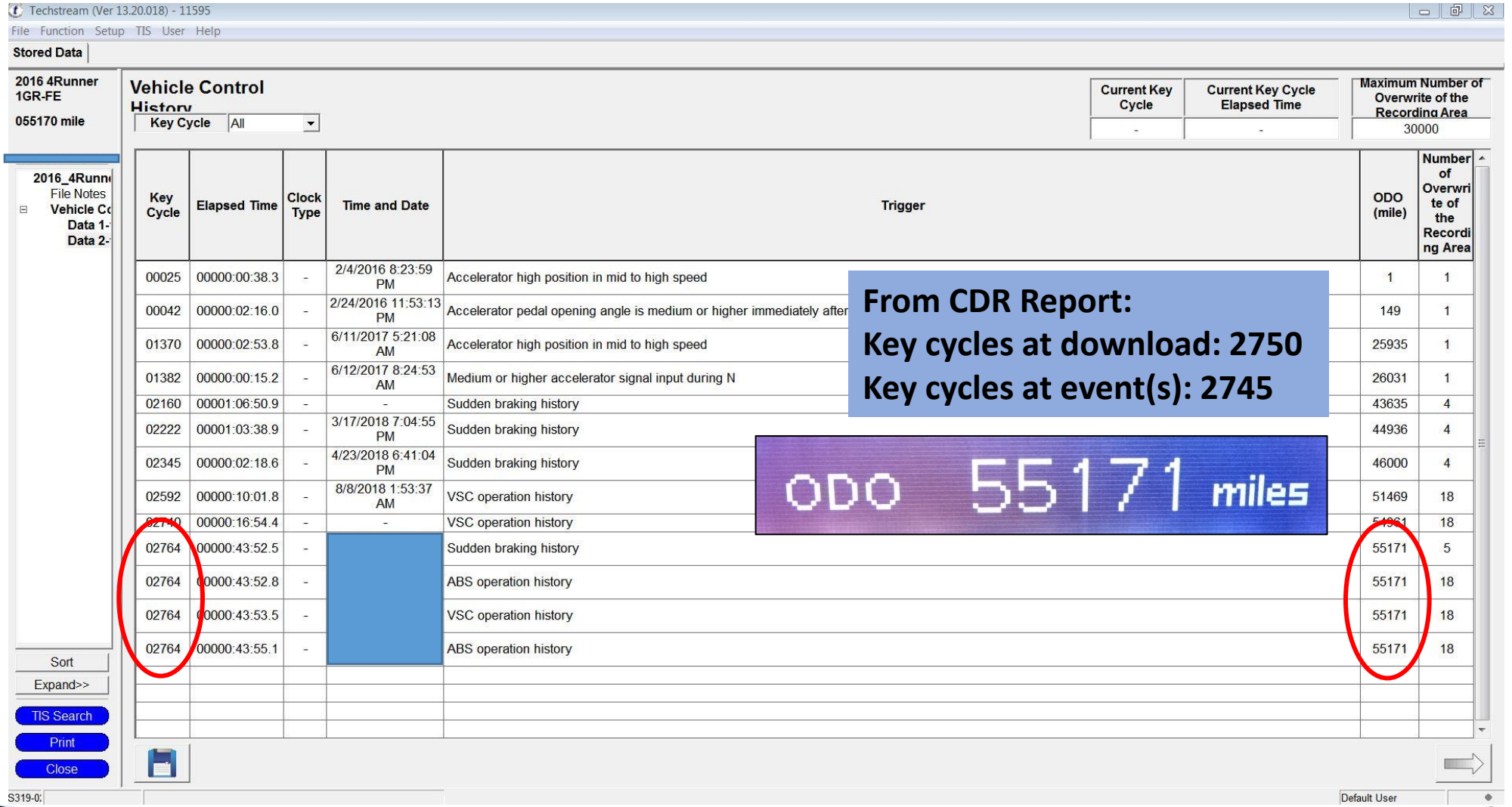


Toyota Travel Path
Approximate vehicle impact
Building Impact
Google Earth
N
200 ft

Toyota Vehicle Control History - Comparison to CDR Data



Control History Validation Example



2020 Toyota CDR shows “Trip Count” which matches Control History Key Cycle

Current Key Cycle
00121

System Status at Event (Most Recent Event, TRG 10)

TRG Count (times)	
Event Type	
Previous Crash Type	
Time from Previous TRG (msec)	
Freeze Signal	
Freeze Signal Factor	
Odometer signal (miles [km])	
Trip count (times)	117
Time count (msec)	12,774,900
Time count input system	Normal

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 4)

Time (sec)	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0 (TRG)
Vehicle Speed (MPH [km/h])	59.7 [96]	59.7 [96]	59.7 [96]	59.7 [96]	59.7 [96]	59.7 [96]	59.7 [96]	59.7 [96]	57.8 [93]	48.5 [78]	38.5 [62]
Accelerator Pedal, % Full (%)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Percentage of Engine Throttle (%)	8.5	8.5	8.5	8.5	8.0	8.0	8.0	8.5	0.5	0.0	0.0
Engine RPM (RPM)	1,700	1,700	1,700	1,700	1,700	1,700	1,700	1,700	1,600	1,200	1,000
Motor RPM (RPM)	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid
Service Brake, ON/OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
Brake Oil Pressure (Mpa)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.84	12.14	12.14
Longitudinal Acceleration , VSC Sensor (m/sec^2)	0.000	-0.072	-0.144	-0.144	0.000	-0.287	-0.072	-0.072	-7.680	-8.111	-8.973
Yaw Rate (deg/sec)	0.49	0.00	-0.49	0.00	0.00	0.49	0.00	-0.49	0.98	2.93	-0.98
Steering Input (degrees)	4.5	1.5	0.0	1.5	3.0	3.0	1.5	0.0	1.5	0.0	-1.5
Shift Position	D	D	D	D	D	D	D	D	D	D	D
Sequential Shift Range	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined	Undetermined
Cruise Control Status	ON	ON	ON	ON	ON	ON	ON	ON	OFF	OFF	OFF
Drive Mode, PWR	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, ECO	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, Sport	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, Snow	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Drive Mode, EV	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid	Invalid

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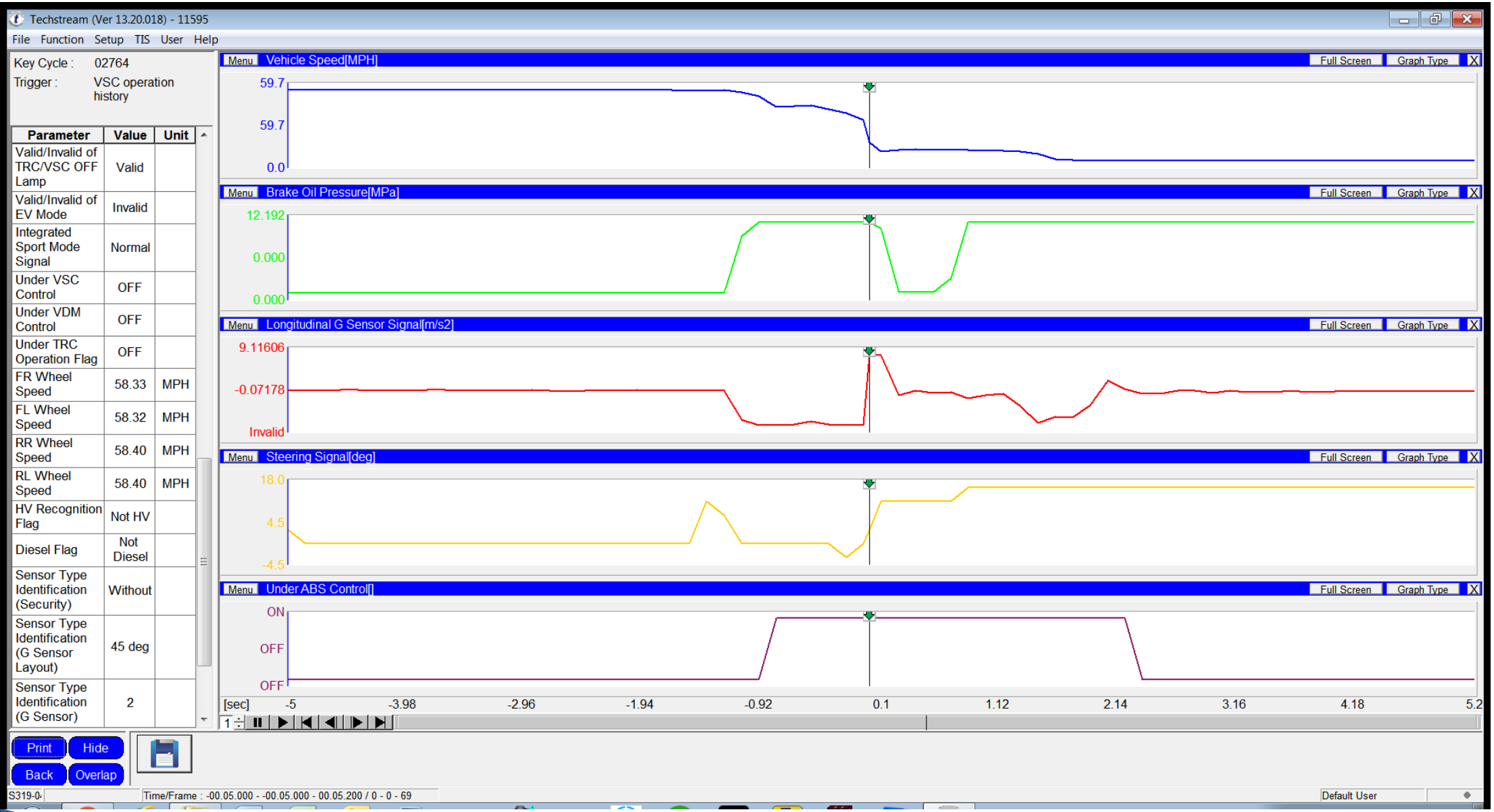
Techstream (Ver 13.20.018) - 11595
File Function Setup TIS User Help
Key Cycle : 02764
Trigger : VSC operation history
Parameter | Value | Unit
Valid/Invalid of TRC/VSC OFF Lamp | Valid
Valid/Invalid of EV Mode | Invalid
Integrated Sport Mode Signal | Normal
Under VSC Control | OFF
Under VDM Control | OFF
Under TRC Operation Flag | OFF
FR Wheel Speed | 58.33 | MPH
FL Wheel Speed | 58.32 | MPH
RR Wheel Speed | 58.40 | MPH
RL Wheel Speed | 58.40 | MPH
HV Recognition Flag | Not HV
Diesel Flag | Not Diesel
Sensor Type Identification (Security) | Without
Sensor Type Identification (G Sensor Layout) | 45 deg
Sensor Type Identification (G Sensor) | 2
Menu Vehicle Speed[MPH] Full Screen Graph Type
59.7
59.7
0.0
Menu Brake Oil Pressure[MPa] Full Screen Graph Type
12.192
0.000
0.000
Menu Longitudinal G Sensor Signal[m/s2] Full Screen Graph Type
9.11606
-0.07178
Invalid
Menu Steering Signal[deg] Full Screen Graph Type
18.0
4.5
-4.5
Menu Under ABS Control[] Full Screen Graph Type
ON
OFF
OFF
[sec] -5 -3.98 -2.96 -1.94 -0.92 0.1 1.12 2.14 3.16 4.18 5.2
Print Hide
Back Overlap
Time/Frame : -00.05.000 - -00.05.000 - 00.05.200 / 0 - 0 - 69
Default User

Control History Validation Example

Managem	3	Maximum	30000	Memory Efficiency In	System Presence/Abs	Number of Overwrite	Time Stamp at PCS Pr	Time Stamp at PCS Pr	Number of Overwrite	Time Stamp at ICS Pr	Time Stamp at ICS Pr	Number of Overwrite	Time Stamp
Current Ke	-	Current ke	-	Current Ti	########								
Time	Data ID	Trigger ID	Trigger ID	Trigger Tin	Detection	Detection	Number o	Time and I	Key Cycle	Clock Type	Elapsed Ti	ODO	Odometer	HV Recogr	Diesel Flag	Writing Co	Successive	Vehicle Sp	Acceleratc	Engine RPI	Cruise Cor	Shift Positi	Br
sec				ms	ms				trip		ms						ms	km/h	%	rpm			
30.85	M0	3																36	0	2000	0	1	
31.35	M1	3																36	0	1600	0	1	
31.85	M2	3																34	0	1400	0	1	
32.35	M3	3																31	0	1400	0	1	
32.85	M4	3																30	0	1400	0	1	
33.35	M5	3																29	0	1300	0	1	
33.85	M6	3																28	19	1200	0	1	
34.35	M7	3																30	92	2000	0	1	
34.85	M8	3																32	95	2400	0	1	
35.35	M9	3																35	84.5	3400	0	1	
35.85	M10	3																41	85	3900	0	1	
36.35	M11	3																46	84.5	4400	0	1	
36.85	M12	3																51	84.5	3100	0	1	
37.35	M13	3																54	84.5	3100	0	1	
37.85	M14	3																57	84.5	3200	0	1	
38.3	T	3	0	50	32766.5	65533	1	########	25	0	38300	1	2	0	0	85		60	85	3300	0	1	
38.35	N0	3																60	84.5	3400	0	1	
38.85	N1	3																63	82	3500	0	1	
39.35	N2	3																66	77.5	3700	0	1	
39.85	N3	3																69	77.5	3900	0	1	
40.35	N4	3																73	68	4000	0	1	
40.85	N5	3																76	52	4200	0	1	
41.35	N6	3																79	37.5	3200	0	1	
41.85	N7	3																79	0	2700	0	1	

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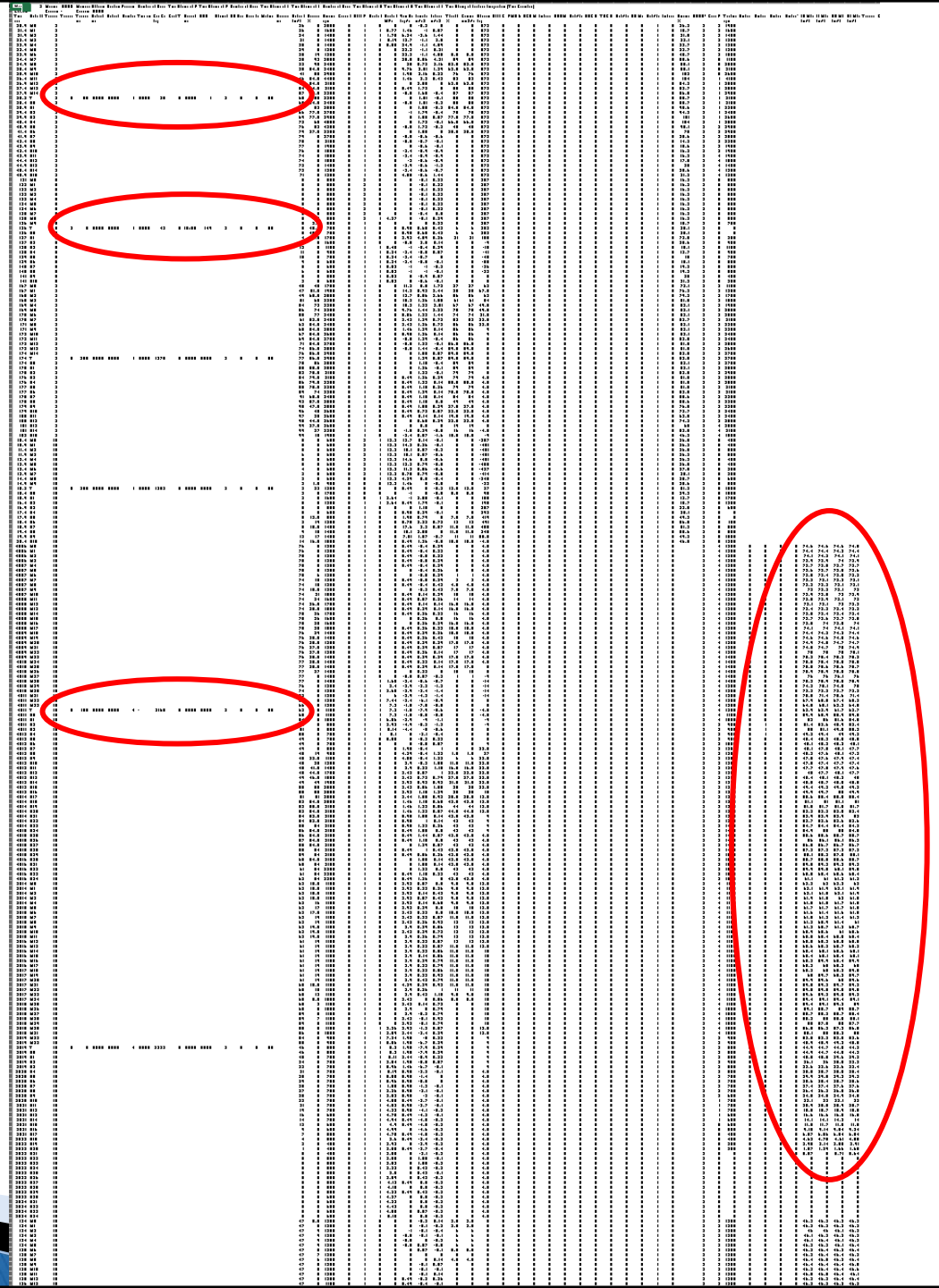

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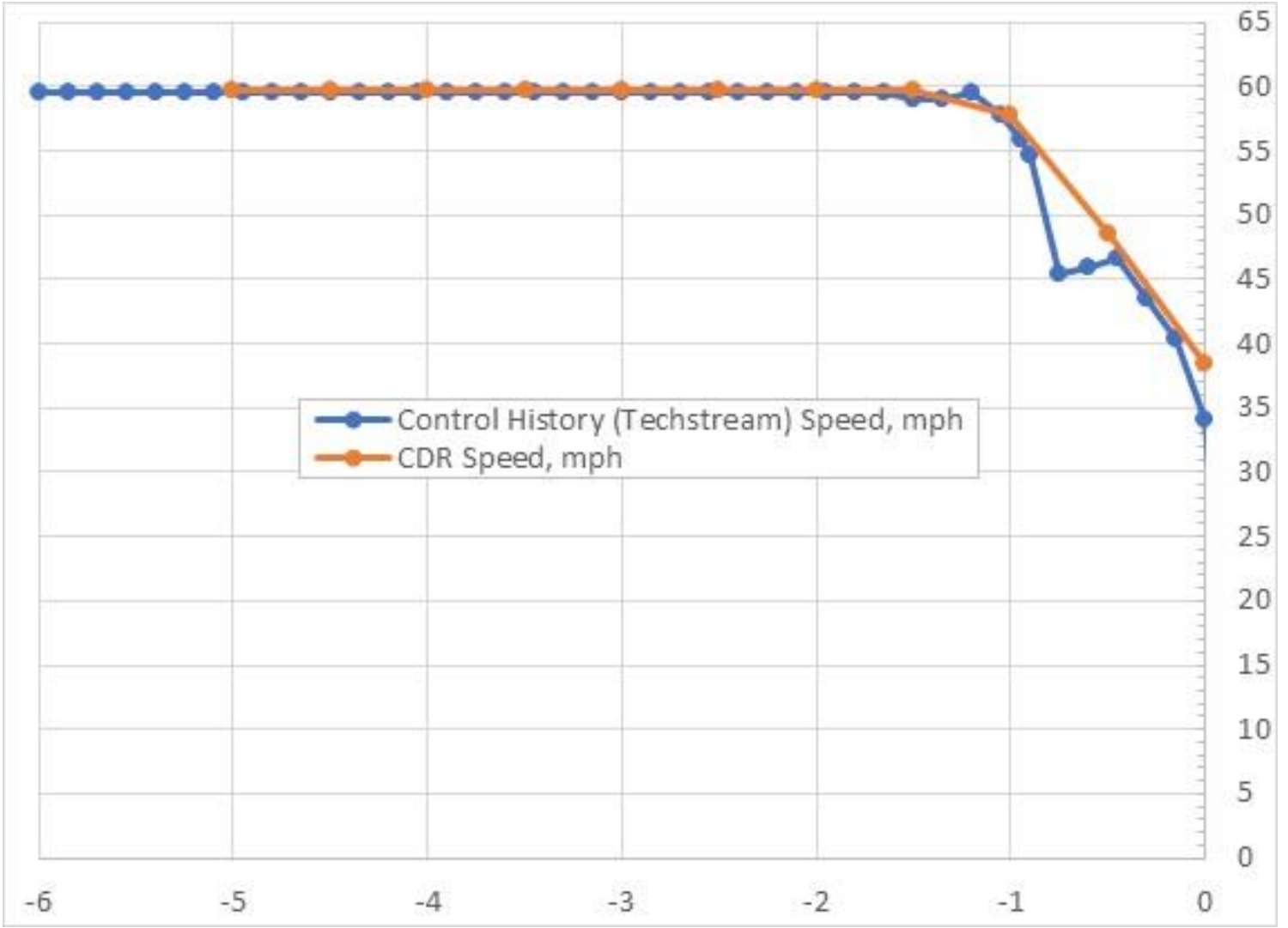
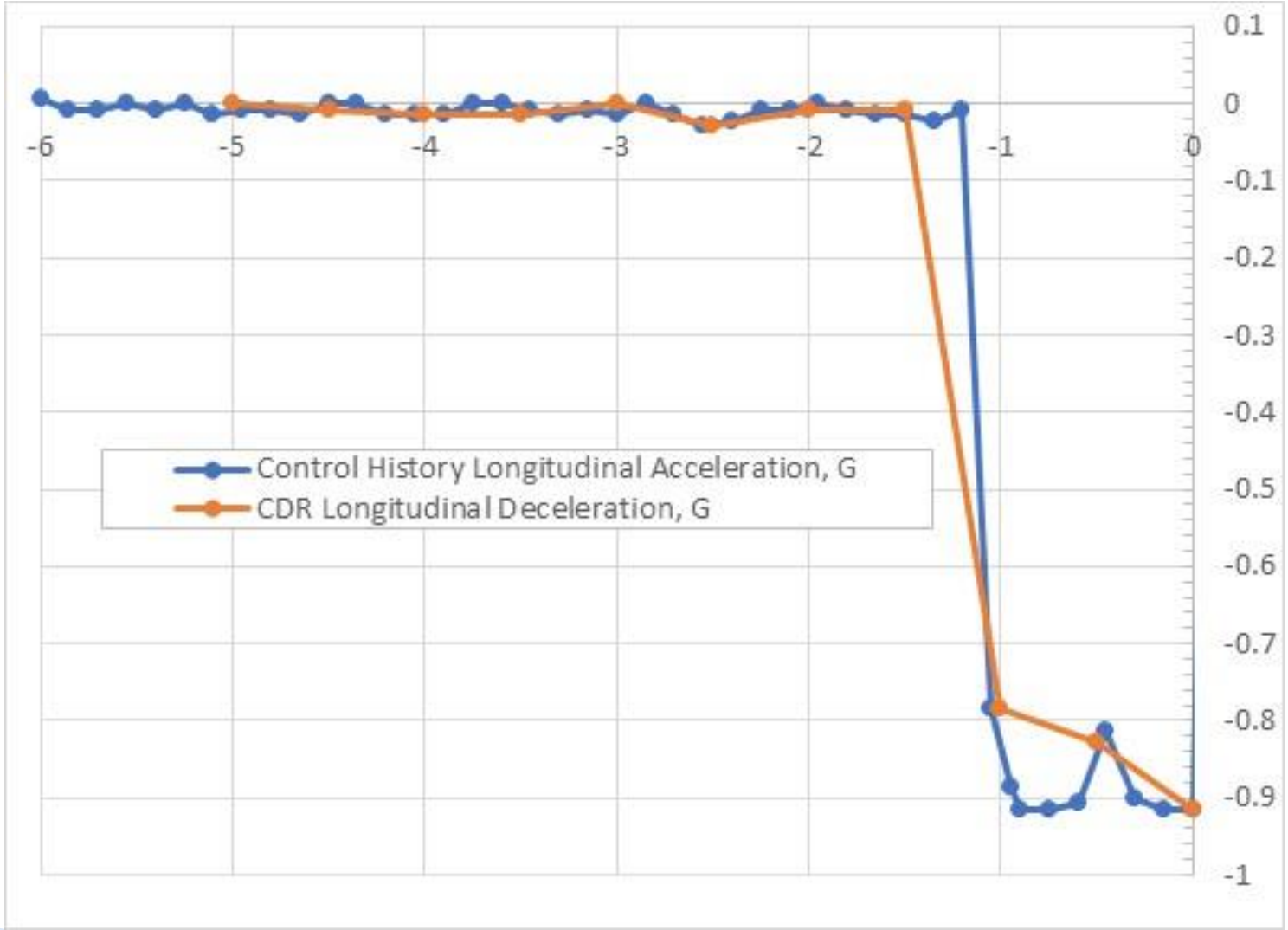
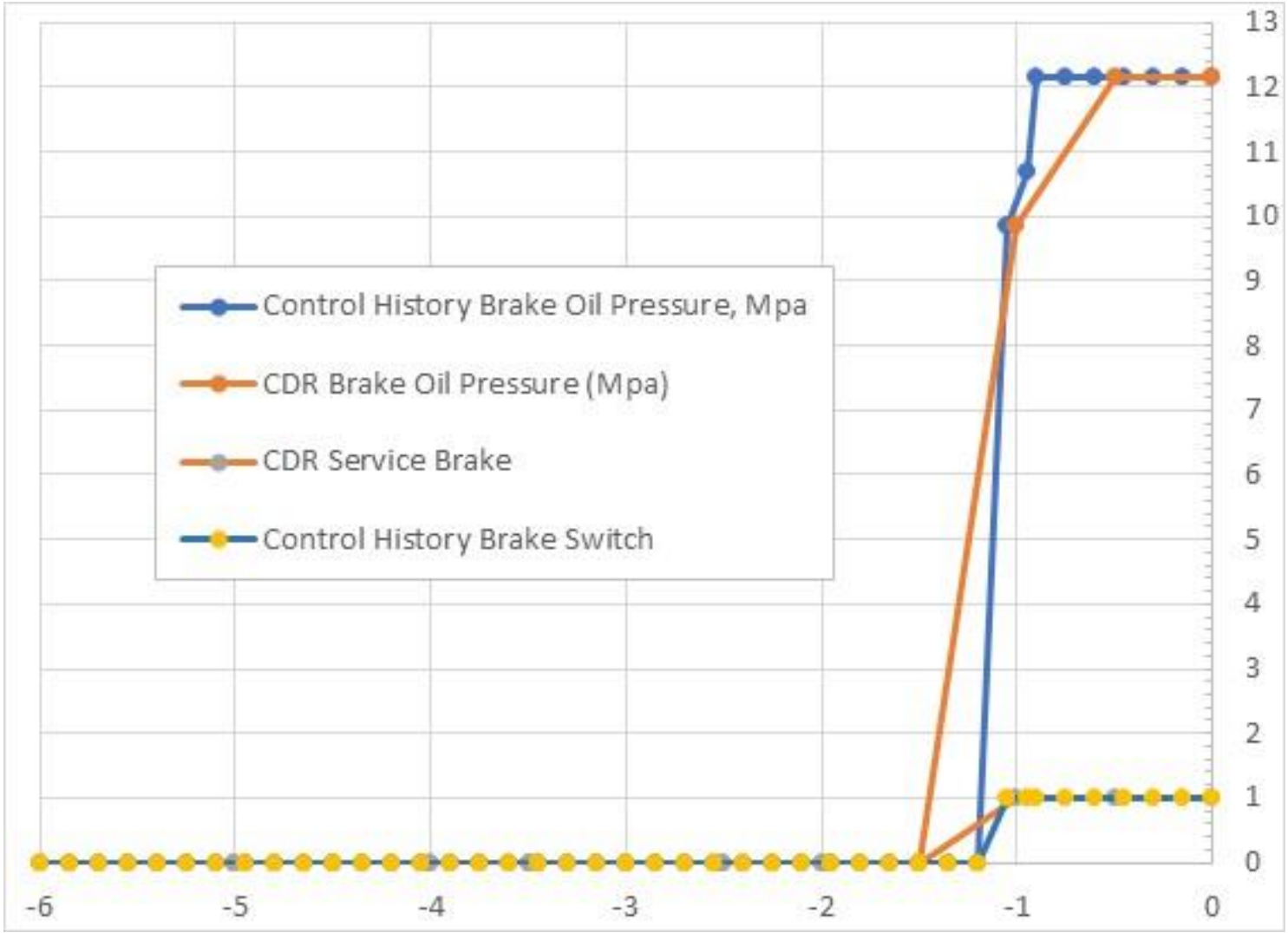
Control History Validation Example



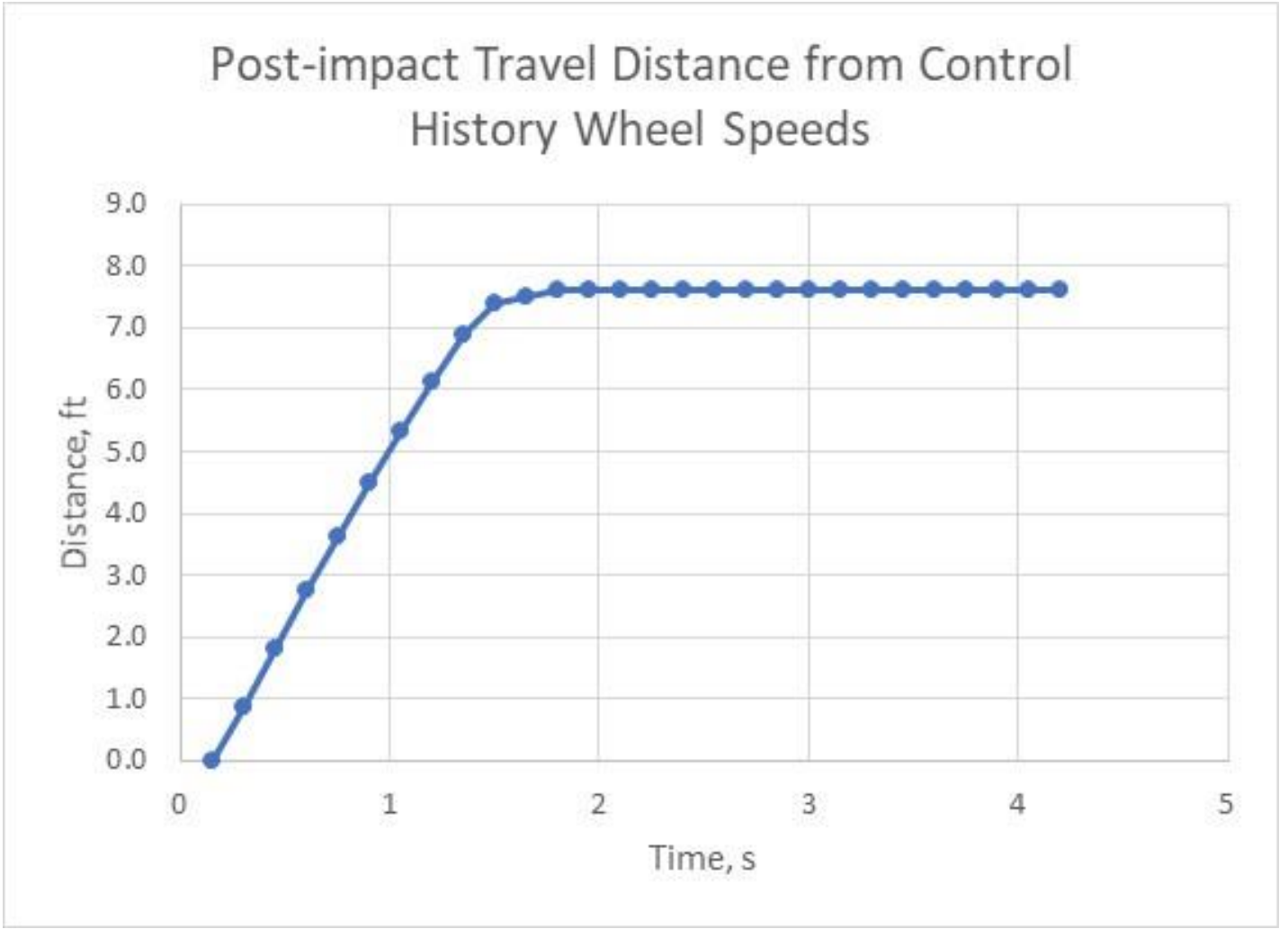
Control History Validation Example



Control History Validation Example



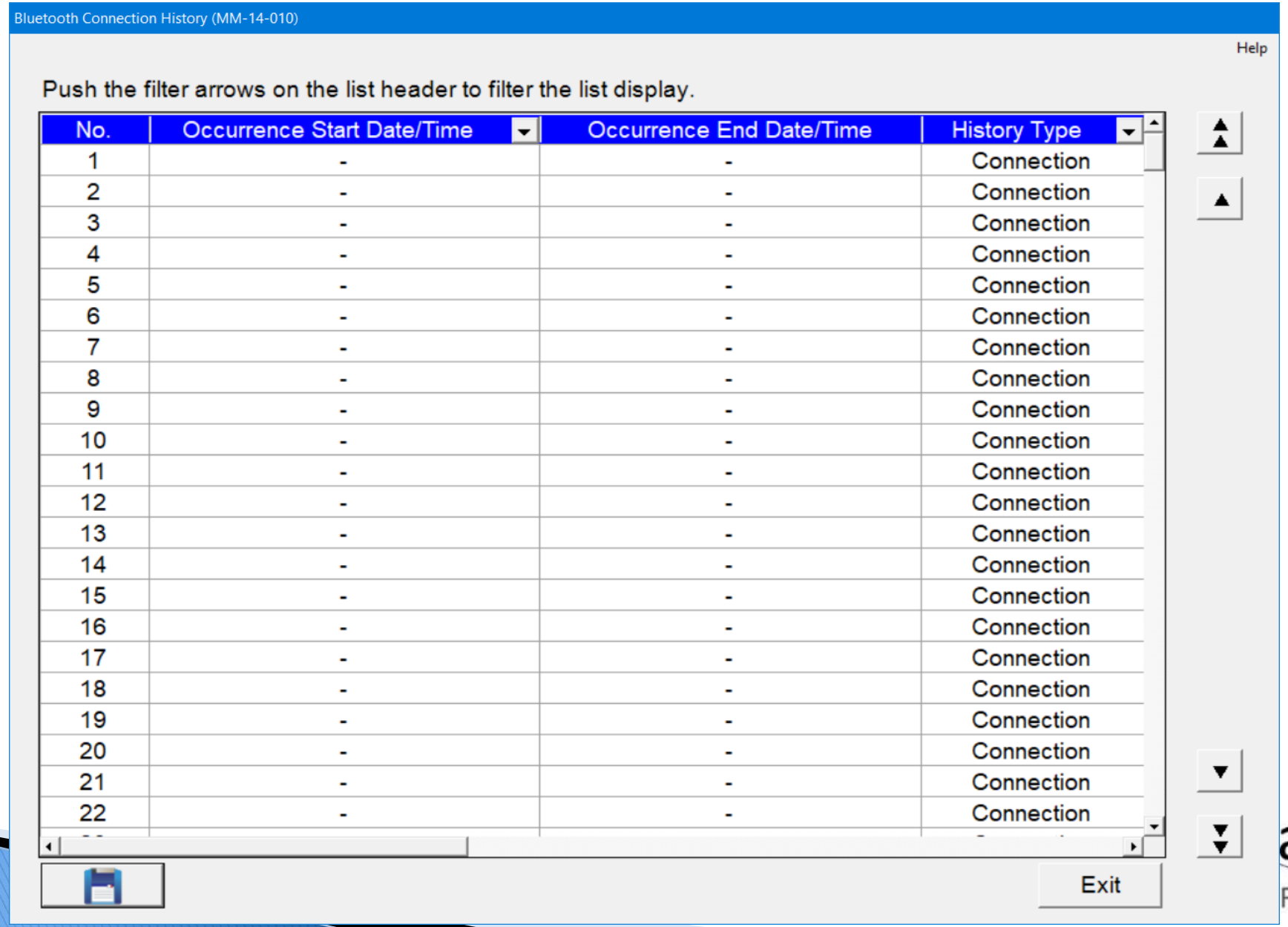
Control History Validation Example

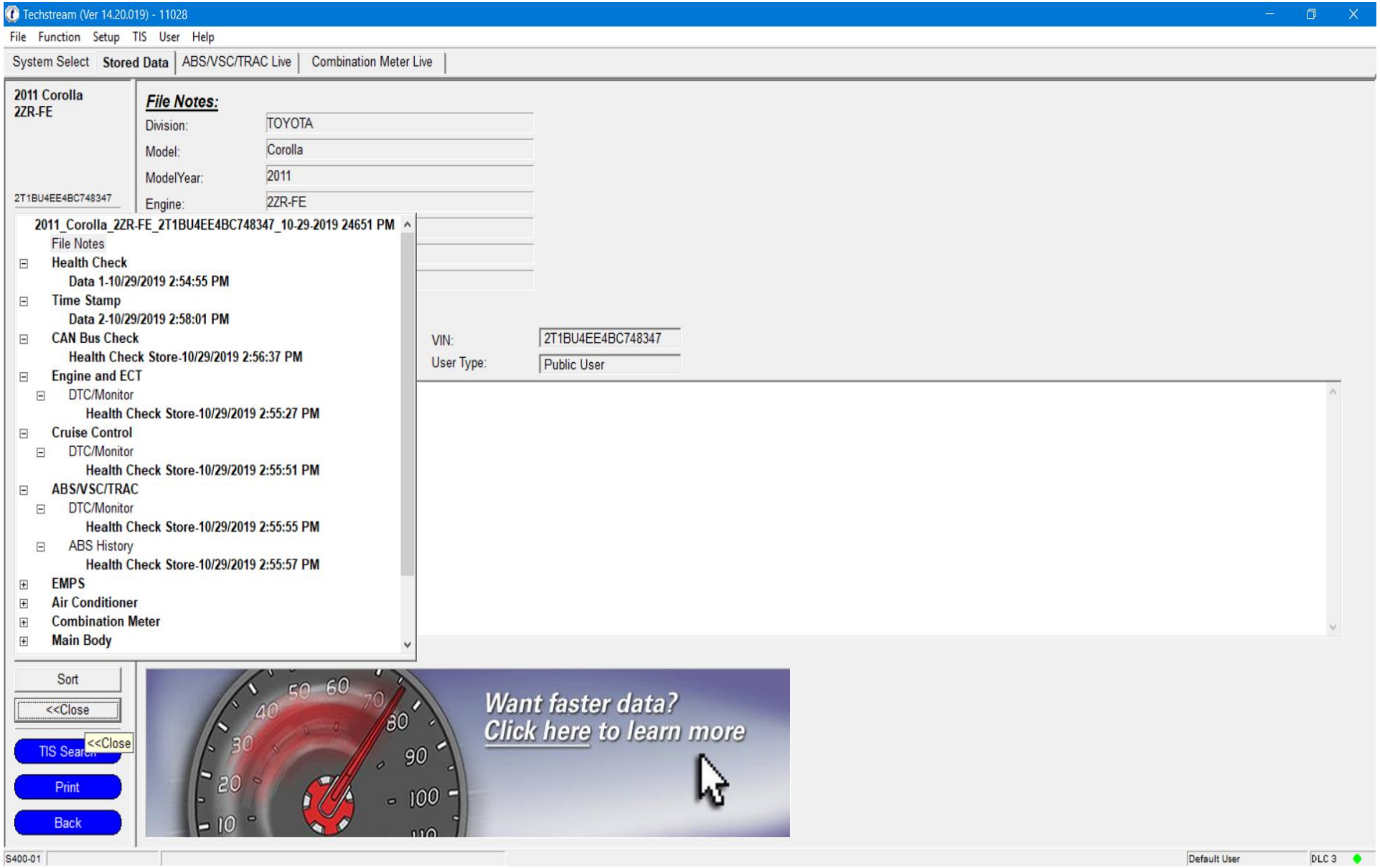


Misc History Data from Techstream

- Bluetooth Connection History
- ABS & Traction Control History (2011+)
  - Possibly only first three events since manufacture date
- Door Lock History

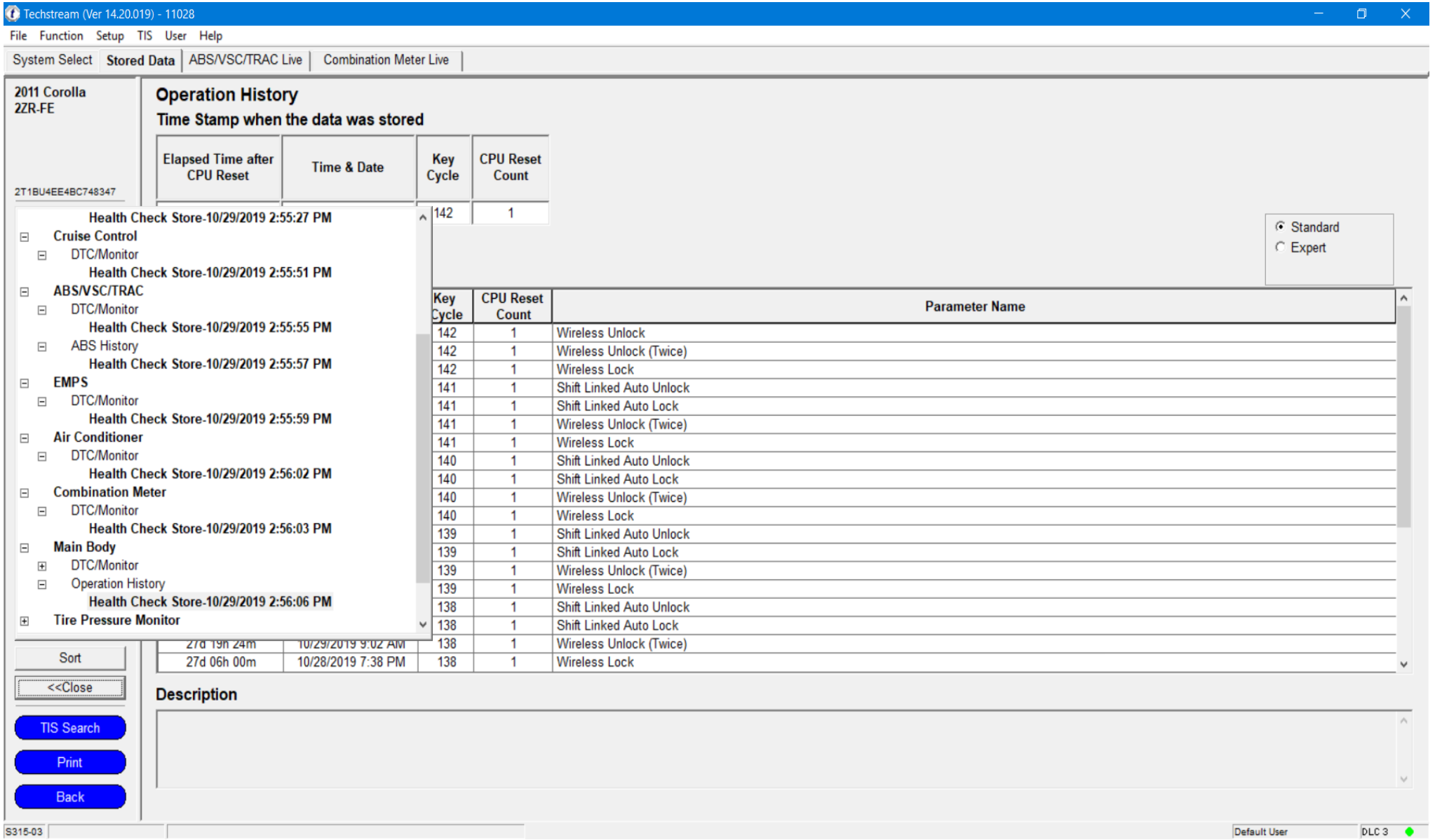
Bluetooth connection history




Techstream (Ver 14.20.019) - 11028
File Function Setup TIS User Help
System Select
Stored Data
ABS/VSC/TRAC Live
Combination Meter Live
2011 Corolla
2ZR-FE
2T1BU4EE4BC748347
File Notes:
Division: TOYOTA
Model: Corolla
ModelYear: 2011
Engine: 2ZR-FE
2011_Corolla_2ZR-FE_2T1BU4EE4BC748347_10-29-2019 24651 PM
File Notes
Health Check
Data 1-10/29/2019 2:54:55 PM
Time Stamp
Data 2-10/29/2019 2:58:01 PM
CAN Bus Check
Health Check Store-10/29/2019 2:56:37 PM
Engine and ECT
DTC/Monitor
Health Check Store-10/29/2019 2:55:27 PM
Cruise Control
DTC/Monitor
Health Check Store-10/29/2019 2:55:51 PM
ABS/VSC/TRAC
DTC/Monitor
Health Check Store-10/29/2019 2:55:55 PM
ABS History
Health Check Store-10/29/2019 2:55:57 PM
EMPS
Air Conditioner
Combination Meter
Main Body
VIN: 2T1BU4EE4BC748347
User Type: Public User
Sort
<<Close
<<Close
TIS Search
Print
Back
Want faster data?
Click here to learn more
S400-01
Default User
DLC 3

ABS Event

Elapsed Time after Freeze Trigger	0	msec
Number of IG ON	31	
Buzzer	OFF	
Stop Light SW	ON	
Parking Brake SW	OFF	
Gear Position	3rd	
Shift Lever Position	D/M	
Operated System	ABS	
Master Cylinder Sensor	0.52	V
M/C Sensor Grade	-3	MPa/s
Yaw Rate Sensor	14	degrees/s
Lateral G	4.51	m/s2
Forward and Rearward G	-1.17	m/s2
FR Wheel Speed	17	MPH
FL Wheel Speed	16	MPH
RR Wheel Speed	17	MPH
RL Wheel Speed	14	MPH
Vehicle Speed	17	MPH
Vehicle Speed Grade	0.78	m/s2
Real Engine Torque	2.4	Nm
Accelerator Opening Angle %	0.0	%
IG1 Voltage Value	13.49	V
Inspection Mode	Other	
Steering Angle Sensor	113.8	degrees
TRAC/VSC Off Mode	Normal	


Techstream (Ver 14.20.019) - 11028
File Function Setup TIS User Help
System Select Stored Data ABS/VSC/TRAC Live Combination Meter Live
2011 Corolla
2ZR-FE
2T1BU4EE4BC748347
Operation History
Time Stamp when the data was stored
Elapsed Time after CPU Reset
Time & Date
Key Cycle
CPU Reset Count
142
1
Health Check Store-10/29/2019 2:55:27 PM
Cruise Control
DTC/Monitor
Health Check Store-10/29/2019 2:55:51 PM
ABS/VSC/TRAC
DTC/Monitor
Health Check Store-10/29/2019 2:55:55 PM
ABS History
Health Check Store-10/29/2019 2:55:57 PM
EMPS
DTC/Monitor
Health Check Store-10/29/2019 2:55:59 PM
Air Conditioner
DTC/Monitor
Health Check Store-10/29/2019 2:56:02 PM
Combination Meter
DTC/Monitor
Health Check Store-10/29/2019 2:56:03 PM
Main Body
DTC/Monitor
Operation History
Health Check Store-10/29/2019 2:56:06 PM
Tire Pressure Monitor
Standard
Expert
Key Cycle
CPU Reset Count
Parameter Name
142 1 Wireless Unlock
142 1 Wireless Unlock (Twice)
142 1 Wireless Lock
141 1 Shift Linked Auto Unlock
141 1 Shift Linked Auto Lock
141 1 Wireless Unlock (Twice)
141 1 Wireless Lock
140 1 Shift Linked Auto Unlock
140 1 Shift Linked Auto Lock
140 1 Wireless Unlock (Twice)
140 1 Wireless Lock
139 1 Shift Linked Auto Unlock
139 1 Shift Linked Auto Lock
139 1 Wireless Unlock (Twice)
139 1 Wireless Lock
138 1 Shift Linked Auto Unlock
138 1 Shift Linked Auto Lock
27d 19h 24m 10/29/2019 9:02 AM 138 1 Wireless Unlock (Twice)
27d 06h 00m 10/28/2019 7:38 PM 138 1 Wireless Lock
Sort
<<Close
TIS Search
Print
Back
Description
S315-03
Default User
DLC 3

Main Body Operation History

Elapsed Time after CPU Reset	Time & Date	Key Cycle	CPU Reset Count
28d 01h 18m	10/29/2019 2:56 PM	142	1

Operation History Report(Main Body)
Door Lock/Unlock(1 of 1)

Elapsed Time after CPU Reset	Time & Date	Key Cycle	CPU Reset Count	Parameter Name
28d 01h 06m	10/29/2019 2:44 PM	142	1	Wireless Unlock
28d 01h 06m	10/29/2019 2:44 PM	142	1	Wireless Unlock (Twice)
27d 22h 57m	10/29/2019 12:35 PM	142	1	Wireless Lock
27d 22h 57m	10/29/2019 12:35 PM	141	1	Shift Linked Auto Unlock
27d 22h 51m	10/29/2019 12:29 PM	141	1	Shift Linked Auto Lock
27d 22h 51m	10/29/2019 12:29 PM	141	1	Wireless Unlock (Twice)
27d 22h 27m	10/29/2019 12:05 PM	141	1	Wireless Lock
27d 22h 27m	10/29/2019 12:05 PM	140	1	Shift Linked Auto Unlock
27d 22h 09m	10/29/2019 11:47 AM	140	1	Shift Linked Auto Lock
27d 22h 09m	10/29/2019 11:47 AM	140	1	Wireless Unlock (Twice)
27d 22h 06m	10/29/2019 11:44 AM	140	1	Wireless Lock
27d 22h 06m	10/29/2019 11:44 AM	139	1	Shift Linked Auto Unlock
27d 21h 48m	10/29/2019 11:26 AM	139	1	Shift Linked Auto Lock
27d 21h 48m	10/29/2019 11:26 AM	139	1	Wireless Unlock (Twice)
27d 19h 54m	10/29/2019 9:32 AM	139	1	Wireless Lock
27d 19h 54m	10/29/2019 9:32 AM	138	1	Shift Linked Auto Unlock
27d 19h 24m	10/29/2019 9:02 AM	138	1	Shift Linked Auto Lock
27d 19h 24m	10/29/2019 9:02 AM	138	1	Wireless Unlock (Twice)

Bendix Wingman

Steering wheel angle is positive in left turns

Time	FLR Audible Alert	Vehicle Speed (mph)	Steering Angle (deg)	Accelerator Pedal Position (%)
0.0	0	49.21	10.74	96.0
0.5	0	49.21	10.74	96.0
1.0	0	49.21	10.74	96.0
1.5	0	51.45	10.74	96.0
2.0	0	51.45	16.12	96.0
2.5	0	51.45	21.49	96.0
3.0	0	51.45	21.49	96.0
3.5	0	51.45	16.12	96.0
4.0	0	51.45	16.12	96.0
4.5	0	53.69	10.74	96.0
5.0	0	53.69	10.74	80.0
5.5	0	53.69	10.74	76.8
6.0	0	53.69	10.74	80.0
6.5	0	53.69	10.74	83.2
7.0	0	53.69	16.12	57.6
7.5	0	53.69	10.74	41.6
8.0	0	53.69	16.12	51.2
8.5	0	53.69	16.12	32.0
9.0	0	53.69	21.49	0.0
9.5	0	53.69	21.49	35.2
10.0	1	53.69	21.49	51.2
10.5	1	51.45	26.86	57.6
11.0	1	51.45	26.86	89.6
11.5	2	53.69	26.86	89.6
12.0	2	53.69	26.86	96.0
12.5	2	53.69	26.86	96.0
13.0	0	53.69	26.86	96.0
13.5	0	53.69	26.86	96.0
14.0	0	53.69	21.49	96.0

Bendix Wingman – False Positive

Time	Vehicle Speed (mph)	Steering Angle (deg)	Accelerator Pedal Position (%)	Brake Light	Driver Brake Application	Radar Collision Warning	ABS Activity
0	56	5	0	2	1	1	0
0.5	56	11	0	2	1	1	0
1	54	5	0	2	1	0	0
1.5	54	0	0	2	1	0	0
2	51	-11	0	2	1	0	0
2.5	51	-11	0	2	0	0	0
3	51	-16	38	0	0	0	0

Bendix Wingman – False Positive

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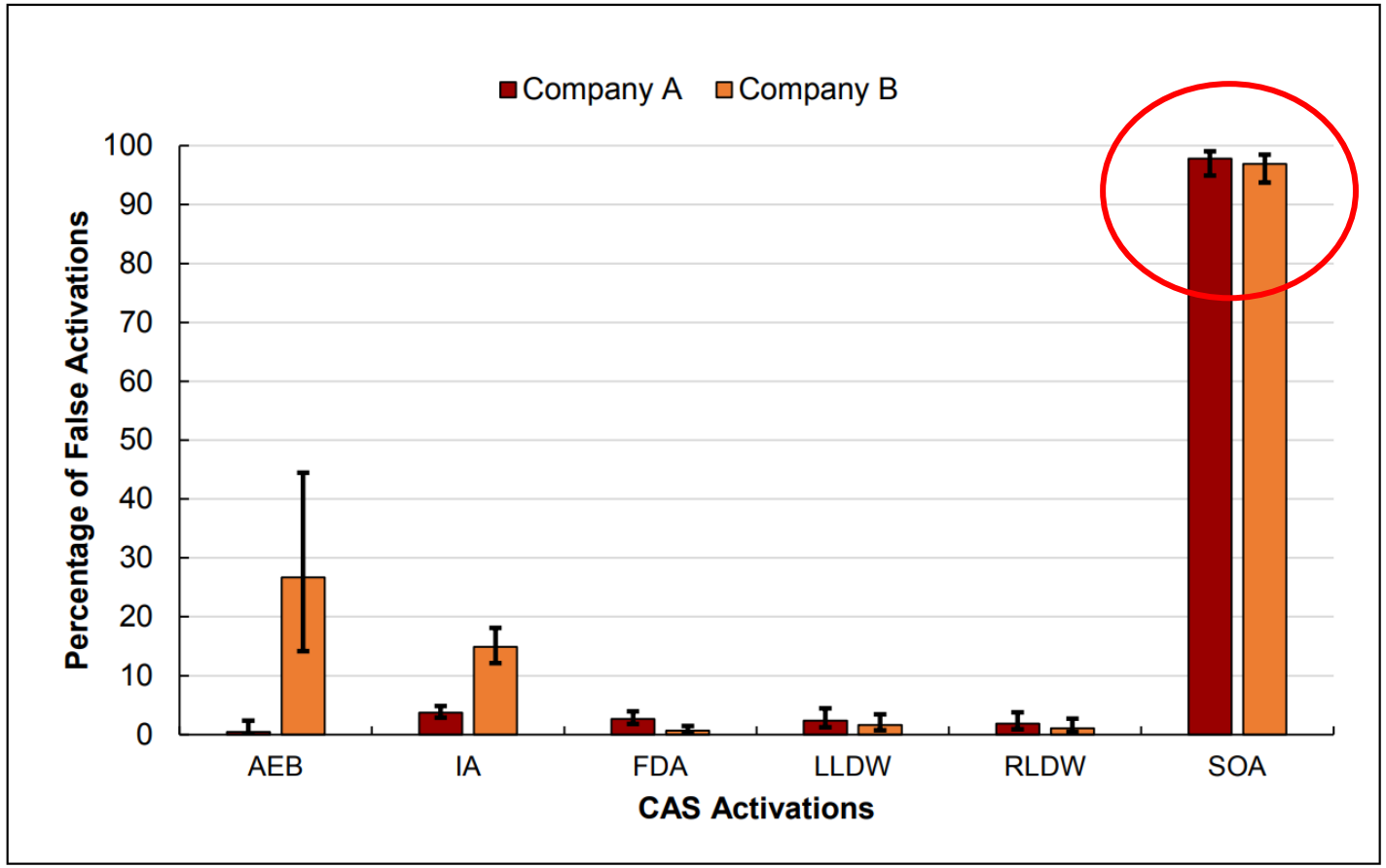
Bendix Wingman – False Positive



CAS – Collision Avoidance System
SOA – Stationary Object Alert

Data is for model year 2013 radar-only heavy truck systems

Field Study of Heavy-Vehicle Crash Avoidance Systems, NHTSA & VTTI, 2016
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812280_fieldstudyheavy-vehiclecas.pdf

Bendix Wingman

Wingman Fusion – includes adaptive cruise control, collision warning and AEB

Safety Direct Processor (SDP) – dashcam function, recorded on an SD card in-cab, and/or sent to a web server.

Bendix Safety Direct Processor (SDP)

FIGURE 1 - *SAFETYDIRECT® WEB PORTAL PROCESSOR*

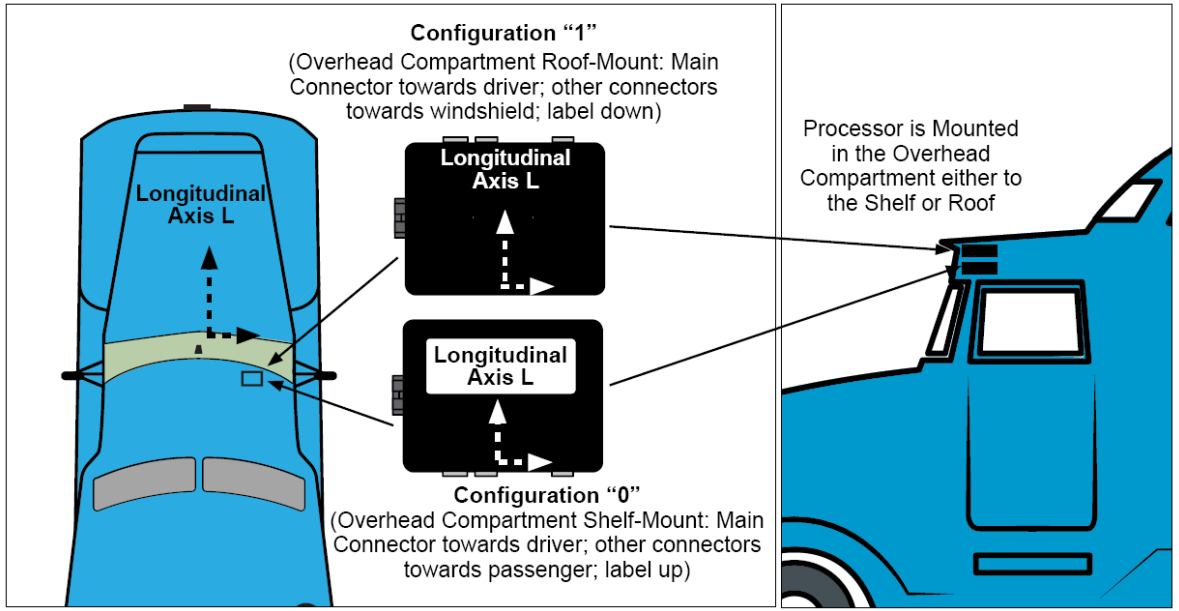


FIGURE 4 - *SAFETYDIRECT® BY BENDIX CVS WEB PORTAL PROCESSOR PERMISSIBLE CONFIGURATIONS*

SafetyDirect CVS Web Portal Processor Service Data, SD-65-21025
https://www.bendix.com/en/servicessupport/documentlibrary/doclib_1.jsp

Bendix Data Recording (2015)

For the Bendix Wingman Fusion system, if an event causes data recording to be triggered in the Bendix® EC-80™ Electronic Control Unit (ECU), as noted in *Section C,* then the camera will record and save five (5) seconds of video prior to the event trigger being activated and five (5) seconds of video after the event trigger is activated, for a total of ten seconds of event video. As noted in B, the system captures trigger information every half (0.5) second.

The AutoVue Lane Departure Warning (LDW) System – linked with Bendix™ SafetyDirect® – operates independently of the Bendix EC-80 system. As such, it may, or may not, capture events or data which are not captured by the Bendix EC-80 data recorder. This system, when triggered, will all also capture 10 seconds of video before and 10 seconds of video after the event trigger is activated.

Bendix EC-80 ECU Data Storage TCH-013-026, 2015
https://www.bendix.com/en/servicessupport/documentlibrary/doclib_1.jsp

Bendix Available Data

- EDR data in EC-80 and EC-60 Advanced ECUs
  - Data is in a hex format, embedded in the HTML report file
  - Send HTML file and/or the ECU to Bendix for interpretation and report
- Safety Direct Processor (SDP)
  - Send module to Bendix for video and event data download
  - Some fleets may have *Safety Direct Event Log Client software*, which allows downloading video at the truck.
- AutoVue FLC-20 Camera
  - May have video recorded; unclear in Bendix literature
  - Send to Bendix

More info in SAE Course: Accessing and Interpreting Heavy Vehicle Event Data Recorders C1022

https://www.sae.org/learn/content/c1022/

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Bendix ECU Data Request

Section 3: Vehicle & Component Information (all fields mandatory, 1 component per form)			
	Vehicle VIN, Make & Model		Photo of Component Sent To Bendix
	Component removed	ABS ECU [ ] FLC20 Camera [ ] AutoVue 3G ECU [ ] SafetyDirect Processor [ ] Other [ ] (check all that apply)	
Component removed from vehicle	Removed from vehicle by		
	Removal date & time		
Component sent to Bendix **MUST BE SENT VIA FEDEX OR UPS FOR TRACKING**	Date sent to Bendix		
	Sender name		
	Sender signature		
	Carrier	FedEx [ ] UPS [ ]	Clearly show label, body and connector with photos. Add as many photos as necessary.
	Tracking number		

REQUEST AND RELEASE OF BENDIX ECU DATA DOWNLOAD Form (A.B. Moore copy):
https://www.dropbox.com/s/kclcltgpl395xyy/Aug%202018%20Y206579_1_US_001_Bendix%20Request%20and%20Release%20ECU%20Data%20Download%20Form.pdf?dl=0

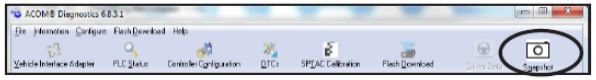
Bendix Wingman Fusion Camera Calibration

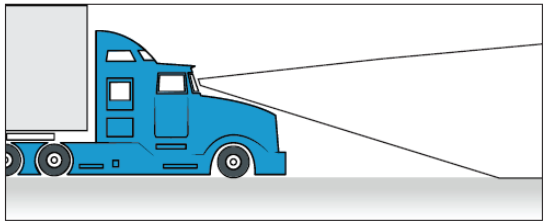
2.3 CAMERA TEST IMAGE

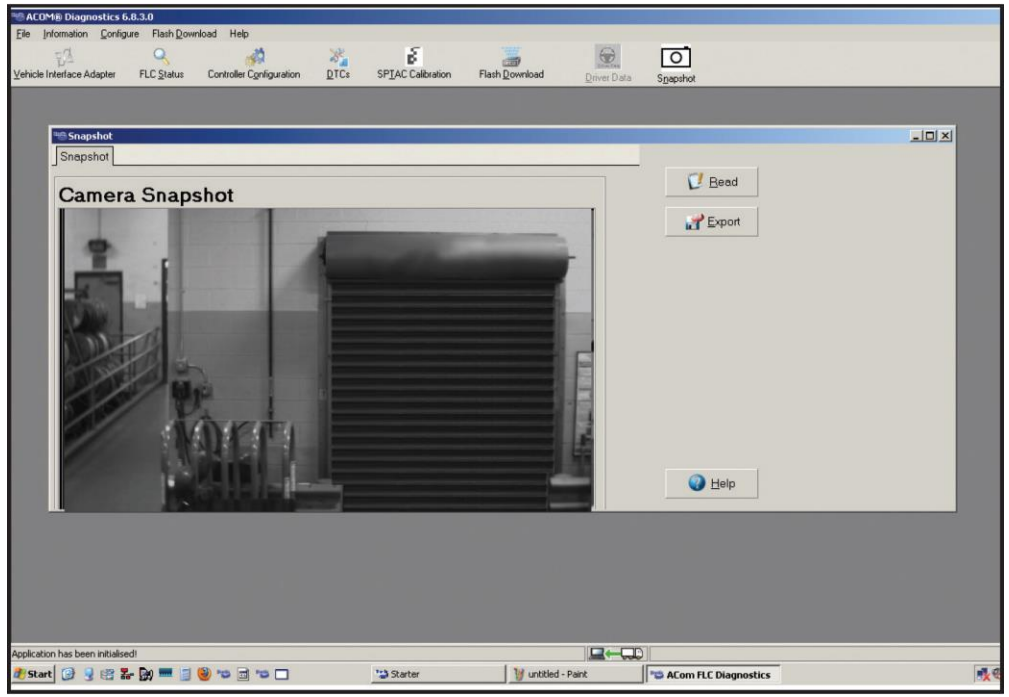
Take a Test Image Using Latest Version of Bendix® ACom® Diagnostic Software

Troubleshooting a camera may be assisted by viewing a test image from the camera to ensure it is not blocked, or has another problem.

- Open Bendix® ACom® Diagnostic Software and select “FLC-20” from the opening page list, and the “Start with ECU” button.
- Select “Snapshot” from the choices.







Bendix™ AutoVue® FLC-20™ Camera, SD-64-20124 Service Data Sheet
https://www.bendix.com/en/servicessupport/documentlibrary/doclib_1.jsp

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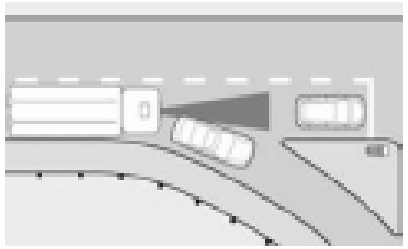
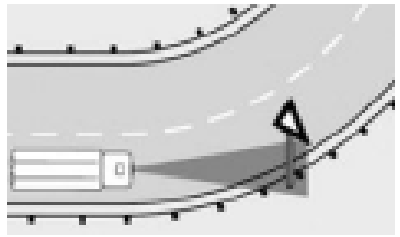
Detroit Diesel Assurance 2 - AEB

2018 Freightliner Cascadia with Detroit Assurance 2

Obstacles and stationary vehicles	 12/18/2014 f040837a	ABA and ACC cannot detect obstacles or stationary vehicles in front of the detected vehicle. ABA and ACC may unexpectedly issue warnings or brake your vehicle. ACC may also accelerate the vehicle unexpectedly.
Stationary objects	 12/18/2014 f040842a	ABA can also unexpectedly issue warnings and brake the vehicle if it detects stationary objects like: • vehicles which have broken down • signs • bridges

Detroit Assurance 2 - AEB

	Object [the Toyota] Detected	Optic Accoustic Warning [Light and Alarm]	Haptic Braking [Brake pulsation]	Emergency Braking [Performed by computer]	Object [the Toyota] lost
Time (in Seconds)	-2.3	0	0.96	2.36	4.49
My Speed (in mph)	69.46	69.47	69.47	69.49	66.5
My acceleration (m/s2)	0	0	0	0	-2.5
My Steering angle (in degrees)	N/A	-5.32	-1.06	4.76	N/A
ABS Activity	0	0	0	0	0
ESP Activity	0	0	0	0	1
Tracked Object Speed (in mph)	25.13	15	10.73	4.11	2
Tracked Object Acceleration (in m/s2)	-2.2	-1.9	-1.9	-2.1	0
Distance to tracked object (in ft)	614.88	443.25	362.35	231.53	18.28
Lateral offset of tracked object (in ft)	0	-3.94	-2.62	-1.31	-7.22
ABA Availability (Y/N)	1	1	1	1	1
Driver Activity Occured	N/A	Yes	Yes	Yes	Yes

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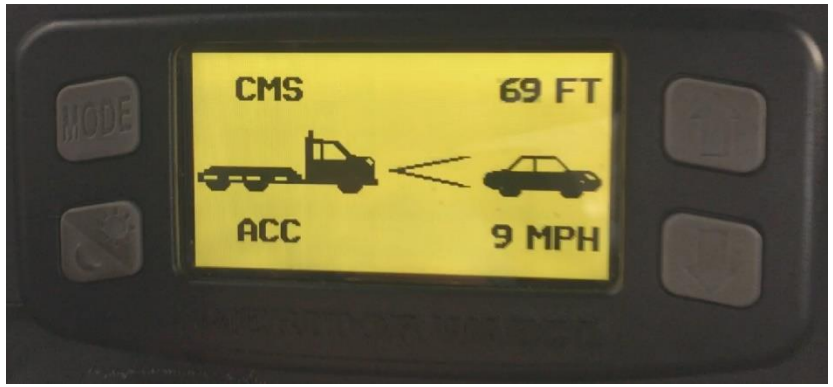
WabCo OnGuard (Radar Only)

- ACC seeks a 3.6s following distance[1]. But if intruding vehicles reduce it, ACC is slow to increase the distance, taking more than 10s to do so in testing.
- CMS will alert, but not brake, for always-stationary objects
- CMS will alert and brake, for moving and was-moving (stopped) objects, using a haptic brake apply (0.5s) followed by a 50% or 0.35g autobrake
- “Should not be relied upon” to detect motorcycles, bicycles and pedestrians
- Active only above truck speed of 15mph



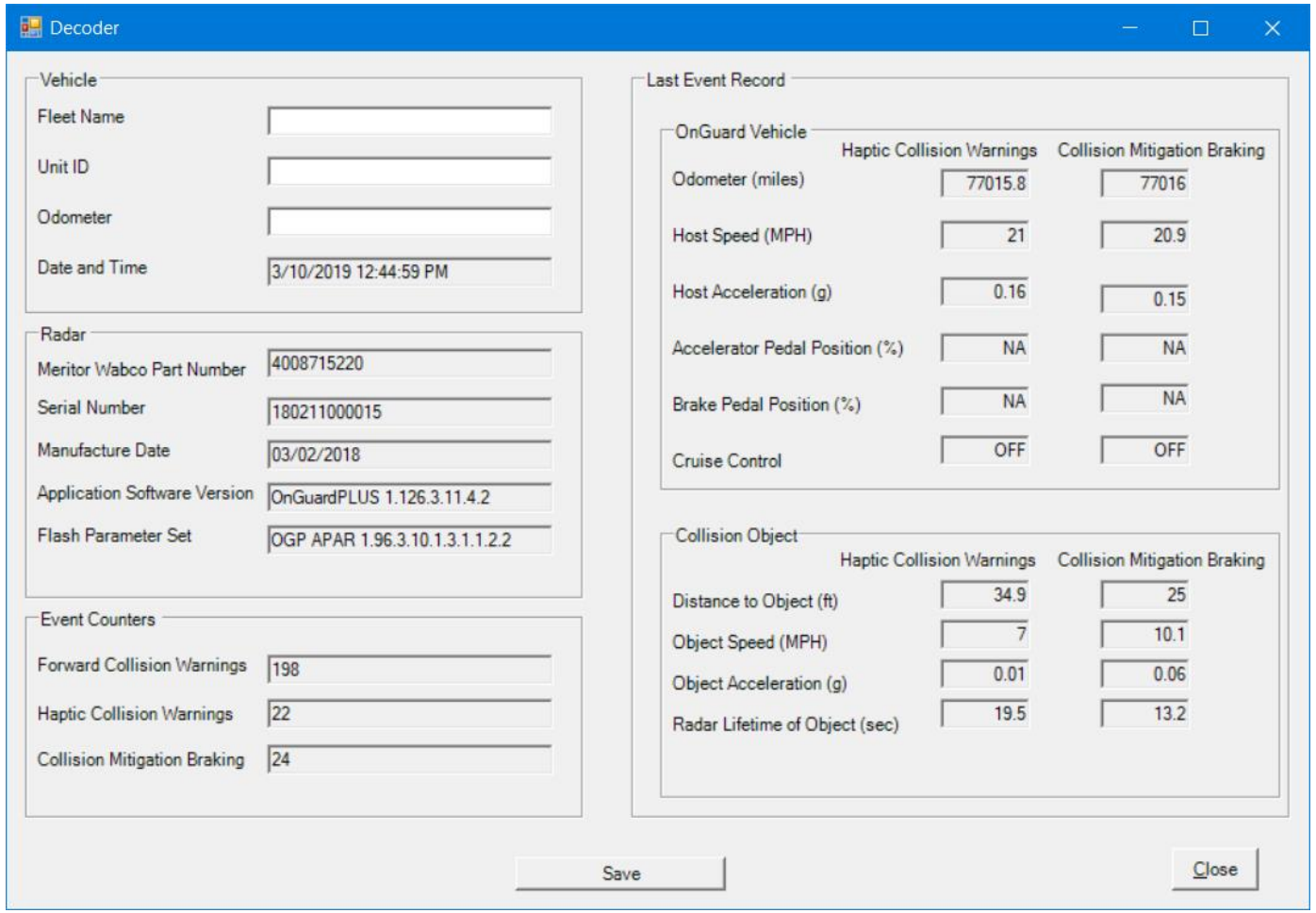
[1]Meritor WabCo OnGuard Collision Mitigation System Maintenance Manual MM-1306, Revised 02-16

Wabco OnGuard (Radar Only)

21mph approach to slow object

WabCo OnGuard (Radar Only)

In WabCo Toolbox, click on OnGuard, then Parameter Decoder



Wabco OnGuard (Radar Only)

OnGuard Event Log Fleet Report

Vehicle:-
Fleet Name: Ryder 2
Unit ID: 030919 1444
Odometer: Dash odo 77016
Date & Time: 3/9/2019 2:45:02 PM

Radar:-
Meritor Wabco Part Number: 4008715220
Serial Number: 180211000015
Manufcture Date: 03/02/2018
Application Software Version: OnGuardPLUS 1.126.3.11.4.2
Flash Parameter Set: OGP APAR 1.96.3.10.1.3.1.1.2.2

Event Counters:-
Forward Collision Warnings: 198
Haptic Collision Warnings: 22
Collision Mitigation Braking: 24

Event Records:-

{OnGuard Vhehicle}	[Haptic Warning]	[Collision Mitigation]
Odometer(miles):	77015.8	77016
Host Speed(MPH):	21	20.9
Host Acceleration(g):	0.16	0.15
Accelerator Pedal Position(%):	NA	NA
Brake Pedal Position(%):	NA	NA
Cruise Control:	OFF	OFF

{Collision Object}	[Haptic Warning]	[Collision Mitigation]
Distance to Object(ft):	34.9	25
Object Speed(MPH):	7	10.1
Object Acceleration(g):	0.01	0.06
Radar Lifetime of Object(sec):	19.5	13.2

Report saved from OnGuard Parameter Decoder

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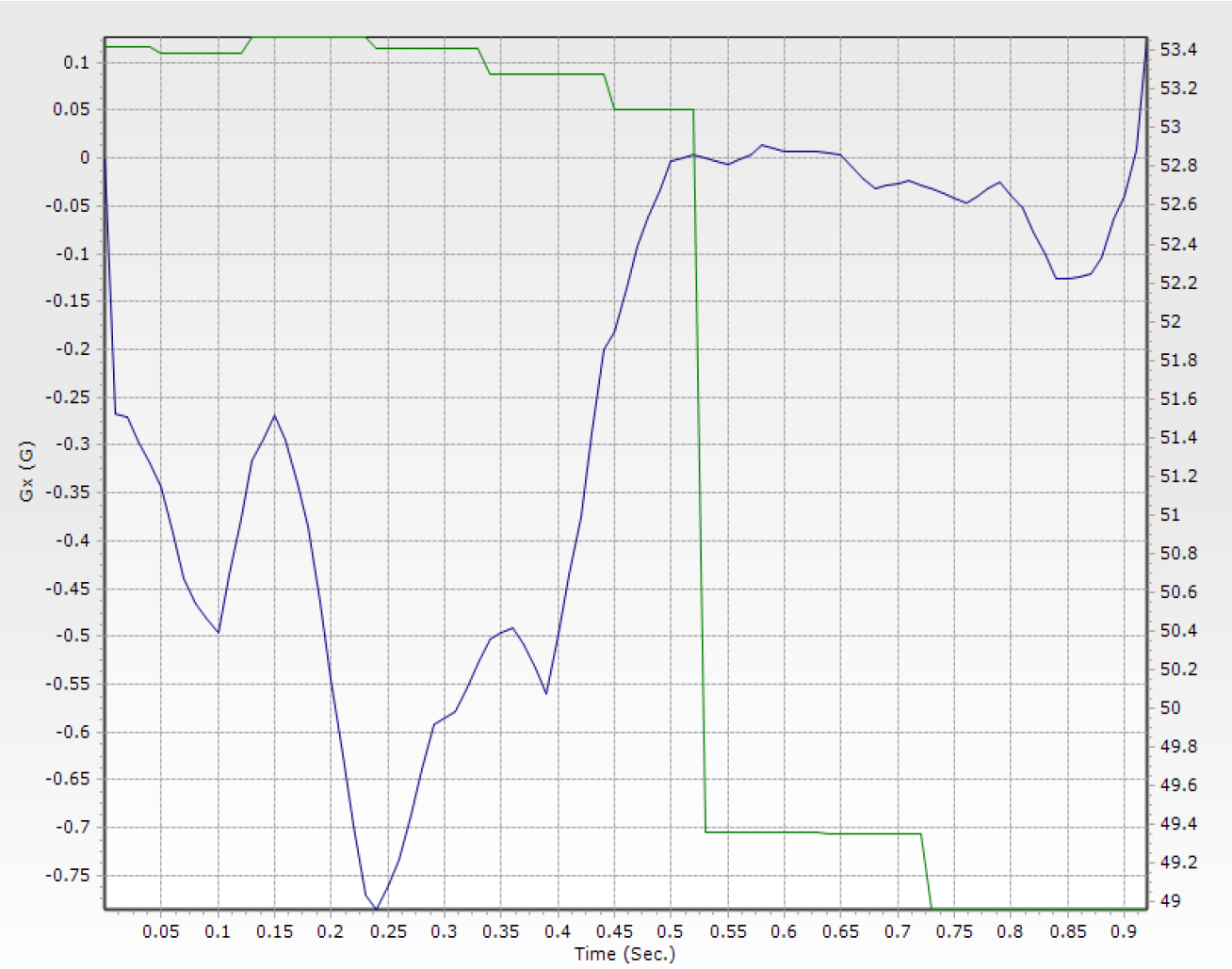
Wabco OnGuard (Radar Only)

53mph approach to slow object, demonstrating challenges of testing at speed

Wabco OnGuard (Radar Only)



0.5s haptic brake apply

WabCo OnGuard (Radar Only)

OnGuard Event Log Fleet Report

Vehicle:-
Fleet Name:
Unit ID: Ryder Mar 2019
Odometer: 77057 odo user-entered
Date & Time: 3/10/2019 8:42:24 PM

Radar:-
Wabco Part Number: 4008715220
Serial Number: 180211000015
Manufcture Date: 03/02/2018
Application Software Version: OnGuardPLUS 1.126.3.11.4.2
Flash Parameter Set: OGP APAR 1.96.3.10.1.3.1.1.2.2

Event Counters:-
Forward Collision Warnings: 200
Haptic Collision Warnings: 23
Collision Mitigation Braking: 26

Event Records:-

{OnGuard Vhehicle}	[Haptic Warning]	[Collision Mitigation]
Odometer(miles):	77055	77056.5
Host Speed(MPH):	23.8	53.5
Host Acceleration(g):	0.17	0.07
Accelerator Pedal Position(%):	NA	NA
Brake Pedal Position(%):	NA	NA
Cruise Control:	OFF	OFF

{Collision Object}	[Haptic Warning]	[Collision Mitigation]
Distance to Object(ft):	37.9	40.6
Object Speed(MPH):	8.1	10.6
Object Acceleration(g):	0.02	0.03
Radar Lifetime of Object(sec):	9	2.6

Report saved from OnGuard Parameter Decoder

Tesla “EDR” Elements – Camera Images of SUA

As part of FCW and AEB event data, the vehicle supports the acquisition and storage of image data captured by the forward facing camera. This system can buffer 8 frames of image data sampled at one second intervals centered[9] upon a triggering FCW / AEB event. These frames are initially captured in volatile memory in the Camera ECU and then stored in non-volatile memory at the end of the drive. From there it is transferred via CAN bus to the Gateway ECU where it is stored on the internal SD card. This image data is not timestamped.

Tesla “EDR” Elements – Camera Images of SUA

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Tesla “EDR” Elements – Camera Images of SUA

Tesla “EDR” Elements – Camera Images of SUA

Tesla “EDR” Elements – Camera Images of SUA

Tesla “EDR” Elements – Camera Images of SUA

Tesla “EDR” Elements – Camera Images of LTAP

Tesla “EDR” Elements – Camera Images of LTAP

Tesla Model 3 DashCam Sample

https://www.youtube.com/watch?v=pa1U2mnTgBk

Tesla Model 3 Crash Video

NTSB Investigation #HWY19FH008, Delray Beach Model 3 crash, March 1, 2019
https://dms.ntsb.gov/pubdms/search/hitlist.cfm?docketID=63437

Tesla Image Recording Summary

2014-2016 – on SD card behind main screen

2017 – Unknown

2018+ - 2 sources

- Dashcam feature, using thumb drive in center console
- Video recorded and accessible by Tesla in a crash event

Subaru Eyesight EDR

Japan-spec vehicle, Subaru Levorg, with Eyesight 3

"Subaru EyeSight Version 3 can only be accessed by Subaru personnel with special software. Subaru's current policy on assisting in data retrieval is only to assist proper legal authority with written owner consent."

Accident Reconstruction with Data Recorded by Electronic Control Units in Vehicles with a Pre-crash Safety System, Oga, et al, SAE 2018-01-1440

Subaru Eyesight EDR

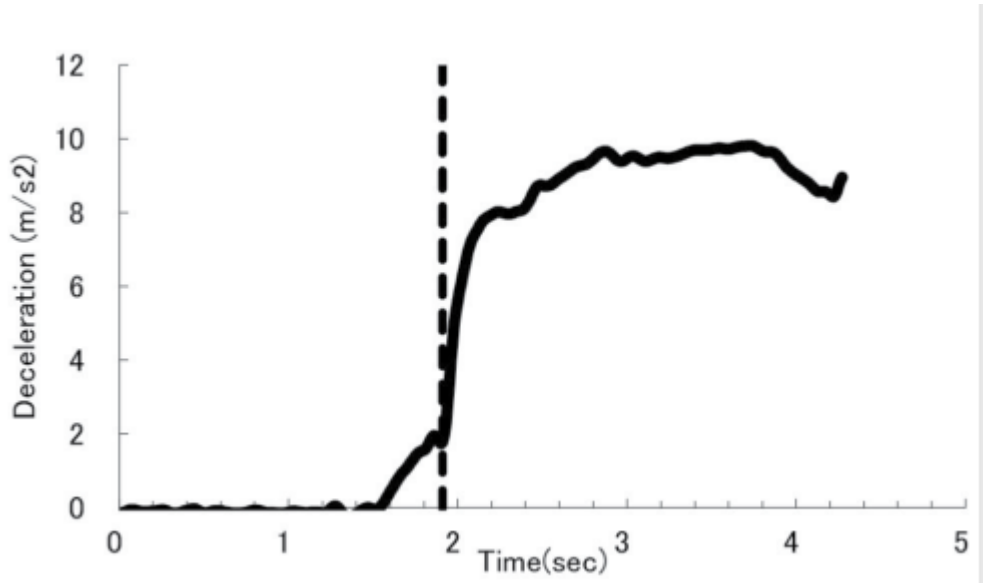
In a 43mph crash test into stopped offset vehicle, the vehicle braked for 2s and reduced speed by 28mph.

Brake was manually applied after warning, so deceleration was a combination of AEB and brake assist.

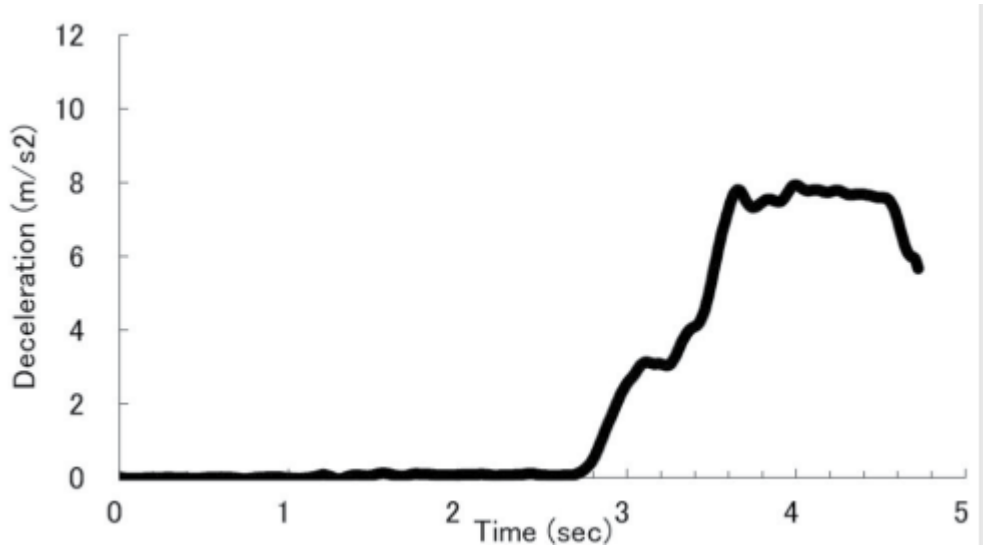
Accident Reconstruction with Data Recorded by Electronic Control Units in Vehicles with a Pre-crash Safety System, Oga, et al, SAE 2018-01-1440

Subaru Eyesight EDR

Brake Assist – driver applied brake after warning



Autobrake – no driver involvement

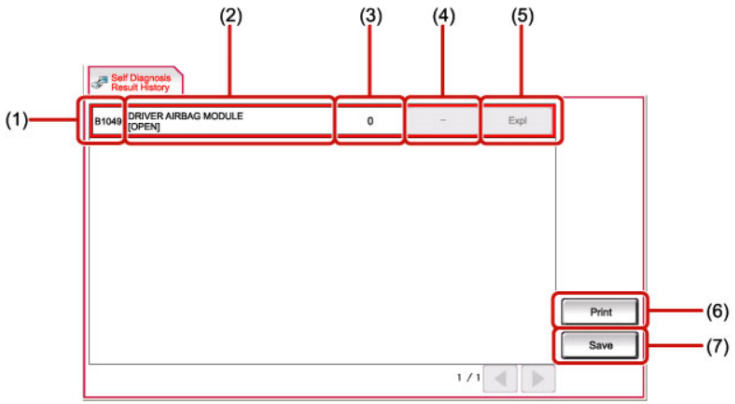


Accident Reconstruction with Data Recorded by Electronic Control Units in Vehicles with a Pre-crash Safety System, Oga, et al, SAE 2018-01-1440

Nissan EDR

- ADAS module can record AEB events
- Access using dealer software Consult III Plus R2R, using Nissan-techinfo.com
- $500 for 12 months, models up to 2018 except GTR
- Requires a specific $1,800 adapter (Car-DAQ)

Nissan EDR



No.	"Button" / Item	Contents
(1)	DTC	Detected DTC is displayed.
(2)	DTC Results	Detected malfunctioning system name is displayed.
(3)	Time (time data of malfunction occurrence)	Number of times vehicle driven since the latest malfunction detection or timing (present/past, or memory) is displayed. When the malfunction is currently detected, "0" (or "present") is displayed. When the first trip DTC is detected at the trip 2 detection diagnosis, "1t" is displayed.
(4)	FFD (Freeze Frame Data)	When the malfunction is detected, the driving date related to the applicable system is displayed.
(5)	Expl	Service Manual (ESM) is started.
(6)	"Print"	The diagnosis results can be printed.
(7)	"Save"	The self-diagnosis results can be saved. Saved results can be browsed in "Recorded Data"

Good ADAS Resource

- https://rts.i-car.com/oem-calibration-requirements-search.html
- $26/day access

Non-OEM Data Interpretation

OEM Data Interpretation

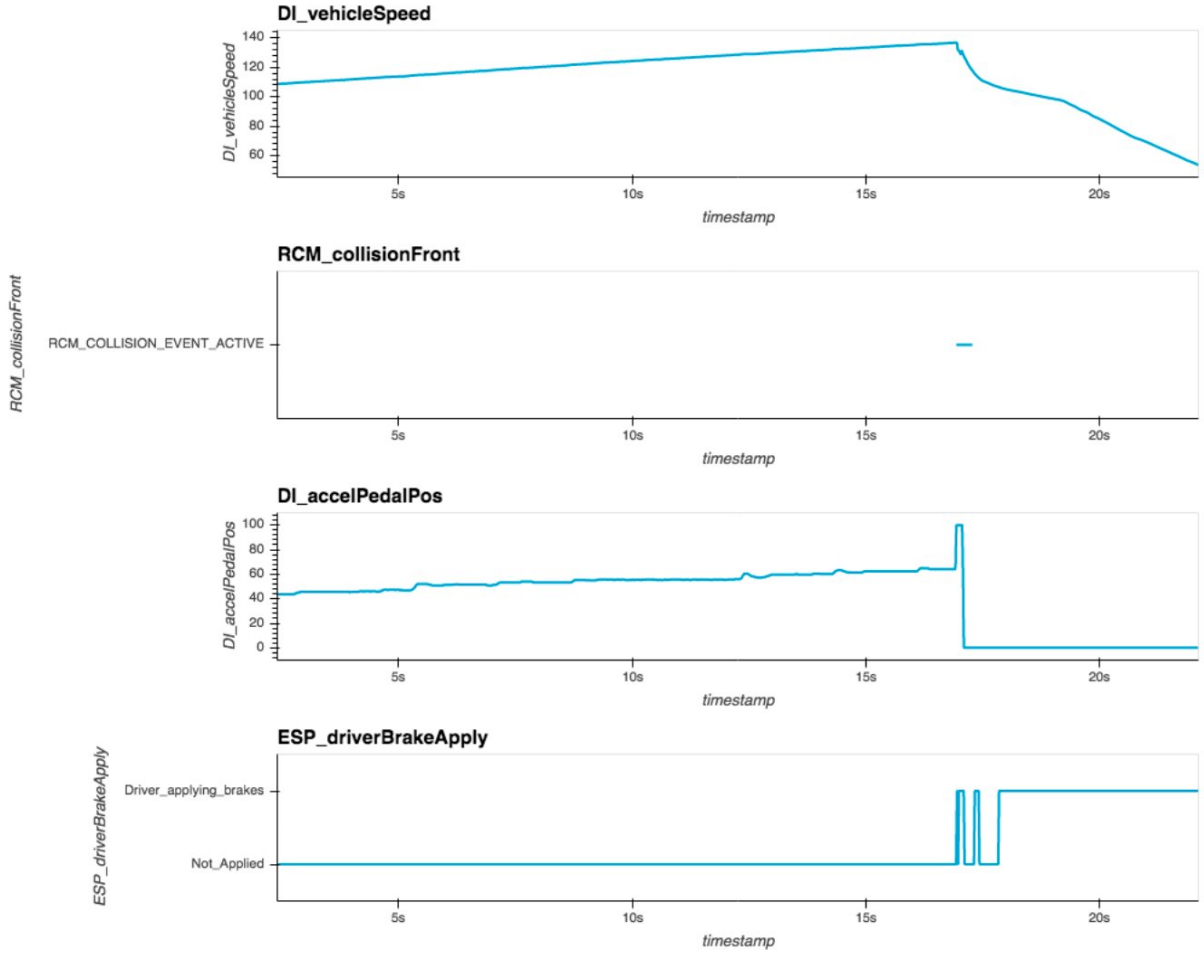
"Tesla was able to communicate with the vehicle via data link to Tesla's virtual private network connection established via Wi-Fi. Tesla provided investigators a 26-page EDR report, a Vehicle Data Log, and forward-facing video leading up to the crash. No further evaluation of the RCM and SD card was required…"

NTSB Investigation #HWY19FH008, Delray Beach Model 3 crash, March 1, 2019
https://dms.ntsb.gov/pubdms/search/hitlist.cfm?docketID=63437

“Security Researchers” – Tesla crash

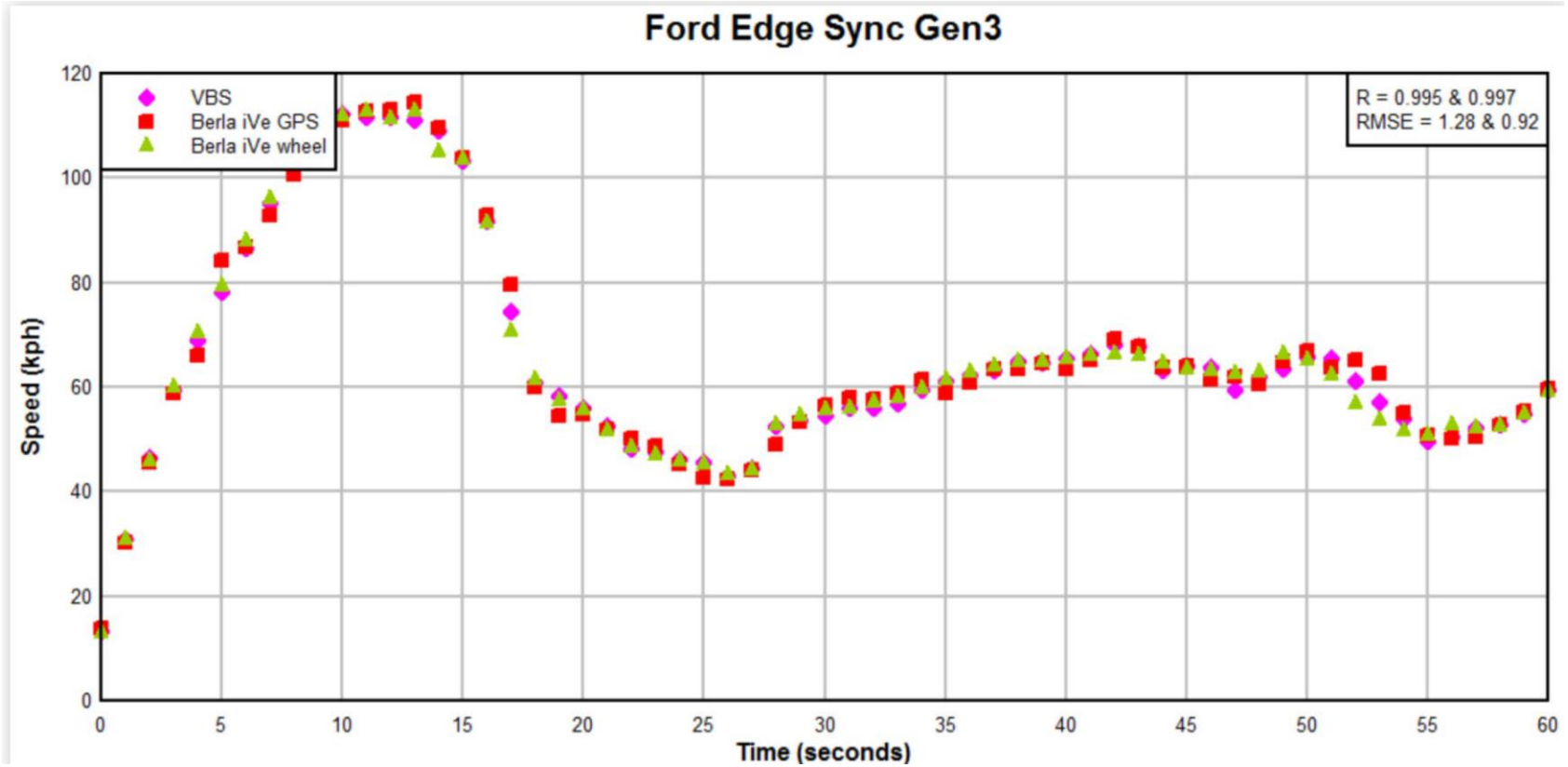
https://twitter.com/greentheonly/status/1326755485610020864

“Security Researchers” – Tesla crash



https://twitter.com/greentheonly/status/1326755485610020864

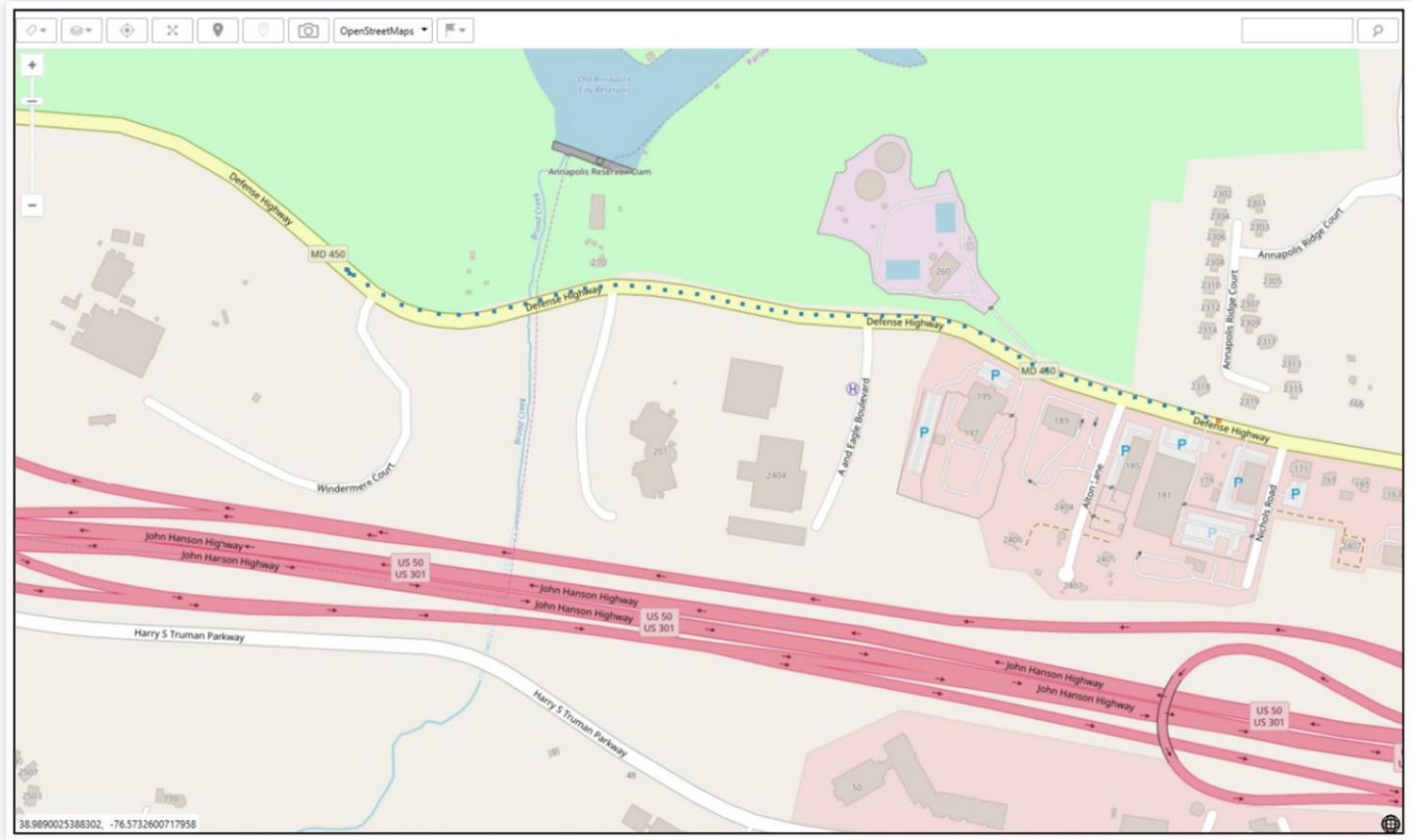
Sync3 Infotainment Data via Berla iVe



Analysis of Berla iVe Acquisitions of Vehicle Speed Data from Ford Sync Systems, Wesley Vandiver and Robert Anderson, SAE Paper #2018-01-1442

Sync3 Infotainment Data via Berla iVe

FIGURE A.11 Test 6 - 2017 Ford Edge, SG3, Berla iVe map of track points.



Analysis of Berla iVe Acquisitions of Vehicle Speed Data from Ford Sync Systems, Wesley Vandiver and Robert Anderson, SAE Paper #2018-01-1442

CDR Hex File Interpretation - VIN

1FT7X2A67FEA79946

CDR Hex File Interpretation - VIN

	F	T		X		A			F	E	A					
31	46	54	37	58	32	41	36	37	46	45	41	37	39	39	34	36

1FT7X2A67FEA79946

ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol
64	40	@	80	50	P	96	60	`	112	70	p
65	41	A	81	51	Q	97	61	a	113	71	q
66	42	B	82	52	R	98	62	b	114	72	r
67	43	C	83	53	S	99	63	c	115	73	s
68	44	D	84	54	T	100	64	d	116	74	t
69	45	E	85	55	U	101	65	e	117	75	u
70	46	F	86	56	V	102	66	f	118	76	v
71	47	G	87	57	W	103	67	g	119	77	w
72	48	H	88	58	X	104	68	h	120	78	x
73	49	I	89	59	Y	105	69	i	121	79	y
74	4A	J	90	5A	Z	106	6A	j	122	7A	z
75	4B	K	91	5B	[	107	6B	k	123	7B	{
76	4C	L	92	5C	\	108	6C	l	124	7C	|
77	4D	M	93	5D	]	109	6D	m	125	7D	}
78	4E	N	94	5E	^	110	6E	n	126	7E	~
79	4F	O	95	5F	_	111	6F	o	127	7F	

Data Interpretation Example – Range Rover

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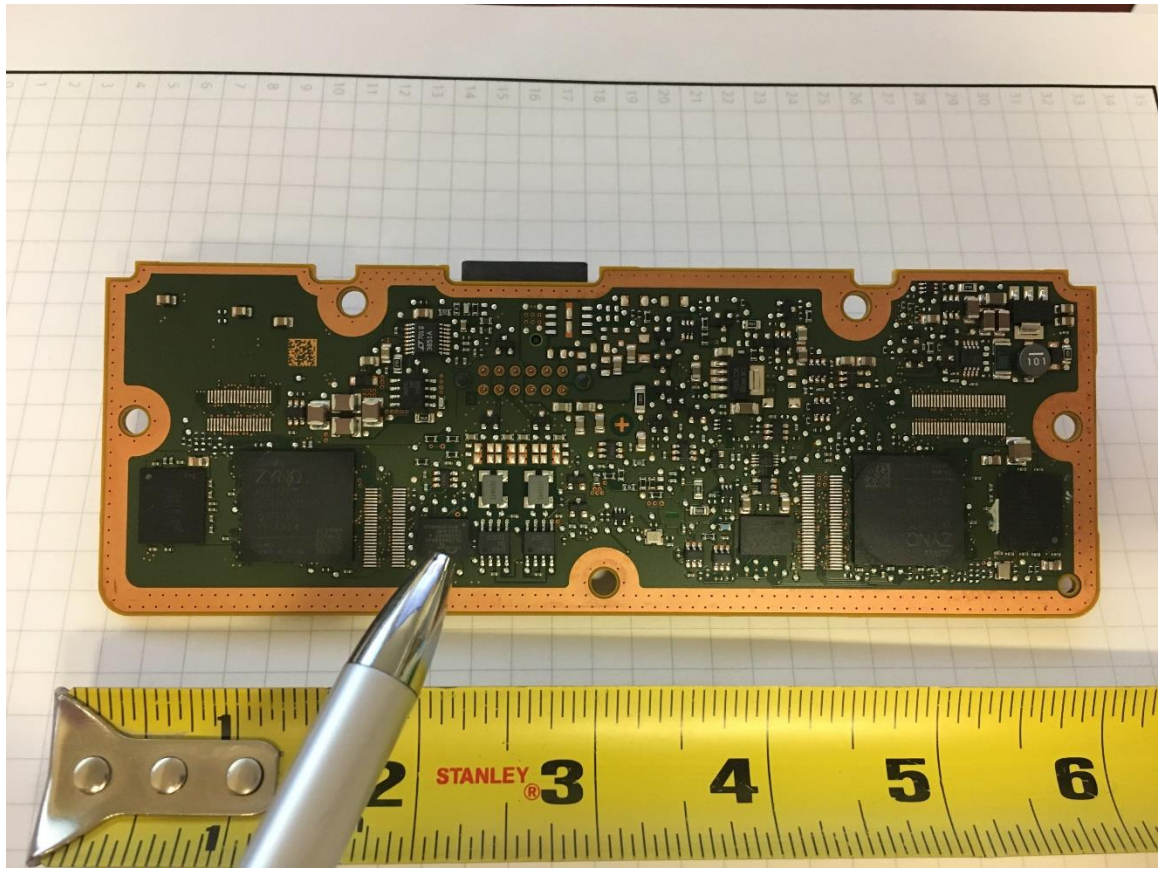
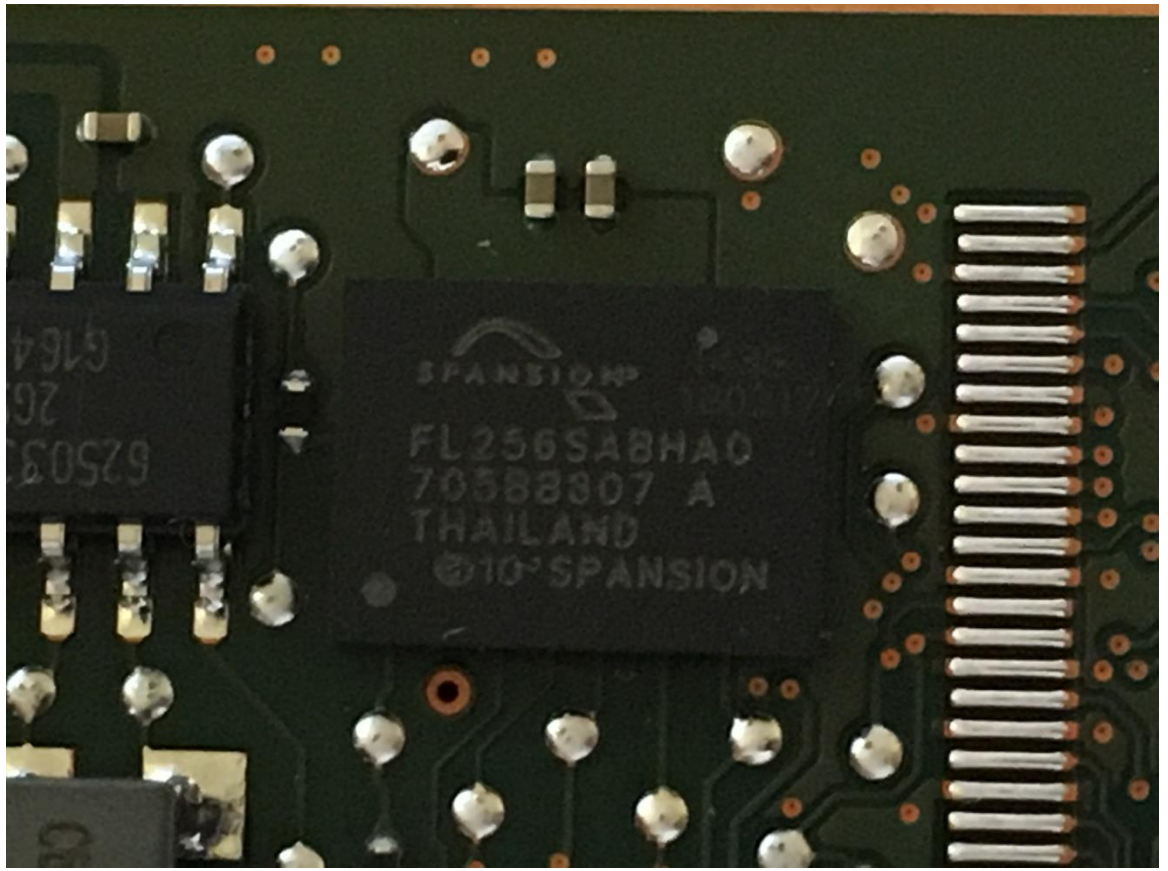
Circuit Board in Range Rover Stereo Camera





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GM Onstar Data From Research Study

"The LDW and FCA warnings are generated by the front camera module (FCM), which both use the same forward-looking camera. The FCM accumulates and stores into its memory a pre-determined set of data during an ignition cycle. The data elements are described in the next subsection and appendix. When an ignition-on event occurs, the previous ignition cycle's data are deleted. For the consenting vehicle owners in this study, OnStar remotely loaded a custom script onto the onboard OnStar module to retrieve the data from the FCM after ignition-off. This data is then sent over the air through the OnStar system during an ignition-off state." – *This applies only to vehicles in the subject study.*

Large-Scale Field Test of Forward Collision Alert And Lane Departure Warning Systems, DOT HS 812 247, 2016
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812247b-lanedepartureWarningsystems.pdf

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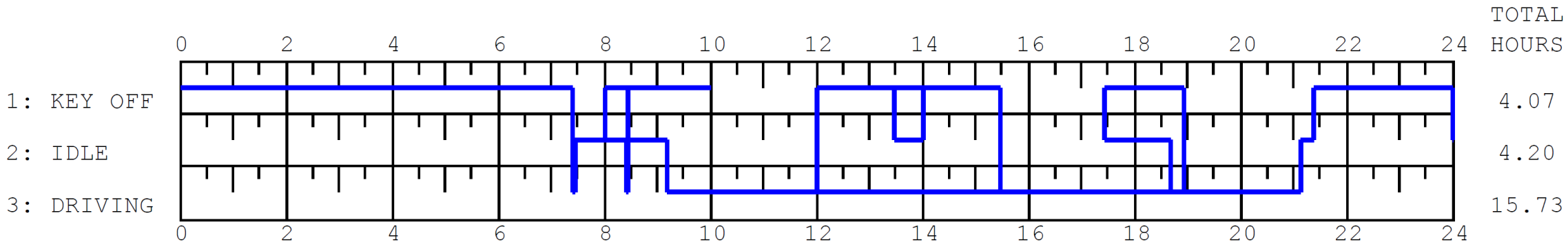
EDR Inconsistencies – Risk of relying on OEMs for data interpretation

Date:	3/27/2018	
Start Time:	09:39:29	EST
Odometer:	6842.30	mi
Distance:	987.00	mi
Fuel:	127.25	gal
Fuel Economy:	7.76	mpg
Average Speed:	62.73	mph
DEF Fluid:	5.03	gal
Trip Economy:	196.36	mpg

Total(hh:mm)	15:44	04:12	04:04
Hour(EST)	Drive(min)	Idle(min)	Off(min)
00:00-02:00	0	0	120
02:00-04:00	0	0	120
04:00-06:00	0	0	120
06:00-08:00	3	34	83
08:00-10:00	2	24	94
10:00-12:00	170	70	-120
12:00-14:00	207	33	-120
14:00-16:00	240	0	-120
16:00-18:00	175	35	-90
18:00-20:00	80	40	0
20:00-22:00	67	15	38
22:00-24:00	0	1	119

(3/27/2018)



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EDR Inconsistencies – Risk of relying on OEMs for data interpretation

	Fuel gal (US)	Time h	Distance miles	Consumption mpg(US)	Percentage of total time %
1. Driving	15 741.6	2 162.2	726 031.0	46.1	83.3
2. Cruise control	752.2	70.5	4 614.7	6.1	2.7
3. Idle	232.0	433.8	–	–	16.7
4. Power take-off	0.0	0.0	–	–	0.0
5. Highest gear	12 810.4	1 221.3	–	–	47.1
6. Total	15 986.5	2 594.9	726 031.0	45.4	–

Trip vehicle speed and engine speed - 11/2/2018 12:38:04 PM

Time	mph - Average speed	r/min - Maximum engine speed
0.2	128	6 046

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EDR Inconsistencies – Risk of relying on OEMs for data interpretation

Life of Vehicle

	Time hh:mm:ss	Fuel	Distance	Consumption	Time %	Fuel %	Distance %
1. Moving	8885 h	69934.7 galUS	154509.8 mi	2.2 mi/galUS	54.9 %	96.3 %	-
2. Cruise control	3258.3 h	32154.2 galUS	221917.2 mi	6.9 mi/galUS	20.1 %	44.3 %	143.6 %
3. Idle (All)	8510.4 h	2742.7 galUS	-	-	52.6 %	3.8 %	-
4. Idle (True)	1438 h	11343745.2 galUS	-	-	8.9 %	15621.4 %	-
5. Qualified Idle	7867.7 h	59634.3 galUS	-	-	48.6 %	82.1 %	-
6. Power Take Off (PTO) 1	0 h	0 galUS	-	-	0 %	0 %	-
7. Power Take Off (PTO) 2	0 h	0 galUS	-	-	0 %	0 %	-
8. PTO1 Moving	0 h	0 galUS	0 mi	0	0 %	0 %	0 %
9. PTO2 Moving	0 h	0 galUS	0 mi	0	0 %	0 %	0 %
10. Hand throttle	6957.9 h	5488.9 galUS	0.1 mi	0.0 mi/galUS	43 %	7.6 %	0.0 %
11. Sweet Spot	6919.3 h	60593.5 galUS	426418.2 mi	7 mi/galUS	42.7 %	83.4 %	276 %
12. Highest gear	1927.4 h	17803.2 galUS	127719.6 mi	7.2 mi/galUS	11.9 %	24.5 %	82.7 %
13. Total	**16186 h**	**72616.66 galUS**	**154509.8 mi**	**2.1 mi/galUS**			

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EDR Inconsistencies – Risk of relying on OEMs for data interpretation

Bendix ABS download (while parked) indicated 78mph.

EVENT HISTORY - INFO Events

Count	Power Up Time	Vehicle Velocity	Event Stamp	Description
1	10s-15m	78 mph	3738.05 (Engine Hrs.)	Diagnostic Mode Entered

NASS ADAS Reports; no related EDR files yet

Special Crash Investigations (Current)

New Search | New Filter Search

Search performed with the following criteria:
Crash Year = 'ALL' AND Crash Avoidance Equipment Activated = ('Blind Spot Detection' OR 'FCW with Auto Braking' OR 'FCW without Auto Braking' OR 'LDW with Lane Keeping' OR 'LDW without Lane Keeping')

Page: 1 ▾ Cases 1-2 of 2

View Case	Technical Report	Crash Date	SCI Case Type	State	Driver Age	Vehicle	Change In Velocity	Fatal
DS16027	DS16027	10/2016	Crash Avoidance Technology	AZ	20	2016 Honda Civic/CRX, del Sol	34 kmph	Non Fatal
DS17016	DS17016	7/2017	Rollover	CA	23	2017 Mercedes-Benz GLA Class	11 kmph	Non Fatal
View Case	Technical Report	Crash Date	SCI Case Type	State	Driver Age	Vehicle	Change In Velocity	Fatal

CISS Crash viewer, 2016; no events with ADAS data available as of Jan 2020

Crash Investigation Sampling System (Current)

New Search | New Filter Search

Search performed with the following criteria:
Crash Year = '2016' AND Crash Avoidance Equipment Activated = ('Blind Spot Detection' OR 'FCW with Auto Braking' OR 'FCW without Auto Braking' OR 'LDW with Lane Keeping' OR 'LDW without Lane Keeping')

No records found for selected criteria!

Electronic Data (EDR)

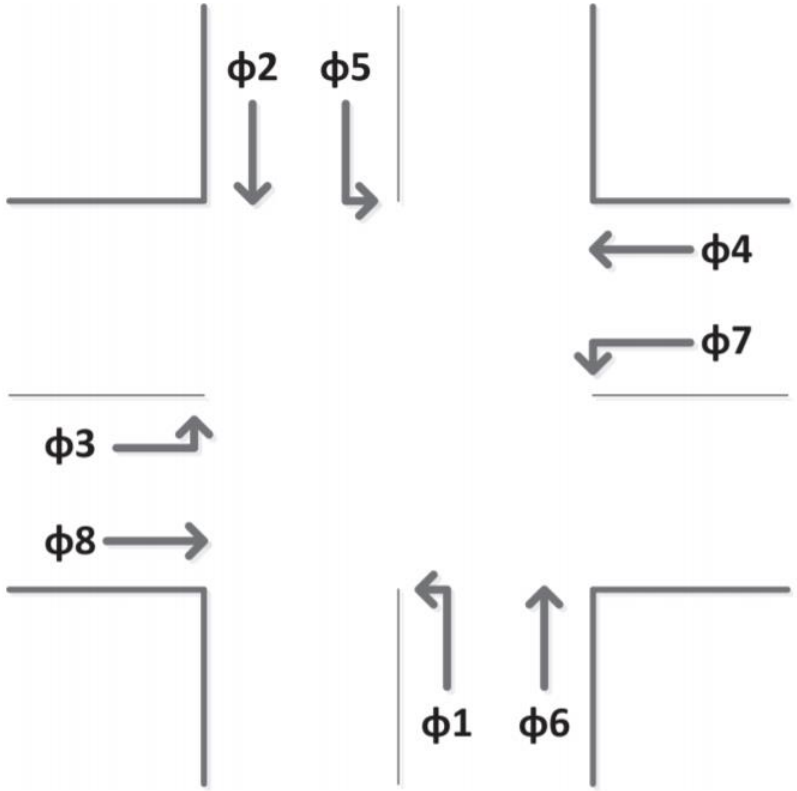
Summary:

- Toyota, Chrysler, Bendix, and Detroit Diesel have an abundance of ADAS-relevant data
- Although the data is useful, there are limitations
- Reverse engineering EDR data is time consuming, case specific, and not feasible for most forensic cases

Brief Overview of Traffic Signal Design

Signal Phasing, Ring Barriers, Phase Diagrams, and Timing Sheets

Traffic Signal NEMA Phase Numbers



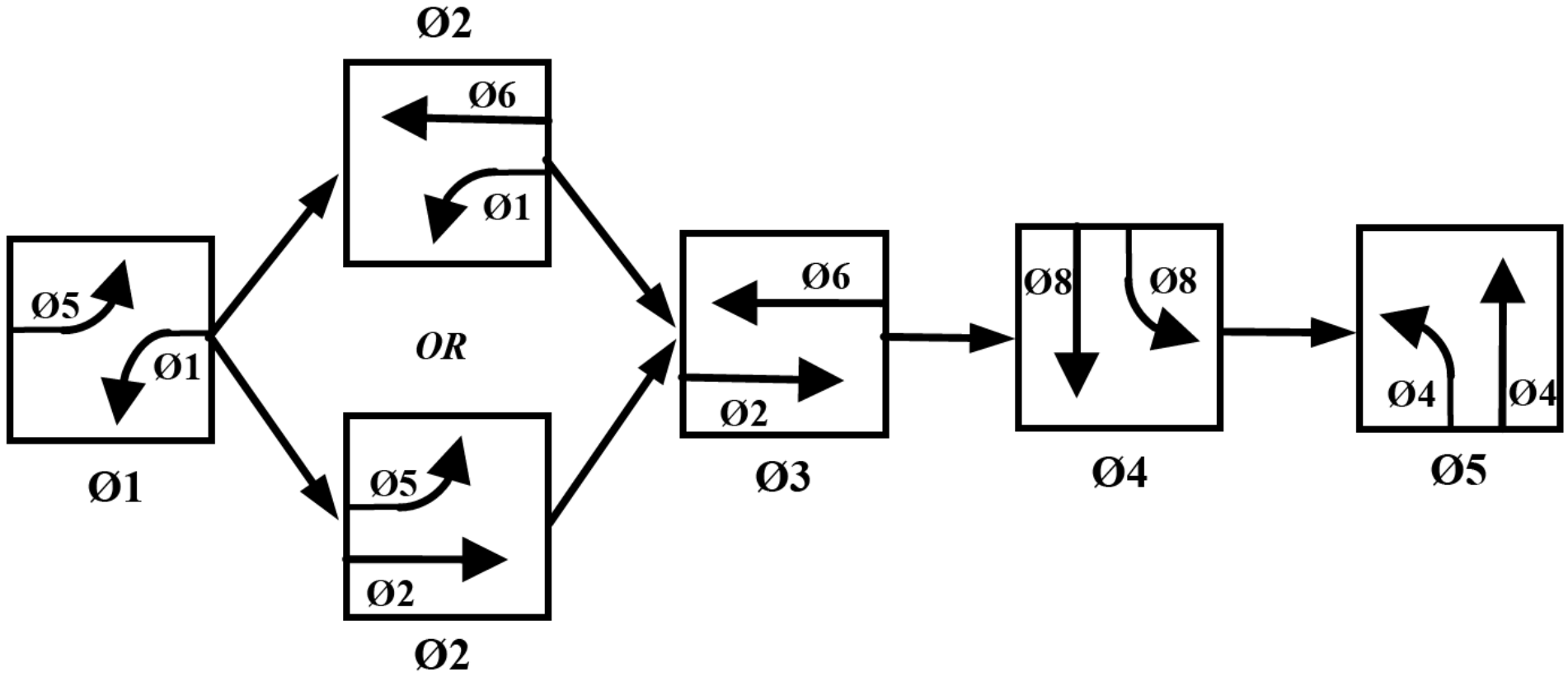
Traffic Signal Systems Operation and Design - Kyte & Urbanik, U. Idaho.
https://www.webpages.uidaho.edu/TrafficSignalSystems/traffic/instructor/ch3a.pdf

Traffic Signal NEMA Phase Numbers

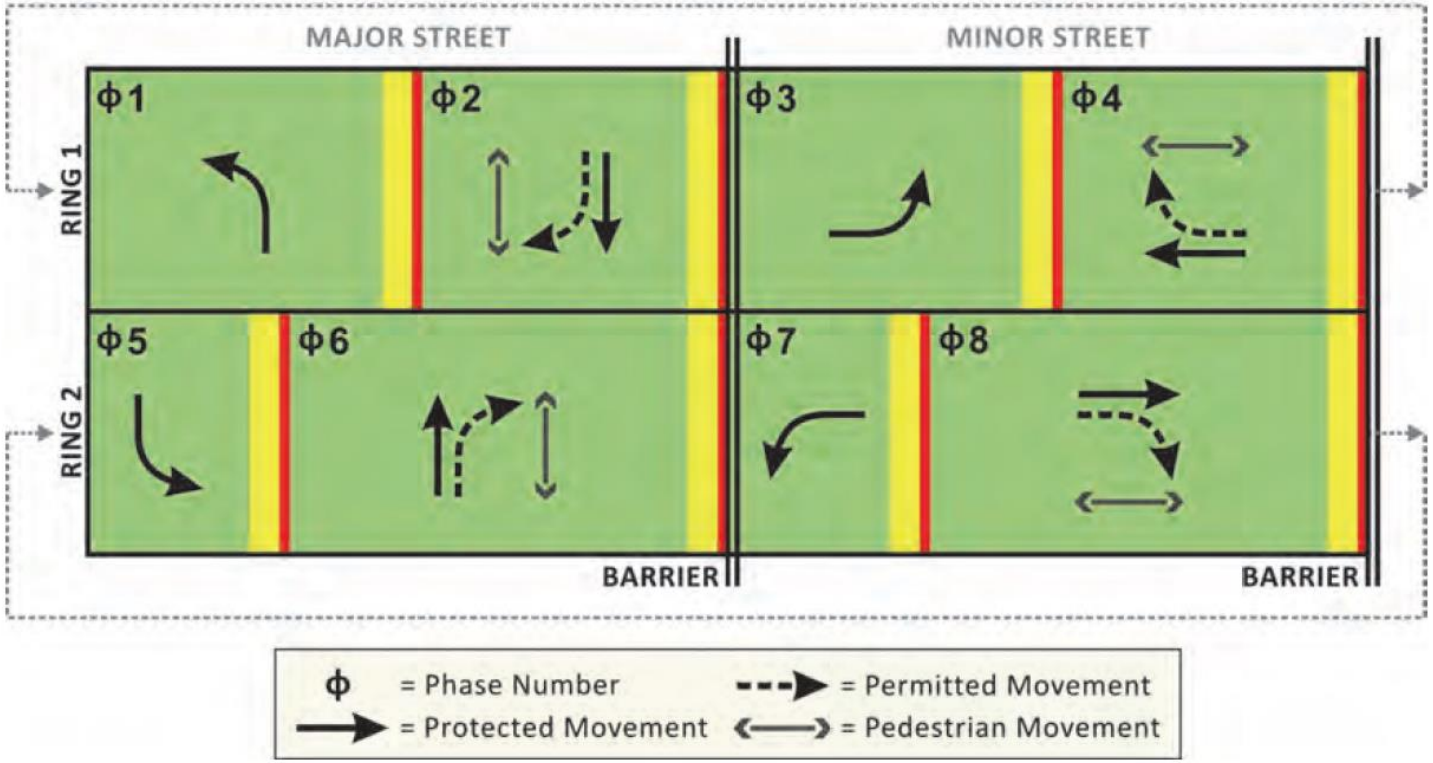
- Even numbers are ***through*** movements
  - 2 & 6 are usually the main street through movements
  - 4 & 8 are usually the side street through movements
- Odd numbers are ***left turns***
  - 1 & 5 are usually the main street left turns
  - 3 & 7 are usually the side street left turns
- Pedestrian phases are usually numbered the same as the parallel movement
- Right turn phase is numbered as the same-direction through movement, plus 10. So a main street right turn phase is usually 12.

Signal Timing Manual, 2nd Edition – NCHRP Report 812, 2015
https://transops.s3.amazonaws.com/uploaded_files/Signal%20Timing%20Manual%20812.pdf

Phase Diagram



Ring and Barrier Diagram



Signal Timing Manual, 2nd Edition – NCHRP Report 812, 2015

https://transops.s3.amazonaws.com/uploaded_files/Signal%20Timing%20Manual%20812.pdf

Ring and Barrier Diagram

- Signal phases on the left of the “barrier” must finish before the signal can cross the barrier
- A phase on one ring can coincide with any phase on the other ring (on its own side of the barrier)

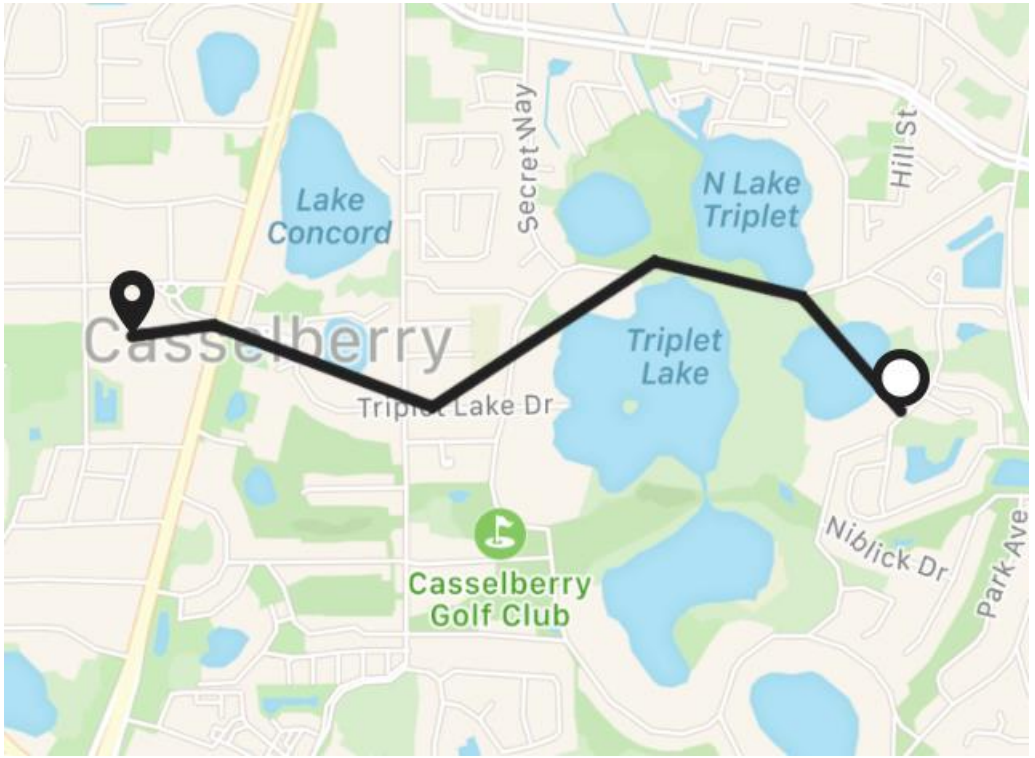
Sample Timing Sheet

Controller Timings (seconds)								
Controller Phase Number	1	2	3	4	5	6	7	8
Direction	EB LT	WB	SB LT	NB	WB LT	EB	NB LT	SB
Minimum Green	5	15	5	10	5	15	5	10
Vehicle Extention	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Yellow Clr/Alt Clr	4.8	4.8	4	4	4.8	4.8	4	4
Red Clr/Alt Red Clr	2.2	2	2.5	3.4	2.4	2	2.1	3.4
Max Green I	15	90	20	20	20	90	20	25
Max Green II	20	120	20	50	35	110	20	50
Walk		7		7		7		7
Walk - XGuard								
FDW		21		36		21		36
FDW - XGuard								
Detector Memory	---	---	---	---	---	---	---	---
Phase Recall	---	MAX	---	---	---	MAX	---	---
Ped Recall	---	ON	---	---	---	ON	---	---
Flash Operation	---	YEL	---	RED	---	YEL	---	RED

Why Record the Signal Operation?

- Monitor and improve signal timing
- Identify traffic backups
- Emergency response
- Evaluate public complaints
- Automate traffic studies
- Research projects
- Accident reconstruction is a side benefit

Kids' bike ride to Target



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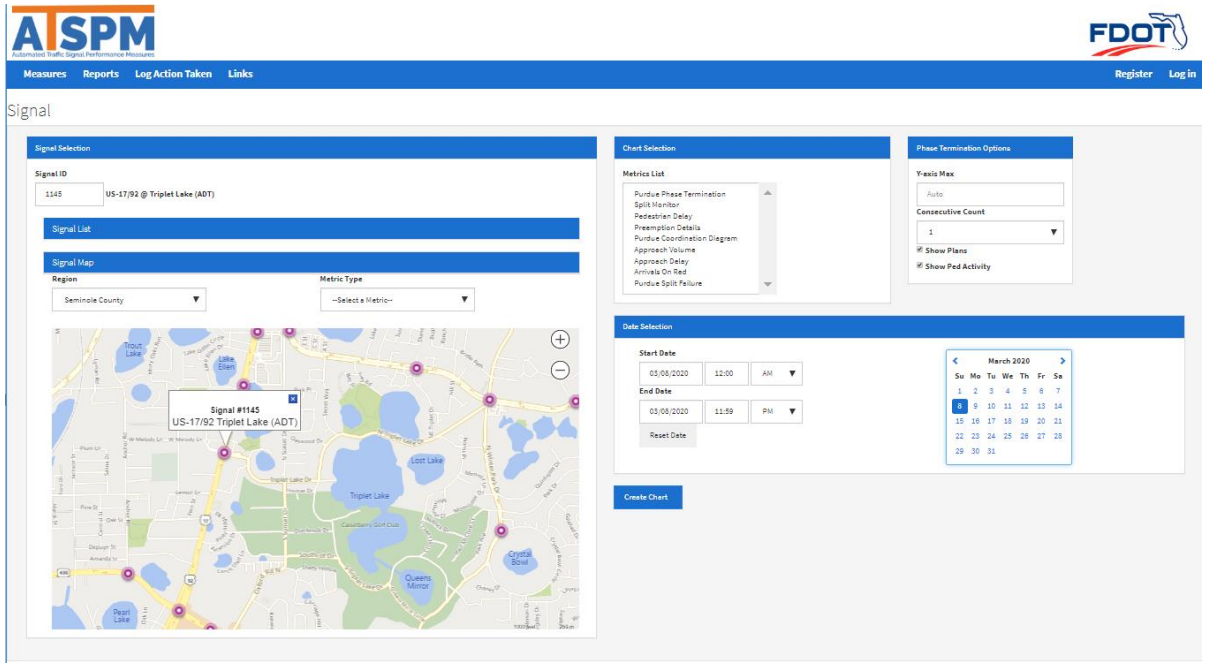
Traffic Signal monitoring using ATSPM

ATSPMs (Automated Traffic Signal Performance Measures) consist of a high-resolution data-logging capability added to existing traffic signal infrastructure and data analysis techniques.



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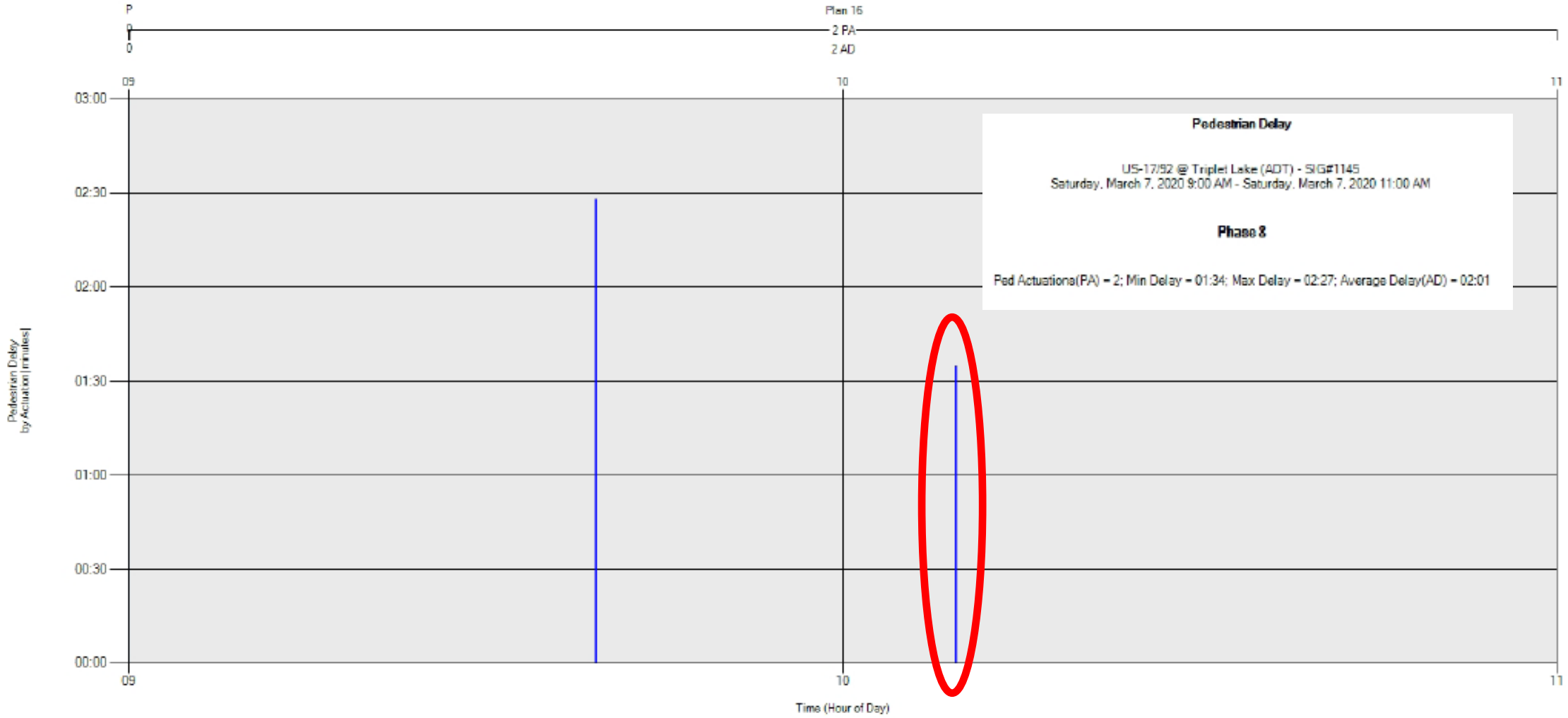
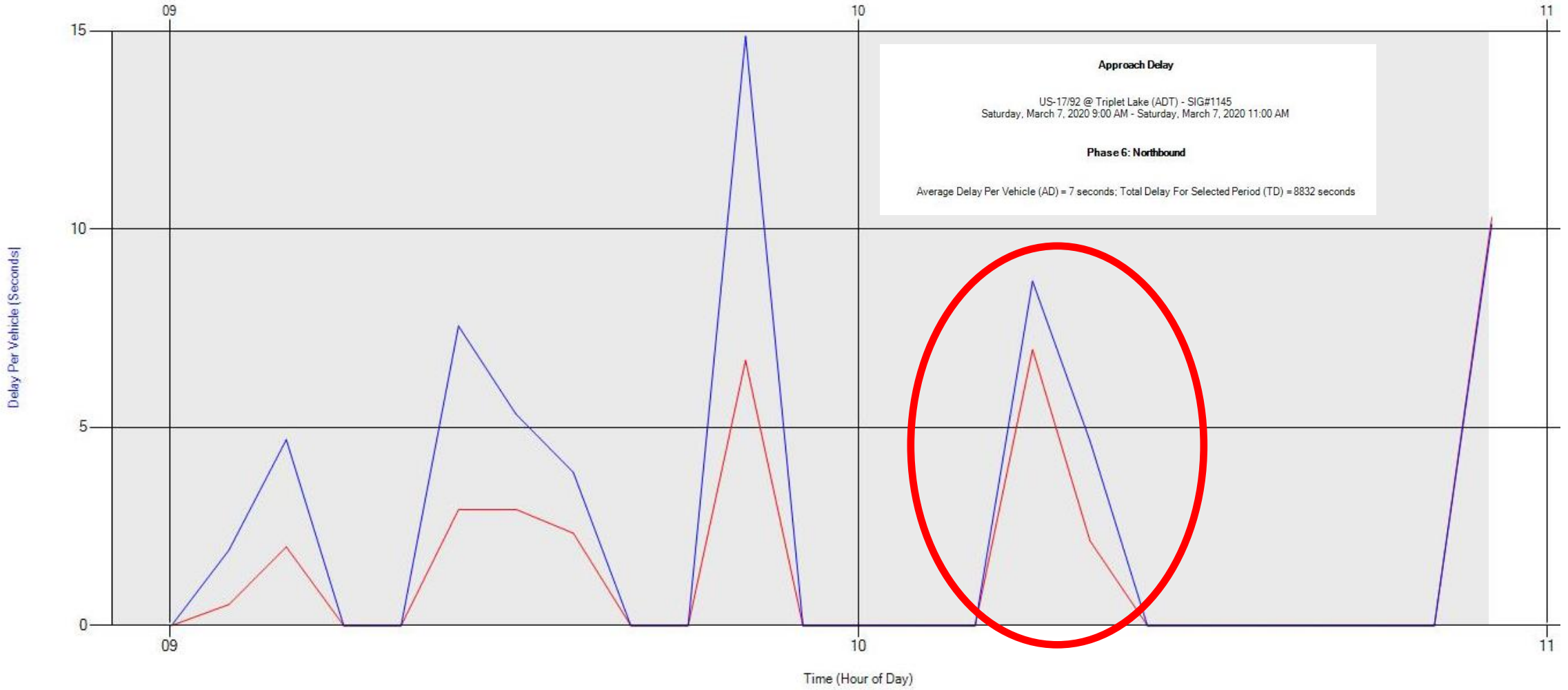
ATSPM – Pedestrian Delay, Crossing Main Street

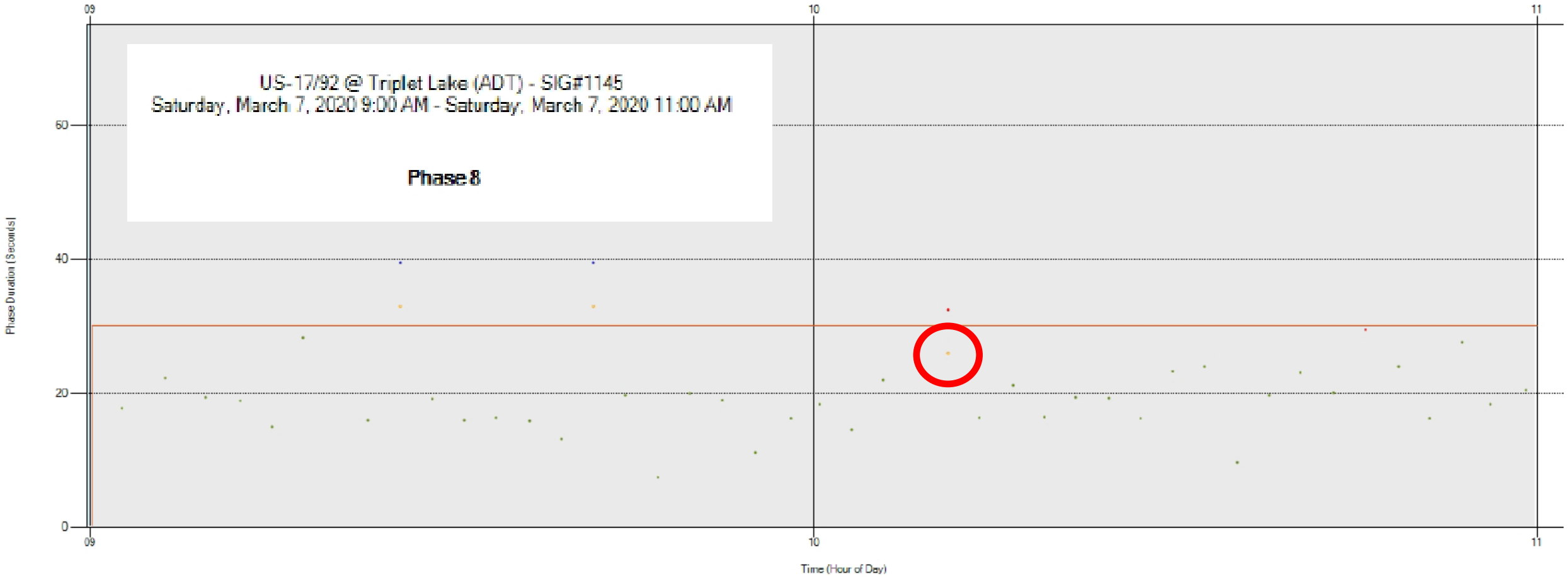


Approach Delay on Main Street



Crossing on the “Walk” signal, 10:10 a.m.

Split history for main street ped crossing



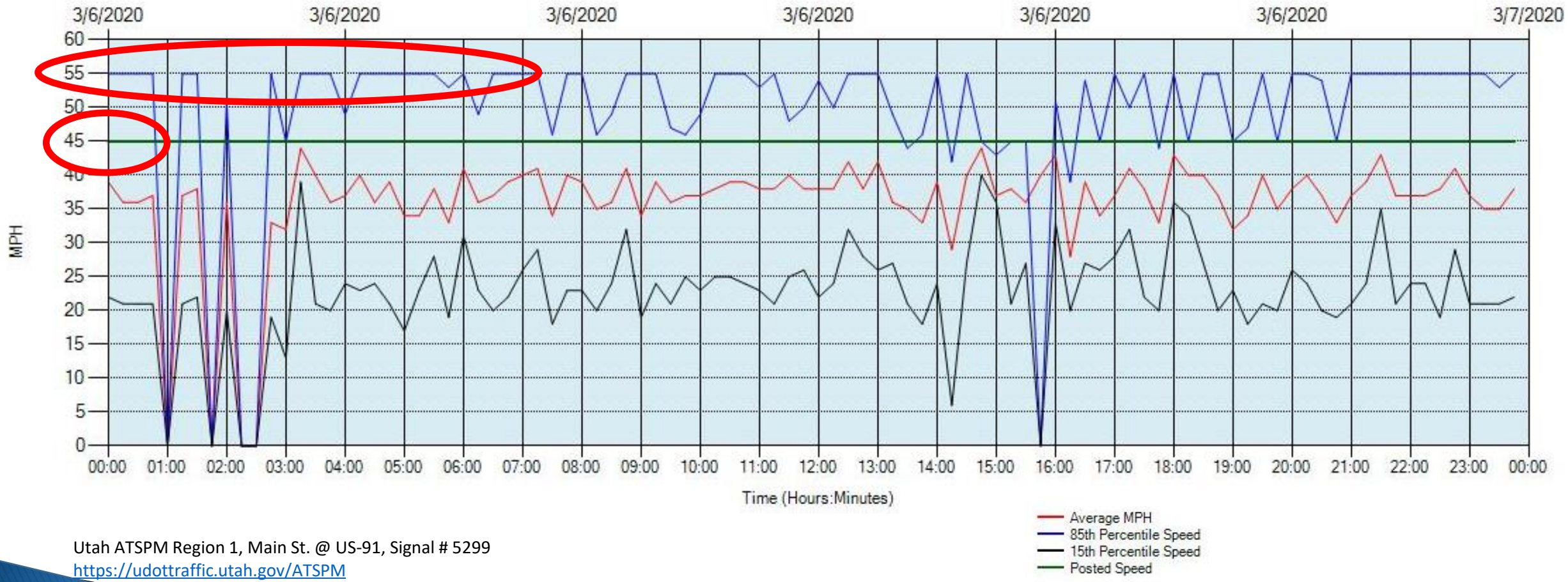
Safely at their destination

Did the eastbound left-turn (Phase 1) driver have a green arrow? No, based on the split monitor.

Date/Time	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Trans	P/R/T
5:52:19 AM		89/89	19/20	41/25		89/89	60/20											
5:54:49 AM		13/89		9/25	12/16		8/20										X	
5:55:11 AM		98/89	11/20	17/25	15/16	82/89		28/25									X	
5:57:19 AM		115/89	15/20	17/25		115/89	32/20											
5:59:49 AM		98/89		17/25		98/89		17/25									X	
6:01:45 AM		189/114	20/17		13/26	175/114		21/53										
6:05:15 AM		145/114	16/17	48/53	23/26	122/114	15/17	50/53										
6:08:45 AM		143/114	14/17	52/53	26/26	117/114	17/17	50/53										
6:12:15 AM		143/114	16/17	50/53	26/26	117/114	17/17	50/53										
6:15:45 AM		172/114	16/17	20/53	25/26	146/114		37/53										
6:19:15 AM		163/114	16/17	30/53	26/26	137/114	30/17	17/53										

(Dates redacted)

Was the speeding vehicle “moving with traffic”? From a radar advance traffic sensor



Utah ATSPM Region 1, Main St. @ US-91, Signal # 5299
https://udottraffic.utah.gov/ATSPM

How To Use Networked Traffic Signal Data

SAE INTERNATIONAL

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Forensic Approach to Networked Traffic Signal Accidents

Investigation: Was the signal able to log events?

Analysis: Is the logging valid?

Evidentiary Value: What does it say about accident causation?

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How To Determine if Signal Is Networked and Has Data

- Look for fiber-optic or network hardware
- Look for high-tech sensors (microwave, radar, camera)
- Request split history from municipality

Public Records Request to Municipality

Request the following:

- Signal timing plans
- Phase diagram
- Timing sheet
- Intersection schematic
- Split history
- High resolution detector data

Municipality may not be familiar with some of the records, and may not be aware that they exist.

Accident Site Inspection

Record a long (~30 minute) video of signal operation with a clock reference occasionally visible in the video (www.time.gov).

Then validate split history data with inspection video

Compare Crash Date/Time to Split History

Split History with Max(M),Gap(G), and Force-off(F)

Report Date: 4/19/2018

Controller: 1115 US 17-92 & 23-SR 419 #4299 Time: 02/28/2017 22:00:00 To 02/28/2017 22:35:59

Date/Time	Pattern	Cycle	SP1	SP2	SP3	SP4	SP5	SP6	SP7	SP8	SP9	SP10	SP11	SP12	SP13	SP14	SP15	SP16
02/28/2017 22:00:16 PM	254	147	0	132/M	15/G	0	29/G	103/M	0	15/G	0	0	0	0	0	0	0	0
02/28/2017 22:02:43 PM	254	152	15/G	105/M	16/G	16/G	17/G	103/M	0	32/G	0	0	0	0	0	0	0	0
02/28/2017 22:05:14 PM	254	137	0	121/M	16/G	0	18/G	103/M	0	16/G	0	0	0	0	0	0	0	0

How To Tell When The Crash Occurred?

- Crash Report Time *Estimate*
- GPS Data in Vehicles
- 911 dispatch records
- Security video
- Emergency vehicle pre-emption at signal

Crash Time From Detector Data

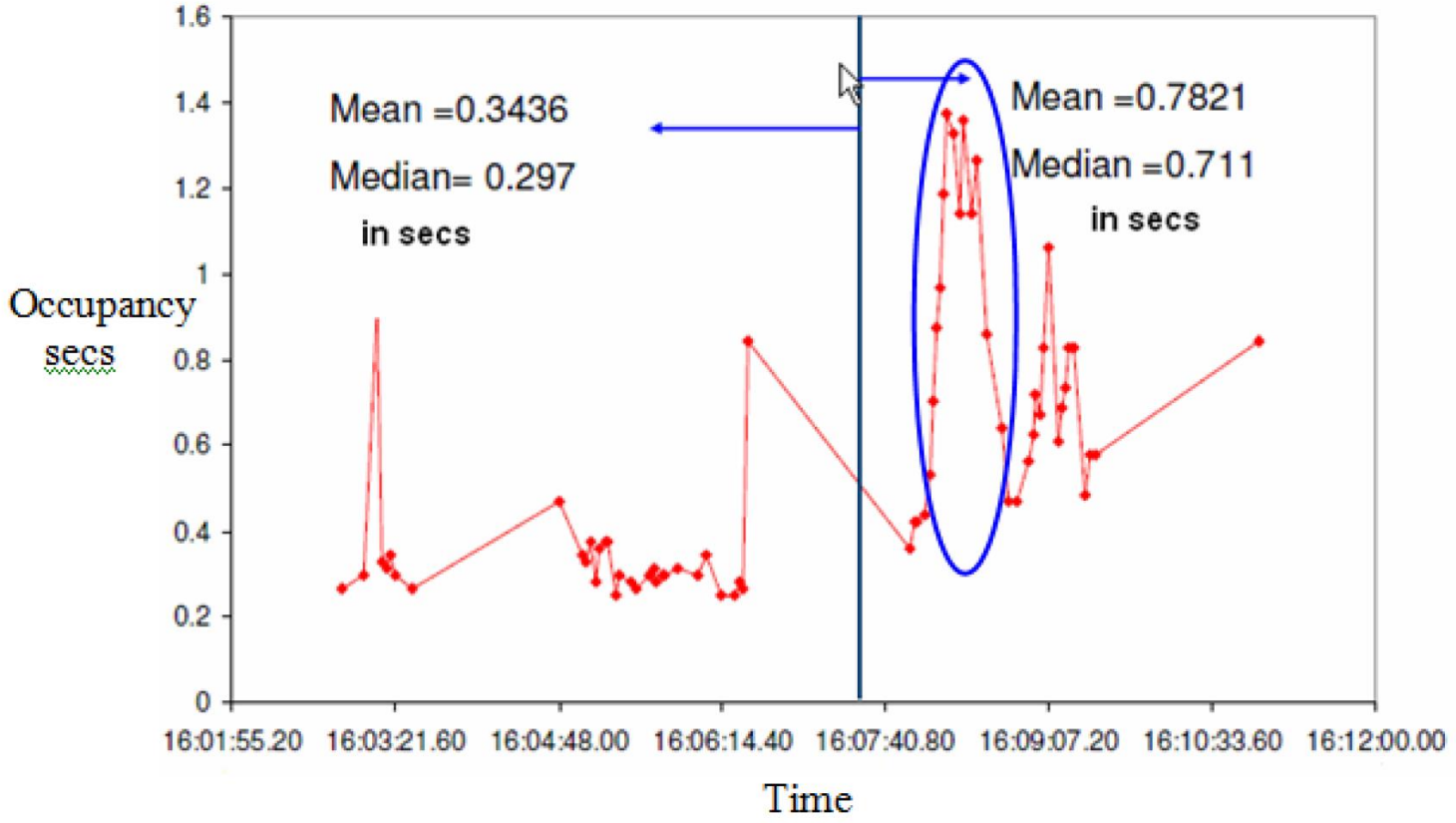


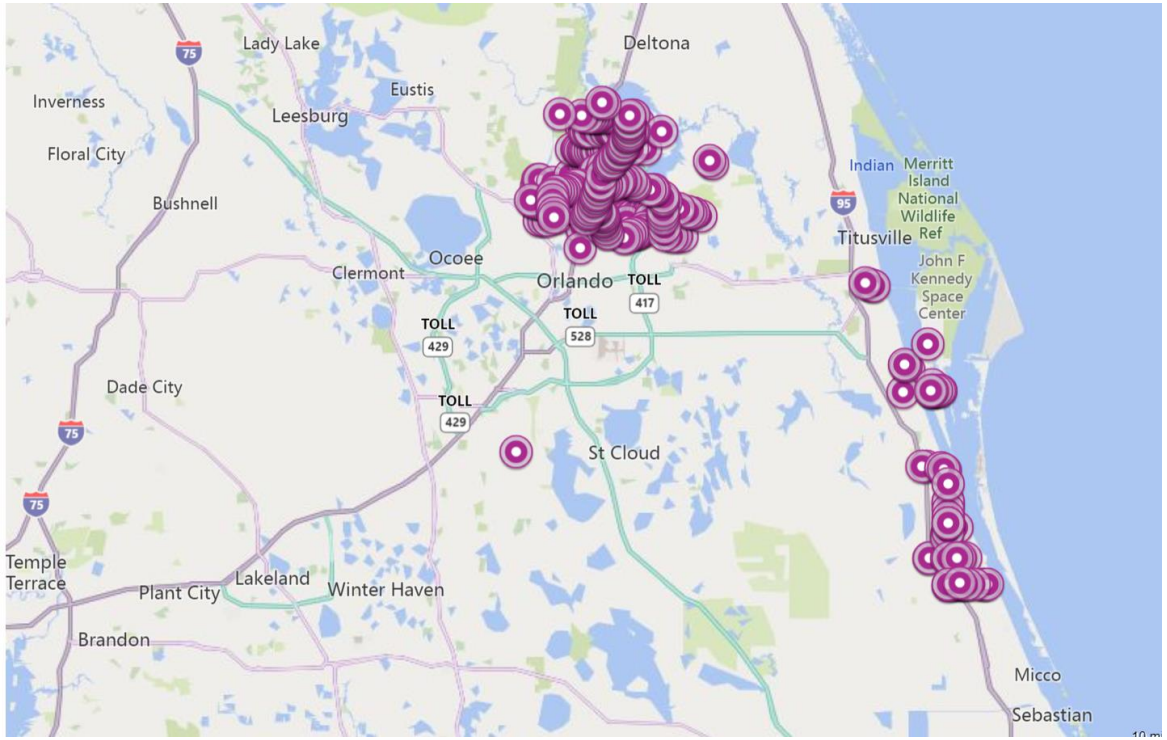
Figure 3.6: Post-incident occupancy pattern at detector 4 during green phase.

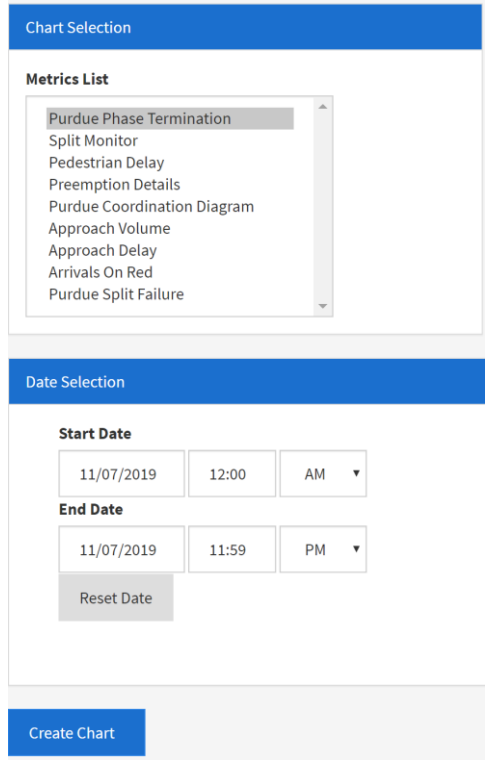
Using Detailed Signal and Detector Data to Investigate Intersection Crash Causation - Final Report
Gary A. Davis , Indrajit Chatterjee - Department of Civil Engineering, University of Minnesota
CTS 13-05 , January 2013

Sample Split History

Time	Plan	Ring 1 Phase	Interval	Grn Dur	Ring 2 Phase	Interval	Grn Dur	Status
3/26 06:00:19	Fr							Free
3/26 06:00:25	41	4-EBT	Green	8	8-WBT	Green	8	Coord
3/26 06:00:33		4-EBT	Clear		8-WBT	Clear		
3/26 06:00:39		2-NBT	Green	235	6-SBT	Green	235	
3/26 06:01:41	41	2-NBT			6-SBT			Coord
3/26 06:03:04	41	2-NBT			6-SBT			Coord
3/26 06:04:34	41	2-NBT	Clear		6-SBT	Clear		Coord
3/26 06:04:41		4-EBT	Green	8	8-WBT	Green	8	
3/26 06:04:49		4-EBT	Clear		8-WBT	Clear		
3/26 06:04:55		2-NBT	Green	521	6-SBT	Green	521	

Data available from Central FL Connected Signals





- https://atspm.cflsmartroads.com/ATSPM/

Split history, on inspection date

Date/Time	1	2	3	4	5	6	7	8
1/30/2020 8:00:45 AM	14/26	143/114	17/17	36/53	13/26	144/114		53/53
1/30/2020 8:04:15 AM		165/114	17/17	28/53	20/26	145/114	17/17	28/53
1/30/2020 8:07:45 AM		140/114	17/17	52/53	22/26	117/114	17/17	52/53
1/30/2020 8:11:15 AM		142/114	17/17	50/53	25/26	116/114	17/17	50/53
1/30/2020 8:14:45 AM		140/114	17/17	52/53	20/26	120/114	17/17	52/53
1/30/2020 8:18:15 AM	13/26	127/114	17/17	52/53	12/26	127/114	17/17	52/53
1/30/2020 8:21:45 AM	12/26	142/114	17/17	38/53	24/26	129/114	16/17	38/53
1/30/2020 8:25:15 AM	14/26	126/114	17/17	52/53	25/26	114/114	16/17	52/53
1/30/2020 8:28:45 AM		156/114	17/17	35/53	18/26	137/114	16/17	36/53
1/30/2020 8:32:14 AM		142/114	17/17	50/53	24/26	117/114	16/17	50/53
1/30/2020 8:35:44 AM		142/114	17/17	50/53	23/26	119/114	16/17	50/53
1/30/2020 8:39:14 AM		169/114	18/17	23/53	19/26	150/114	16/17	24/53
1/30/2020 8:42:44 AM	15/26	25/114	17/17	53/53	19/26	121/114	16/17	53/53
1/30/2020 8:46:14 AM		142/114	17/17	50/53	25/26	116/114	16/17	50/53
1/30/2020 8:49:44 AM		159/114	17/17	34/53	25/26	134/114	16/17	34/53

08:42:44

00:00:00
08:45:46
08:45:46 a.m.

Network time vs GPS time

GPS time can vary from “network” time by 16~52 seconds, due in part to leap seconds.

Does the GPS correct for this error? Some do, and a correction signal is broadcast

https://timetoolsltd.com/gps/gps-ntp-server/

https://timetoolsltd.com/gps/what-is-the-gps-clock/

https://en.wikipedia.org/wiki/Leap_second

https://tycho.usno.navy.mil/leapsec.html

Possible causes of time discrepancies

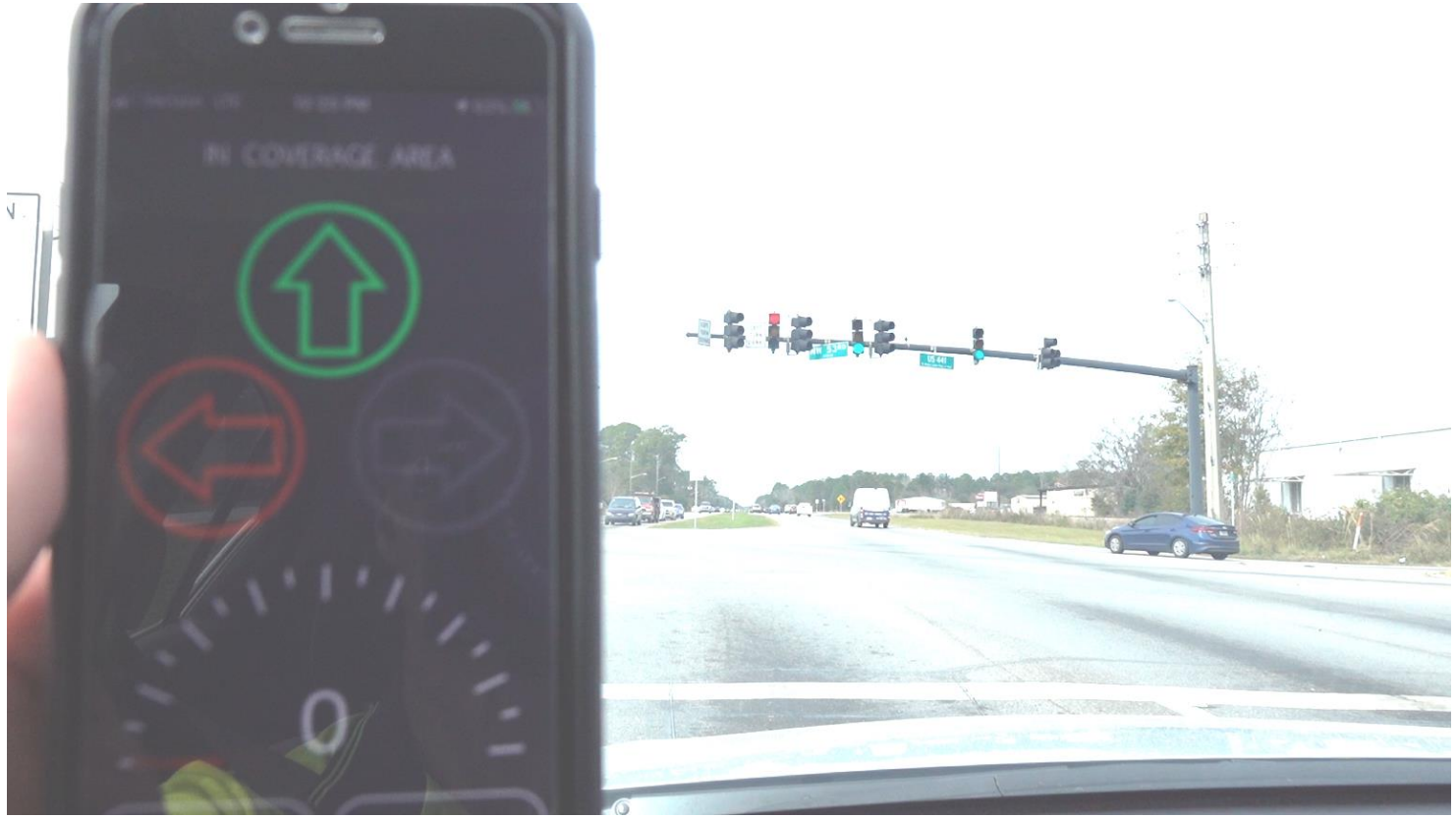
- GPS time vs network time
- Asynchronous data logging
- Reference point of timestamp may not be the start of Phase 1 green
- Error in parsing of data from individual events to split monitor reports
- ?

Event Logging Raw Data

Timestamp	Event Code ID	Parameter	Description
2012/10/17 13:38:25.6	81	43	Detector 43 Off
2012/10/17 13:38:26.2	82	30	Detector 30 On
2012/10/17 13:38:26.3	82	32	Detector 32 On
2012/10/17 13:38:26.4	81	32	Detector 32 Off
2012/10/17 13:38:26.4	81	30	Detector 30 Off
2012/10/17 13:38:27.7	82	28	Detector 28 On
2012/10/17 13:38:27.8	81	28	Detector 28 Off
2012/10/17 13:38:28.5	44	2	Phase 2 Call Dropped
2012/10/17 13:38:28.5	81	26	Detector 26 Off
2012/10/17 13:38:30.3	82	17	Detector 17 On
2012/10/17 13:38:30.3	43	6	Phase 6 Call Registered
2012/10/17 13:38:30.6	6	2	Phase 6 Force-off
2012/10/17 13:38:30.6	8	6	Phase 6 Begin Yellow Clearance
2012/10/17 13:38:30.6	6	6	Phase 6 Force-off
2012/10/17 13:38:30.6	8	2	Phase 2 Begin Yellow Clearance
2012/10/17 13:38:30.6	7	6	Phase 6 Green Termination
2012/10/17 13:38:30.6	7	2	Phase 2 Green Termination
2012/10/17 13:38:30.7	2	3	Phase 3 Check
2012/10/17 13:38:30.7	2	4	Phase 4 Check
2012/10/17 13:38:30.7	2	8	Phase 8 Check
2012/10/17 13:38:30.7	43	2	Pedestrian Call Phase 2

Performance Measures for Traffic Signal Systems, Day et al, Purdue University, 2014
https://t2center.uconn.edu/pdfs/PerformanceMeasuresTrafficSignalSystemsPurdue%202014.pdf

V2I – Networked Signal Timing & Split History

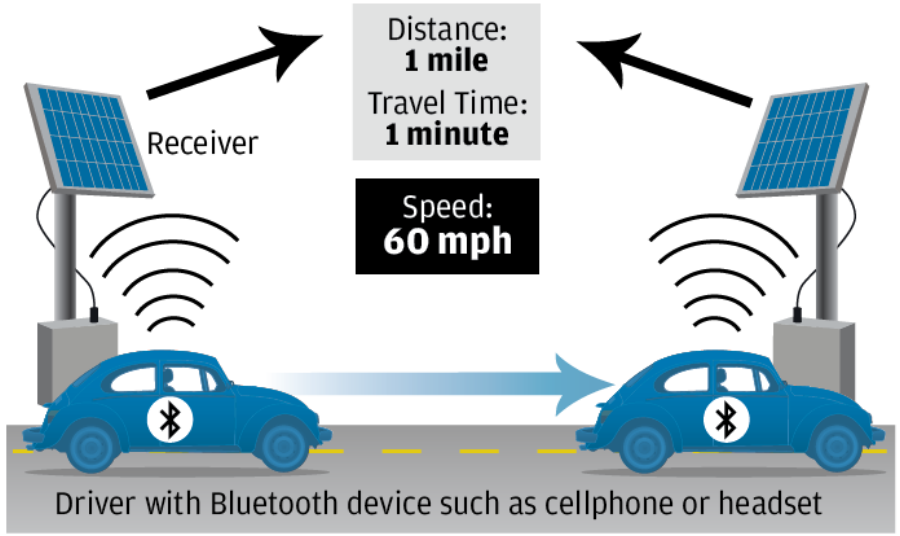


Connected Signal's Enlighten app, BMW and Audi for predictive signal timing

BlueTOAD

TrafficCast's BlueTOAD

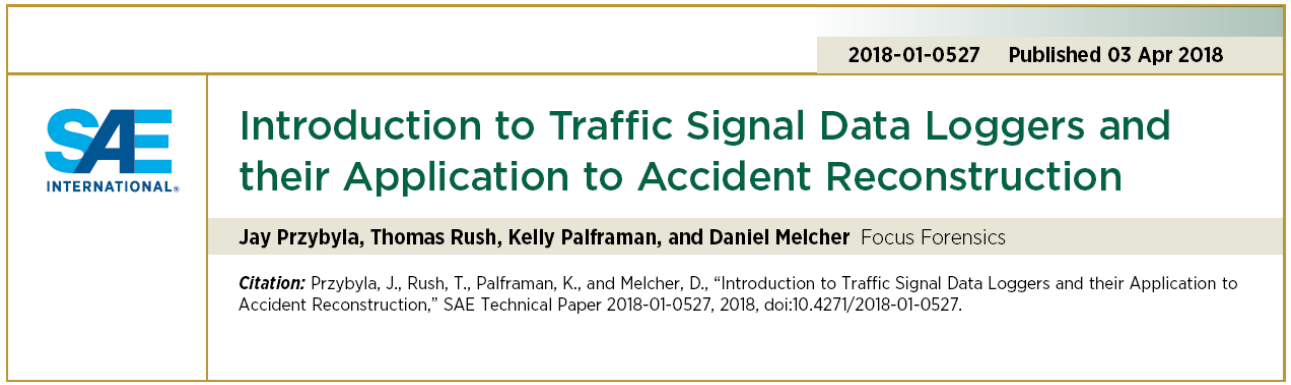
The BlueTOAD system consists of solar-powered receivers, attached to existing road signs, that read the unique signals from Bluetooth devices. The information is sent to servers where it is computed and in seconds converted into traffic flow data and used by navigation companies like TomTom and Google.



SOURCE: TrafficCast State Journal

- http://www.trafficcast.com/index.html

Further Info



2018-01-0527 Published 03 Apr 2018

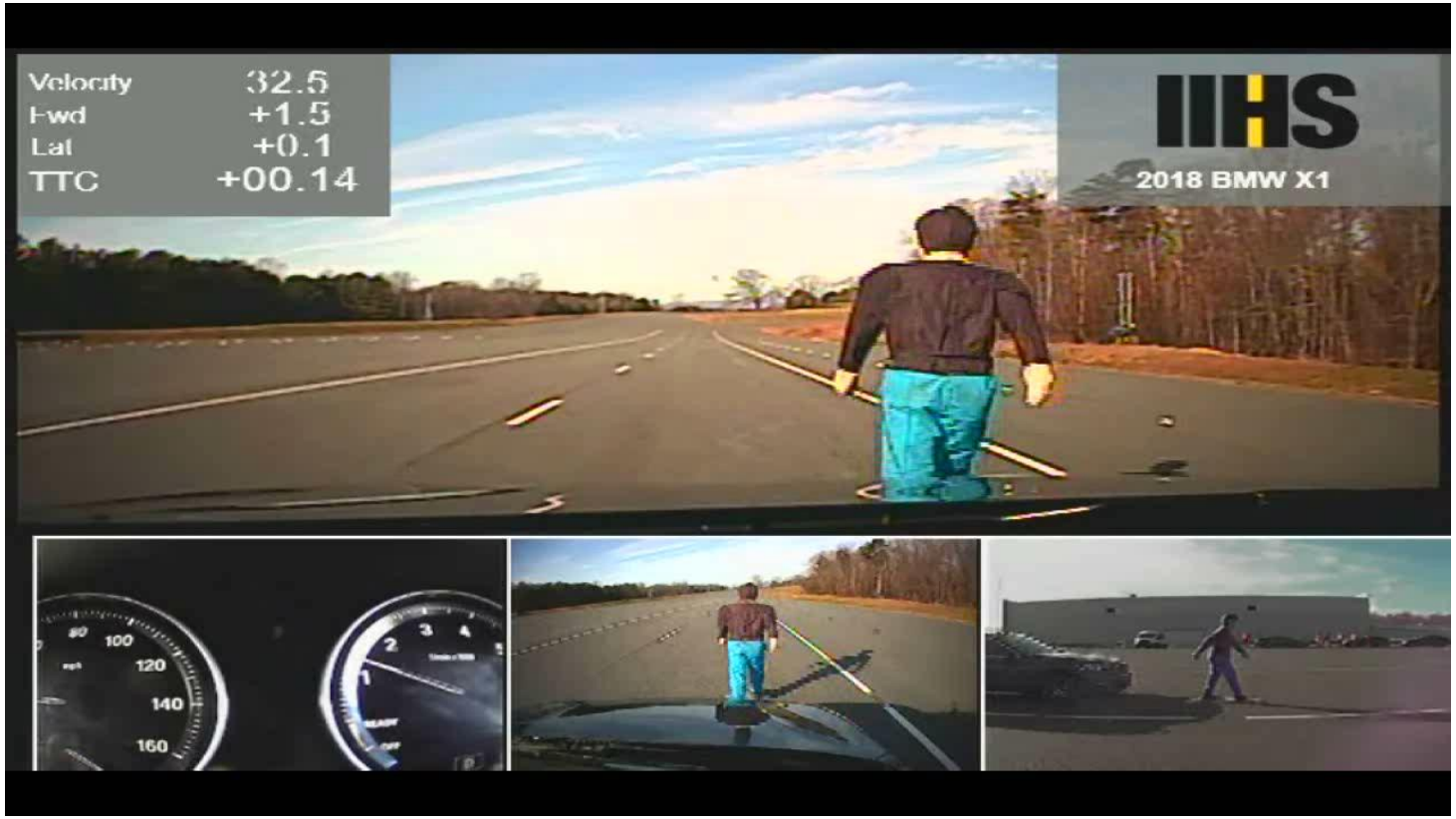
Introduction to Traffic Signal Data Loggers and their Application to Accident Reconstruction

Jay Przybyla, Thomas Rush, Kelly Palframan, and Daniel Melcher Focus Forensics

Citation: Przybyla, J., Rush, T., Palframan, K., and Melcher, D., "Introduction to Traffic Signal Data Loggers and their Application to Accident Reconstruction," SAE Technical Paper 2018-01-0527, 2018, doi:10.4271/2018-01-0527.

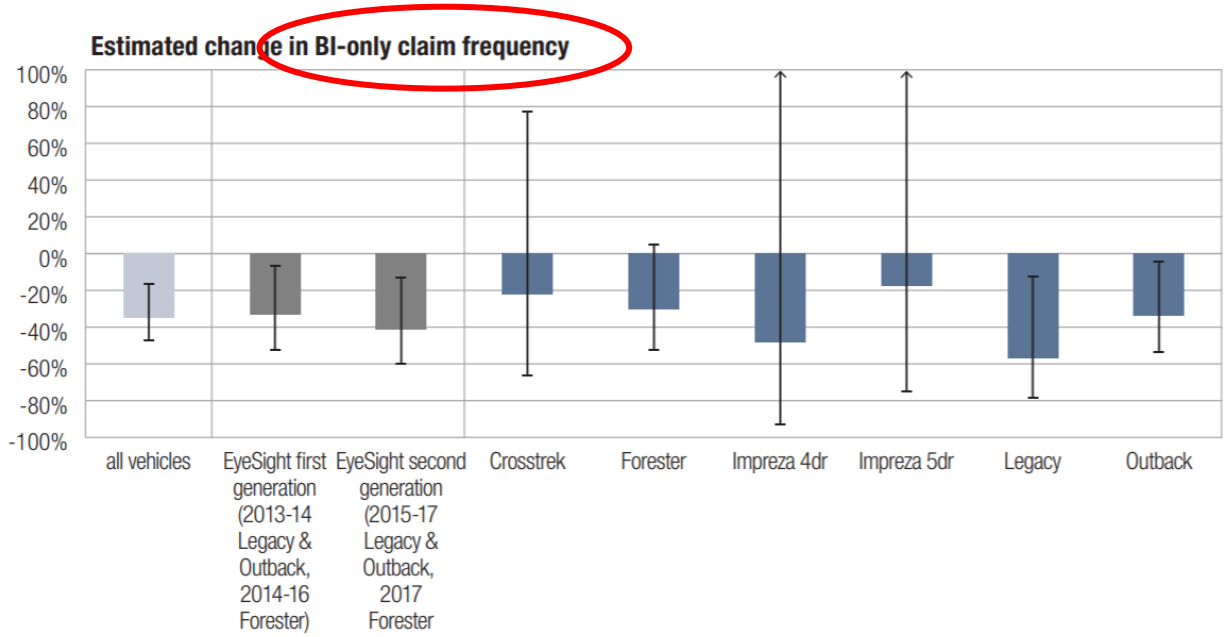
Pedestrian Automatic Emergency Braking (P-AEB)

IIHS Blooper Reel



Effectiveness of Subaru Pedestrian Detection

Subaru vehicles equipped with EyeSight reduced pedestrian-related claim frequency by 35 percent compared to Subaru vehicles without EyeSight



Effect of Subaru EyeSight on pedestrian-related bodily injury liability claim frequencies, Bulletin Vol. 34, No. 39 : December 2017
https://www.iihs.org/media/e8f617a5-b8f8-42e6-8b0f-9f56f9cb4208/W63Mlg/HLDI%20Research/Bulletins/hldi_bulletin_34.39.pdf

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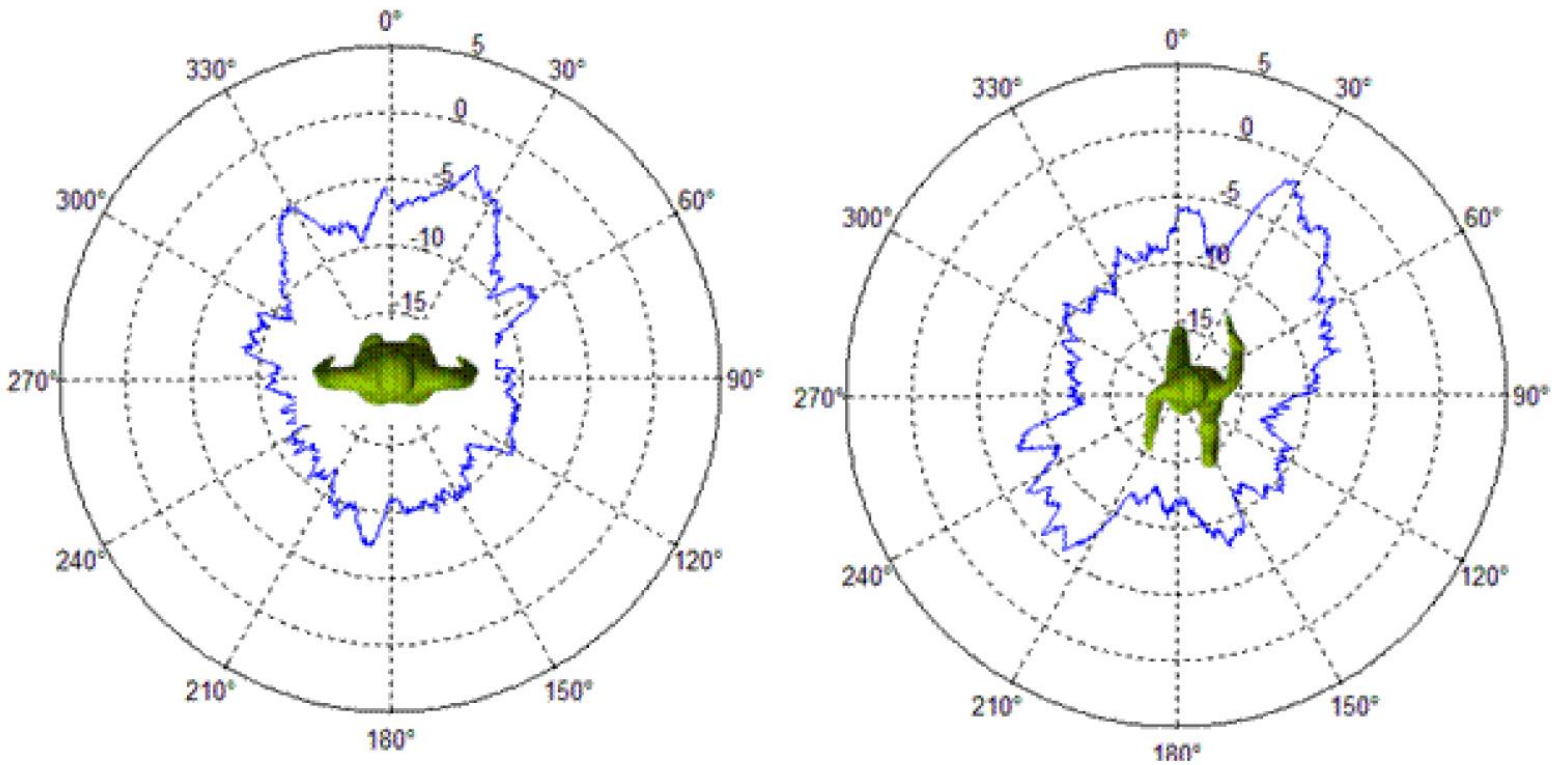
What is P-AEB?

- Forward Collision Warning
- Brake Assist
- Automatic Emergency Braking
- Balancing risk of false positives and false negatives

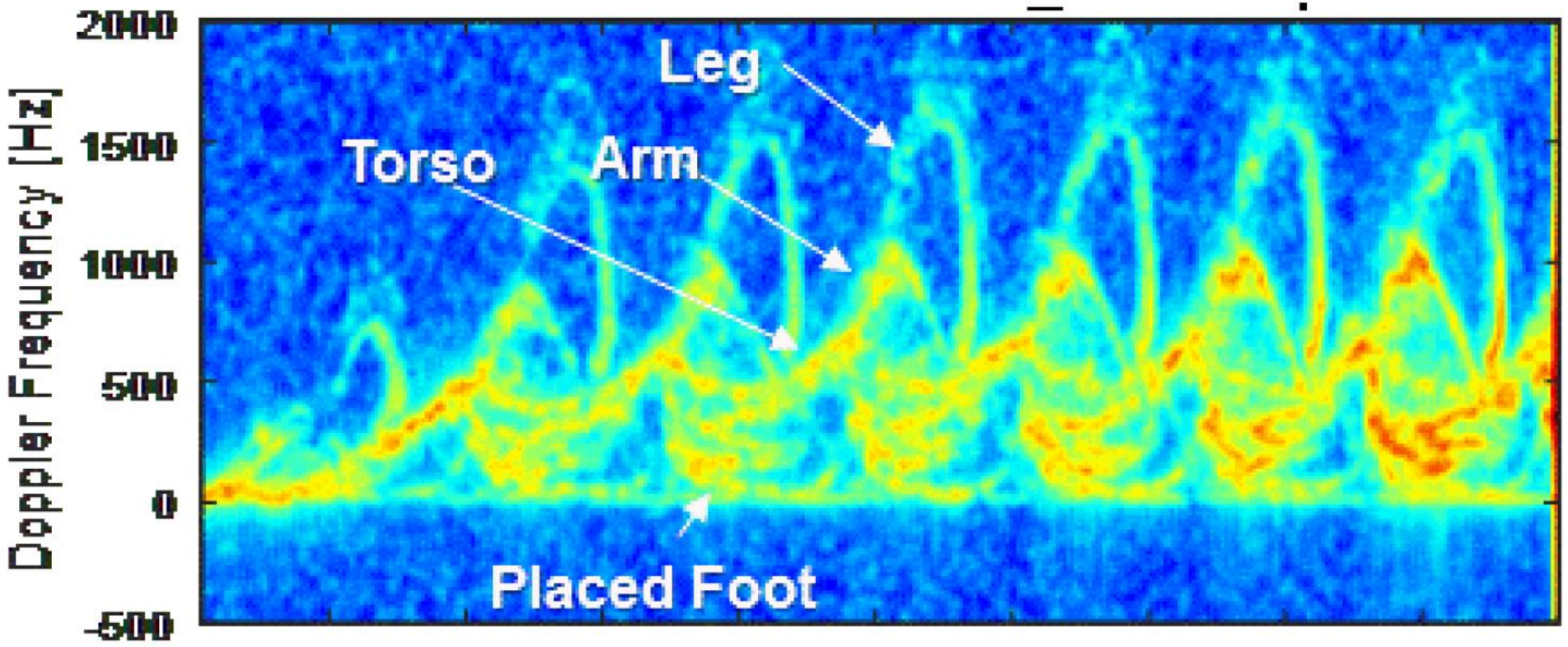
2019 Ford Edge Pedestrian Detection – Camera Only

Radar Pedestrian Detection



SAE J3116, Surface Vehicle Recommend Practice, June 2017

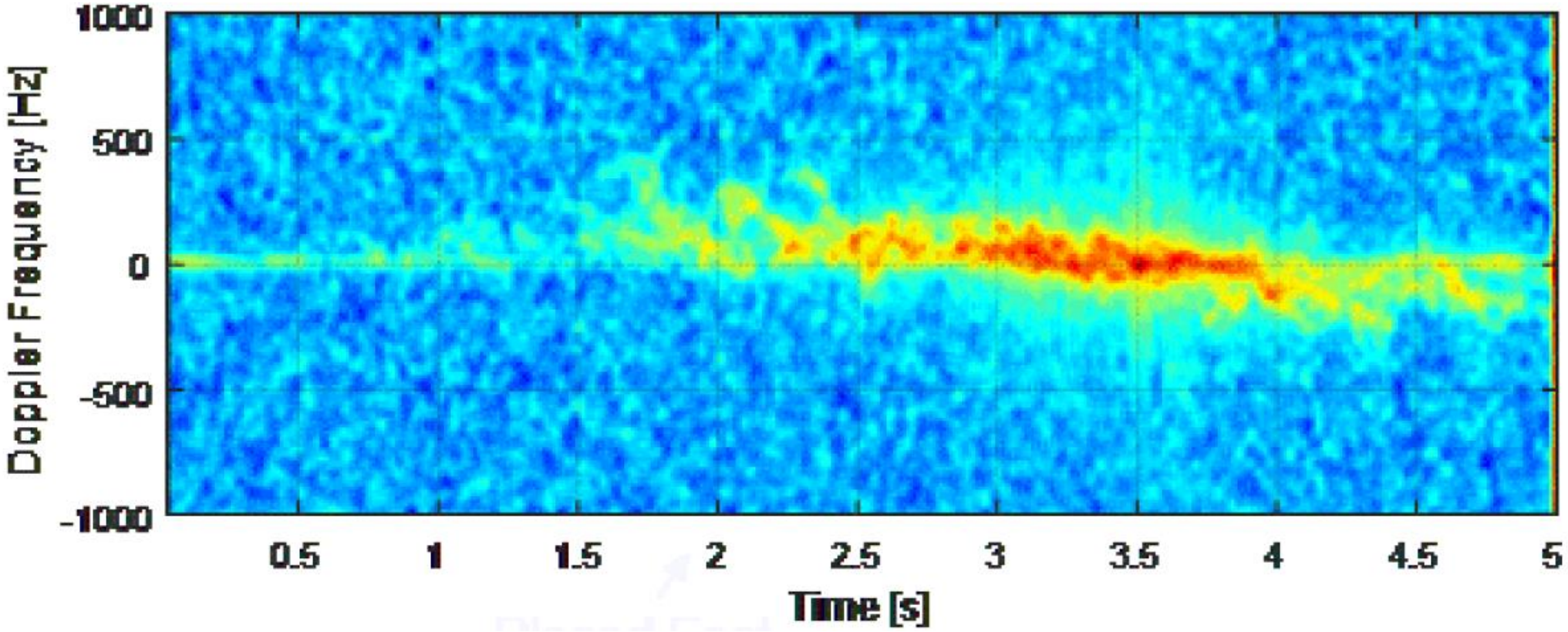
Radar Pedestrian Detection



Micro-Doppler effect of a ped walking toward a 77GHz radar.

SAE J3116, Surface Vehicle Recommend Practice, June 2017

Radar Pedestrian Detection



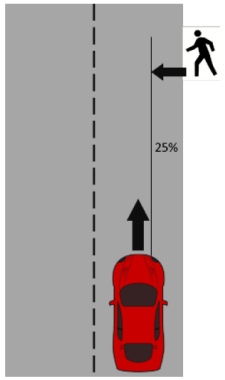
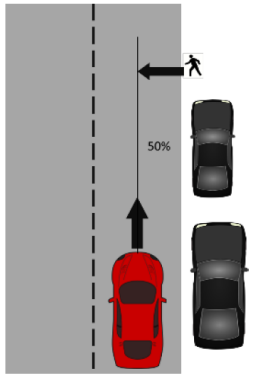
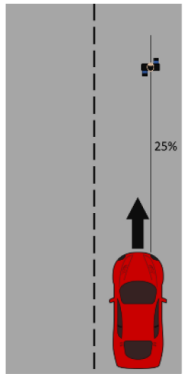
Micro-Doppler effect of a ped walking across a 77GHz radar.

SAE J3116, Surface Vehicle Recommend Practice, June 2017

Why is Pedestrian Detection Challenging?

- Short TTC
- Multiple trajectories – perpendicular, parallel, both moving, not in lane
- Sensing limitations
- Image classification
- Balancing risk
- Edge cases

IIHS Pedestrian Test Scenarios

Parameter	*Scenario*		
	Perpendicular adult (CPNA-25)	**Perpendicular child (CPNC-50)**	**Parallel adult (CPLA-25)**
Test vehicle speed	20, 40 km/h	20, 40 km/h	40, 60 km/h
Pedestrian target speed	5 km/h	5 km/h	0 km/h
Target direction	Crossing (R-to-L)	Crossing (R-to-L)	Facing away
Target path (relative to test vehicle)	Perpendicular	Perpendicular	Parallel
Pedestrian dummy size	Adult	Child	Adult
Dummy articulation (fixed rate)	Yes	Yes	No
Overlap	25%	50%	25%
Obstructed	No	Yes	No
Number of valid runs	5	5	5
Test diagram			

Pedestrian Autonomous Emergency Braking Test Protocol (Version II), February 2019, IIHS
https://www.iihs.org/media/f6a24355-fe4b-4d71-bd19-0aab8b39aa7e/TfEBAA/Ratings/Protocols/current/test_protocol_pedestrian_aeb.pdf

Pedestrian Targets – 4activesystems

4activePS pedestrian static

4activeSystems produces dummies, which replicate human properties in size and shape. The child dummy represents a child at the age of 7 years. 4a dummies can be recognized by mono and stereo cameras as well as radar and infrared systems. In infrared is 50% refl ectivity, between 850nm and 910nm. There is no danger of crashes for the dummy, respectively the test vehicle, during a test.

- static EPT - Euro NCAP Pedestrian Target Static
- homogeneous distribution of radar cross section (RCS)
- visual signature - mono and stereo camera
- infrared signature (IR)
- crashable up to 60km/h
- adult weight < 4kg - child weight < 2kg

http://www.4activesystems.at/en/products/dummies/4activeps.html

AAA Testing – Adult Crossing Path

https://www.youtube.com/watch?v=eEnklD0Vs4A

AAA Ped Testing – 2019 Toyota Camry, 20mph

20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
1.255	1.079	1.229	1.330	1.339	**1.246**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
35.83	30.12	34.89	38.40	38.12	**35.47**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
30.15	26.15	30.88	28.71	29.34	**29.05**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
0.897 @ 23.81	1.024 @ 20.16	1.096 @ 23.14	0.842 @ 21.70	1.053 @ 22.41	**0.982 @ 22.24**
Impact Speed (mph)					
0.0	0.0	0.0	0.0	0.0	**0.0**
Seperation Distance At End of Test (ft)					
4.67	5.25	4.26	4.27	4.54	**4.60**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Ped Testing – 2019 Honda Accord, 20mph

20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
0.023	1.153	0.865	1.000	0.534	**0.715**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
0.92	33.71	25.48	29.39	16.04	**21.11**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	31.71	24.32	27.62	15.15	**24.70**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
N/A	0.883 @ 5.62	0.999 @ 14.29	1.018 @ 13.90	1.950 @ 4.92	**1.213 @ 9.68**
Impact Speed (mph)					
20.1	0.0	0.0	0.0	0.6	**4.1**
Seperation Distance At End of Test (ft)					
0.00	3.91	3.57	4.10	0.00	**2.32**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

AAA Ped Testing – 2019 Chevrolet Malibu, 20mph

20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
2.480	2.410	2.366	2.31	1.064	**2.126**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
74.12	69.54	71.71	69.50	31.53	**63.28**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	N/A	N/A	N/A	N/A	**N/A**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
N/A	N/A	N/A	N/A	N/A	**N/A**
Impact Speed (mph)					
19.0	19.5	18.3	19.7	19.7	**19.2**
Seperation Distance At End of Test (ft)					
0.00	0.00	0.00	0.00	0.00	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

AAA Ped Testing – 2019 Chevrolet Malibu, 30mph

30 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
2.103	DNT	DNT	DNT	DNT	**2.103**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
93.77	DNT	DNT	DNT	DNT	**93.77**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
27.1	DNT	DNT	DNT	DNT	**27.1**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
0.776 @ 18.34	DNT	DNT	DNT	DNT	**0.776 @ 18.34**
Impact Speed (mph)					
25.5	DNT	DNT	DNT	DNT	**25.5**
Seperation Distance At End of Test (ft)					
0.00	DNT	DNT	DNT	DNT	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

AAA Ped Testing – 2019 Tesla Model 3, 20mph

20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
0.807	0.935	3.636	0.792	1.057	**1.445**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
23.41	27.40	105.38	22.91	29.71	**41.76**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
4.96	0.74	1.15	5.62	3.82	**3.26**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
1.083 @ 0.52	0.108 @ 0.74	0.110 @ 1.15	0.991 @ 1.67	0.868 @ 1.10	**0.632 @ 1.04**
Impact Speed (mph)					
17.3	19.6	19.0	16.8	17.5	**18.0**
Seperation Distance At End of Test (ft)					
0.00	0.00	0.00	0.00	0.00	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

AAA Testing – Child darting out from between cars

https://www.youtube.com/watch?v=eEnklD0Vs4A

AAA Testing – 2019 Toyota Camry – Child darting out from between cars

20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Impact Speed (mph)					
20.2	20.0	19.8	20.2	DNT	**20.1**
Seperation Distance At End of Test (ft)					
0.0	0.0	0.0	0.0	DNT	**0.0**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

AAA Testing – 2019 Chevrolet Malibu – Child darting out from between cars

20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
1.303	0.583	1.047	1.225	0.901	**1.012**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
37.10	17.06	29.92	38.89	26.41	**29.88**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
19.01	10.09	20.63	N/A	N/A	**16.58**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
0.391 @ 13.40	0.312 @ 7.22	0.353 @ 13.67	N/A	N/A	**0.352 @ 11.43**
Impact Speed (mph)					
17.1	19.9	16.6	20.5	19.5	**18.7**
Seperation Distance At End of Test (ft)					
0.0	0.0	0.0	0.0	0.0	**0.0**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

AAA Testing – All vehicles at night

Run 1	Run 2	Run 3	Run 4	Average
Notification TTC (s)				
N/A	N/A	N/A	N/A	**N/A**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)				
N/A	N/A	N/A	N/A	**N/A**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)				
N/A	N/A	N/A	N/A	**N/A**
Max Deceleration (G)/Associated Longitudinal Distance (ft)				
N/A	N/A	N/A	N/A	**N/A**
Impact Speed (mph)				
24.1	25.3	24.0	24.7	**24.5**
Seperation Distance At End of Test (ft)				
0.00	0.00	0.00	0.00	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

AAA - Nighttime Pedestrian Detection

"75% percent of pedestrian fatalities occurred in the dark
as compared to daylight (21%), dusk (2%) and dawn (2%)."

"Drivers must not rely on assistance from current pedestrian detection systems during nighttime driving or other environments with reduced visibility"

"The system must not brake for pedestrians alongside the vehicle in an adjacent sidewalk."

"Drivers must never rely on these systems to prevent a collision with a pedestrian; instead only considering them as a backup of last resort."

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

IIHS Testing – 2019 Subaru Forester

2019 Subaru Forester

Test Speed (km/h)	TTC (s)		Max Decel (m/s^2)	Initial Speed (km/h)	Impact Speed (km/h)
	FCW	AEB			
20	0.81	0.69	9.13	20.15	0.00
20	1.00	0.85	9.06	19.85	0.00
20	0.77	0.66	11.20	20.70	0.00
20	0.89	0.74	9.61	20.09	0.00
20	0.74	0.71	11.06	20.10	0.00
40	1.57	1.39	8.73	39.25	0.00
40	1.50	1.40	9.62	39.90	0.00
40	1.49	1.31	10.23	39.95	0.00
40	1.61	1.49	10.81	39.45	0.00
40	1.47	1.36	9.19	40.16	0.00
20	1.06	1.00	7.27	20.53	0.00
20	1.18	1.03	9.52	19.98	0.00
20	1.05	0.98	8.77	20.52	0.00
20	1.11	0.97	8.50	19.75	0.00
20	1.06	0.87	8.61	20.50	0.00
40	1.08	0.93	11.14	39.47	0.00
40	1.08	0.97	7.97	40.05	0.00
40	1.01	0.86	10.38	40.11	0.00
40	1.08	0.96	9.90	39.95	0.00
40	0.95	0.79	9.63	40.11	0.00
Average	**1.13**	**1.01**			

Adult Crossing Scenarios

www.techdata.iihs.org

IIHS Testing – 2018 BMW X1

2018 BMW X1

Test Speed	TTC (s)		Impact Speed (km/h)
	FCW	AEB	
20	0.08	0.02	20.30
20	0.00	0.00	20.71
20	0.00	0.00	19.59
20	0.14	0.10	19.73
20	0.14	0.19	18.83
40	0.44	0.35	36.66
40	0.47	0.44	35.90
40	0.46	0.45	35.16
40	0.48	0.44	35.75
40	0.32	0.39	35.87
20	0.49	0.16	18.37
20	0.00	0.00	20.38
20	0.59	0.60	12.26
20	0.49	0.00	19.95
20	0.81	0.72	8.96
40	0.53	0.45	35.71
40	0.66	0.52	34.50
40	0.49	0.35	36.68
40	0.47	0.51	34.65
40	0.57	0.63	32.26
Average	**0.38**	**0.32**	

Adult Crossing Scenarios

www.techdata.iihs.org

Volvo P-AEB Failure at Car Dealership

Test variables to watch for, in both published and exemplar testing

- Time of day and lighting
- Background objects and visual noise
- Driver inputs, which can inhibit systems
- Biofidelic motion
- Contrast
- Structure from motion

P-AEB EDR - Toyota

Vehicle Control History via Techstream can provide:

- Target type
- Target speed toward vehicle, and across path
- Target distance
- Warnings and autobrake triggers
- Driver response

P-AEB EDR – General Motors

CDR 900 can provide:

- Some pedestrian classification
- Target speed toward vehicle, and across path
- Target distance
- Warnings and autobrake triggers
- Driver response
- Override/Inhibit Reasons

Examples of ADAS – successes and the blooper reel

By the end of this module, you will:

- Recognize strengths and weaknesses of ADAS, and how it can influence a crash

Ford Steering Assist

Evasive Steering Assist (If Equipped)

If your vehicle is rapidly approaching a stationary vehicle or a vehicle traveling in the same direction as your vehicle, the system is designed to help you steer around the vehicle.

The system only activates when all of the following occur:

- The Pre-Collision Assist system detects a vehicle ahead and starts to apply Active Braking.
- You turn the steering wheel in an attempt to steer around the vehicle.

After you turn the steering wheel, the system applies additional steering torque to help you steer around the vehicle. After you pass the vehicle, the system applies steering torque in the opposite direction to encourage you to steer back into the lane. The system deactivates after you fully pass the vehicle.

Note: *The system does not automatically steer around a vehicle. If you do not turn the steering wheel, the system does not activate.*

Note: *The system does not activate if the distance to the vehicle ahead is too small and a crash cannot be avoided.*

2019 Ford Edge Owner Manual

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Ford Steering Assist

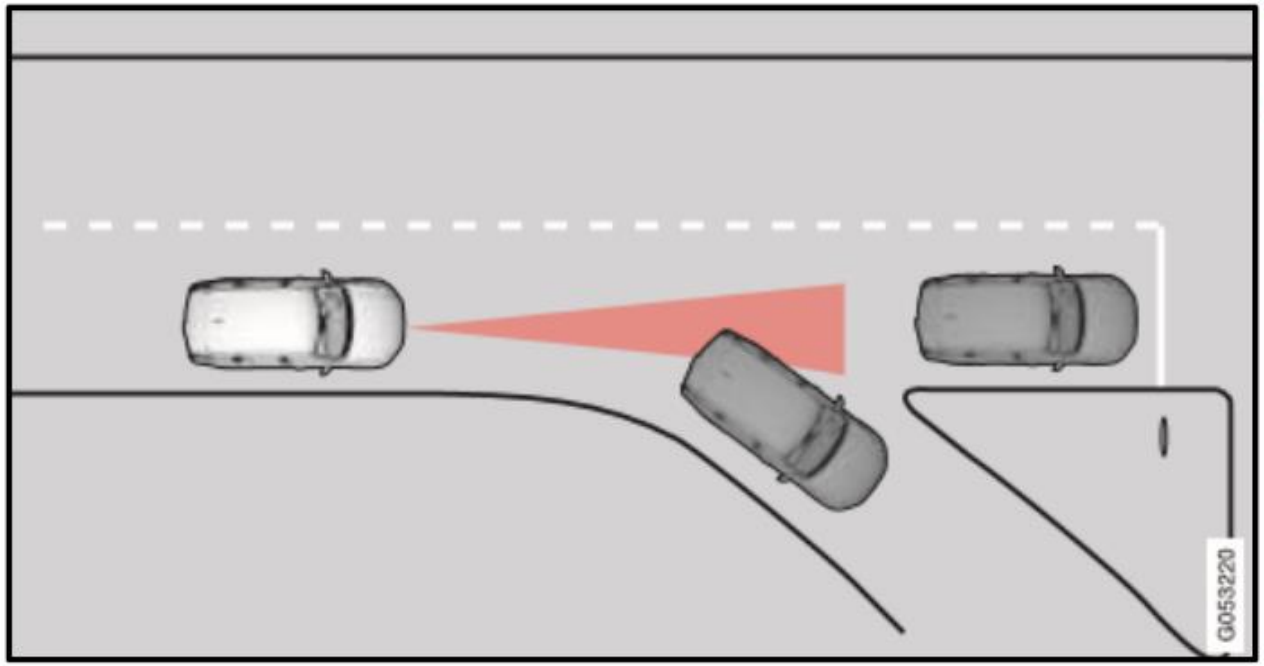
https://www.westernslopeauto.com/blog/how-ford-evasive-steering-assist-works/

Examples of ADAS – Adaptive cruise

- 51st red light
- Cut-out scenarios
- Stopped vehicles with partial speed range ACC
- Stopped vehicles above ~50mph
- Radar tracking lead vehicle, but camera obscured by rain/snow

Examples of ADAS – Adaptive cruise

- 2016 Volvo XC90 Owner's Manual



WARNING

If ACC changes targets from a moving vehicle to a stationary one at speeds **above** 20 mph (30 km/h), the system will **not** react to the stationary vehicle and will accelerate to the previously set speed.

The driver must actively apply the brakes to slow/stop the vehicle.

Tesla radar skip

Examples of ADAS – Automatic Emergency Braking

- Balancing risk of false positives and false negatives
  - Frontal vs rear impact probability
- Good performance only to classified objects
- Forward collision *mitigation*
- Distance used to alert driver by audible/visual/haptic warnings

Examples of ADAS – Automatic Emergency Braking

Ford F-150 recall due to false positives when passing a wide, reflective vehicle

Recall #15V614000, 09/30/2015, FORWARD COLLISION AVOIDANCE: ADAPTIVE CRUISE CONTROL

2014-2015 Jeep Grand Cherokee – NHTSA investigation for unintended braking with AEB

- Haptic alert, or “brake jerk”, is less than 400ms and <3mph speed change
- False positives vary greatly by software version

Assistance Feature Release Number	Production Range	Vehicles Sold	Vehicle Exposure (VehYrs)	Unintended Braking Incidents	Failure Rate (IPTV)	Exposure Adjusted Failure Rate (IPTVY)
68109355AF	07/12 to 02/14	17,263	46,057	69	4.0	1.5
68223771AD	07/13 to 08/13	3,657	8,880	14	3.8	1.6
68223771AE	08/13 to 01/14	15,309	32,652	39	2.5	1.2
68223771AF	01/14 to 04/14	8,922	16,098	17	1.9	1.1
68109355AG	02/14 to 04/14	688	1,276	1	1.5	0.8
68109355AI	04/14 to 07/14	1,158	1,864	1	0.9	0.5
68223771AH	04/14 to 07/14	7,349	11,778	8	1.1	0.7
68259548AA	07/14 to 12/15	40,832	32,192	27	0.7	0.8
Grand Total	07/12 to 12/15	95,178	150,798	176	1.8	1.2

Table 2. Field Experience by Software Version.

NHTSA Investigation #PE 15-021 - 2014-2015 Jeep Grand Cherokee Unintended Braking
https://static.nhtsa.gov/odi/inv/2015/INCLA-PE15021-6366.PDF

Mazda Automatic Emergency Braking

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2018 Toyota Camry Automatic Emergency Braking

SAE INTERNATIONAL

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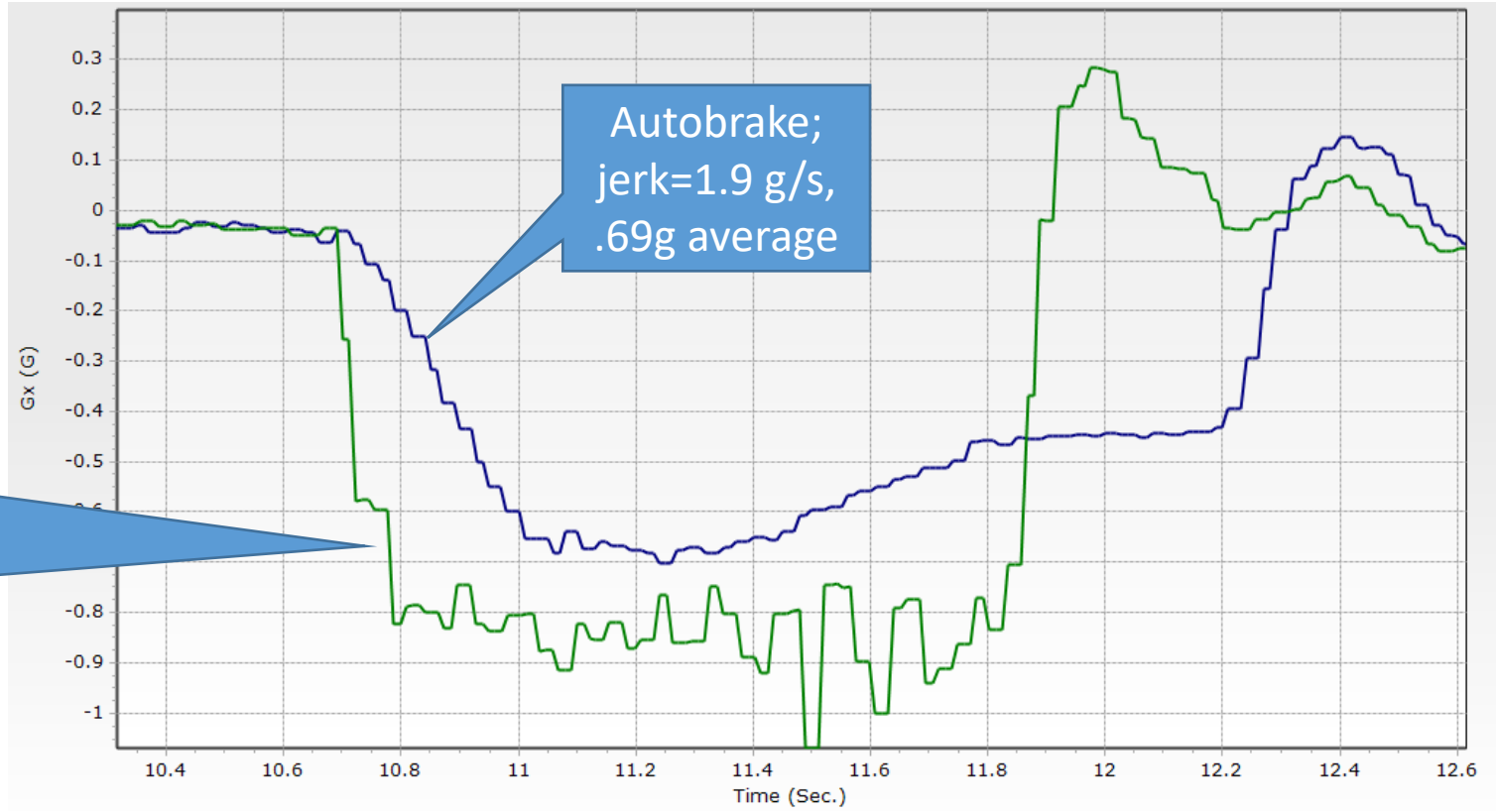
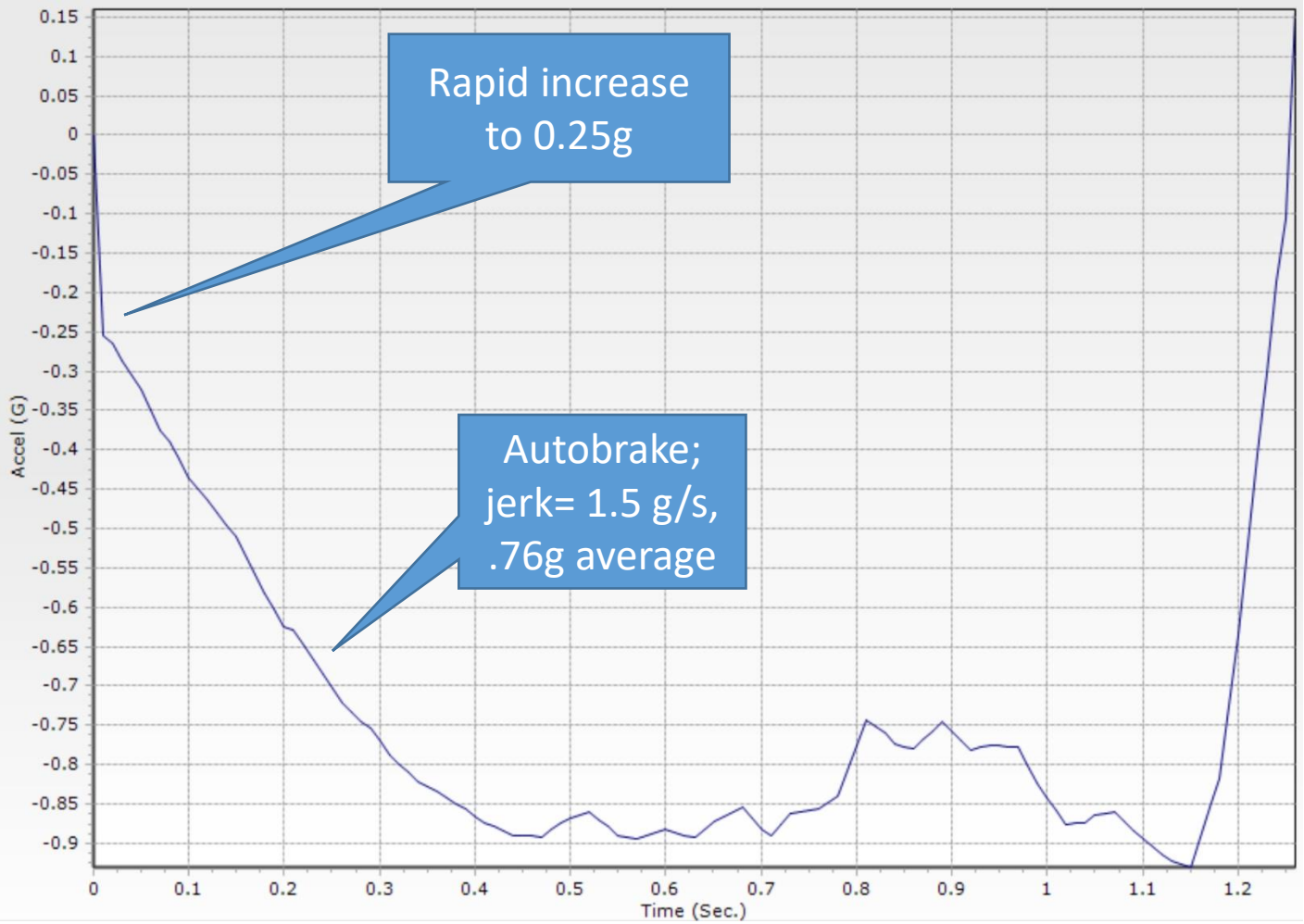
2018 Toyota Camry – Brake Assist v Autobrake



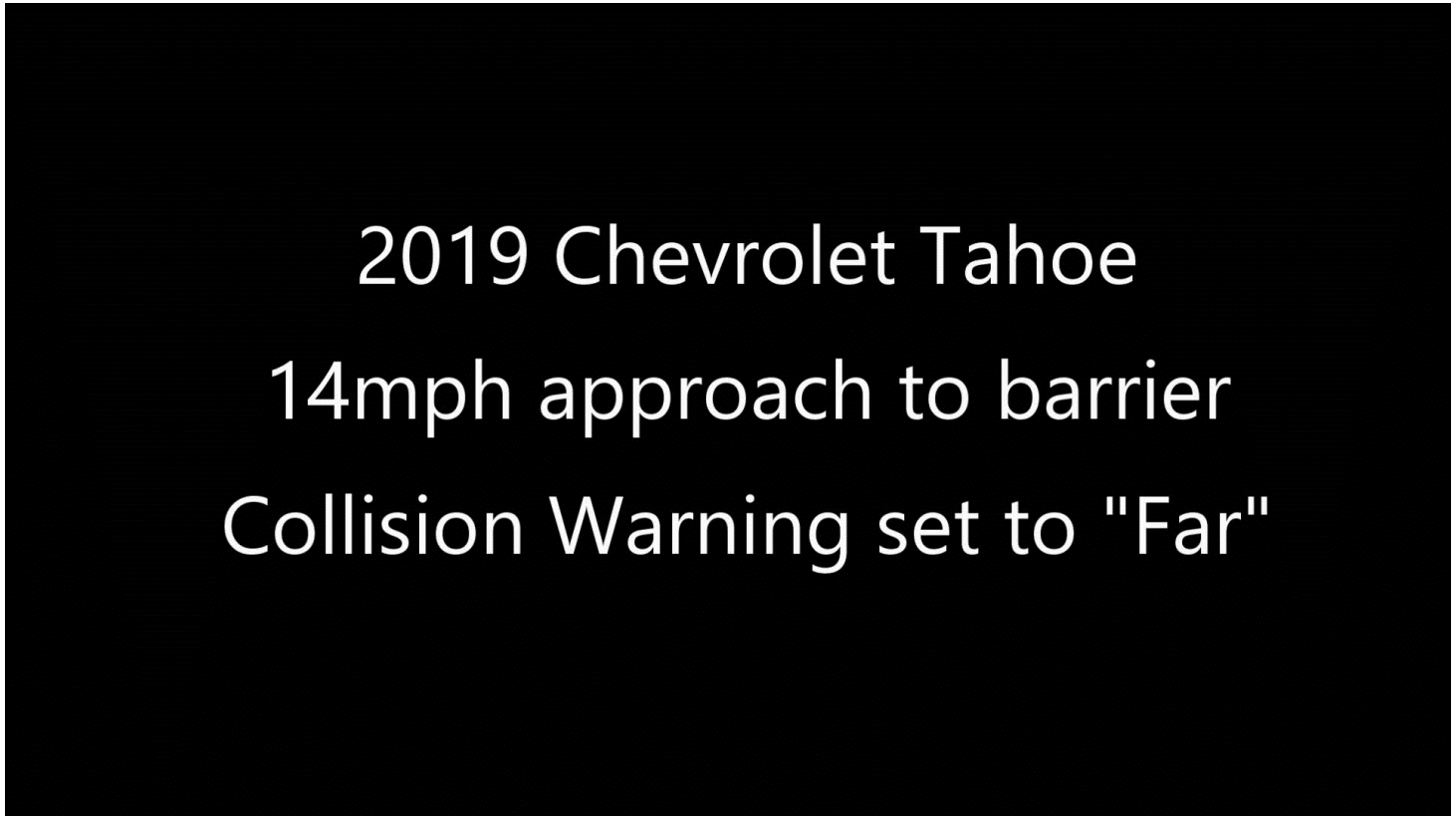
18mph. 0.9s from warning to autobrake

Automatic Braking – 2018 Toyota RAV4



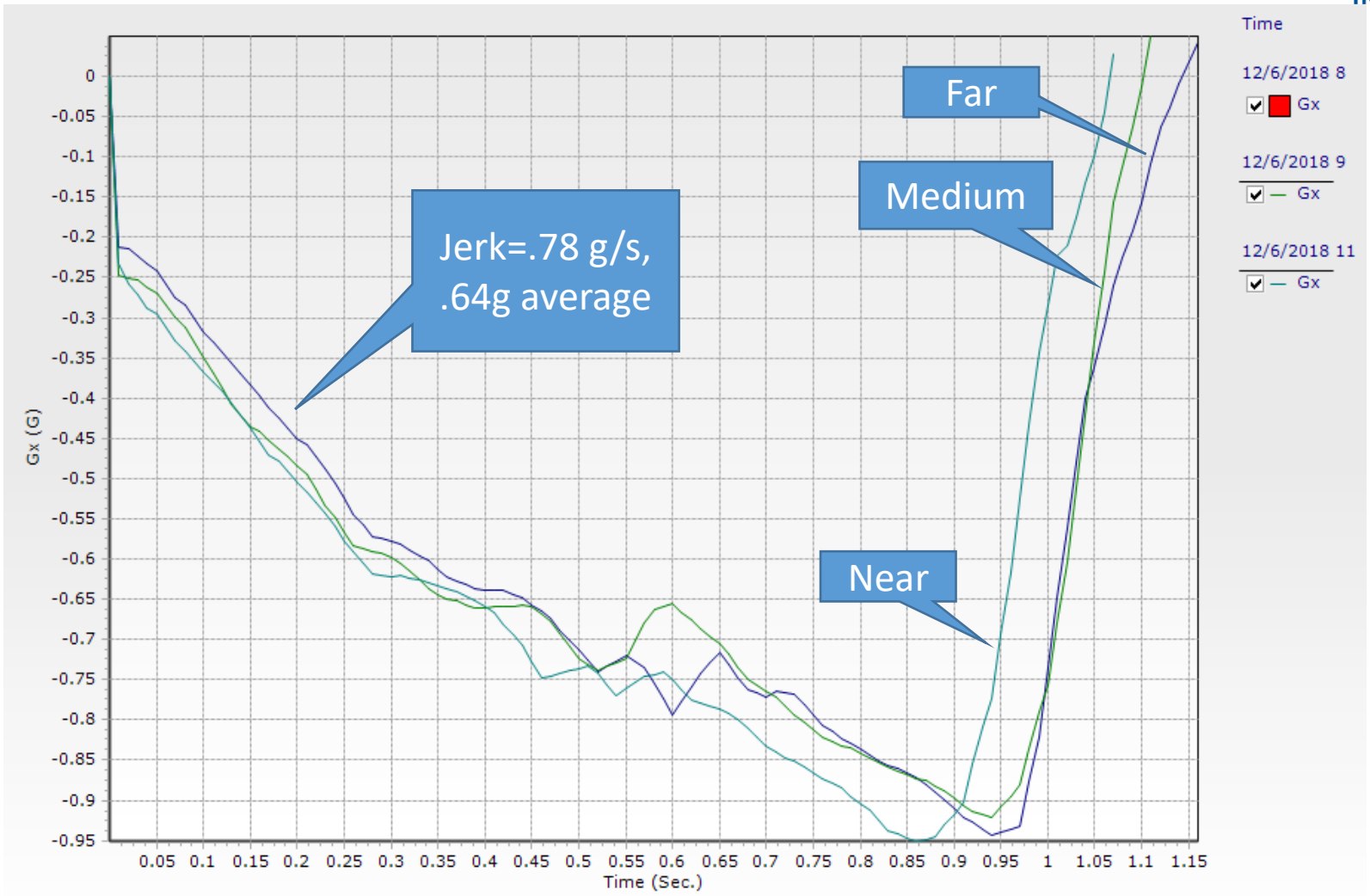
20mph approach

2019 Chevrolet Tahoe Automatic Emergency Braking



2019 Chevrolet Tahoe Automatic Emergency Braking

Collision Warning Sensitivity Settings Indicated



2019 Chevrolet Tahoe Automatic Emergency Braking

	Warning light to:	
FCW Setting	Brake light	Full stop
Far	0.30	1.47
Medium	0.33	1.43
Near	0.30	1.23

Braking begins approximately 0.5s after warning sounds

Tesla Collision Warning

Examples of ADAS – Automatic Emergency Braking

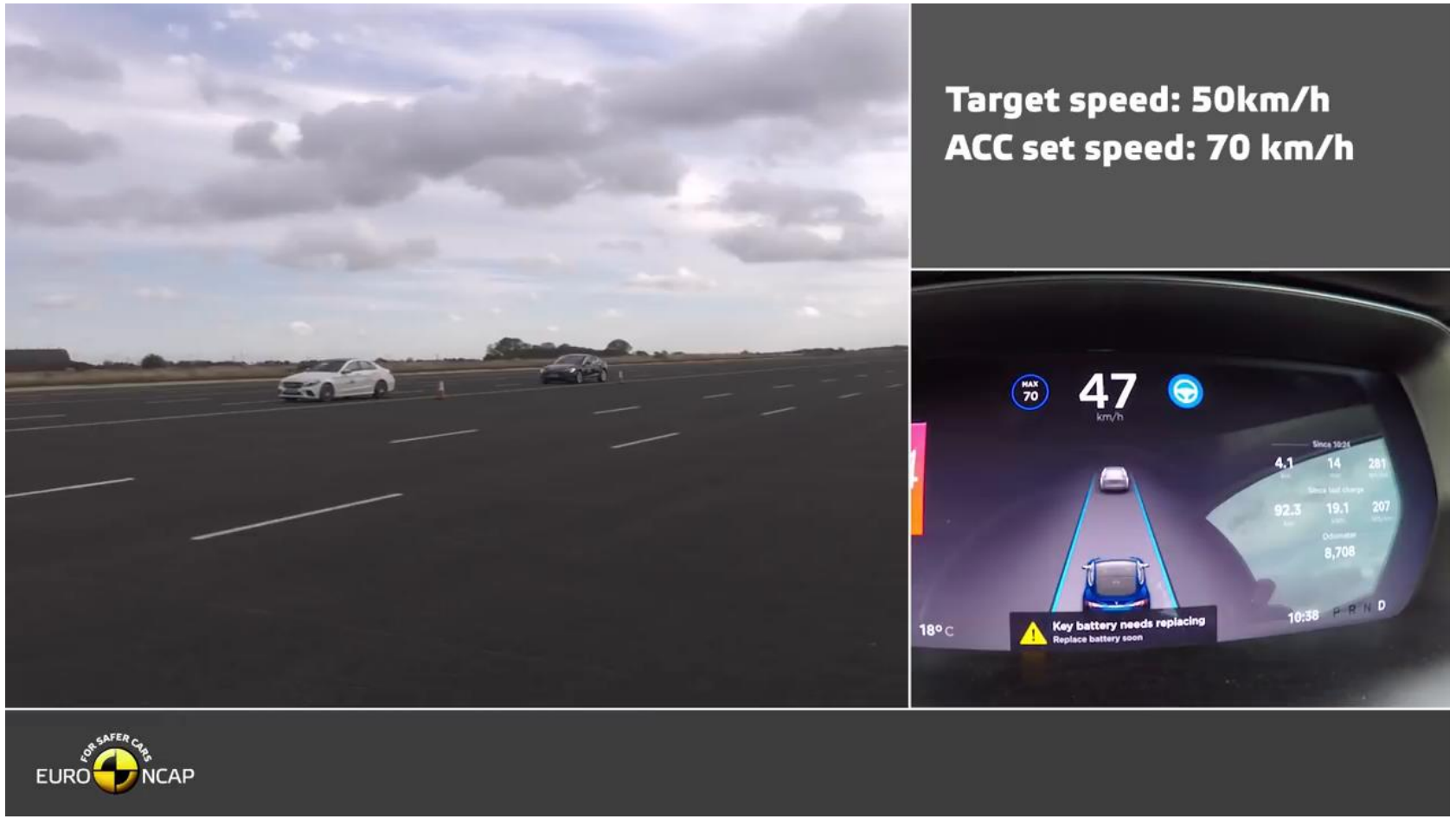
Failure to classify and respond to an unusual vehicle

Examples of ADAS – Automatic Emergency Braking

How did this happen?

Examples of ADAS – Automatic Emergency Braking

NTSB Photo



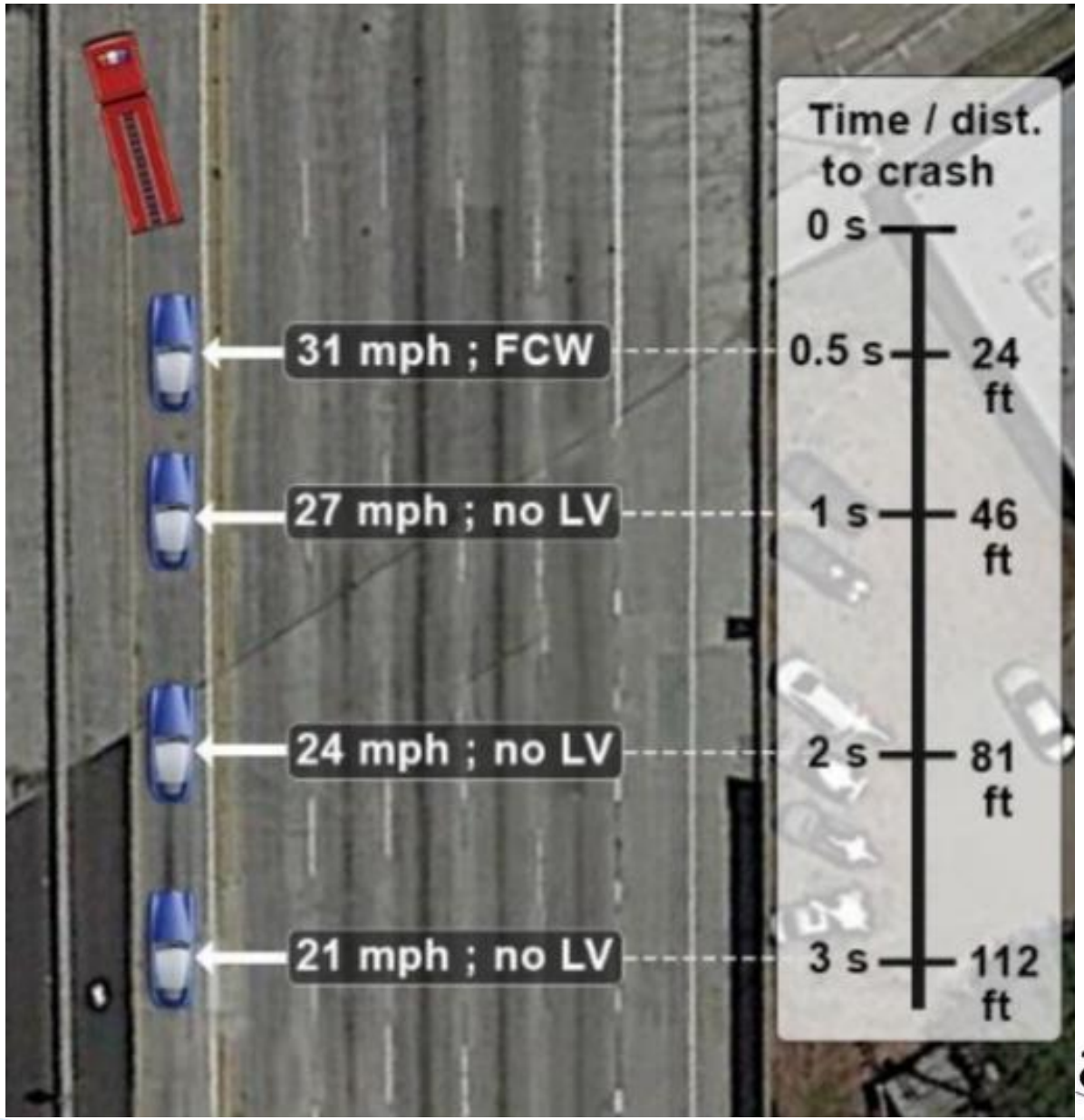
Example cut-out scenario from EuroNCAP testing.

EuroNCAP ADAS Testing of 2018 Tesla Model S (different autopilot than Culver City crash)
https://www.euroncap.com/en/vehicle-safety/safety-campaigns/2018-automated-driving-tests/

Examples of ADAS – Automatic Emergency Braking

2018 Tesla vs Fire Truck Crash

Culver City, CA



NTSB HWY18FH004 Highway Accident Brief
https://www.ntsb.gov/investigations/AccidentReports/Reports/HAB1907.pdf

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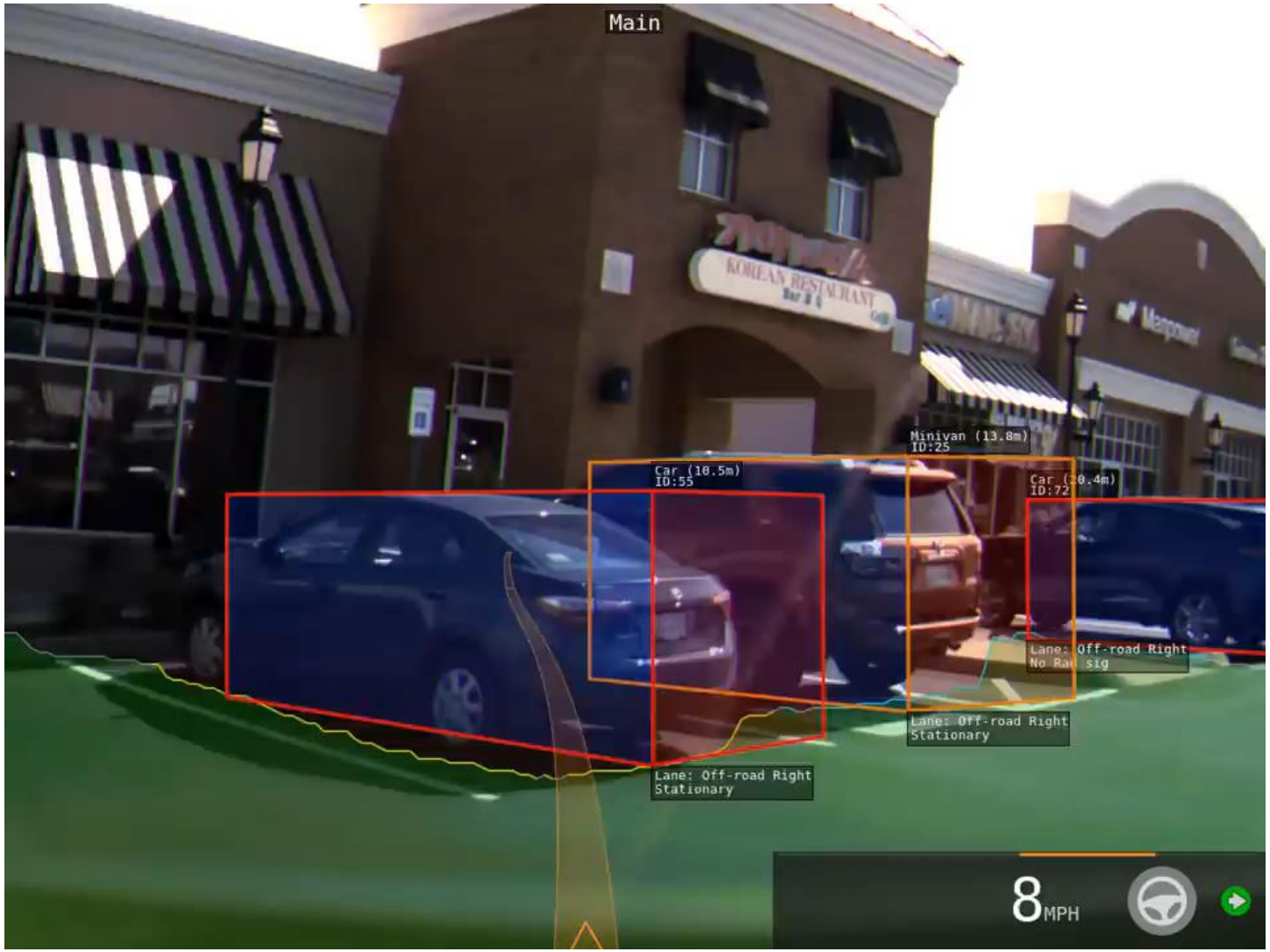


Tesla debug view of a firetruck – version 19.12
https://www.youtube.com/watch?v=9YBBc2eg27g

Examples of ADAS – Automatic Emergency Braking

- The radar sensor has some performance limitations. If a stationary vehicle is in the vehicle's path, the AEB system will not function when the vehicle is driven at speeds over approximately 50 mph (80 km/h).

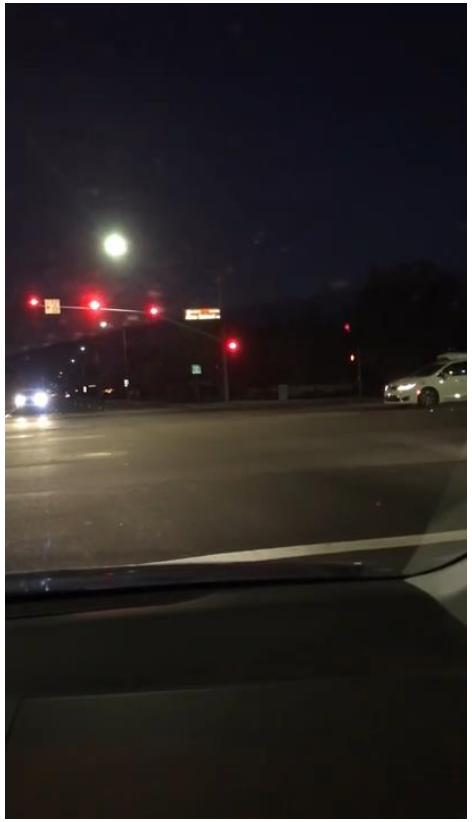
2019 Nissan Sentra Owner's Manual and Maintenance Information, June 2019
https://owners.nissanusa.com/content/techpub/ManualsAndGuides/Sentra/2019/2019-Sentra-owner-manual.pdf

Tesla Autopilot crash - Handan, China

NHTSA ODI Consumer Complaint

Uber red light running*

Waymo red light running*



Mobileye red light running*

Uber Pedestrian Crash – Tempe, AZ

Las Vegas Autonomous Shuttle Crash

NTSB investigation HWY18FH001: https://www.ntsb.gov/investigations/AccidentReports/Reports/HAB1906.pdf

Tesla Florida fatal crash

Examples of ADAS – Automatic Emergency Braking (Tesla Florida fatal crash)

ISO 22839 – Automatic Emergency Braking

6.3.4.3 Detection width and detection height

Table 4 — Detection width and detection height requirements

Distance	Minimum detection width	Minimum detection height
d_{max}	W_L (lane width) meters	h_1 (lower detection height from ground) = 0,2 m and h (upper detection height from ground) = 1,1 m
d_2	W_V (subject vehicle width) meters	h_1 (lower detection height from ground) = 0,2 m and h (upper detection height from ground) = 1,1 m
d_1	not specified	not specified
d_0	not specified	not specified

Tesla Florida fatal crash

Tesla Florida fatal crash

Extraction of SD card from MCU (Media Control Unit)

Tesla Florida fatal crash– Logged Data

5/7/2016 13:30:01.5 Insufficient steering wheel torque to detect hands on

5/7/2016 13:30:01.5 Sufficient steering wheel torque to detect hands on
5/7/2016 13:30:01.5
5/7/2016 13:30:01.5 Autosteer engages
5/7/2016 13:30:01.5 Autosteer cruise speed restricted due to detected road class.
Target speed limited to at most 45mph
5/7/2016 13:30:02.5
5/7/2016 13:30:03.5
5/7/2016 13:30:04.1 Accelerator pedal no longer overriding cruise control.

5/7/2016 13:30:04.5
5/7/2016 13:30:04.7 Insufficient steering wheel torque to detect hands on

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Tesla Florida fatal crash– Logged Data

5/7/2016 13:34:04.0	The Autopilot system is tracking a lead vehicle 18.6 meters away
5/7/2016 13:34:34.0	The Autopilot system is no longer tracking a lead vehicle
5/7/2016 13:34:21.0	Since the last reported PT_DI_CRUISESET the driver increased the cruise control set speed from 73 mph to 74 mph
5/7/2016 13:36:12.7	Vehicle alert consistent with collision damage

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Tesla Florida fatal crash– Logged Data

alert.id_text	PT_DI_VEHICLESPEED	PT_DI_PEDALPOS	CH_ESP_DRIVERBRAKEAPPLY.text	CH_EPAS_INTERNALSAS	CH_DAS_AUTOPIL
	47.55	17.6		-0.2	
	73.95	0		-0.5	
	73.9	0		-0.9	
DIS_f037_busUV: Drive Inverter Slave Fault: Battery under voltage					
DI_f037_busUV: Drive Inverter Fault: Battery under voltage					
	73.95	0		-0.8	

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Tesla Florida Fatal Crash

Crash Date Drive Rail Cycle Data				
Drive Rail Cycle [*at least 0.16 km (0.1 mi) traveled*]	**DRC1**	**DRC2**	**DRC3**	**Total**
Autopilot activations	22	7	3	32
Driving time (hh:mm:ss)	2:04:47	1:20:16	0:40:47	4:05:50
Autopilot time (hh:mm:ss)	1:40:56	1:09:19	0:37:26	3:27:41
Percent time in Autopilot	81%	86%	92%	84%
Trip distance	169.3 km (105.2 mi)	117.5 km (73.0 mi)	65.2 km (40.5 mi)	352.0 km (218.7 mi)
Autopilot distance	152.0 km (94.5 mi)	112.5 km (69.9 mi)	63.0 km (39.1 mi)	327.5 km (203.5 mi)
Percent distance in Autopilot	90%	96%	97%	93%
Hands Not Detected Visual alerts	19	10	7	36
Hands Not Detected First Chime alerts	9	6	6	21
Hands Not Detected Enhanced Chime alerts	0	0	0	0
Hands Not Detected Slowing alerts	0	0	0	0
TOTAL HANDS NOT DETECTED ALERTS	28	16	13	57
Avg. response time to initial hands not detected alert (sec)	11.9	12.6	16.0	13.0
Hand Detections	43	17	10	70
Autopilot time with Hands Detected (h:mm:ss)	0:01:45	0:00:38	0:00:26	0:02:49
Autopilot time with Hands Not Detected (h:mm:ss)	1:35:55	1:08:05	0:37:00	3:21:00
Percent of Autopilot time with Hands Detected	1.8%	0.9%	1.1%	1.4%
Number of brake applications	37	14	5	56
Number of Steering Overrides	2	1	0	3

Table 4: Trip summary and ADAS operational data from the Tesla on the date of the crash.

DOT HS 812 481: NHTSA Special Crash Investigations: On-Site Automated Driver Assistance System Crash Investigation of the 2015 Tesla Model S 70D, January 2018

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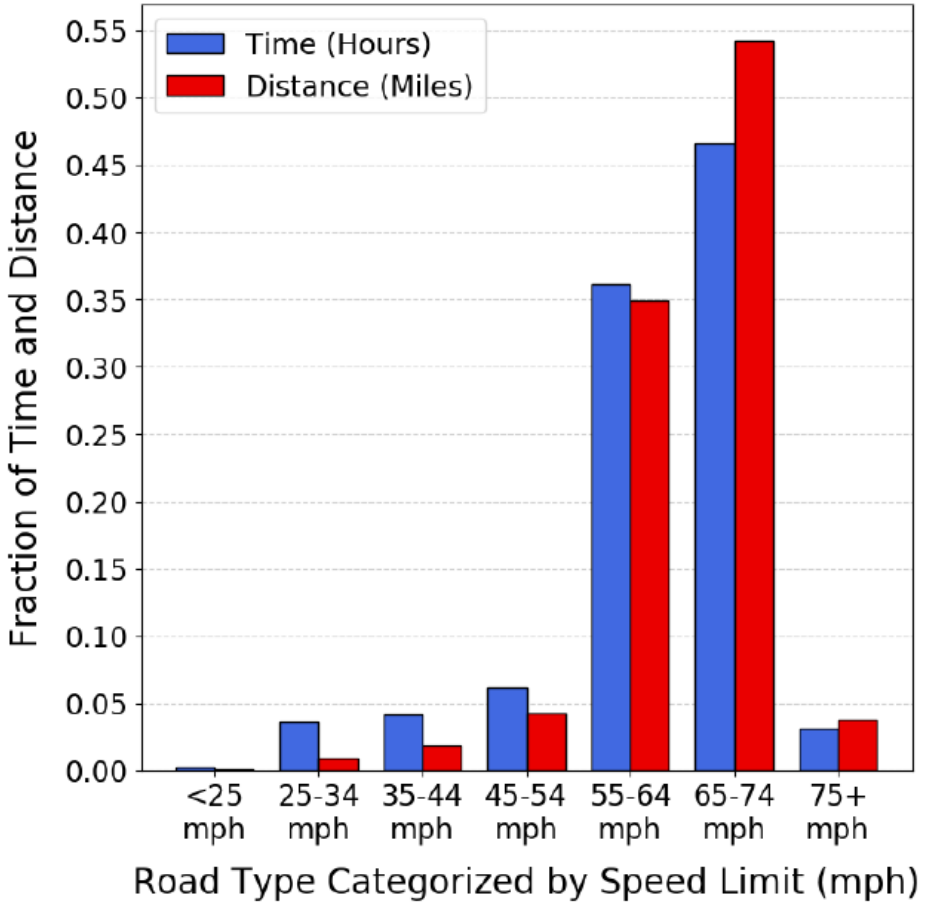
Where do drivers use Tesla Autopilot?



(b) Vehicle under Autopilot control.

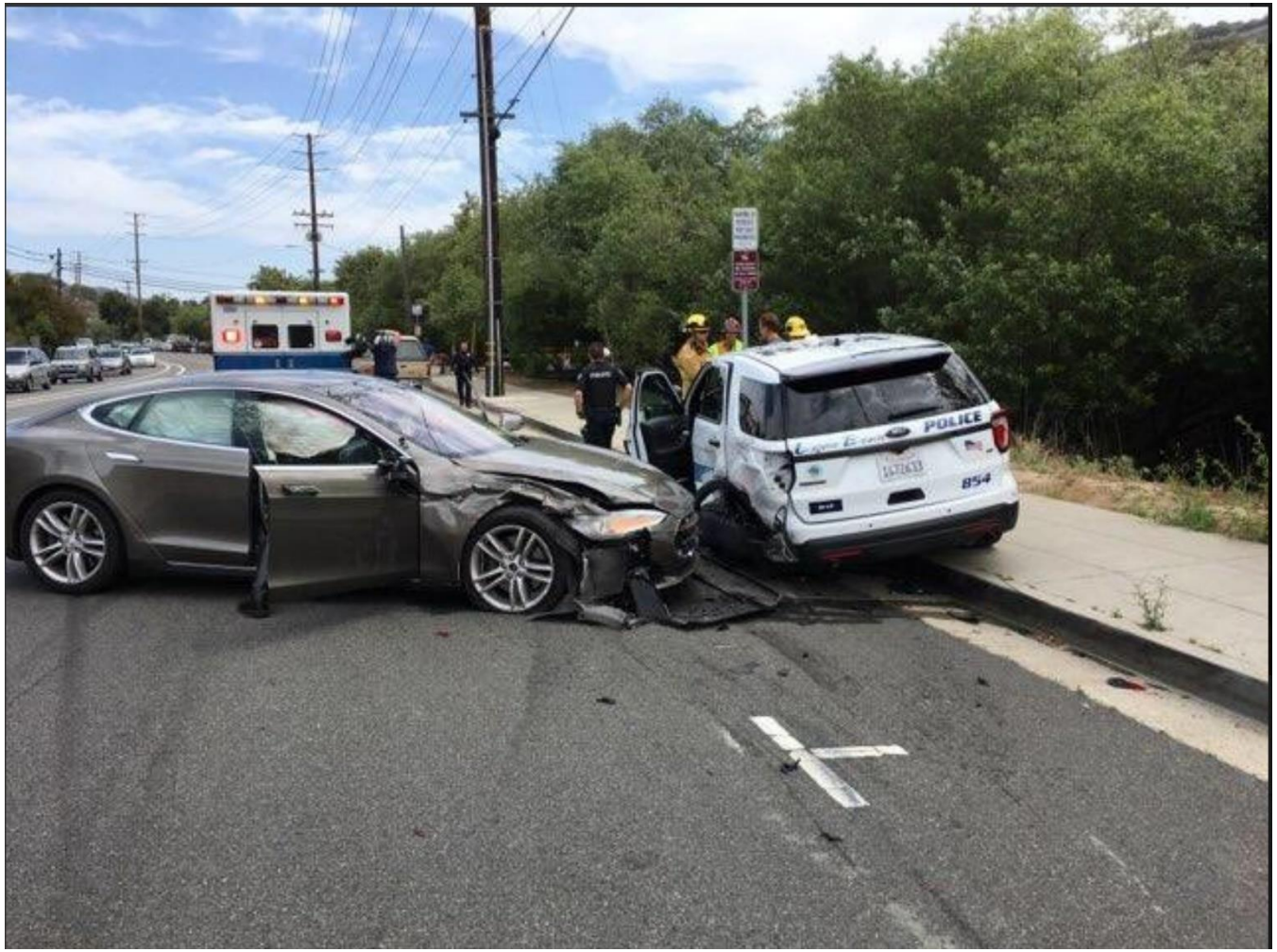
Human Side of Tesla Autopilot:Exploration of Functional Vigilance
in Real-World Human-Machine Collaboration, Lex Fridman et al, , Massachusetts Institute of Technology (MIT)

Examples of ADAS – Lane Centering

Tesla S Construction Site

Tesla autopilot barrier impact

Tesla – Police Car Crash, Laguna Beach, CA



Tesla – Police Car Crash, Laguna Beach, CA

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Tesla Impact with guardrail end treatment

NTSB Presentation on Mountain View, CA Crash – Crash Investigation HWY18FH011
https://www.ntsb.gov/news/events/Documents/2020-HWY18FH011-BMG-presentations.pdf#search=carlog

Tesla Impact with guardrail end treatment

A

Level 2/3 Autonomy* on a traffic circle

Hyundai Ghost Ride

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Examples of ADAS – successes and the blooper reel

Summary:

- ADAS technologies have strengths and weaknesses
- ADAS can prevent a crash, but there are scenarios where it can contribute to a crash

Media and Public Perception

By the end of this module, you will:

- Understand that public perception is dependent on the media, and is influenced by coverage of isolated events
- Be aware of road rage tendencies against autonomous vehicles

Media and Public Perception

- Inflation of incidents
- Most drivers, when interviewed, claim they won't rely on ADAS technology.
- Studies show that drivers adopt and rely on ADAS tech too quickly and place too much trust[1].
- Ethics - "Self-driving cars programmed to decide who dies in a crash" [2]
- Using augmented reality to improve driver acceptance and Human-Machine Interface

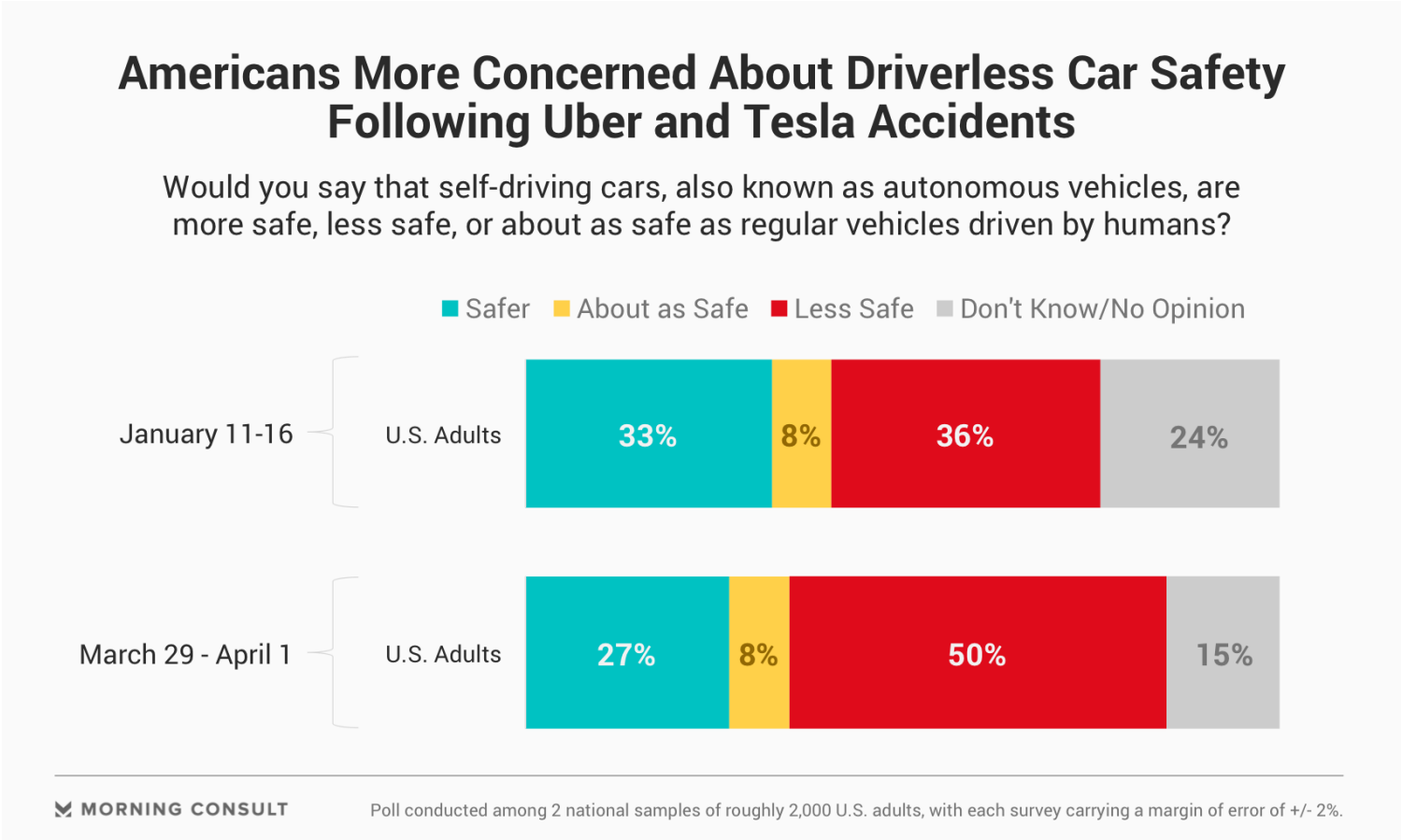
[1] Waymo Safety Report, 2017

[2] Self-driving cars programmed to decide who dies in a crash, 2017, Todd Spangler, Detroit Free Press

Ethics – The Vulnerable Road User

- The trolley problem is unlikely
- The most common ethics decision is brake vs not-brake
- But the VRU, specifically a child dart-out from between cars, is a unique case.

Media and Public Perception



Americans Less Trusting of Self-Driving Safety Following High-Profile Accidents, Edward Graham, MorningConsult, 2018

Media and Public Perception

California public road "disengagements" by autonomous car prototypes
Dec. 2016 - Nov. 2017

Waymo	0.18 disengagements per 1K miles driven [largely suburban], 75 vehicles, 352,545 total miles
Cruise:	0.8 disengagements per 1K miles [urban driving, San Francisco], 90+ vehicles, 125,000 total mi
Nissan:	4.8 disengagements per 1K miles, 5 cars, 5,007 total miles
Zoox:	6.2 disengagements per 1K miles [primarily urban driving in San Francisco], 7 vehicles, 2,244 tc
Drive.ai:	15.4 disengagements per 1K miles, 7 vehicles, 6,015 total miles
Baidu:	24 disengagements per 1K miles, 4 cars, 1,971 total miles
Telenav:	31.3 disengagements per 1K miles [highway primarily], 1 vehicle, 1,824 total miles
Aptiv/Delphi:	45.3 disengagements per 1K miles [mainly highway miles], 1 car, 1810.6 total miles
Nvidia:	215.8 disengagements per 1K miles, 2 cars, 505 total miles
Bosch:	412 disengagements per 1K miles [almost entirely highway miles], 3 cars, 1,454 total miles
Valeo:	375 disengagements per 1K miles [curved sections of highways], 1 vehicle, 574 total miles
Mercedes:	774.6 disengagements per 1K miles, 3 vehicles, 1,087 total miles

5,555 miles per driver intervention

Media and Public Perception

By the end of this module, you will:

- Understand that public perception is dependent on the media, and is influenced by coverage of isolated events
- Be aware of road rage tendencies against autonomous vehicles

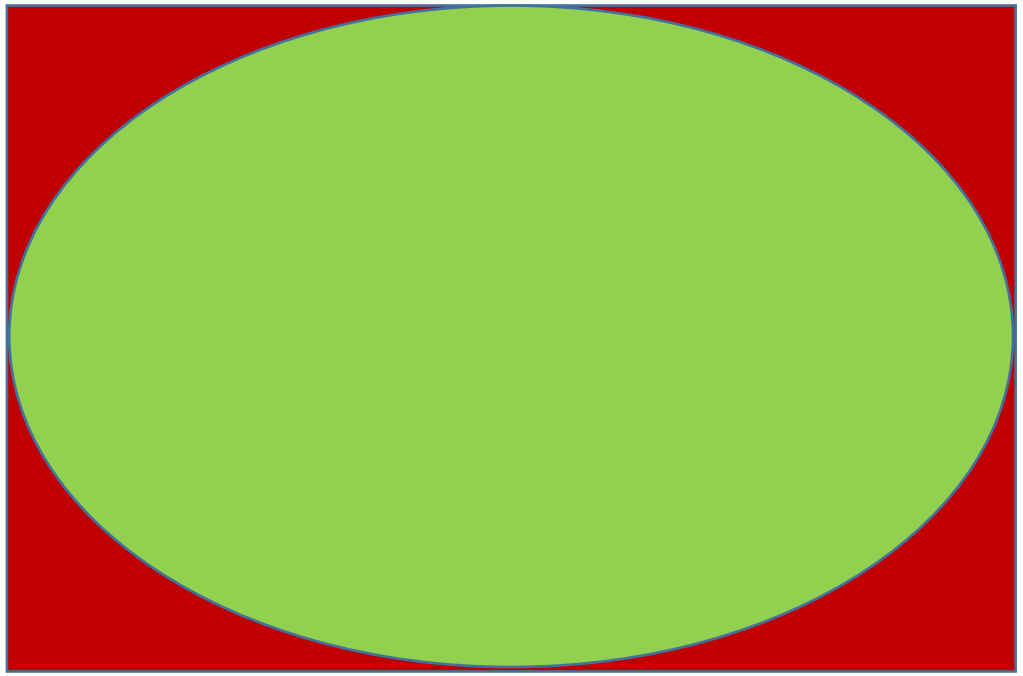
Looking into the future

By the end of this module, you will understand the significance of :

- Edge cases
- Fallback behaviors
- Limiting Operational Design Domains (ODD)

Looking into the future – edge cases

- Unusual and unpredictable scenarios
- Fallback behaviors
- Limiting Operational Design Domains (ODD)



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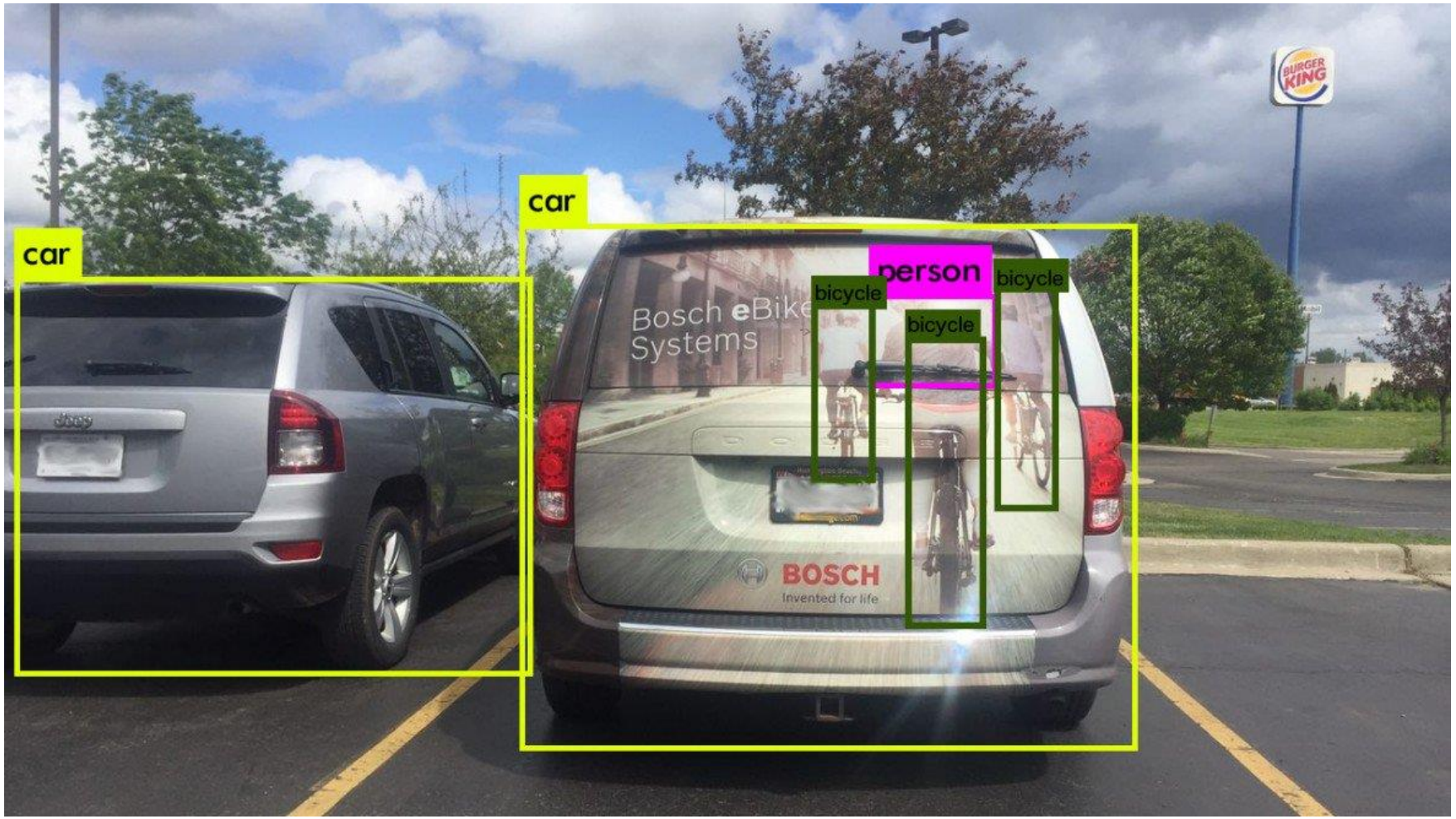
Edge case – wheelchair chasing a chicken

Volvo’s Kangaroo Detection Project

Edge case - Machine vision inadvertent image classification



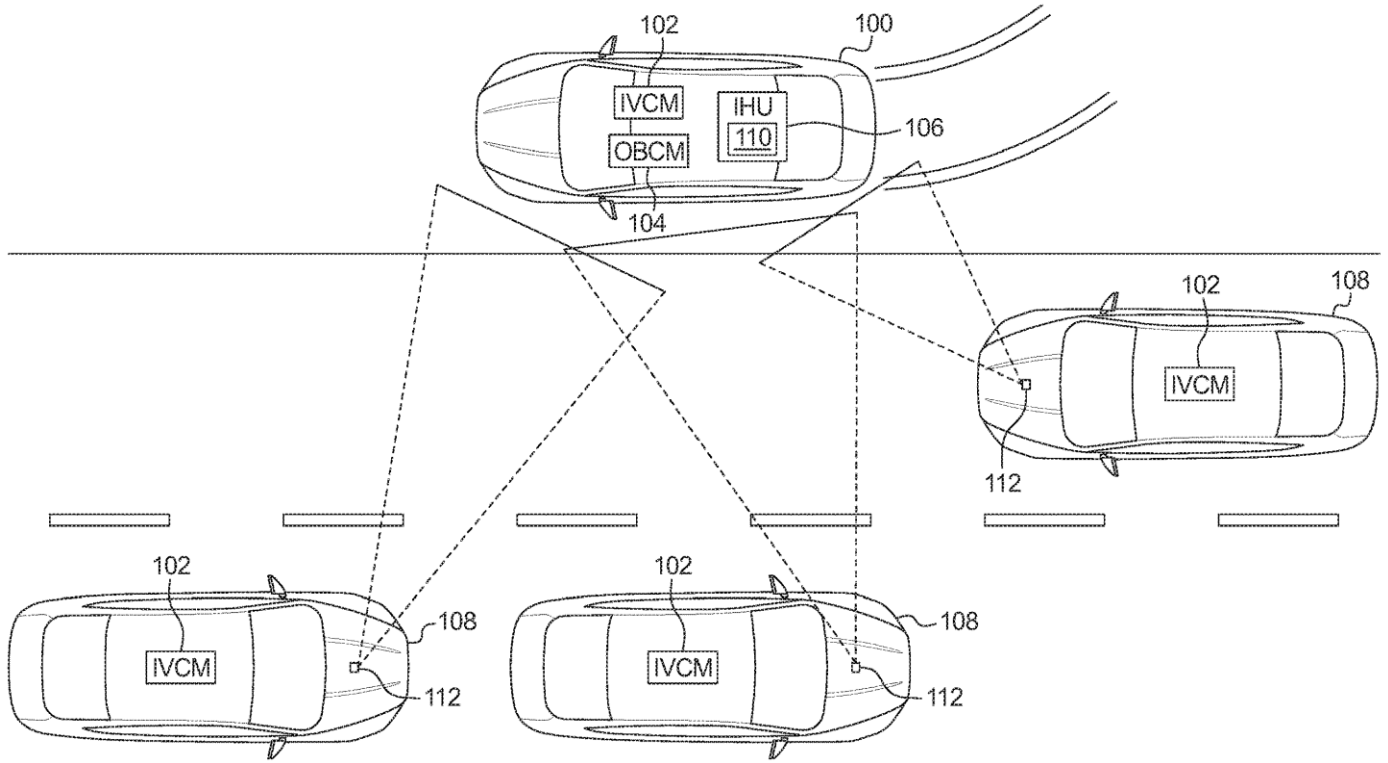
Future Work

- NHTSA CISS EDR file review, 2016+
- Fingerprinting of AEB performance in EDR pre-crash data

Future Work – Unusual Test Methods

Water cannon used to emulate PIT maneuver for post-collision automatic braking

Networked “Selfie” images for damage assessment



A pair of Ford patents suggesting using V2V to request other vehicles to provide exterior photographs of the subject vehicle, which can be assembled into a 3D model for exterior damage assessment. Useful perhaps for unattended parked vehicles, or autonomous vehicles that are not frequently inspected by humans.

Inter-Vehicle Cooperation for Physical Exterior Damage Detection, U.S. Patents US20190188901A1 and US15846148A1

Looking into the future

Summary: The future of ADAS accident reconstruction will involve developments in:

- Edge cases
- Fallback behaviors
- Limiting Operational Design Domains (ODD)

Further Learning

- SAE Reconstruction of Autonomous Vehicle Accidents
- SAE Introduction to Highly Automated Vehicles C1603
- SAE ADAS Application: Automatic Emergency Braking C1704
- Udacity.com – Self-Driving Cars “Nanodegree”
- MIT’s Artificial Intelligence Lectures – Lex Fridman on Youtube
- Innovators’ blogs on Medium.com – Sebastian Thrun, comma.ai, Andrew Ng
- Good List of publications: http://www.cvlibs.net/projects/autonomous_vision_survey/

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Wrap up and Questions

Acronym	Description of Acronym	Definition
5G	Fifth-generation mobile network	Upcoming telecommunications standard expected to deliver 10 Gbps and ultra-low latency.
ACC	Adaptive Cruise Control	Cruise control system that automatically adapts speed to maintain safe distance from vehicles in front
ADAS	Advanced driver assistance systems	Vehicle systems designed to improve driving safety
AEB	Automatic Emergency Braking	System that detects an impending forward crash with another vehicle in time to avoid or mitigate the crash.
AI	Artificial Intelligence	Intelligence and decision-making that come from a machine
AV	Autonomous Vehicle	Vehicle that is capable of driving itself, typically classified at levels 3, 4, or 5 for driving automation
BA	Brake Assist	Helps in detection of attempted panic braking based on the force that is applied to the brake pedal and how fast the driver is stepping on the pedal
BSM	Blind Spot Monitoring	ADAS applications that monitors the driver's blind spots at the rear quarters of the car and provides visual, audible, and/or tactile alerts when a vehicle is present in a situation requiring such alerts
BSM	Basic Safety Messages	DSRC data broadcasted from vehicles through V2V and V2I at a frequency of 10Hz. Core contents are data elements that describe a vehicle's position (latitude, longitude, elevation) and motion (heading, speed, and acceleration)
CAMP	Collision Avoidance Metrics Partnership	Partnership formed by vehicle OEMs to accelerate the implementation of crash avoidance countermeasures in passenger vehicles to improve traffic safety

CIB	Crash Imminent Braking	Systems automatically apply the brakes in a crash imminent situation if the driver does not respond to warnings
DBS	Dynamic Brake Support	Supplements a driver's commanded brake input by increasing the output of the foundation brake system, for the purpose of mitigating the severity of, or , in some cases avoiding , rear-end collisions
DDT	Dynamic Driving Task	Various activities required to operate a moving vehicle in an environment that changes due to alignment, weather, roadside conditions, etc. Tasks include steering, braking, and accelerating.
Deep Learning	Deep Learning	Subset of machine learning that involves many layers of processing, massive amount of data, and enormous computing capacity. These algorithms can facilitate computer vision, natural language processing, driving strategy, personalization, and decision-making.
DSRC	Dedicated Short Range Communication	Communications protocol developed to address the safety critical issues associated with sending and receiving data among vehicles and between moving vehicles and fixed roadside access points.
FCW	Forward Collision Warning	System comprising of sensors, computational capability, software and alert transducers which function to warn the driver of potential forward collision
Fuse	Sensor Fuse	Second of the 3 stages of in-vehicle compute required for autonomous driving (sense, fuse, decide). The vehicle correlates and fuses sensor data to create a model of its environment.

Haptic Alert	Haptic Alert	Warning modality in which a transducer produces a tactile sensation to the driver. Implemented as vibration in either the driver's seat pan or the steering wheel.
HAV	Highly Automated Vehicles	SAE Levels 3-5 vehicles with automated systems that are responsible for monitoring the driving environment, in order to draw a distinction between Levels 0-2 and 3-5 based on whether the human operator or the automated system is primarily responsible for monitoring the driving environment
Headway	Headway	Longitudinal distance from the front most location of the SV to rear most point on the POV or to the fixed point on the ground longitudinally coincident with the rear most point on a stationary POV.
HMI	Human-machine Interface	Interface responsible for two-way communication between a vehicle and its occupants. Touchscreen displays, voice recognition, or integration with mobile devices.
IoT	Internet of Things	Network of physical objects - devices, vehicles, buildings, machines, and other items - embedded with electronics, software, sensors, and network connectivity that enables these objects to collect and exchange data.
LCA	Lane Centering Assist	Combines a forward-facing camera to detect lane markings with an electric steering system, keeping the vehicle in the center of the lane.
LDW	Lane Departure Warning	Uses a forward-facing camera to detect lane markings, warning the driver in case the vehicle leaves the lane without proper use of the turn signal.
LIDAR	Light Detection and Ranging	Laser based session of Radar and used in autonomous vehicles. Sits on top of the vehicle roof to enable unhindered 360-degree view of the area surrounding the vehicle. 64 lasers spin at about 900 rpm and create a detailed 360-degree 3D map of the surrounding environment in order to view all obstacles in real time.
LKA	Lane Keeping Assist	Combines a forward-facing camera to detect lane markings with an electric steering system, keeping the vehicle in the center of the lane.
LV	Lead Vehicle	Principal Other Vehicle (POV) when the potential collision is a Rear-end collision with the POV.
Machine learning	Machine Learning	Subset of AI that gives machines the ability to learn on their own, resulting in algorithms that make data-driven decisions.

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MV	Machine Vision	Machine-assisted moderation of images and the ability to extract rich information from images to categorize and process visual data.
ODD	Operational Design Domain	Domain describing the specific conditions under which a given automated driving system or feature is intended to function, in the automated driving environment. Exp: Freeways, downtown areas, roadways with speed limits <35 mph.
OEDR	Object and Event Detection and Response	Detection by the driver or automated driving systems of any circumstance that is relevant to the immediate driving task, as well as the implementation of the appropriate driver or system response to such circumstance.
OEM	Original Equipment Manufacturer	
OTA	Over-the-air Updates	Software or firmware updates to a vehicle that are downloaded from the cloud.
PAEB	Pedestrian Automatic Emergency Braking	System that performs automatic braking in cases a pedestrian is detected in front of the vehicle.
Park Assist	Park Assist	System of ultrasonic sensors on the front and/or rear bumpers that provide the driver with visual, audible, and/or tactile alerts as their vehicle approaches a stationary object.
Platooning	Platooning	Method of grouping vehicles together, primarily to increase the road capacity, which will lead up to fully automated vehicles on a wide scale. Will decrease the distances between cars or trucks and would allow vehicles to accelerate or brake simultaneously.
Rain-Sensing Wipers	Rain-Sensing Wipers	Windshield wipers whose rate of operation is electronically controlled based on the amount of moisture on the windshield.
RCTA	Rear Cross-Traffic Alert	Systems let you know if you're about to back into oncoming cross traffic. Multiple sensors or wide angles cameras are located near the rear of the vehicle, detecting traffic that comes from the side, typical parking lot situations.
RTK	Real Time Kinematics	Technique is based on the use of carrier measurements and the transmission of corrections from the base station, whose location is well known, to the rover, so that the main errors that drive the stand-along positioning cancel out.
SPaT	Signal Phase and Timing	Signal state of the intersection and how long this state will persist for each approach and lane that is active.

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SPaT	Signal Phase and Timing	Signal state of the intersection and how long this state will persist for each approach and lane that is active.
TTC	Time to Collision	Time interval, usually measured in seconds, required for one vehicle to strike another object.
V2X	Vehicle-to-everything	Communication between a vehicle and the cloud, other cars (vehicle-to-vehicle or V2V) and infrastructure (vehicle-to-infrastructure, V2I)
PIB	Post-Impact Automatic Braking	Decelerates the vehicle after a collision occurs.